



(12) **United States Patent**
Bu et al.

(10) **Patent No.: US 12,410,240 B2**
(45) **Date of Patent: *Sep. 9, 2025**

(54) **ANTIBODIES AND METHODS FOR THE
DIAGNOSIS AND TREATMENT OF EPSTEIN
BARR VIRUS INFECTION**

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

(21) Appl. No.: **18/440,344**

(22) Filed: **Feb. 13, 2024**

(65) **Prior Publication Data**
US 2024/0174735 A1 May 30, 2024

Related U.S. Application Data

(60) Continuation of application No. 17/553,139, filed on
Dec. 16, 2021, now Pat. No. 11,926,656, which is a
division of application No. 16/608,386, filed as
application No. PCT/US2018/029463 on Apr. 25,
2018, now Pat. No. 11,236,151.

(60) Provisional application No. 62/490,023, filed on Apr.
25, 2017.

(51) **Int. Cl.**
C07K 16/08 (2006.01)
A61K 38/00 (2006.01)
C07K 14/05 (2006.01)

(52) **U.S. Cl.**
CPC **C07K 16/085** (2013.01); **C07K 14/05**
(2013.01); **A61K 38/00** (2013.01); **C07K**
2317/10 (2013.01); **C07K 2317/21** (2013.01);
C07K 2317/565 (2013.01)

(58) **Field of Classification Search**
CPC .. **C07K 16/085; C07K 14/05; C07K 2317/10;**
C07K 2317/21; C07K 2317/565; A61K
38/00

See application file for complete search history.

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(57) **ABSTRACT**

Antibodies and compositions of matter useful for the detec-
tion, diagnosis and treatment of Epstein Barr Virus (EBV)
infection in mammals, and methods of using those compo-
sitions of matter for the detection, diagnosis and treatment of
EBV infection are described. Also described are proteins,
referred to as anti-gp350 antibody probes, and anti-gp350
B-cell probes, that maintain the epitope structure of the
CR2-binding region of gp350, but do not bind CR2.

16 Claims, 23 Drawing Sheets

Specification includes a Sequence Listing.

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FIG. 1

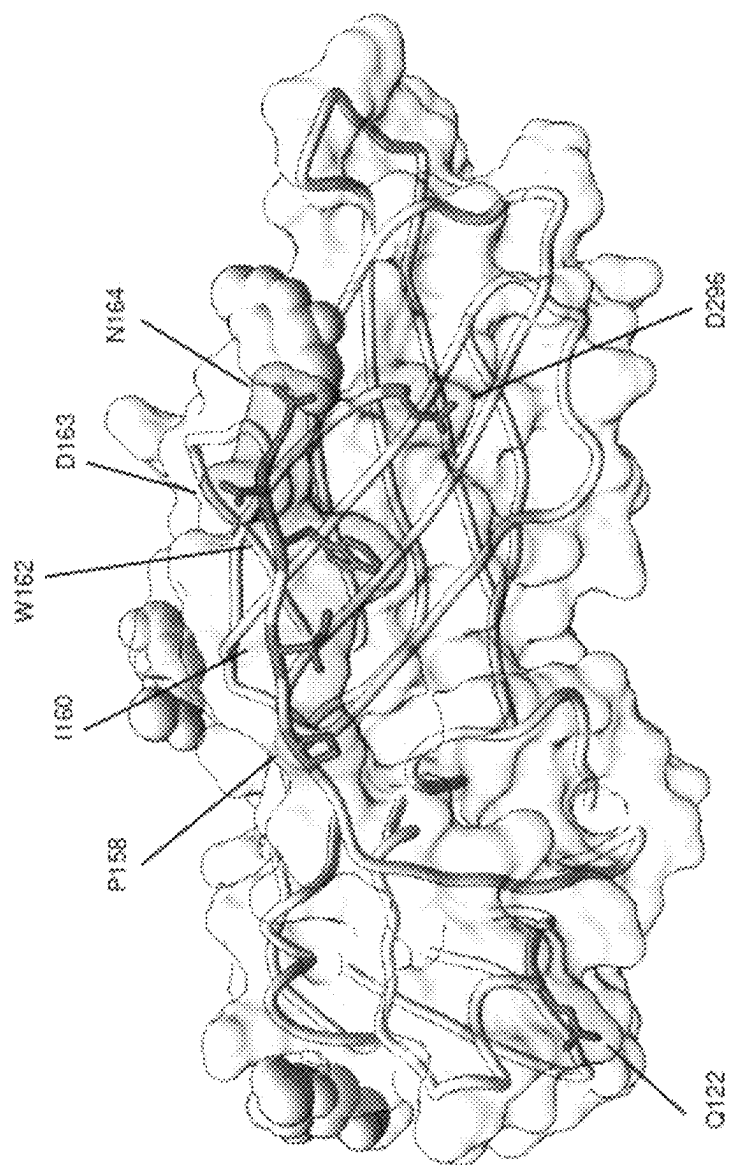


FIG. 2

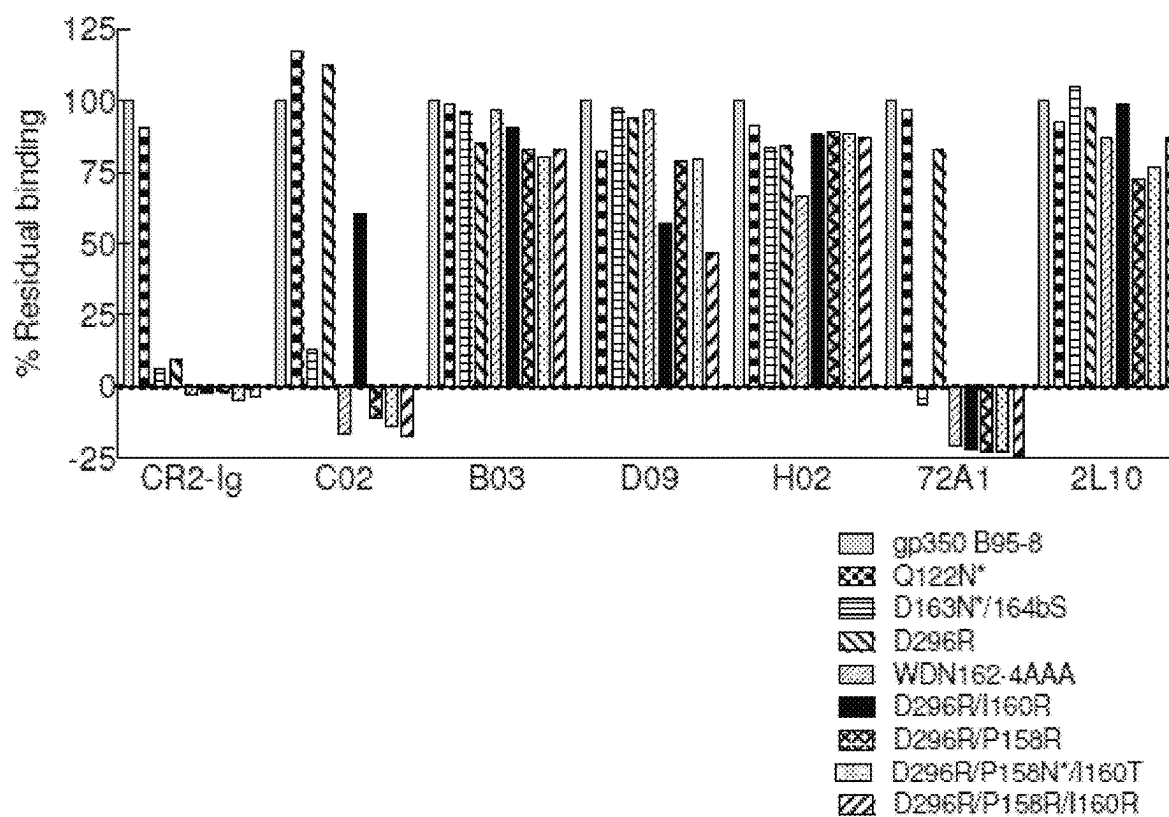


FIG. 3A

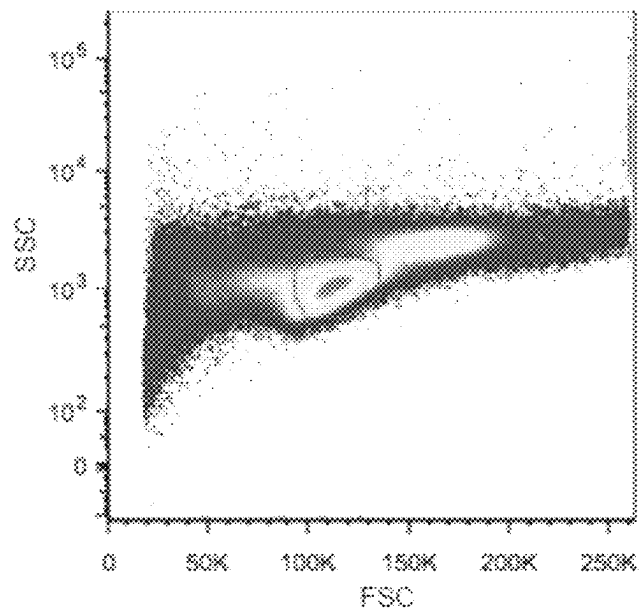


FIG. 3B

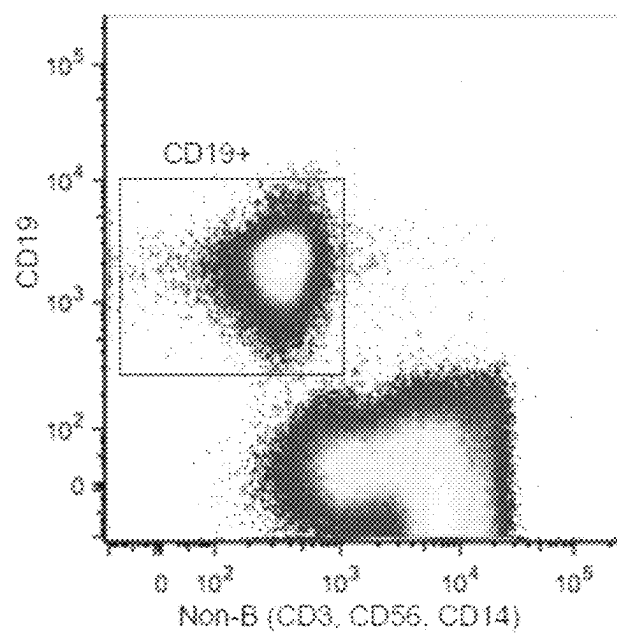


FIG. 3C

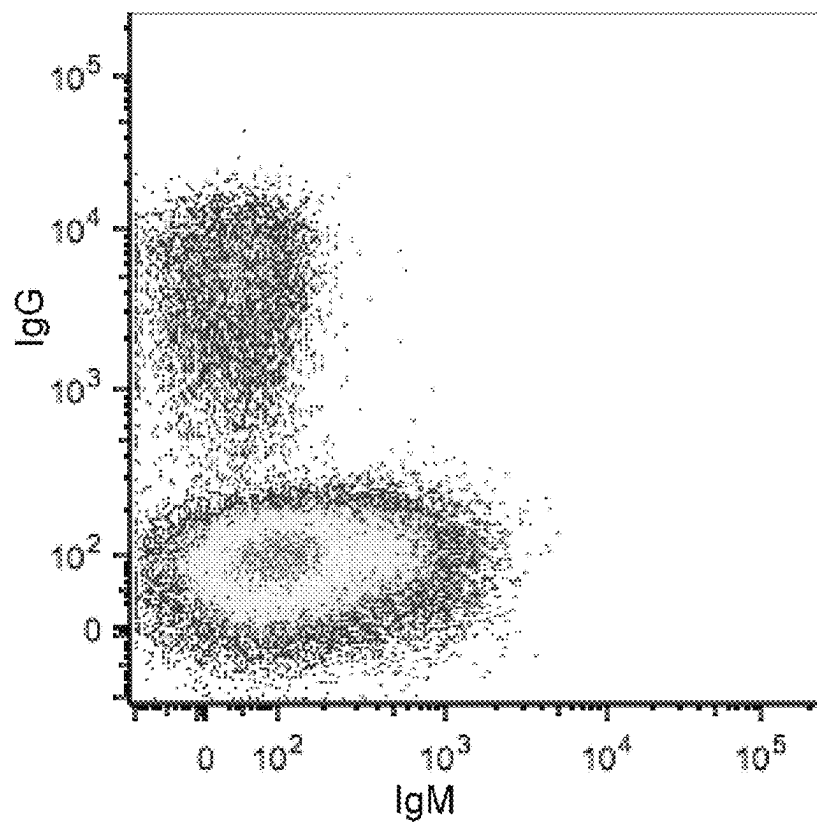


FIG. 3D

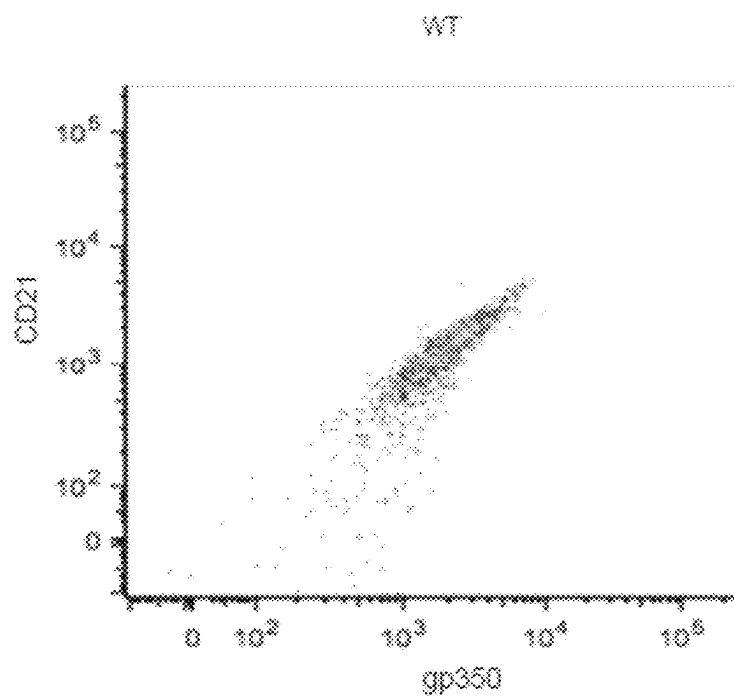


FIG. 3E

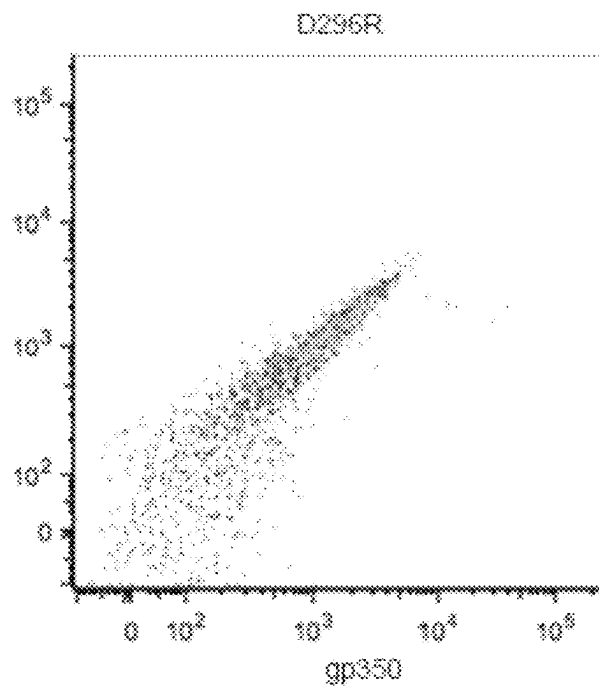


FIG. 3F

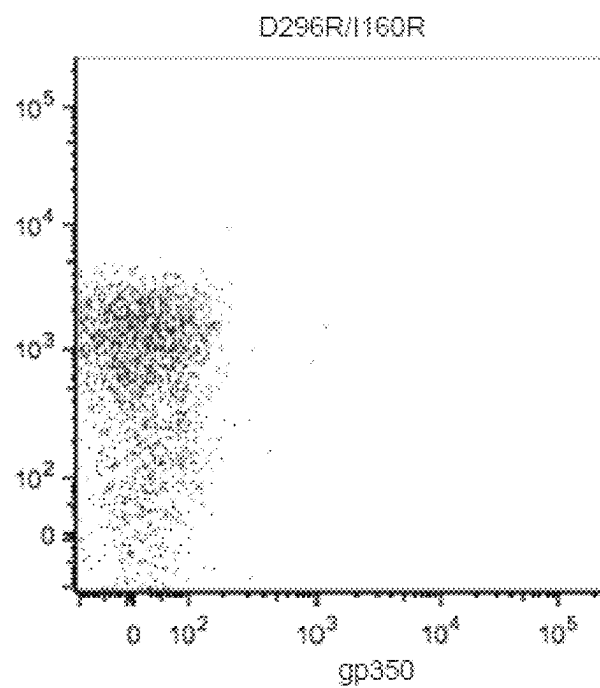


FIG. 4A

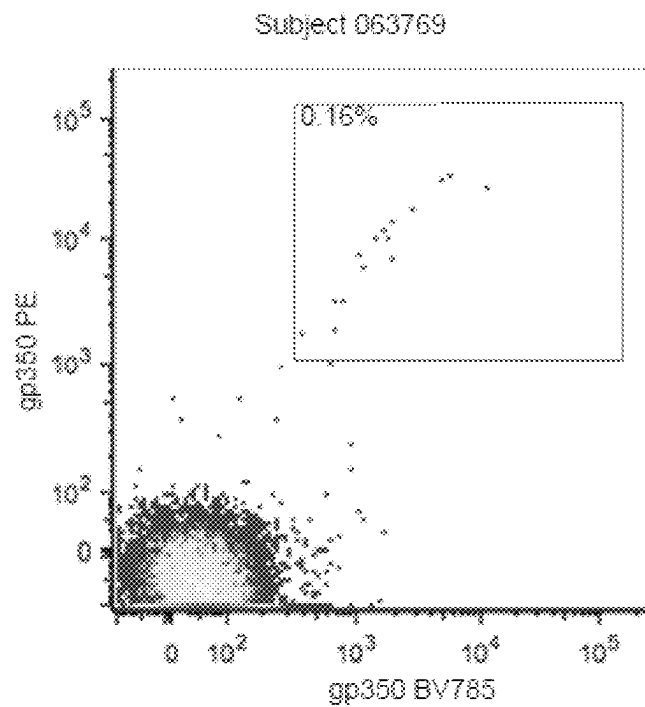


FIG. 4B

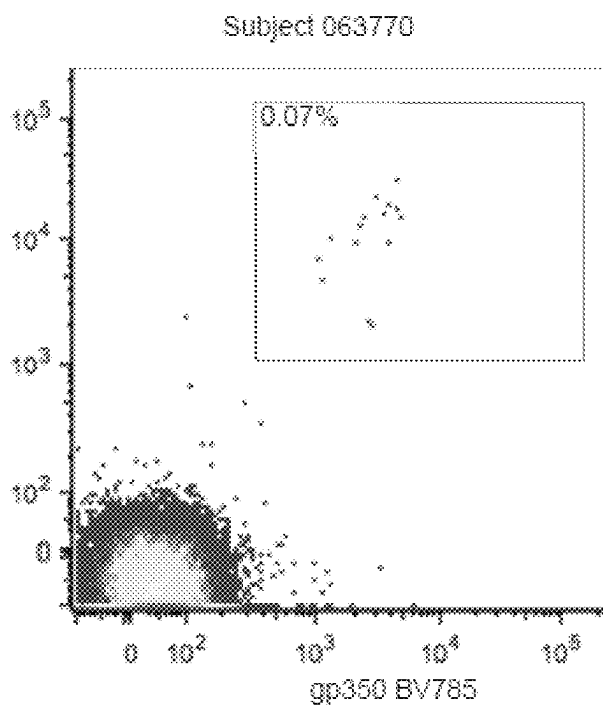


FIG. 5A

Isolation of gp350-specific B cells from Immunized Macaques

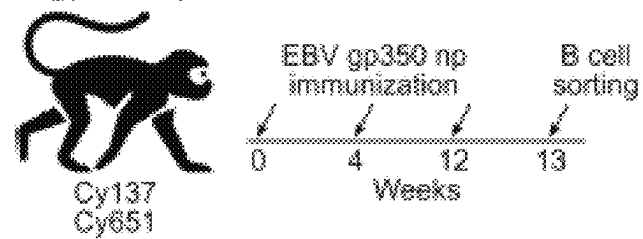


FIG. 5B

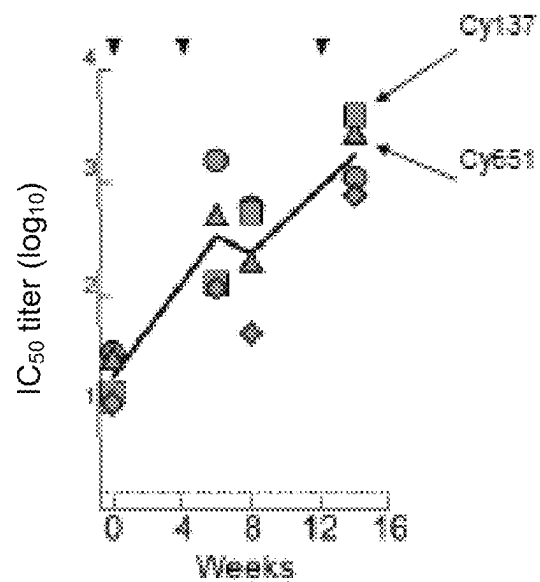


FIG. 5C

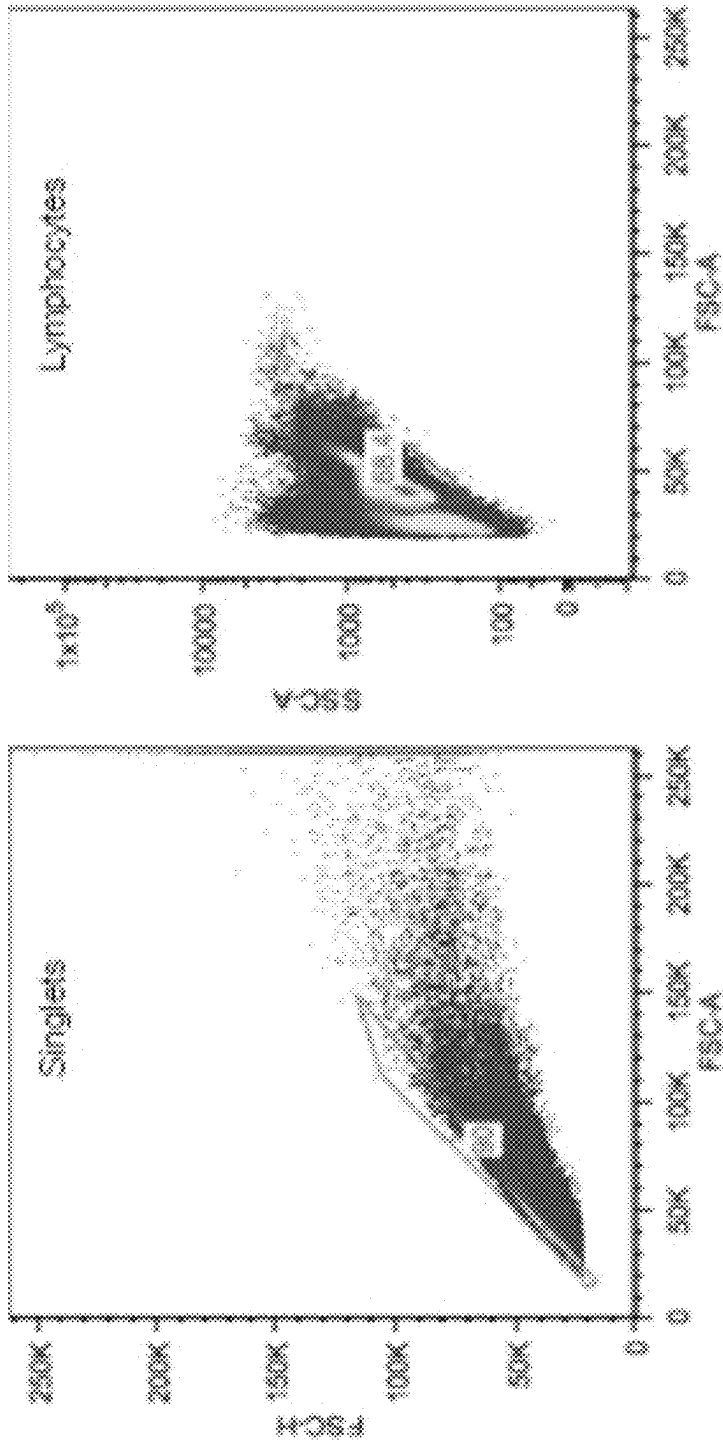


FIG. 5D

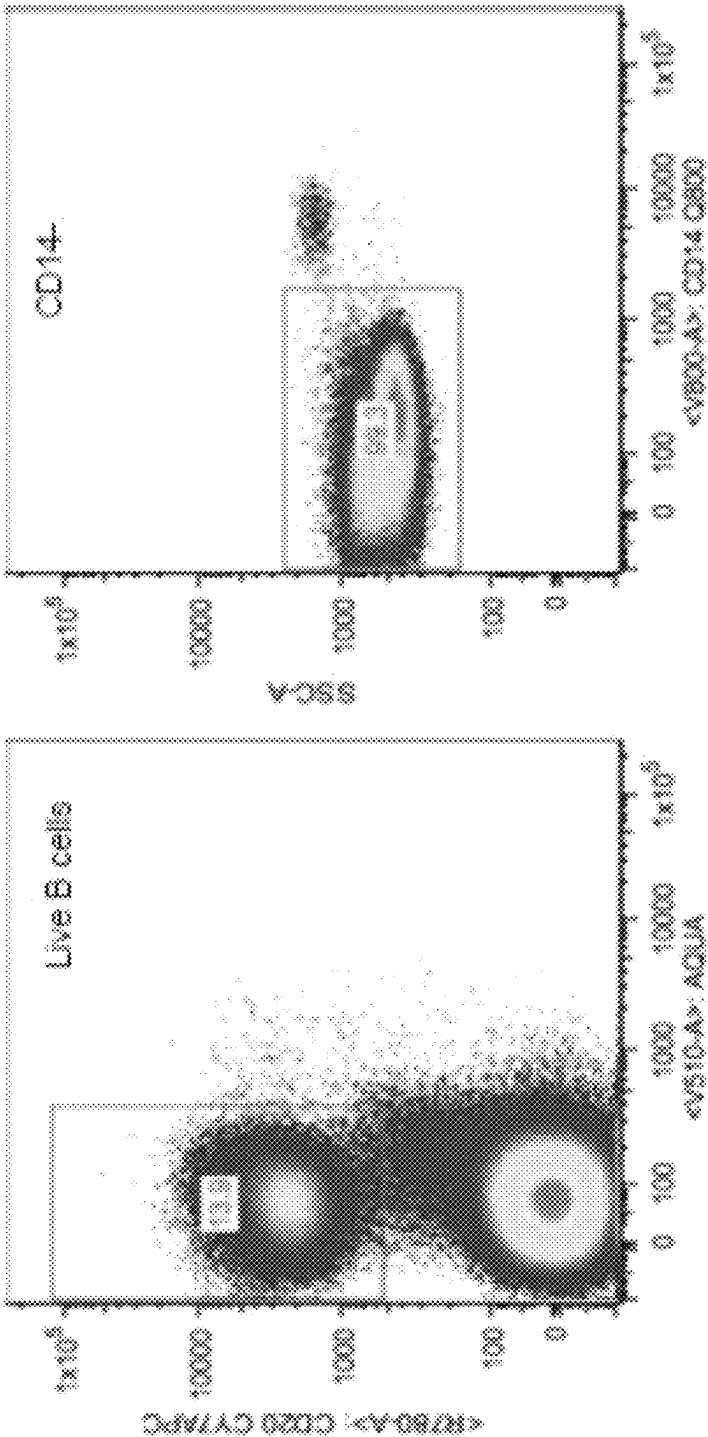


FIG. 5E

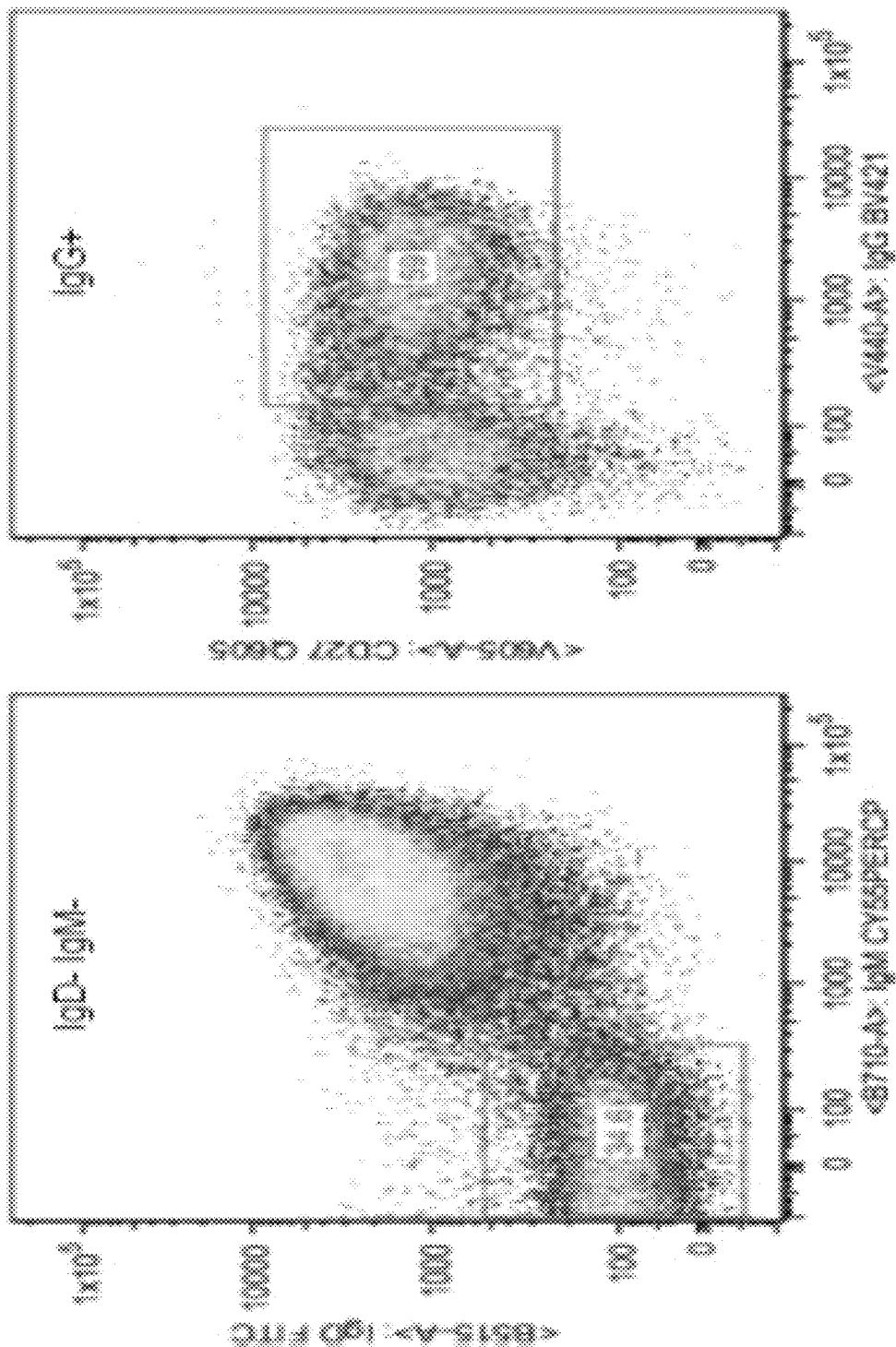


FIG. 5F

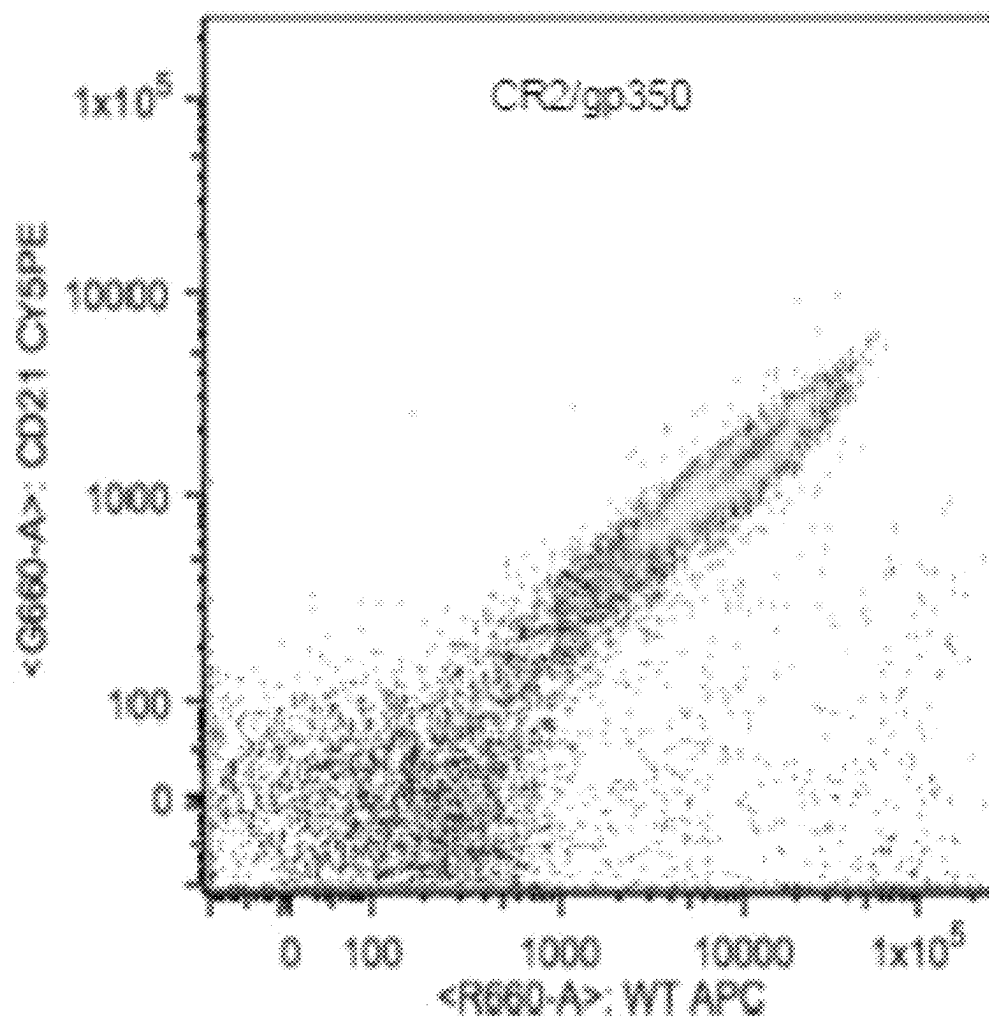


FIG. 6A

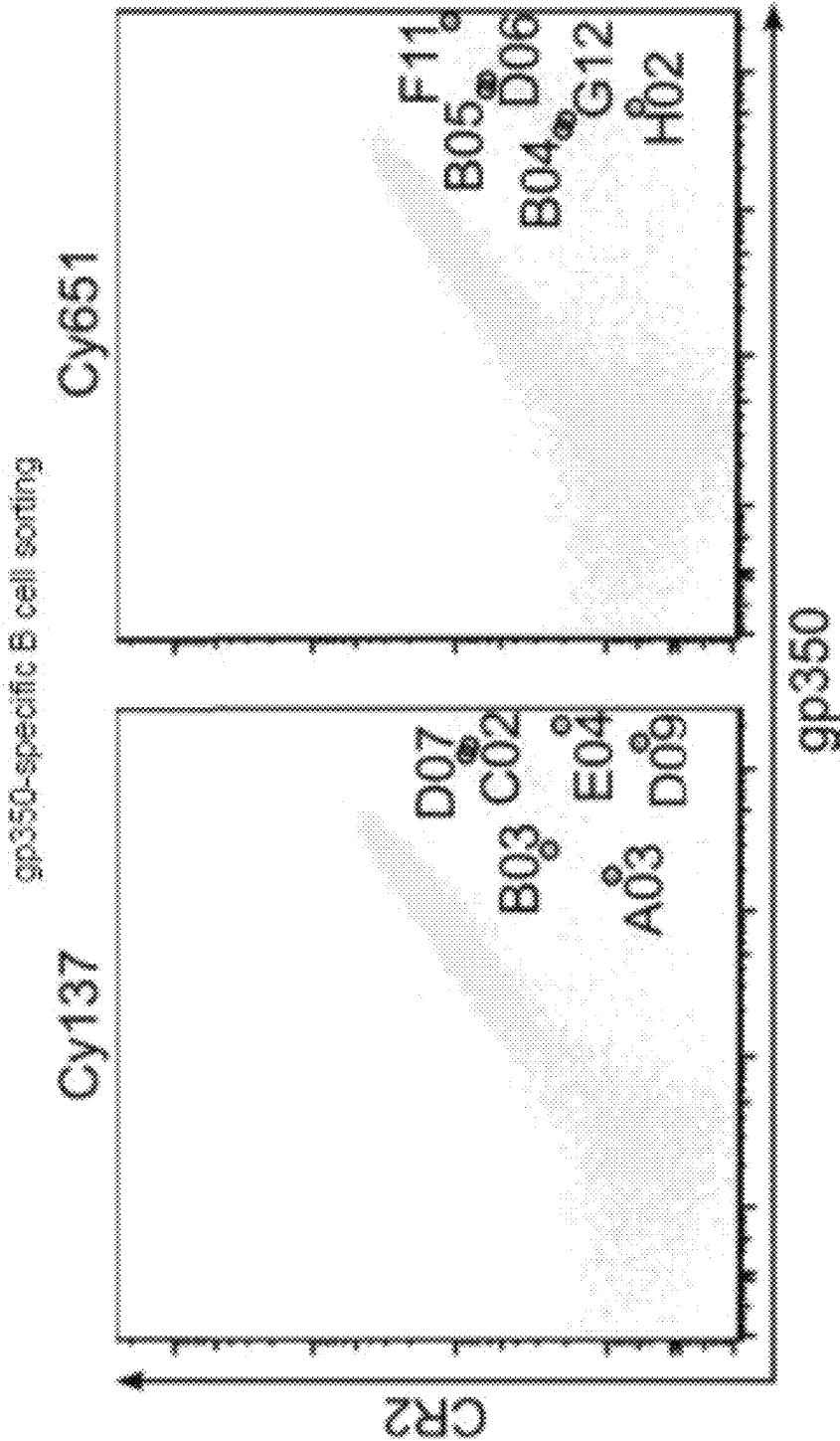


FIG. 6B

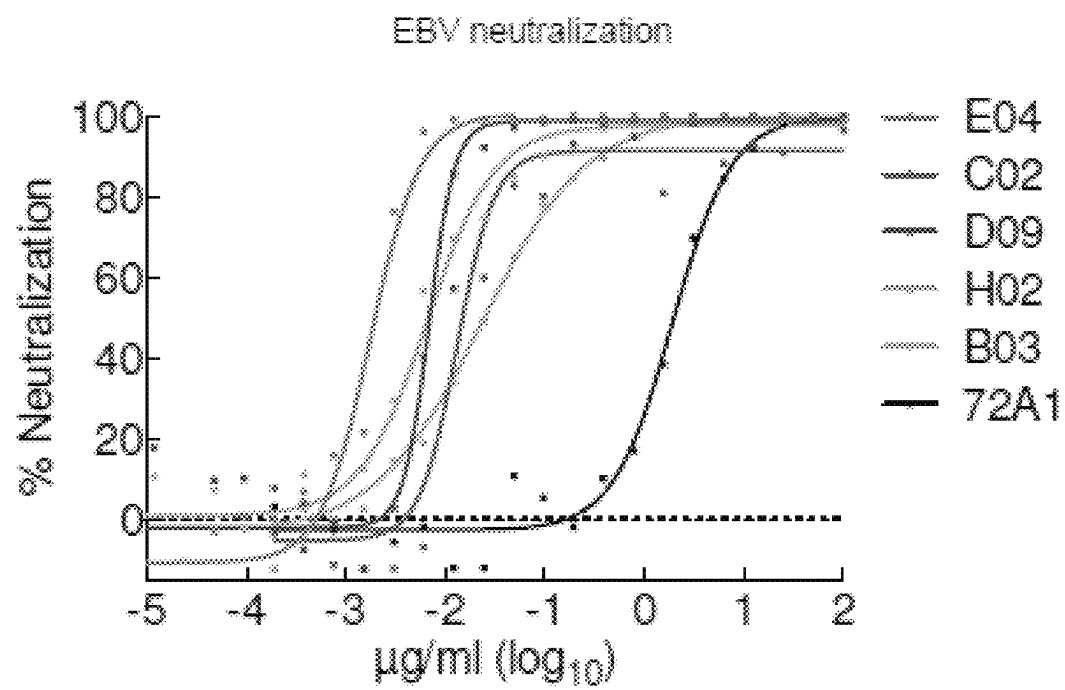


FIG. 7

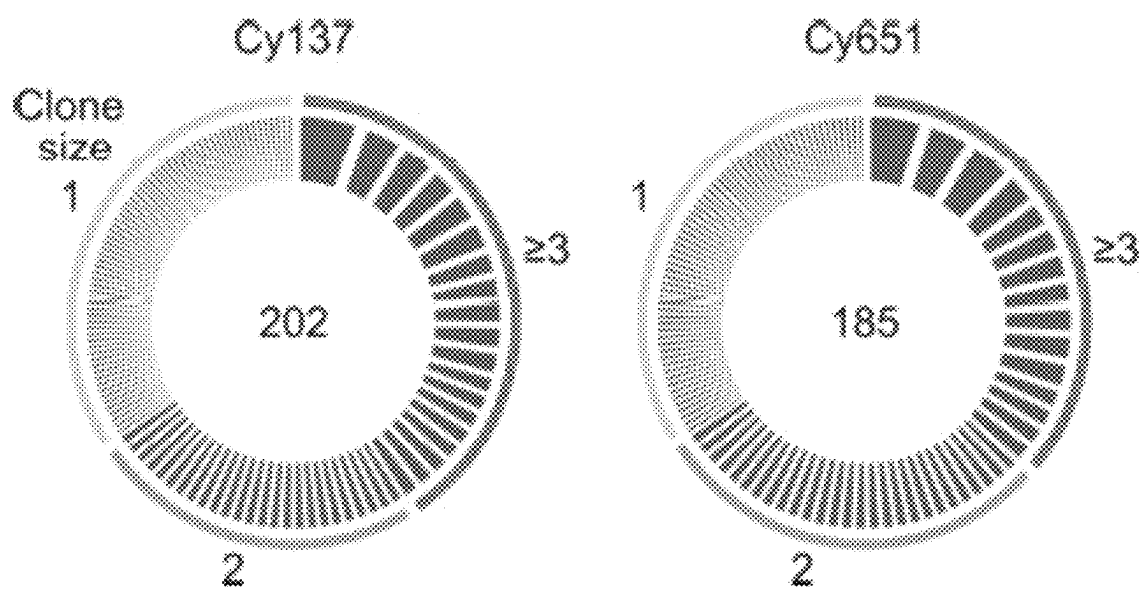


FIG. 8A

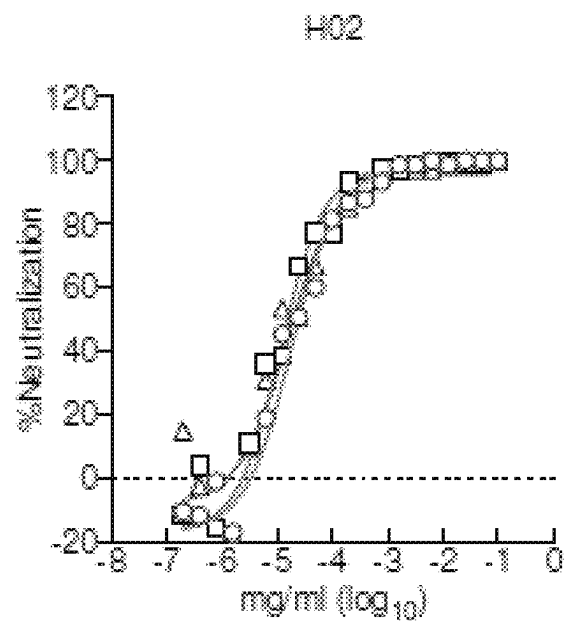


FIG. 8B

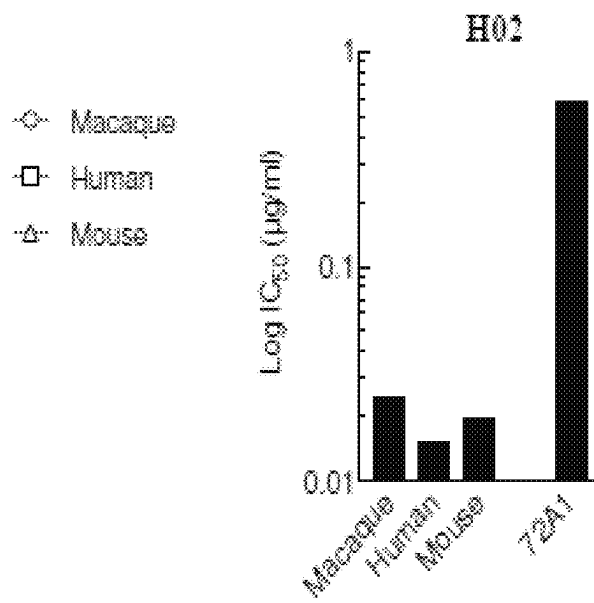


FIG. 9A

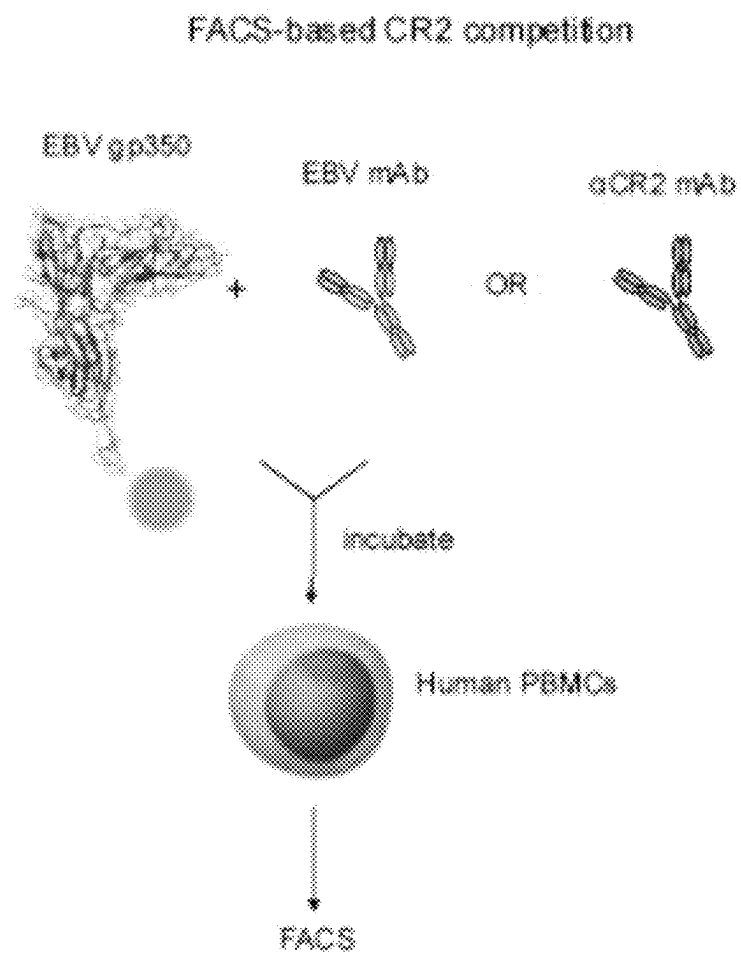


FIG. 9B

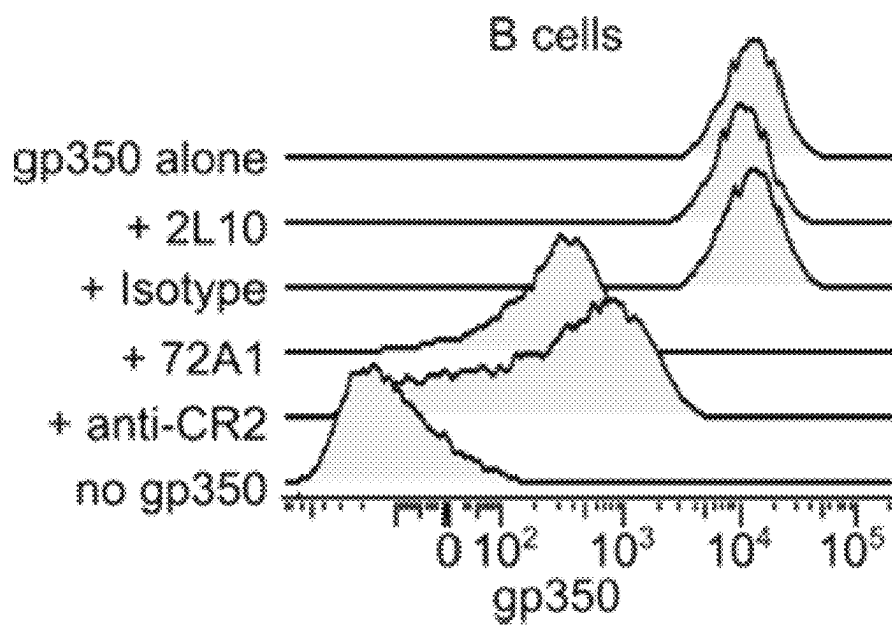
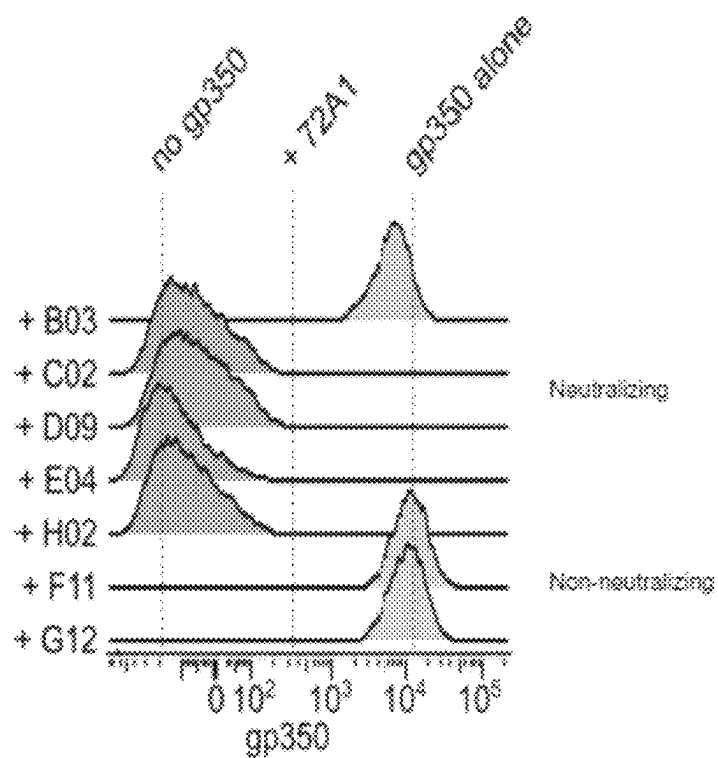


FIG. 9C



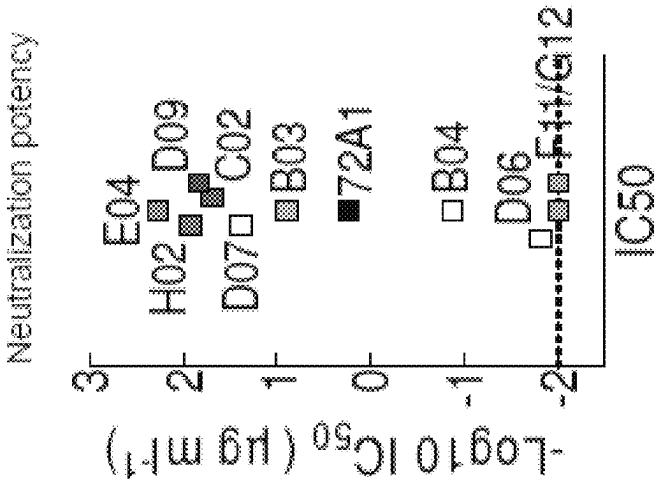


FIG. 10A

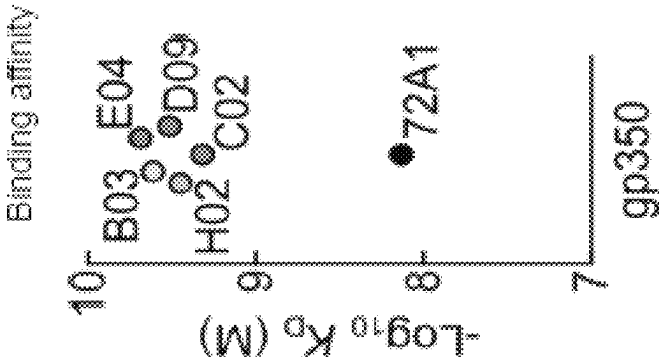


FIG. 10B

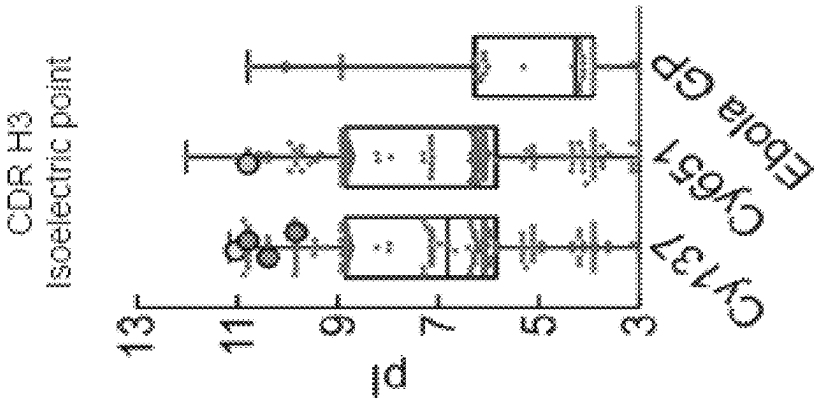


FIG. 10C

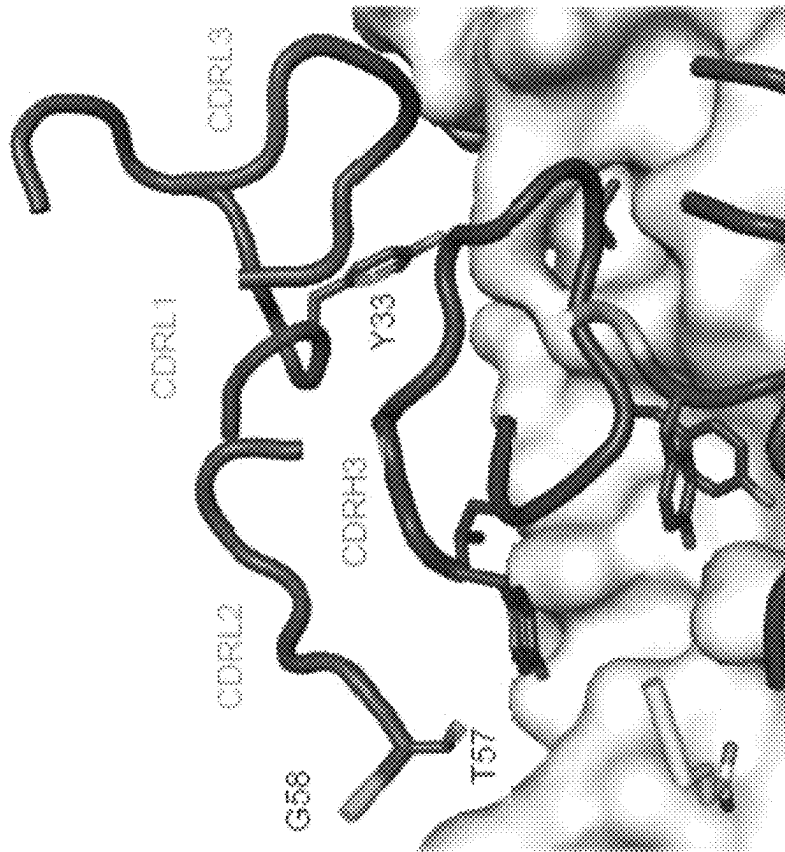
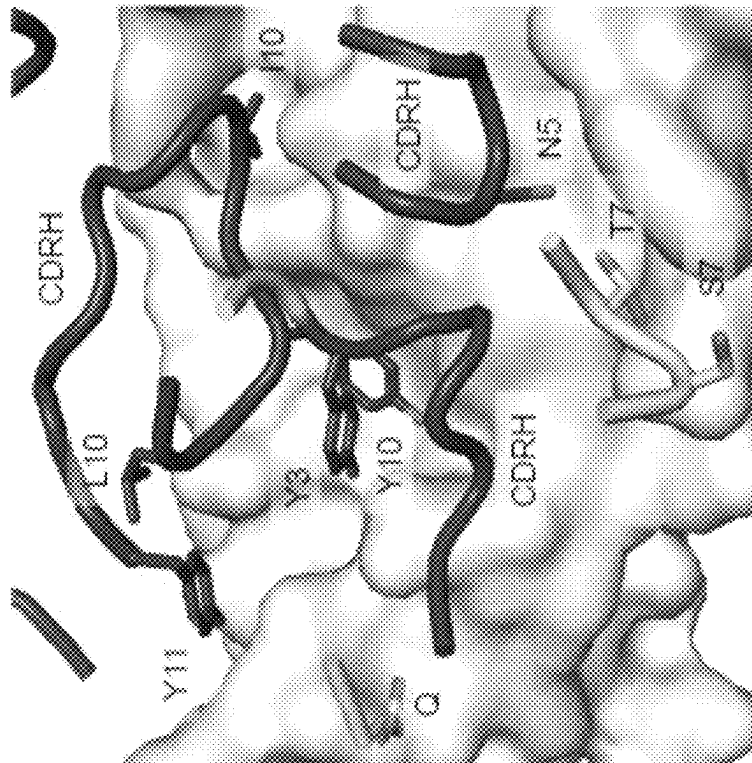


FIG. 12



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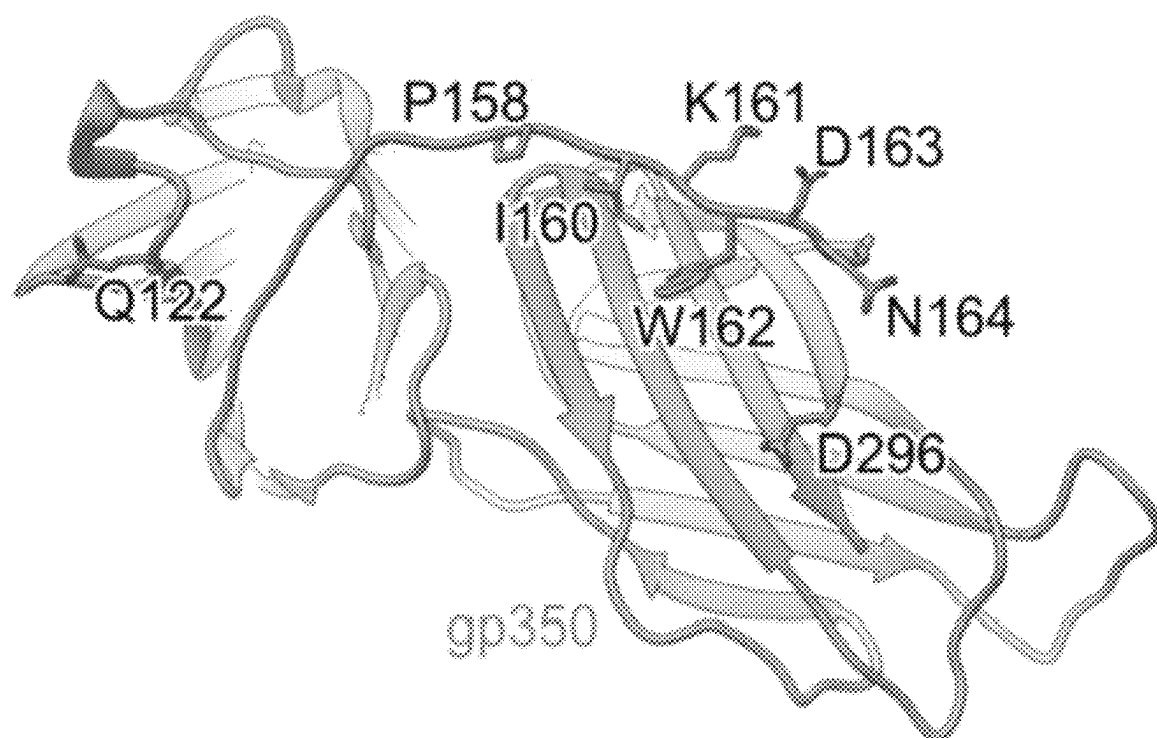


FIG. 13

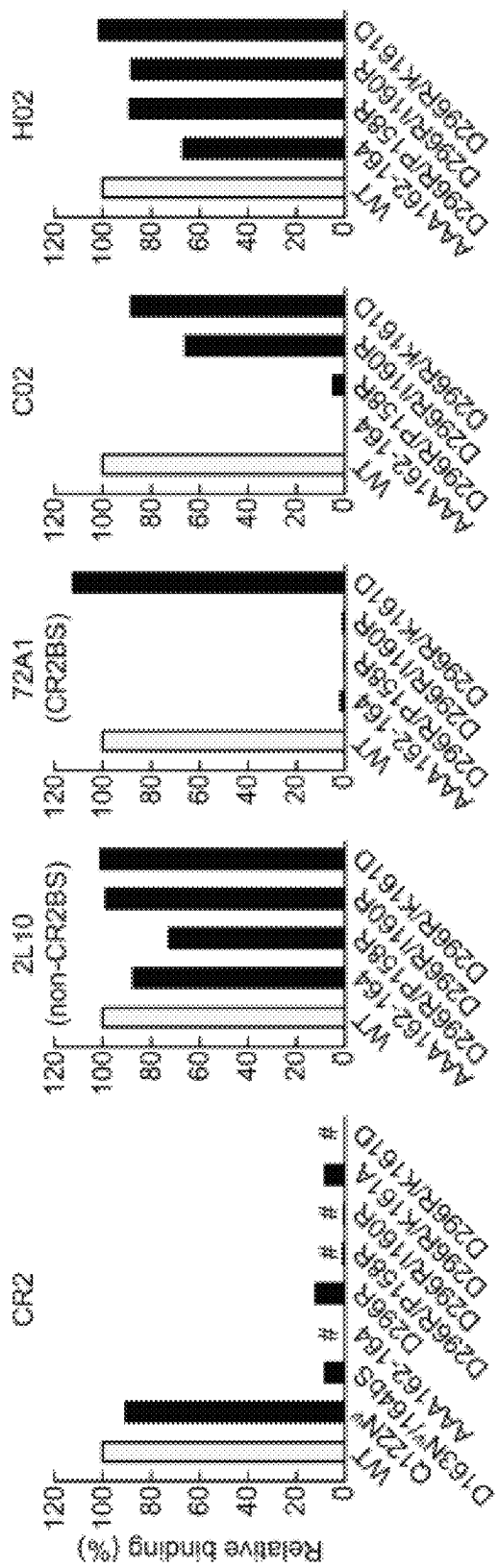


FIG. 14

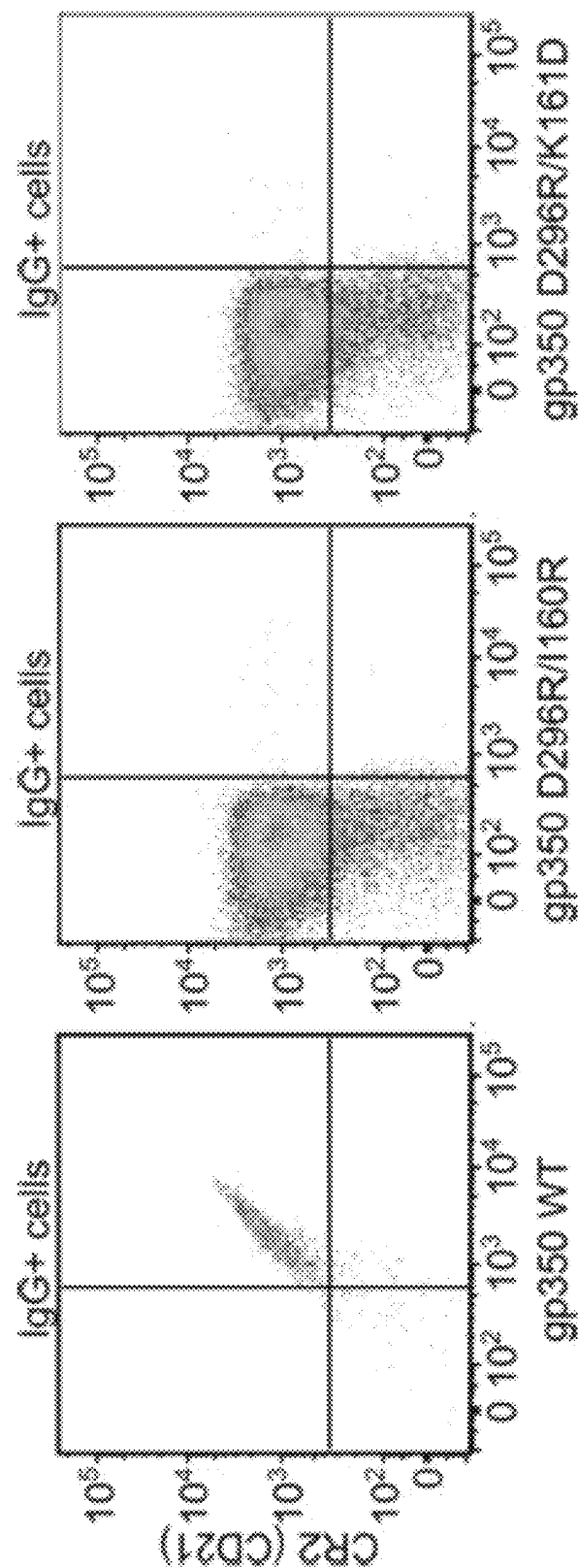
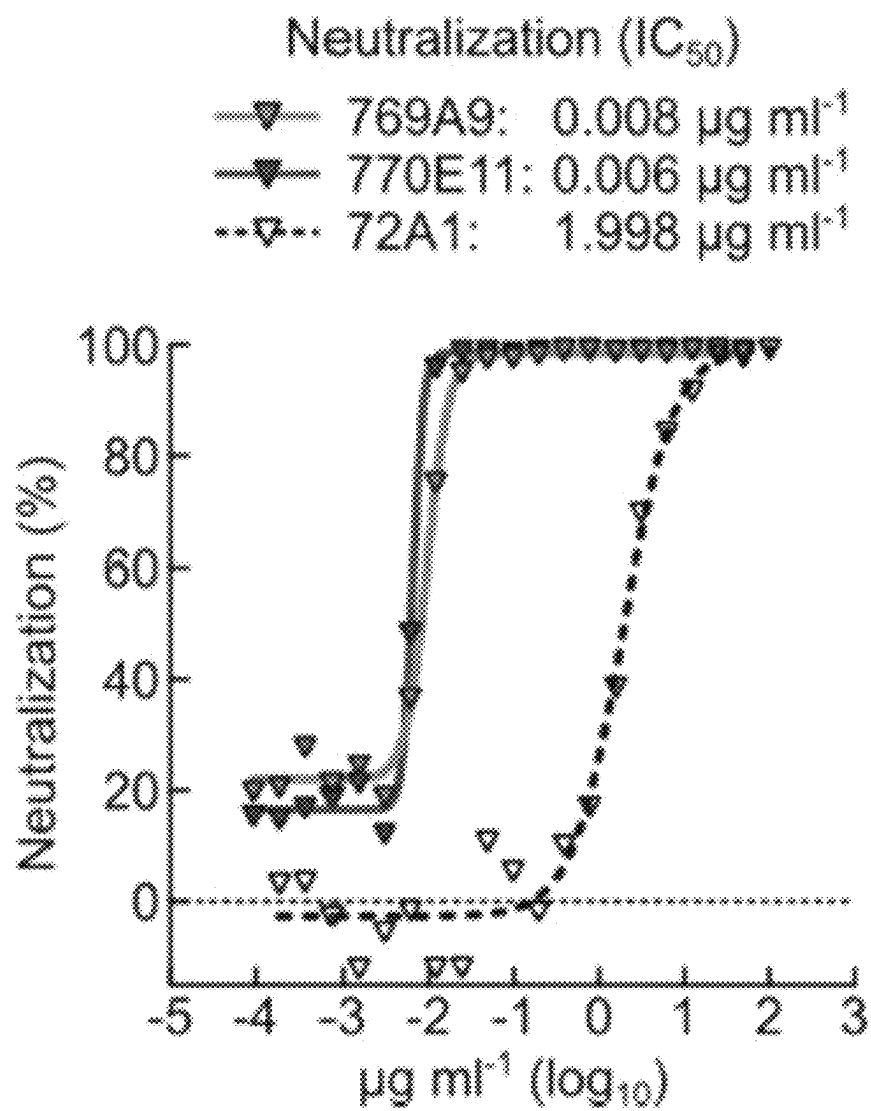


FIG. 15

FIG. 16



1

ANTIBODIES AND METHODS FOR THE DIAGNOSIS AND TREATMENT OF EPSTEIN BARR VIRUS INFECTION

CROSS-REFERENCE TO RELATED APPLICATIONS

This is a continuation application of U.S. application Ser. No. 17/553,139, filed Dec. 16, 2021, issued as U.S. Pat. No. 11,926,656 on Mar. 12, 2024, which is a divisional application of U.S. application Ser. No. 16/608,386, filed Oct. 25, 2019, issued as U.S. Pat. No. 11,236,151 on Feb. 1, 2022, which is a national stage application under 35 U.S.C. § 371 and claims the benefit of PCT Application No. PCT/US2018/029463 having an international filing date of Apr. 25, 2018, which designated the United States, which PCT application claims the benefit of priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 62/490,023, filed Apr. 25, 2017. The entire disclosures of the above-listed applications are incorporated herein by reference.

FIELD OF THE INVENTION

The present invention is directed to compositions of matter useful for the diagnosis and treatment of Epstein Barr Virus infections in mammals and to methods of using those compositions of matter for the same.

INCORPORATION OF ELECTRONIC SEQUENCE LISTING

The electronic sequence listing, submitted herewith as an XML file named Sequence.xml (343,206 bytes), created on Apr. 17, 2024, is herein incorporated by reference in its entirety.

BACKGROUND

Epstein-Barr-Virus (EBV) is a human herpes virus that infects over 90% of the population world-wide with a life-long persistence in its host. In most cases, primary infection occurs during early childhood and is usually asymptomatic. In contrast, if infection is retarded and happens during adolescence or adulthood, it is regularly symptomatic, causing a benign, normally self-limiting, lymphoproliferative syndrome termed infectious mononucleosis (IM) in up to 50% of cases. Although the disease is normally self-limiting, prolonged forms of chronic active EBV infection (CAEBV) with fatal outcome have been reported. EBV infection also significantly increases the risk of developing Hodgkin disease and other types of lymphoma later in life. EBV infection is also an independent risk factor for multiple sclerosis later in life. In addition, EBV is causally associated with a heterogeneous group of malignant diseases like nasopharyngeal carcinoma, gastric carcinoma, and various types of lymphoma, and the WHO classifies EBV as a class I carcinogen.

Besides the above described medical conditions caused by EBV, patients with primary or secondary immune defects, like transplant recipients, are at elevated risk for EBV-associated diseases because of the detrimental effect of immunosuppressive agents on the immune-control of EBV-infected B-cells. EBV-associated post-transplant lymphoproliferative disorder (PTLD) is an important form of post-transplant complications, occurring in up to 20% of organ recipients. Importantly, immunocompromised transplant

2

recipients who are immunologically naive for EBV at the onset of immunosuppression are at a particular high risk of developing life-threatening EBV positive PTLD due to a primary EBV infection, e.g. often caused after transplantation via transmission of the virus through a donor organ due to the high prevalence of EBV. Due to impaired T-cell immunity that results from exposure to immunosuppressive drugs, these patients are unable to effectively prime EBV-specific T-cells that play a critical role in controlling proliferation of EBV-infected B-cells. In contrast, patients who are EBV-seropositive at transplant have a much lower risk for developing PTLD, demonstrating the essential role of EBV-specific T-cells in eliminating virally infected cells. In general, patients who are EBV-seronegative before transplantation are at a much higher risk to develop EBV-associated diseases, since transmission of donor EBV in transplanted organs or natural infection with the virus causes lymphoproliferative disease in EBV-seronegative recipients after transplantation. As with many virus-associated diseases a promising approach for diagnosing and/or treating virus infection and its consequences in the host is the use of antibodies that specifically recognize the virus. This is also true in the case of reducing the high risk of PTLD in seronegative patients by identifying them and treating them prior to the transplantation.

These EBV-associated diseases highlight the need for a better understanding of herpes viruses and their role in mammalian diseases. As part of this understanding, there is a great need for additional diagnostic and therapeutic agents capable of detecting the presence of EBV in a mammal and effectively inhibiting EBV infection and replication. Accordingly, it is an objective of the present invention to specifically identify EBV-associated polypeptides and to use that identification specificity to produce compositions of matter useful in the therapeutic treatment and diagnostic detection of EBV in mammals.

SUMMARY

The invention is in part based on a variety of antibodies to Epstein Barr virus (EBV) glycoprotein 350 (gp350 protein; viral envelope glycoprotein that initiates EBV infection by binding to the B cell surface receptor, CR2) and their use in the detection and diagnosis during active EBV infection. The inventors isolated monoclonal antibodies (mAbs) from macaque monkeys immunized with a gp350 nanoparticle vaccine. Characterization using multiple methods with active virus and gp350 revealed the mAbs of this disclosure (including those monoclonal antibodies produced by the B03, E04, D09, C02, H02, B04, B05, D06, D07, A03, and G12 clones) were EBV-specific, and have as much as 100-fold greater EBV neutralizing activity than 72A1, a murine antibody that targets the putative CR2-binding site on gp350 and is the most potent EBV-neutralizing antibody reported to date. Thus, the mAbs of this disclosure can recognize both cell-associated and secreted forms of native gp350 from infected cell culture and are therefore high value mAbs with potential uses in immunoassay development and as immunodiagnostic reagents for clinical sample and tissue confirmation of EBV, and treatment or prevention of EBV-associated diseases and disorders.

This disclosure provides an antibody which binds, preferably specifically, to an EBV gp350 protein. Optionally, the antibody is a monoclonal antibody, antibody fragment, chimeric antibody, humanized antibody, single-chain antibody or antibody that competitively inhibits the binding of an anti-EBV gp350 protein antibody to its respective antigenic

epitope. The antibodies of this disclosure may optionally be produced in CHO cells or bacterial cells and preferably inhibit the growth or proliferation of or induce the death of a cell to which they bind. For diagnostic purposes, the antibodies of this disclosure may be detectably labeled, attached to a solid support, or the like, such as a lateral flow assay device which provides for point-of-care detection and/or diagnosis of EBV infection.

This disclosure also provides vectors comprising DNA encoding any of the herein described antibodies. Host cells comprising any such vector are also provided. By way of example, the host cells may be CHO cells, *E. coli* cells, or yeast cells. A process for producing any of the herein described antibodies is further provided and comprises culturing host cells under conditions suitable for expression of the desired antibody and recovering the desired antibody from the cell culture.

The disclosure also provides a composition of matter comprising an anti-EBV gp350 antibody as described herein, in combination with a carrier. Optionally, the carrier is a pharmaceutically acceptable carrier.

This disclosure also provides an article of manufacture comprising a container and a composition of matter contained within the container, wherein the composition of matter may comprise an anti-EBV gp350 antibody as described herein. The article may optionally comprise a label affixed to the container, or a package insert included with the container, that refers to the use of the composition of matter for the therapeutic treatment or diagnostic detection of an EBV infection.

This disclosure also provides the use of an anti-EBV gp350 polypeptide antibody as described herein, for the preparation of a medicament useful in the treatment of a condition which is responsive to the anti-EBV gp350 protein antibody.

This disclosure also provides any isolated antibody comprising one or more of the complementary determining regions (CDRs), including a CDR-L1, CDR-L2, CDR-L3, CDR-H1, CDR-H2, or CDR-H3 sequence disclosed herein, or any antibody that binds to the same epitope as such antibody.

This disclosure also provides a method for inhibiting the growth of a cell that expresses an EBV gp350 protein, including contacting the cell with an antibody that binds to the EBV gp350 protein, wherein the binding of the antibody to the EBV gp350 protein causes inhibition of the growth of the cell expressing the EBV gp350 protein. In these methods, the cell may be one or more of a B-lymphocyte and an epithelial cell. Binding of the antibody to the EBV gp350 protein causes death of the cell expressing the EBV gp350 protein. Optionally, the antibody is a monoclonal antibody, antibody fragment, chimeric antibody, humanized antibody, or single-chain antibody. Antibodies employed in the methods of this disclosure may optionally be conjugated to a growth inhibitory agent or cytotoxic agent such as a toxin. The antibodies employed in the methods of this disclosure may optionally be produced in CHO cells or bacterial cells.

This disclosure also provides a method of therapeutically treating a mammal having an EBV infection by administering to the mammal a therapeutically effective amount of an antibody that binds to the EBV gp350 protein, thereby resulting in the effective therapeutic treatment of the infection in the mammal. In these therapeutic methods, the antibody may be a monoclonal antibody, antibody fragment, chimeric antibody, humanized antibody, or single-chain antibody. Antibodies employed in these methods may optionally be conjugated to a growth inhibitory agent or

cytotoxic agent such as a toxin. The antibodies employed in these methods of this disclosure may optionally be produced in CHO cells or bacterial cells.

This disclosure also provides a method of determining the presence of an EBV gp350 protein in a sample suspected of containing the EBV gp350 protein, by exposing the sample to an antibody that binds to the EBV gp350 protein and determining binding of the antibody to the EBV gp350 protein in the sample, wherein the presence of such binding is indicative of the presence of the EBV gp350 protein in the sample. Optionally, the sample may contain cells (which may be fibroblasts, keratinocytes, or dendritic cells) suspected of expressing the EBV gp350 protein. The antibody employed in these methods may optionally be detectably labeled, attached to a solid support, or the like.

This disclosure also provides methods of diagnosing the presence of an EBV infection in a mammal, by detecting the level of an EBV gp350 protein in a test sample of tissue cells obtained from the mammal, wherein detection of the EBV gp350 protein in the test sample is indicative of the presence of EBV infection in the mammal from which the test sample was obtained.

This disclosure also provides methods of diagnosing the presence of an EBV infection in a mammal, by contacting a test sample comprising tissue cells obtained from the mammal with an antibody that binds to an EBV gp350 protein and detecting the formation of a complex between the antibody and the EBV gp350 protein in the test sample, wherein the formation of a complex is indicative of the presence of an EBV infection in the mammal. Optionally, the antibody employed is detectably labeled, attached to a solid support, or the like. In these methods, the test sample of tissue cells may be obtained from an individual suspected of having a viral infection.

This disclosure also provides a method of treating or preventing an EBV infection-related disorder by administering to a subject in need of such treatment an effective amount of an antagonist of an EBV gp350 protein. The EBV infection-related disorder may be infectious mononucleosis (glandular fever), particular forms of cancer, such as Hodgkin's lymphoma, Burkitt's lymphoma, gastric cancer, nasopharyngeal carcinoma, and conditions associated with human immunodeficiency virus (HIV), such as hairy leukoplakia and central nervous system lymphomas, autoimmune diseases, including dermatomyositis, systemic lupus erythematosus, rheumatoid arthritis, Sjögren's syndrome, and multiple sclerosis, and post-transplant lymphoproliferative disorder (PTLD). In these methods, the antagonist of the EBV gp350 protein is an anti-EBV gp350 protein antibody of this disclosure. Effective treatment or prevention of the disorder may be a result of direct killing or growth inhibition of cells that express an EBV gp350 protein or by antagonizing the production of EBV gp350 protein.

This disclosure also provides methods of binding an antibody to a cell that expresses an EBV gp350 protein, by contacting a cell that expresses an EBV gp350 protein with the antibody of this disclosure under conditions which are suitable for binding of the antibody to the EBV gp350 protein and allowing binding therebetween. The antibody may be labeled with a molecule or compound that is useful for qualitatively and/or quantitatively determining the location and/or amount of binding of the antibody to the cell.

This disclosure also provides for the use of an EBV gp350 protein, a nucleic acid encoding an EBV gp350 protein, or a vector or host cell comprising that nucleic acid, or an anti-EBV gp350 protein antibody in the preparation of a medicament useful for (i) the therapeutic treatment or diag-

nostic detection of an EBV infection, or (ii) the therapeutic treatment or prevention of an EBV infection-related disorder.

This disclosure also provides a method for inhibiting the production of additional viral particles in an EBV-infected mammal or cell, wherein the growth of the EBV infected cell is at least in part dependent upon the expression of an EBV gp350 protein (wherein the EBV gp350 protein may be expressed either within the infected cell itself or a cell that produces polypeptide(s) that have a growth potentiating effect on the infected cells), by contacting the EBV gp350 protein with an antibody that binds to the EBV gp350 protein, thereby antagonizing the growth-potentiating activity of the EBV gp350 protein and, in turn, inhibiting the growth of the infected cell. Preferably the growth of the infected cell is completely inhibited. More preferably, binding of the antibody to the EBV gp350 protein induces the death of the infected cell. Optionally, the antibody is a monoclonal antibody, antibody fragment, chimeric antibody, humanized antibody, or single-chain antibody. Antibodies employed in these methods may optionally be conjugated to a growth inhibitory agent or cytotoxic agent such as a toxin, or the like. The antibodies employed in the methods of this disclosure may optionally be produced in CHO cells or bacterial cells.

This disclosure also provides methods of treating a viral infection in a mammal, wherein the infection is at least in part dependent upon the expression of an EBV gp350 protein, by administering to the mammal a therapeutically effective amount of an antibody that binds to the EBV gp350 protein, thereby antagonizing the activity of the EBV gp350 protein and resulting in the effective therapeutic treatment of the infection in the mammal. Optionally, the antibody is a monoclonal antibody, antibody fragment, chimeric antibody, humanized antibody, or single-chain antibody. Antibodies employed in these methods may optionally be conjugated to a growth inhibitory agent or cytotoxic agent such as a toxin, or the like. The antibodies employed in the methods of this disclosure may optionally be produced in CHO cells or bacterial cells.

This disclosure also provides a protein useful for identifying an anti-gp350 antibody, or an anti-gp350 B-cell. Such a protein comprises: a) a polypeptide sequence at least 80%, at least 85%, at least 90%, at least 95%, at least 97% or at least 99% identical to the amino acid sequence of the CR2-binding region of EBV gp350; or b) a polypeptide sequence at comprising at least 100, at least 150, at least 200, at least 250, at last 300, at least 350, at least 400, or at least 425 contiguous amino acids from the amino acid sequence of the CR2-binding region of EBV gp350; wherein the protein can bind an anti-gp350 antibody, or a B-cell expressing an anti-the gp350 B-cell receptor (BCR), wherein binding of the B-cell is through the BCR, and wherein the protein is unable to bind CR2. In certain aspects, at least one amino acid residue corresponding to amino acid Q122, P158, I160, K161, W162, D163, N164, or D296, of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the amino acid sequence of the CR2-binding region comprises SEQ ID NO:125.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue is selected from the group consisting of methionine, leucine, isoleucine, glycine, alanine, valine, phenylalanine, tyrosine, tryptophan, lysine, arginine,

histidine, glutamate, and aspartate. In certain aspects, the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 has been substituted with asparagine.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid P158 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residues is selected from the group consisting of methionine, leucine, isoleucine, glycine, alanine, valine, phenylalanine, tyrosine, tryptophan, lysine, arginine, histidine, glutamate, and aspartate. In certain aspects, the amino acid corresponding to amino acid P158 of SEQ ID NO:124 has been substituted with arginine.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid I160 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue is selected from the group consisting of serine, threonine, cysteine, proline, asparagine, glutamine, tyrosine, phenylalanine, tryptophan, glutamate, aspartate, arginine, glutamine, histidine and lysine. In certain aspects, the amino acid corresponding to amino acid I160 of SEQ ID NO:124 has been substituted with arginine.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid W162 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue is selected from the group consisting of serine, threonine, cysteine, proline, asparagine, glutamine, aspartate, glutamate, asparagine, arginine, glutamate, histidine, glycine, alanine, valine, leucine, methionine, isoleucine, and lysine. In certain aspects, the amino acid corresponding to amino acid W162 of SEQ ID NO:124 has been substituted with alanine.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue is selected from the group consisting of tyrosine, phenylalanine, tryptophan, serine, threonine, cysteine, proline, asparagine, glutamine, lysine, arginine, histidine, glycine, alanine, valine, leucine, methionine, and isoleucine. In certain aspects, the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been substituted with asparagine. In certain aspects, at least one amino acid has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124. In certain aspects, the at least one amino acid is selected from the group consisting of serine, threonine, cysteine, proline, asparagine, and glutamine. In certain aspects, a serine residue has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid N164 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue is selected from the group consisting of methionine, leucine, isoleucine, glycine, alanine, valine, phenylalanine, tyrosine, tryptophan, lysine, arginine, histidine, glutamate, serine, and aspartate. In certain aspects, the amino acid corresponding to amino acid N164 of SEQ ID NO:124 has been substituted with serine.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue is selected from the group consisting of tyrosine, phenylalanine, tryptophan, serine, threonine, cysteine, proline, asparagine, glutamine, lysine, arginine, histidine, glycine, alanine, valine, leucine, methionine, and isoleucine. In certain aspects, the amino acid corresponding to amino acid D296 of SEQ ID NO:124 has been substituted with arginine. In certain aspects, the amino acid corresponding to amino acid I160 of SEQ ID NO:124 has been deleted, or has been substituted with at least one amino acid residue. In certain aspects, the at least one amino acid residue is selected from the group consisting of serine, threonine, cysteine, proline, asparagine, glutamine, tyrosine, phenylalanine, tryptophan, glutamate, aspartate, arginine, glutamine, histidine and lysine. In certain aspects, the amino acid corresponding to amino acid I160 of SEQ ID NO:124 has been substituted with arginine. In further aspects, the amino acid corresponding to amino acid P158 of SEQ ID NO:124 has been deleted, or has been substituted with at least one amino acid residue. In certain aspects, the at least one amino acid residue is selected from the group consisting of methionine, leucine, isoleucine, glycine, alanine, valine, phenylalanine, tyrosine, tryptophan, lysine, arginine, histidine, glutamate, and aspartate. In certain aspects, the amino acid corresponding to amino acid P158 of SEQ ID NO:124 has been substituted with arginine or asparagine. In certain aspects, the amino acid corresponding to amino acid I160 of SEQ ID NO:124, has been substituted with a threonine or an arginine.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein one or more amino acids has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124. In certain aspects, the one or more amino acids are selected from the group consisting of serine, threonine, cysteine, proline, asparagine, and glutamine. In certain aspects, a serine residue has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124.

This disclosure also provides a comprising an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acids corresponding to amino acids W162-N164 of

SEQ ID NO:124 have been deleted, or have been substituted with one or more amino acid residues. In certain aspects, the one or more amino acid residue, is selected from the group consisting of serine, threonine, cysteine, proline, asparagine, glutamine, aspartate, glutamate, asparagine, arginine, glutamate, histidine, glycine, alanine, valine, leucine, methionine, isoleucine, and lysine. In certain aspects, the amino acids corresponding to amino acids W162-N164 of SEQ ID NO:124 have each been substituted with an alanine residue.

This disclosure also provides a protein having an amino acid sequence at least 85%, at least 90%, at least 95%, at least 97% or at least 99% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:127, SEQ ID NO:129, SEQ ID NO:131, SEQ ID NO:133, SEQ ID NO:135, SEQ ID NO:137, SEQ ID NO:139, SEQ ID NO:141, SEQ ID NO:145, SEQ ID NO:147, SEQ ID NO:149, SEQ ID NO:151, SEQ ID NO:153, SEQ ID NO:155, SEQ ID NO:157, and SEQ ID NO:159. In certain aspects, the protein comprises an amino acid sequence selected from the group consisting of SEQ ID NO:127, SEQ ID NO:129, SEQ ID NO:131, SEQ ID NO:133, SEQ ID NO:135, SEQ ID NO:137, SEQ ID NO:139, SEQ ID NO:141, SEQ ID NO:145, SEQ ID NO:147, SEQ ID NO:149, SEQ ID NO:151, SEQ ID NO:153, SEQ ID NO:155, SEQ ID NO:157, and SEQ ID NO:159.

This disclosure also provides a nucleic acid molecule encoding a protein having an amino acid sequence at least 85%, at least 90%, at least 95%, at least 97% or at least 99% identical to an amino acid sequence selected from the group consisting of SEQ ID NO:127, SEQ ID NO:129, SEQ ID NO:131, SEQ ID NO:133, SEQ ID NO:135, SEQ ID NO:137, SEQ ID NO:139, SEQ ID NO:141, SEQ ID NO:145, SEQ ID NO:147, SEQ ID NO:149, SEQ ID NO:151, SEQ ID NO:153, SEQ ID NO:155, SEQ ID NO:157, and SEQ ID NO:159. In certain aspects, the nucleic acid molecule encodes a protein comprises an amino acid sequence selected from the group consisting of SEQ ID NO:127, SEQ ID NO:129, SEQ ID NO:131, SEQ ID NO:133, SEQ ID NO:135, SEQ ID NO:137, SEQ ID NO:139, SEQ ID NO:141, SEQ ID NO:145, SEQ ID NO:147, SEQ ID NO:149, SEQ ID NO:151, SEQ ID NO:153, SEQ ID NO:155, SEQ ID NO:157, and SEQ ID NO:159. In certain aspects, the nucleic acid molecule comprises a sequence selected from the group consisting of SEQ ID NO:128, SEQ ID NO:130, SEQ ID NO:132, SEQ ID NO:134, SEQ ID NO:136, SEQ ID NO:138, SEQ ID NO:140, SEQ ID NO:142, SEQ ID NO:146, SEQ ID NO:148, SEQ ID NO:150, SEQ ID NO:152, SEQ ID NO:154, SEQ ID NO:156, SEQ ID NO:158, and SEQ ID NO:160.

This disclosure also provides a method of identifying an anti-gp350 antibody, comprising contacting a protein of the invention with a solution comprising antibodies, and isolating antibodies that specifically bind to the protein. In certain aspects, the protein comprises a His-tag, an epitope tag, or a detectable label.

This disclosure also provides a method of identifying an anti-gp350 B-cell, comprising contacting a protein of the invention with a solution comprising B-cells, and isolating B-cells that specifically bind to the protein. In certain aspects, the protein comprises a His-tag, an epitope tag, or a detectable label.

Further embodiments will be evident to the skilled artisan upon a reading of the present specification.

This disclosure contains the following Sequences:

SEQ ID NO	Description	Sequence
1	B03 clone Heavy Chain aa sequence	QVQLQESGPGLVKPAETLSLTCTVSGGSFSSYWGW WIRQSPGKLEWIGHISSGGNNYLNPSLKSRTLSD TSKNQFSLKLSVTAADSAVYYCARAPRIVVRGRYF DQWGGQGLVTVSS
2	B03 clone Heavy Chain nucleotide sequence	caggtgcagctgcaggagtcgggcccaggactggtagaagcctgcggagaccct gtccctcacctgcactgtctctgggtgctctttcagcagttactggtaggggctggatc cgtcagtcctccagggaaggagctggagtgattgggcataatcagtagtggtggaa acaactaccttaatccgtccctcaagagtcgagtcaccctgtcactagacacgtcca agaaccagttctccctgaagctgaactctgtgaccgcccggactcggccgtgtat tactgtgccagagccccctgattgttggtagaggccgatactttgaccaatggggc cagggagtcctgggtcaccgtctcctca
3	B03 clone Heavy Chain CDR1	GGSFSSYW
4	B03 clone Heavy Chain CDR2	ISSGGNN
5	B03 clone Heavy Chain CDR3	APAPRIVVRGRYFDQW
6	B03 clone Kappa Chain aa sequence	DIQMTQSPSSLSASVGDVTITCRASQGGINIYLNWFQ QRPKAPKLLIYAATTLQSGVPSRFSGSGSDFTLTI SSLQPEDFATYYCLQCESYPLTFGGGKVEIK
7	B03 clone Kappa Chain nucleotide sequence	gacatccagatgaccagtcctccatctccctgtctgcatctgtaggagacactgtc accatcacttgcgggcaagtcagggtatcaatatctacctgaattggtttcagcag agaccagggaagccccctaaactcctgatctatgtgcgaccactttacaaagtgg ggtcccatcaagattcagcggcagtggtctgggacagatttcactctcaccatcag cagcctgcagcctgaagatttcgcaacttatctgtctacagtgtagaagttatccg ctcactttcggcggaggaccgaaggtggagatcaaa
8	B03 clone Kappa Chain CDR1	QGGINIY
9	B03 clone Kappa Chain CDR2	AAT
10	B03 clone Kappa Chain CDR3	LQCESYPLT
11	E04 clone Heavy Chain aa sequence	QVQLQESGPGLVKPSSETLSLTCTVSGGSISGAYYYW SWIRQPPGKGLDWIGYIYGSFSAAYNPSLKSRTAIS KDTPKNQFSLKLSVTAADTAVYYCARGRLGYSN WFDVWGPGLVTVSS
12	E04 clone Heavy Chain nucleotide sequence	caggtgcaactgcaggagtcgggcccaggactggtagaagccttcggagaccctg tccctcacctgcactgtctctgggtgctccatcagcgggtGCTTACTactactgg agctggattcgacagcccccggggaaggagctggactggattggatatactatg gaagt tttgggagtgctactacaacccctccctcaagagtcgagccaccatttcaa aagacacgcccagaaccagttctccctgaactgagctctgtgaccgcccggga cacggcctgtattactgtgcgagaggaaggcgactaggctattcgaaactggttcg atgtctggggccgggagtcctgggtcaccgtctcctca
13	E04 clone Heavy Chain CDR1	GGSISGAYYY
14	E04 clone Heavy Chain CDR2	IYGSFGSA
15	E04 clone Heavy Chain CDR3	ARGRRLGYSNWFDVW
16	E04 clone Kappa Chain aa sequence	DIQMTQSPSSLSASVGDKVTITCRTSQDVSSYLAWY QQKPGKAPQLLIYAASSLQSGVPSRFTGSGSGAEFTL TISSLQPEDFASYYCQYKNLPLTFGGGKVEIK
17	E04 clone Kappa Chain nucleotide sequence	gacatccagatgaccagtcctccatctccctgtctgcatctgtaggagacaaagtc accatcacttgcggacaagtcaggacgttagcagttatttagcctggatcagcag aaaccagggaagccctcagctcctgatctatgtgcacacagtttgcgaagtgg ggtcccatcaaggttcacggcagtggtctggggcagaattcactctcaccatca gcagccttcagcctgaagattttgcatcatattactgtcaacagtataaaaatctccc ctcactttcggcggaggaccgaaggtggagatcaaa
18	E04 clone Kappa Chain CDR1	QDVSSY
19	E04 clone Kappa Chain CDR2	AAS
20	E04 clone Kappa Chain CDR3	QYKNLPLT
21	D09 clone Heavy Chain aa sequence	QVQLQESGPGLVKPSSETLSLTCDVSGGSFSGDFYWS WIRQPPGKGLDWIGNIHGSSAGTKYKPSLKSRTISK DTSKNQFSLKLSVTAADTAVYYCTRGPLSRIVAGF GRGINWFDVWGPGLVTVSS

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SEQ ID NO	Description	Sequence
22	D09 clone Heavy Chain nucleotide sequence	caggtgcagctgcaggagtcgggcccaggactggggaagccttcggagaccctg tccctcacctgcgatgtctctgggtgctccttcagcggatTTCTactggagctg gatccgccagccccaggggaaggactggactggattgggaatatccatggcag cagtgcggggaccacaaatacaagccctccctcaagagtcgagtcaccatttcaaaa gacacgtccaagaaccagttctccctgaaactgagctctgtgaccgccgggaca cggccgtctattactgtacgagaggcccccttagtaggatagtagctgggtttggga gggggattaaactgggtcgatgtctggggccgggagtcctggtaaccgtctcctca
23	D09 clone Heavy Chain CDR1	GGSFSGDFY
24	D09 clone Heavy Chain CDR2	IHGSSAGT
25	D09 clone Heavy Chain CDR3	TRGPLSRIVAGFGRGINWFDVW
26	D09 clone Lambda Chain aa sequence	QPVLTQPTSLSPGASVRLSCTLSSGINVGSYSIFWY QQKPGSPPRYLLFYFSDSSKHQSGVPSRFSKSDTS ANAGLLLIISGLQSEDEADYYCAIWHSSASVLFGGGT RLTVL
27	D09 clone Lambda Chain nucleotide sequence	cagcctgtgctgaccagccaacctccctctcagcatctccgggagcatcagtcag actcagctgcaccttgagcagtggtcatcaatgttggtagtacagcatattctggtag cagcagaagccaggagtcctcccccggtagctctctgttctatttctcagactcaagta agcaccagggtctctggagtcctccagccgttctctggatccaaggatacttcagcc aatgcagggtctttactgatctctgggtccagctctgaagatgaggtctgactattact gtgccatatggcacagcagcgcttctgtgttatttcggaggagggaacccggctgaca gtacta
28	D09 clone Lambda Chain CDR1	TLSSGINVGSYSIF
29	D09 clone Lambda Chain CDR2	YFSDSSK
30	D09 clone Lambda Chain CDR3	AIWHSSASVL
31	C02 clone Heavy Chain aa sequence	QLQLQESGPGLVKPSSETLSLTCVSGGSISGYYSWI RQPPGKGPEWIGFIDGNTVGTNYPNLSKSRVTLTKDT SKNQFSLKSSVTAADTAVYYCARKPLRRYFWFDV WGPGLVTVSS
32	C02 clone Heavy Chain nucleotide sequence	cagctgcagctgcaggagtcgggcccaggactggggaagccttcggagaccctg tccctcacctgcgatgtctctgggtgctccatcagcgggtactactggagtggattc gccagcccccagggaagggaacgggagtggttgggtttattgatggtaatactgtg ggcaccaactacaacccctccctcaagagtcgagtcaccattcaaaagacacgtc caagaatcagttctcctgaaggtagttctgtgaccgcgcggacacggccgtgtg attactgtgcgaggaagccgctacgccgttatttctgggtcgatgtctggggcccg gagtcctggtaaccgtctcctca
33	C02 clone Heavy Chain CDR1	GGSISGY
34	C02 clone Heavy Chain CDR2	IDGNTVGT
35	C02 clone Heavy Chain CDR3	ARKPLRRYFWFDVW
36	C02 clone Lambda Chain aa sequence	QSVLTQPPSVSGDPGQRTISCTGSSSNIGAGYYVYW YQFPFGTAPKLLIYQDNKRPSGVSDRFSKSGTSAS LTITGLQPGDEADYYCSAWDSSLSAVMFGRGRLTV L
37	C02 clone Lambda Chain nucleotide sequence	cagtctgtgctgacgcagccgcctcagtgctctggggaccccgaggaggggtc accatctcgtgcactgggagcagctccaacatcgggGCGggtattatgtatact ggtagcagcagttccagggaacggcccccactcctcatctatcaagataataag cgacctcaggggtttctgaccgattctctgggtccaagtcgtgtacccagcctcc ctgaccatcactgggtccagcctggggatgaggctgattattactgctcagcatgg gatagcagcctgagtgctgttatgttcggaagaggcaccaggtgacagtacta
38	C02 clone Lambda Chain CDR1	SSNIGAGY
39	C02 clone Lambda Chain CDR2	QDN
40	C02 clone Lambda Chain CDR3	SAWDSSLSAVM

-continued

SEQ ID NO	Description	Sequence
41	H02 clone Heavy Chain aa sequence	QLQLQESGPGVVKPSETLSLTCTISGGSFSTYYWTWI RQPPGKGLEWVGVIIGNGGRSLNYPNPSLKSRITLSVD ASKNQFSLKVTSVTAADTAVYYCGRARGLRGNWFD VWGPGLVLTVSS
42	H02 clone Heavy Chain nucleotide sequence	caactgcagttgcaggagtcgggcccaggagtggtgaagccttcggagaccctgt ccctcacctgcactatctctgggtggctccttcagtaacttactactggacctggattcgc cagccccagggaaggagctggagtggttggttatatcggtaatggtggtcgta gcctcaactacaacccctccctcaagagtcgcatcacctgtcagtagacgcgtcc aagaaccagttctccctgaaggtagctctgtgaccgccgggacacggccgtct attactgtgggagagccaggggactccgcggaaactggttcgatgtctggggccc gggagtcctggtcaccgtctcctca
43	H02 clone Heavy Chain CDR1	GGSFSTYY
44	H02 clone Heavy Chain CDR2	IGNGGRSL
45	H02 clone Heavy Chain CDR3	GRARGLRGNWFDVW
46	H02 clone Lambda Chain aa sequence	QAALTQPPSVSGSPQSVTISCTGTSSDIGGYNYVSW YQQHPGKAPKVMIIYEVSKRPSGVSDRFSGSKSGNIA SLTISGLQAEADYYCSSYAGSNTFLFGGGTRLTVL
47	H02 clone Lambda Chain nucleotide sequence	caggctgccctgactcagcctccctctgtgtctgggtctcctggacagtcggtcacc atctcctgcactggaaccagcagtgacatcggtggtataactatgtctcctggtagc aacaacacccaggcaagcccccaagtcagatgattatgaggtcagtaagcggcc ctcaggggtctctgatcgcttctctggttccaaatctggcaacatagcctccctgacc atctctgggctccaggctgaggacgaggtgatattactgcagtcacatgcaggc agcaacactttcttattcggaggaggaccggctgacagtacta
48	H02 clone Lambda Chain CDR1	SSDIGGYNY
49	H02 clone Lambda Chain CDR2	EVS
50	H02 clone Lambda Chain CDR3	SSYAGSNTFL
51	A03 clone Heavy Chain aa sequence	QVQLQESGPGVLVKPSETLSLTCAVSGGSISSNYWSWI RQPPGKGLEWIGRIYGGGGTDYNPSLKSRVTISTDT SKNQFSLKVSSVTAADTAVYYCARVRIQWVQLRGW FDVWGPGLVLTVSS
52	A03 clone Heavy Chain nucleotide sequence	caggctgcagctgcaggagtcgggcccaggactggtgaagccttcggagaccctg tccctcacctgcgctgtctctgggtggctccatcagcagtaactactggagctggatc cgccagccccagggaaggagctggagtggttagcagtcacatggttagtggtg ggagcaccgactacaacccctccctcaagagtcgagtcaccatttcaacagacac gtccaagaaccagttctccctgaaggtagctctgtgaccgccgggacacggcc gtgtattactgtgcgagagtgccgatcacagtgggtacagttgcgaggctggttcgat gtctggggcccgaggctcctggtcaccgtctcctca
53	A03 clone Heavy Chain CDR1	GGSISSNYWS
54	A03 clone Heavy Chain CDR2	IYGGGGST
55	A03 clone Heavy Chain CDR3	ARVRIQWVQLRGWFDVW
56	A03 clone Kappa Chain aa sequence	YIQMTQSPSSLSASVGDVTFITGPASQSFSSSLAWYQ QKPGKAPNLLIYSASSLQCGVRSRFSGSKSGTDFTLTIT SSLQPEDIASYYCQQYYSYPFTFGPGTKLDIK
57	A03 clone Kappa Chain nucleotide sequence	Tacatacagatgacgcagtcctccatccctcctgtctgcatctgtaggagacacagtc accttcactggccccgcaagtcagagctttagtagtagtttagcctgggtatcagcag aaaccagggaagccccctaacctcctgatctatagtgcatccagtttgcaatgtggg gttcgttcgaggttcagtgccagtaagtcctgggacagatttactctcaccatcagca gcctgcagcctgaagatattgctagttattactgtcaacagattacagttatccattca cttccgccccgggaccaaactggatatcaaa
58	A03 clone Kappa Chain CDR1	QSFSSS
59	A03 clone Kappa Chain CDR2	SAS
60	A03 clone Kappa Chain CDR3	QQYYSYPFT

-continued

SEQ ID NO	Description	Sequence
61	B04 clone Heavy Chain aa sequence	QVQLQESGPGLVRPSETLSLTCAVSGGSISSNYWSWI RQPPGKGLEWIGYISGSTGSTYQNP SLKSRVTVSKDT SKNQFSLKLN SVTAADTAVYYCARSGRGSSLDLW GRGVLVTVSS
62	B04 clone Heavy Chain nucleotide sequence	Caggtgcagctgcaggagtcgggcccaggactggtagggccttcggagaccct atccctcacctgcgctgtctctggtaggctccatcagcagtaactactggagctggatt cgccagccccaggggaaggggctggagtgattgggtatatctctggtagtactg ggagcacctaccagaacccctccctcaagagtcgagtcaccgtttcaaaagacac gtctaagaaccagttctccctgaagctgaattctgtgaccgcgcggacacggccg tgtattactgtgcgagaagtgggagaagaggcagctcattggatttgtggggccg ggagttctggtcacctctctca
63	B04 clone Heavy Chain CDR1	GGSISSNY
64	B04 clone Heavy Chain CDR2	ISGSTGST
65	B04 clone Heavy Chain CDR3	ARSGRGSSLDLW
66	B04 clone Lambda Chain aa sequence	QSVLTQPPSVSGDPGQRVTISCTGSSSNIGAGYYVYW YQQFPGTAPKLLIYQDNKRPSGVSDRFSGSKSGTSAS LTIITGLQPGDEADYYCSAWDSSLSAVFFGGGTRLTV L
67	B04 clone Lambda Chain nucleotide sequence	cagctctgtctgacgcagccgcctcagtgctctgggacccgggcagagggtc accatctcgtgcactgggagcagctccaacatcgggGCGggttattatgtatact ggtagcagcagttcccaggaacggccccaaactcctcatctatcaagataataag cgaccctcaggggtttctgaccgattctctggctccaagtcgtgacctcagcctcc ctgaccatcactgggctccagcctggggatgaggctgattattactgctcagcatgg gatagcagcctgagtgctgtgtcttcggaggaggacccggctgacagtacta
68	B04 clone Lambda Chain CDR1	SSNIGAGYY
69	B04 clone Lambda Chain CDR2	QDN
70	B04 clone Lambda Chain CDR3	SAWDSSLSAVF
71	B05 clone Heavy Chain aa sequence	QLQESGPGLVKPSETLSLTCTVSGGSISDITYRWSWIR QSPGKGLEWIAIYGTSTSTNYNPSLKSRLTISKDTS KNQFSLNLR SVTAADTAVYYCARGDSGGRSAHV FH FWGQGLRVTVSS
72	B05 clone Heavy Chain nucleotide sequence	cagctgcaggagtcgggcccaggactggtaggccttcggagaccctgtccctca cctgcactgtctctggtaggctccatcagcgatacttacCGGtgagctggattcgc cagtcgccaggggaaggagctggagtgattgcctacatctatggtagtactacgag caccaactacaacccctccctcaagagtcgactcaccatttcaaaagacacgtcca agaaccagttctccttgaacctgaggtctgtgaccgcgcggacacggccgtgtat tactgtgcgagaggggatagcgtggccggtcagcgcgatgttttctattctggggc caagggtcaggggtcacctctcttca
73	B05 clone Heavy Chain CDR1	GGSIDTYR
74	B05 clone Heavy Chain CDR2	IYGTSTST
75	B05 clone Heavy Chain CDR3	ARGDSGGRSAHVFW
76	B05 clone Lambda Chain aa sequence	QSVLTQPPSVFGDPGQRITISCTGSSSNIGAGYYVYW YQQFPGTAPKLLIYQDNKRPSGVSDRFSGSKSGSSAS LTIITGLQPGDEADYYCSAWDSSLSVRVFGGTRLTV L
77	B05 clone Lambda Chain nucleotide sequence	cagagtggtctgacgcagccgcctcagtggttggggacccgggcagaggatca ccatctcgtgcactgggagcagctccaacatcgggGCGggttattatgtatactg gtaccagcagttcccaggaacggccccaaactcctcatctatcaagataataagc gacctcaggggtttctgaccgattttctggctccaagtcgtgtctcagcctccct gaccatcactgggctccagcctggggatgaggctgattattactgctcagcatggg atagcagcctgagtgtagcgggttttcggaggagggaacccggctgacagtacta
78	B05 clone Lambda Chain CDR1	SSNIGAGYY
79	B05 clone Lambda Chain CDR2	QDN

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SEQ ID NO	Description	Sequence
80	B05 clone Lambda Chain CDR3	SAWDSSLSVRV
81	D06 clone Heavy Chain aa sequence	EVQLVESGGGLVQPGGSLRLSCAASGFTFSSYGMN WVRQAPGKGLEWVSFIINTGKTTYADSVRGRFTIS RDNAKKSIVSLQMSSLRAEDTAVYYCTRGRGRHWGS SGVDFWQGQLRVTVS
82	D06 clone Heavy Chain nucleotide sequence	Gaggtgcaactggtgagctctggaggaggcttggtccagcctggagggtccctg agactctcctgtgcagcctctggattcaccttcagtagttacggcatgaactgggtcc gccaggctccgggaagggtggtggtctcattcattactaacactggtaaa accacatactacgtgactctgtgaggggcccattcaccatctccagagacaaagc caagaagtcggtgtctctacaaatgagtagcctgagagccgaggacagggccgtc tattactgtactaggggaaggtagacacggctggtccagtggtgtttttgatttctg gggccaaggtctcagggtcacctgtctcttc
83	D06 clone Heavy Chain CDR1	GFTFSSYG
84	D06 clone Heavy Chain CDR2	INTGKTT
85	D06 clone Heavy Chain CDR3	TGRGRHWSSGVDFW
86	D06 clone Lambda Chain aa sequence	QSVLTQPPSVFGDPGQRITISCTGSSSNIGAGYVYW YQFPFGTAPKLLIYQDNKRPSGVSDRFSGSKFGSSAS LTITGVQRGDEGDYIYCSAWDSSLSVRVLGGGTRLTV L
87	D06 clone Lambda Chain nucleotide sequence	cagtctgttctgacgcagccgcctcagtggtcggggaccccgaggatca ccatttcgtgactgggagcagctccaacatcgggGCGggttatatgtatactg gtaccagcagttcccaggaacggcccaactcctcatctataagataataagc gacctcaggggtttctgaccgattctctggtccaaagtttggttcctcagcctccct gaccatcactgggtccagcgtgggatgaggggtgattattactgctcagcatggg atagcagcctgagtgtagcggttttgggaggaggacccggctgacagtacta
88	D06 clone Lambda Chain CDR1	SSNIGAGY
89	D06 clone Lambda Chain CDR2	QDN
90	D06 clone Lambda Chain CDR3	SAWDSSLSVRV
91	D07 clone Heavy Chain aa sequence	EVQLVESGGGLVQPGGSLRLSCVASGFTFSDRYIDW VRQAPGKGLEWVSTISTGSGDTALYSDSVKGRFTISR DNAKNTLYLQMNSLRAEDTAVYYCARHSGTFYTHF DYWGQGLVTVSS
92	D07 clone Heavy Chain nucleotide sequence	gaagtgcagttggtgagctctgggggaggcttggtacagccgggggggtccctg agactctcctgtgtagcctctggattcaccttcagtgaccgtacatagactgggtcc gccaggctccagggaaggcctggagtggtctcaactattagcactggtAGT ggtgataccgcattgtactcagactctgtcaaggccgattcaccatctccagagac aacgccaagaacacactgtatctgcaaatgaacagcctgagagccgaagacagc gctgtctattactgtgagagacatagtggtactttttacaccactttgactactgggg ccaggagtgctcgtcaccgtctcctca
93	D07 clone Heavy Chain CDR1	GFTFSDRY
94	D07 clone Heavy Chain CDR2	ISTGSGDTA
95	D07 clone Heavy Chain CDR3	ARHSGTFYTHFDYW
96	D07 clone Kappa Chain aa sequence	DIQMTQSPSSLSASVGDVTVFTCRASRSISSWLAWYQ QKPGRAPKVLIIYKASSLQSGVPSRFSGSGSDFTLTI SSLQSEDFATYYCQYSSRPPTFGQGTKVEIR
97	D07 clone Kappa Chain nucleotide sequence	Gacatccagatgaccagctctccatcctcctgtctgcatctgtaggagacacagt caccttcacctgccggcgagtcggagtagtagcagctggttagcctggtatcagc agaaaccaggagagccctaaagtctgtatctataaggcgtccagtttgcaaaagt ggggttccttcaagggtcagcggcagtggtctgggacagacttcactctcaccatc agcagcctacagctctgaagattttgcaacatattattgtcaacagtagtagtcgcc ctccgacgttcggccaagggaaggtggaatcaga
98	D07 clone Kappa Chain CDR1	RSISSW

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SEQ ID NO	Description	Sequence
99	D07 clone Kappa Chain CDR2	KAS
100	D07 clone Kappa Chain CDR3	QQYSSRPPT
101	G12 clone Heavy Chain aa sequence	QVQLQESGPGLLKPSETLSLTCAVSGGSFSSFWNSW LRQPPEKGLEWIGEINGDSGSTNYPNPSLKSRVTISKD ASKNQFSLKLTSTVTAADTAVFYCARVRRILRSLDVW GRGLVTVSS
102	G12 clone Heavy Chain nucleotide sequence	Caggtgcagctgcaggagtcgggcccaggactgctgaagccttcggagaccct gtccctcacctgcgctgtctctgggtggctccttcagtagttctgggtggagctggctc cgccagccccagaaaaggagctggagtgattggggagatcaatggatgag gggagcaccaactacaaccctccctcaagagtcgagtcaccatttcaaaagacg cgtccaagaaccagttctccctgaaactgacctctgtgaccgcccggacacggc cgtttttactgtgtagagattcggcgaattctgaggtcattggatgtctggggcgg ggagttctgggtcaccgtctcctca
103	G12 clone Heavy Chain CDR1	GGSFSSFW
104	G12 clone Heavy Chain CDR2	INGDSGST
105	G12 clone Heavy Chain CDR3	ARVRRILRSLDVW
106	G12 clone Kappa Chain aa sequence	DIQMTQSPFSLFAFVGDRVTITCQASQGISHLLAWYQ QKPGKAPKLLIYSASTLQSGVPSRFSGSGFGTEFTLTI SSLQPEDFATYYCQQHNSYPRTFGQGTKVEIK
107	G12 clone Kappa Chain nucleotide sequence	Gacatccagatgacccagtcctcttttctgttttgattttagtagagacagagtcac catcacttgccaagccagtcagggtattagccacttgtagcttggtatcagcagaa accagggaagccctaaagtccttatttattctgcatccactttgcaagtggggtc ccatcaagggtcagcggcagtgatttgggacggaattcactctcaccatcagcag cctgcagcctgaagattttgcaacttattactgtcaacagcataatagttaccctcgga cgttcggccaagggaacaggtggaaatcaaa
108	G12 clone Kappa Chain CDR1	QGISHL
109	G12 clone Kappa Chain CDR2	SAS
110	G12 clone Kappa Chain CDR3	QQHNSYPRT
111	Macaque HC backbone Nucleotide sequence	tcgcgcgttt cggatgatgac ggtgaaaacc tctgacacat gcagctcccg gagacgggtca cagcttgctc gtaagcggat gccgggagca gacaagcccg tcaggggcgcg tcagcgggtg ttggcgggtg tcggggctgg cttactatg cgccatcaga gcagattgta ctgagagtgcc accatagcgc gtgtgaaata ccgcacagat gcgtaaggag aaaaatccgc atcagattgg ctattggcca ttgcatacgt tgtatccata tcataaatatg tacatttata ttggctcatg tccaacatta ccgccatggt gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc catatatgga gttccgcgtt acataactta cggtaaatgg cccgcctggc tgaccgccca acgacccccc cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga ctttcattg acgtcaatgg gtggagattt tacggtaaac tgcccacttg gcagtacatc aagtgatca tatgccaagt acgcccccta ttgacgtcaa tgacggtaaa tggcccgctt ggcattatgc ccagtcacatg accttatggg actttcctac ttggcagtac atctacgtat tagtcacgcg tattaccatg gtgatcggtt ttggcagta catcaatggg cgtggatagc gggttgactc acggggattt ccaagtctcc accccattga cgtcaatggg agtttggttt ggcacaaaa tcaacgggac ttccaaaaat gtcgtaacaa ctccgccccca ttgacgcaaa tgggcggtag gcgtgtacgg tgggaggtct atataagcag agctcgttta gtgaaccgtc agatcgccctg gagacgccat ccacgctggt ttgacctcca tagaagacac cgggaccgat ccagcctcca tcggctcgca tctctcctc acgcgcccgc cgcctacct gagggcccca tccacgcggg ttgagtcgag ttctgcccgc tcccgcctgt ggtgcctcct gaactgcgtc cgcctctag gtaagtttaa agctcaggtc gagaccgggc ctttgtccgg cgctcccttg gagcctacct agactcagcc ggctctccac gctttgctcg accctgcttg ctcaactcta gttaacgggt gagggcagtg tagtctgagc agtactcgtt gctgccgcgc gcgccaccag acataatagc tgacagacta acagactggt cctttccatg ggtcttttct gcagtcacgg tcgctcgacac gtgtgatcag atatcgcggc cgctctagac caccatggga tggtcatgta tcatcctttt tctagtagca actgcaaccg gtgtacattc ccagggtcag ctgggtgcagc tgcacgagtc gggcccagga ctgggtgaagc cttcggagac cctgtccctc acctgcgctg tctctggtgg cttcatcagc agtagctact ggagctggat ccgccaggcc ccagggaagg gactggagtg gattgggtat gtctatggta gtggtcgtga caccaacgac aacccctccc tcaagagtcg agtcaccctg tcagtagaca cgtccaagaa ccagctctcc ctgaagctga gatctgtgac cgcgcgggac acggccgtgt attactgtgc gagcagcggc tggcctcctg ggttggacta ctggggccag ggagtcacgg

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112	Macaque heavy chain backbone aa sequence	MGWSCIILFLVATATGVHSQVLVQLHESGPGLVKP SETLSLTCAVSGGSISSSYWSWIRQAPGKLEWIGYV YGSGRDTNDNPSLKSRLTSLVDTSKNQLSLKLSVT AADTAVYYCASSGWPPGLDYWGQGVTVTVSSASTK GPSVFLPLAPSSRSTSESTAALGCLVKDYFPEPVTVSW NSGSLTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGT QTYVCNVNHKPSNTKVDKRVETKCGGSKPPTCPP CTSPELLGGPSVFLFPPKPKDITLMI SRTPEVTCVVVD V SQEDPDVFNWYVNGAEVHHAQTKPRETQYNSTY RVVSVLTVTHQDWLNGKEYTCKVSNKALPAPIQKTI SKDKGQPREPQVYTLPPSRREELTKNQVSLTCLVKGF YPSDIVVEWESSGPENTYKTTTPVLDSGYSFLYSK LTVDKSRWQQGNVPSCSVMHEALHNHYTQKSLSLS PGK
113	Macaque Kappa chain backbone nucleotide sequence	tcgcgcgttt cggatgatgac ggtgaaaaacc tctgacacat gcagctcccg gagacggtca cagcttctgtc gtaagcggat gccgggagca gacaagcccg tcaggcgcgcg tcagcgggtg ttggcgggtg tcggggctgg cttaactatg cggcatcaga gcagattgta ctgagagtgc accatagcg gtgtgaataa ccgcacagat gcgtaggag aaaaaccgc atcagattgg ctattggcca ttgcatacgt tgtatccata tcataatag tacatttata ttggctcatg tccaacatta ccgccatgtt gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc catatatgga gttccgcgtt acataactta cggtaaatgg ccgcctggc tgaccgcccc acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga ctttccattg acgtcaatgg gtggagattt tacgggtaaac tgcccacttg gcagtagatc aagtgtatca tatgccaaagt acgcccccta ttgacgtcaa tgacggtaaa tggcccgctt ggcattatgc ccagtagatg acctatggg actttctctac ttggcagtag atctacgtat tagtcatcgc tattaccatg gtgatgcggt ttggcagta catcaatggg cgtggatagc ggtttgactc acgggggattt ccaagtctcc accccattga cgtcaatggg agtttgtttt ggcacaaaaa tcaacgggac tttccaaaaa gtcgtaacaa ctccgccccca ttgacgcaaa tgggcggtag cgtgtgacgg tgggaggtct ataaagcag agctcgttta gtgaaccgtc agatcgctcg gagacgcat ccacgctgtt ttgacctcca tagaagacac cgggaccgat ccagcctcca tcggctcgca tctctccttc acgcgccccg cgcctacact gaggccgcca tccacgcccg ttgagtcgcg ttctgcgcgc tcccgcctgt ggtgcctcct gaactgcgtc cgccgtctag gtaagtttaa agctcaggtc gagaccgggc ctttgtccgg cgctcccttg gagcctaact agactcagcc ggctctccac gctttgcctg acctgcttg ctcaactcta gttaacggtg gagggcagtg tagtctgagc agtacacggt gctgcgcgcg gcgccaccag acataatagc tgacagacta acagagctgt cctttccatg ggtcttttct gcagtcaccg tcgtcgacac gtgtgatcag atatcgcggc cgtctagac caccatggga tggctcatgta tcacccctttt tctagtagca actgcaacgg gtgtacattc agaaattgtg ttgacacagt ctccaggcac cctgtctttg tctccagggg aaacagccat catctcttgt cggaccagtc agtatgggtc cttagcctgg tatcaacaga ggcccgccca ggcgccagg ctcgtcatct attcgggctc tactcggggc gctggcatcc cagacagggt cagcggcagt cgggtggggc cagactacaa tctcaccatc agcaacctgg agtcgggaga ttttgggtgt tattattgccc agcagtatga attttttggc cagggggacca aggtccagggt cgacattaaa cgtacgggtg ctgcaccatc tgtcttcac tttccgccat ctgaggatca ggtgaaatct ggaactgtct ctgttgtgtg cctgtgtaat aacttctatc ccagagaggc cagcgtaaaag tggaaaggtag atggtgccct caaaacgggt aactcccagg agagtgtcac agagcaggac agcaaggaca acacctacag cctgagcagc accctgacgc tgagcagcac agagtaccag agtcacaaag tctatgcctg cgaagtcacc catcagggcc tagctcggc cgtcaccaag agcttcaaca gaggagagtg ttagggatcc agatctgctg tgccctctag ttgccagcca tctgttgttt gccctcccc cgtgccttc ttgacctgg aagggtgccac tcccactgtc ctttctaat aaaatgagga aattgcatcg cattgtctga gtagggtgta ttctattctg gggggtgggg tggggcagga cagcaagggg gaggattggg aagacaatag caggcatgct ggggatgcgg tgggctctat gggtaaccag gtgctgaaga attgaccggt ttccctcctg gccagaaaaga agcaggcaca tccccctctc tgtgacacac cctgtccacg cccctgggtc ttagtccag cccactcat aggacactca tagctcagga gggctccgccc ttcaatccca ccgctaaaag tacatggagc ggtctctccc tccctcatca gccaccacaa ccaaacctag cctccaagag tgggaagaaa ttaagcaag ataggctatt aagtgcagag ggagagaaaa tgccccaac atgtgaggaa gtaatgagag aaatcataga attttaaggc catgatttaa ggccatcatg gccttaactc tccgcttctc

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116	Macaque Heavy chain backbone nucleotide sequence-H02	tcgcgcggtt cgggtgatgac ggtgaaaaacc tctgacacat gcagctcccc gagacggtca cagcttgtct gtaagcggat gccgggagca gacaagcccg tcagggcgcg tcagcgggtg ttggcgggtg tcggggctgg cttaaactatg cggcatcaga gcagattgta ctgagagtgc accatattgc gtgtgaaata ccgcacagat gcgtaaggag aaaataccgc atcagattgg ctattggcca ttgcatacgt tgtatccata tcataaatatg tacatttata ttggctcatg tccaacatta ccgcatggt gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc catataggga gttccgcgtt acataactta cggtaaatgg cccgcctggc tgaccgccc

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SEQ ID NO	Description	Sequence
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		gcccagacatt atcgcgagcc catttatacc catataaatc agcatccatg
		ttggaattta atcgcgccct cgagcaagac gtttccggtt gaatatggct
		cataacaccc cttgtattac tgtttatgta agcagacagt tttattgttc
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		acaacgtggc tttccccccc cccccattat tgaagcattt atcagggtta
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SEQ ID NO	Description	Sequence
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119	Macaque Light chain backbone aa sequence-H02	MGWSCIIFLVATATGVHSAALTPPSVSGSPGQSV TISTGTSSDIGGYNYSWYQHPGKAPKVMIEVS KRPSGVSDRFSGSKSGNIALTISGLQAEDEADYYCS SYAGSNTFLPGGGTRLTVLGQPKAAPSVTLFPPSSEE LQANKATLVCLISDFYPGAVEVANKADGSAVNAGV ETTKPSKQSNKYAASSYLSLTSDQWKSHKSYSCQV THEGSTVEKTVAPAECS
120	Humanized H02 heavy chain backbone nucleotide sequence	tcgcgcgttt cggatgatgac ggtgaaaacc tctgacacat gcagctcccc gagacgggtca cagcttgtct gtaagcggat gccgggagca gacaaagccc tcaggggcgcg tcagcgggtg ttggcgggtg tcggggctgg cttaaactatg cggcatcaga gcagattgta ctgagagatgc accatagcgc gtgtgaaata ccgcacagat gcgtaaggag aaaataccgc atcagattgg ctattggcca ttgcatacgt tgtatccata tcataaatatg tacatttata ttggctcatg tccaaacatta ccgccatgtt gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc catatatgga gttccgcgtt acataactta cggtaaatgg ccgcctggc tgaccgcccc acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga ctttccattg acgtcaatgg gtggagtatt tacggtaaac tgcccacttg gcagtacatc aagtgtatca tatgccaaagt acgcccccta ttgacgtcaa tgacggtaaaa tggcccgctt ggcattatgc ccagtacatg acctatggg actttcctac ttggcagtae atctacgtat tagtcacgcg tattaccatg gtgatgcggt tttggcagta catcaatggg cgtggatagc ggtttgactc acggggattt ccaagtctcc accccattga cgtcaatggg agtttgtttt ggcaccataa tcaacgggac tttccaaaat gtcgtaacaa ctccgcccc ttgacgcaaa tgggcggtag gcgtgtacgg tgggaggtct atataagcag agctcgttta gtgaaccgtc agatcgccgt gagacggcat ccacgctgtt ttgacctcca tagaagacac cgggaccgat ccagcctcca tcggctcgca tctctccttc acgcgccccg cgcctacact gaggcgcccc tccacgcctg ttgagtcgcg tttctgcccgc tcccgcctgt ggtgcctcct gaactgcgtc cgccgtctag gtaagtttaa agctcaggtc gagaccgggc ctttgcctgg cgctcccttg gagcctacct agactcagcc ggctctccac gctttgcctg acctcgttg ctcaactcta gttaacggtg gagggcagtg tagtctgagc agtactcgtt gctgcgcgcg cgcgccaccag acataatagc tgacagacta acagactggt cctttccatg ggtcttttct gcagtcaccg tcgtcgacac gtgtgatcag atatcgccgc cgctctagac caccatggga tgggtcatgta tcactccttt tctagtagca actgcaaccg gtgtacattc ccaactcgag ttgcaggagt cgggcccagg agtggtgaag ccttcggaga cctgtccct cacctgcact atctctggtg gctccttcag tacttactac tggacctgga ttcggcagcc cccagggaag ggaactggag gggttgggta tatcggtaat gggtggtcgta gcctcaacta caaccctcc ctcaagagtc gcatcacct gtcagtagac gcgtccaaga accagttctc cctgaagggt acctctgtga ccgcgcgga caggcccgct tattaacttg ggagagccag gggactccgc ggaaactggt tcgatgtctg gggcccggga gtctggtca ccgtctctc agctagcacc aagggcccat cggctctccc cctggcacc tccccaaga gcacctctgg gggcacagcg gccctgggt gcctgggtcaa ggaactctc ccgaaccgg tgacggtgtc gtggaactca ggccctctga ccagcggcgt gcacaccttc ccggctgtcc tacagtcctc aggaactctac tccctcagca gcgtggtgac cgtgcctcc agcagcttgg gcaccagac ctacatctgc aacgtgaatc acaagcccag caacaccaag gtggacaaga aagttgagcc caaactctgt gacaaaactc acacatgccc accgtgcccc gcacctgaac tccgtggggg accgtcagtc ttcctcttcc ccccaaaacc caaggacacc ctcatgatct cccggacccc tgaggtcaca tgcgtggtgg tggacgtgag ccacgaagac cctgaggtca agttcaactg gtacgtggac ggcgtggagg tgcataatgc caagacaaag ccgcgggagg agcagtacaa cagcacgtac cgtgtggtca gcgtctcac cgtcctgcac caggactggc tgaatggcaa ggagtacaag tgcaaggtct ccaacaaagc cctcccagcc cccatcgaga aaacctctc caaagccaaa gggcagcccc gagaaccaca ggtgtacacc ctgcccccat cccgggatga gctgaccaag aaccaggtca gctgacctg cctggtcaaa ggcttctatc ccagcgacat cgcctgggag tgggagagca atgggcagcc ggagaaccaac tacaagacca cgcctcccg gctggactcc gacggctcct tcttctctca cagcaagctc accgtggaca agagcagggtg gcagcagggg aacgtcttct catgctccgt gatgcagag gctctgcaca accactacac gcagaagagc ctctccctgt ctccgggtaa atgatagga tccagatctg ctgtgccttc tagttgccag ccatctgttg tttgcctctc ccccgctgct tccttgacct tggaggtgac cactccact gtccttctct aataaaatga ggaattgca tcgcatgtc tgagtgggtg tcattctatt ctgggggggtg ggggtgggca ggacagcaag ggggaggatt gggaagacaa tagcaggcat gctggggatg cgggtgggctc tatgggtacc caggtgctga agaattgacc cggttcctcc tgggccagaa agaagcaggc acatccccct ctctgtgaca caccctgtcc acgccccctg ttcttagttc cagccccact cataggacac tcatagctca ggagggtcc gccttcaatc ccaccgccta aagtacttgg agcgggtctc cctccctca tcagccacc

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121	Humanized H02 heavy chain backbone aa sequence	MGWSCIIILFVATATGVHSQLQLQESGPGVVKPSET LSLTCTISGGSFSTYYWTWIRQPPGKLEWVGIGN GGRSLNYPNLSKSRITLSVDASKNQFSLKVTSVTAAD TAVYYCGRARGLRGNWFDVWGPGLVTVSSASTK GPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSW NSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGT QTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPA PELLGGPSVFLFPPKPKDTLMISRTPEVTCVVVDVSH EDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVV SVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKA KGQPREPQVYTLPPSRDELTKNQVSLTCLVKGFYPS DIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLT VDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPG K
122	Humanized H02 light chain backbone aa sequence	tcgcgcgttt cggatgatgac ggtgaaaacc tctgacacat gcagctcccg gagacggtca cagcttgtct gtaagcggat gccggggagca gacaagcccc tcaggcgcg cgacggggtg ttggcgggtg tcggggctgg cttactatg cgccatcaga gcagattgta ctgagagtg accatatgag gtgtgaata ccgcacagat cgttaaggag aaaataccgc atcagattgg ctattggcca ttgcatacgt tgtatccata tcataaatat tacatttata ttggctcatg tcccaacta ccgccatgtt gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc catatatgga gttccgcgtt

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123	Humanized H02 light chain backbone aa sequence	MGWSCIIILFLVATATGVHSQAALTQPPSVSGSPGQSV TISCTGTSSDIGYNYVSWYQQHPGKAPKVMIEVS KRPSGVSDRFGSGKSGNIALSTISGLQAEDEADYYCS SYAGSNTFLFGGGTRLTVLGQPKAAPSVTLFPPSSEE LQANKATLVCLISDFYPGAVTVAKADSSPVKAGV ETTTPSKQSNKYAASSYLSLTPEQWKSHRSYSCQV THEGSTVEKTVAPECS
124	EBV gp350 GenBank: AIM62208.1	MEAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTC NVCTADVNTINFDVGGKKHQLDLDFGQLTPHTKAVYQP RGAFGGSENATNLFLELLGAGELALTMRSKKLPINVT GEEQQVSLESVDVYFQDVFMTMCHHAEMQNPVYLIPET VPIKWDNCNSTNITAVVRAQGLDVTLPPLSLPTSAQDSN FSVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSG YESHVPSSGILSTSPVATPIPGTGYAYSLRLTPRPVSR FLGNNSILYVYFSGNGPKASGGDYCIQSNIVFSDEIPAS QDMPTNTDITYVGDNATYSVPMVTSEDANSPNVTVTA WAWPNTTETDFKCKWLTLSGTPSGCENISGAFASNRFTD ITVSGLTAPKTLIIITRTATNATTTTHKVFPSKAPESST TSPTSNTTGFAAPNTTTGLPSSTHVPNTLTAPASTGPTV STADVTSPTPAGTTSASPVTSPSPRDNGETESKAPDMT SPTSAVTTPTPNATSPSAVTTPTPNATSPTPAVTTPTP NATSPTLGKTSPTS AVTTPTPNATSPTLGKTSPTS AVTT PTPNATSPTLGKTSPTS AVTTPTPNATSPTVGETSPQAN TTNHTLGSTSPVVTSPPKNATSAVTTGQHNTSSSTS SMSLRPSSISSETLSPSTSDNSTSHMPLLTSAHPTGGENI TQVTPASTSTHHVSTSSPAPRPGTTSQASGPGNSSTSK PGEVNVTKGTPPKNATSPQAPSGQKTAVPTVTSGGKAN STTGKHTTGHGARTSTEPTDYGSDSTTPRTRYNATY LPPSTSSSELPRWTFSTSPVTTAQATVPVPPTSQPRFSN LSMLVLQWASLAVLTLLLLVMADCAFRRLSTSHTYTT PPDDAETTYV
125	gp350 B95-8 amino acids 2-425 Avi	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLESVDVYFQDVFMTMCHHAEMQNPVYLIPETV PYIKWDNCNSTNITAVVRAQGLDVTLPPLSLPTSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVYFSGNGPKASGGDYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTA AWPNTTETDFKCKWLTLSGTPSGCENISGAFASNRFTDI TVSGLTAPKTLIIITRTATNATTTTHKVFPSKAPGS
126	NAM encoding SEQ ID NO: 125	gaagctgccctgctcgtgtgccagtaacacatccagagcctgatccacctgaccg gcgaggaccccgctctctcaacgtggaaatccccagagttccccctctacccctacct gcaacgtgtgcaccgcgacgtgaacgtgacctcaacttcgacgtggcgcgga agaagcaccagctggacctggatttcggccagctgacccctcacaccaaggccgt gtatcagcccgagggcgcttctggcgcgagcgagaaacccaccaatctgtttctgc tggaaactcaggcgccggcgagctggccctgacctgagaagcaagaaactgc ccatcaatgtgaccacaggcgagggaacagcaggtgtccctggaaagcgtgg acg tgtacttcaagacgtgttcggcaccatgtggtgccaccacgcccagatgcagaa cccgtgtacctgatccccgagacagtgccctacatcaagtgggacaactgcaaca gcaccaacatcaccgcgctcgtgcgggcccagggaactggatgtgacactgcctct gagcctgcctaccagcgcacaggacagcaacttcagcgtgaaaaccgagatgct

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127	gp350 B95-8 (2-425) Q122N*Avi	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVITINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQNVSLVESVDVYFQDVFGTMWCHHAEMONPVYLIPETV PYIKWDNCNSTNITAVVRAQGLDVTLPPLSLPTSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGDYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTAFW AWPNTTETDFKCKWTLTSGTPSGCENISGAFASNRTFDI TVSGLGTAPKTLIIITRTATNATTTTHKVI FSKAPGS
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131	gp350 B95-8 (2-425) D296R Avi	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLESVDVYFQDVFGTMWCHHAEMQNPVYLIPETV PYIKWDNCNSTNITAVVRAQGLDVTLPPLSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTAFW AWPNNTEPDFKCKWLTSTGTPSGCENISGAFASNRTFDI TVSGLTAPKTLII TRTATNATTTTHKVI FSKAPGS
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137	gp350 B95-8 (2-425) D296R/P158R Avi	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVITINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLESVDVYFQDVFGTMWCHHAEMQNPVYLIPETV RYIKWDNCNSTNITAVVRAQGLDVTLPPLSLPTSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNISILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTAFW AWPNNTEPDFCKWTLTSGTPSGCENISGAFASNRTFDI TVSGLGTAPKTLIIITRTATNATTTTHKVI FSKAPGS
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139	gp350 B95-8 (2-425) D296R/P158N*/I160T Avi	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLSESDVYFQDVFGTMWCHHAEMONPVYLIPETV NYTKWDNCNSTNITAVVRAQGLDVTLP LSLP TSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTTDTITYVDNATYSVPMVTS EDANSPNVTVTAFW AWPNNTE TDFKCKWTLTSGTPSGCENISGAFASNRTFDI TVSGLGTAPKTLII TRTATNATTTTHKVI FSKAPGS
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141	gp350 B95-8 (2-425) D296R/P158R/I160R Avi	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLSESDVYFQDVFGTMWCHHAEMONPVYLIPETV RYRKWDNCNSTNITAVVRAQGLDVTLP LSLP TSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTTDTITYVDNATYSVPMVTS EDANSPNVTVTAFW AWPNNTE TDFKCKWTLTSGTPSGCENISGAFASNRTFDI TVSGLGTAPKTLII TRTATNATTTTHKVI FSKAPGS
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143	gp350 B95-8 (2-425) Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLESVDVYFQDVFGTMWCHHAEMQNPVYLIPETV PYIKWDNCNSTNITAVVRAQGLDVTLPPLSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGDYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTAFW AWPNNTEPDFKCKWLTSGTPSGCENISGAFASNRFTDI TVSGLTAPKTLII TRTATNATTTTHKVI FSKAPGSGLN DIFEAQKIEWHEHHHHHH
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145	gp350 B95-8 (2-425) Q122N* Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQNVLESVDVYFQDVFGTMWCHHAEMQNPVYLIPETV PYIKWDNCNSTNITAVVRAQGLDVTLPPLSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGDYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTAFW AWPNNTEPDFKCKWLTSGTPSGCENISGAFASNRFTDI TVSGLTAPKTLII TRTATNATTTTHKVI FSKAPGSGLN DIFEAQKIEWHEHHHHHH
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147	gp350 B95-8 (2-425) D163N*/164bS Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVITINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLESVDVYFQDVFGTMWCHHAEMQNPVYLIPETV PYIKWNNSCNSTNITAVVRAQGLDVTLPPLSLPTSAQDSN FSVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSG YESHVPSSGILTSTSPVATPIPGTGYAYSLRLTPRPVSR FLGNNSILYVFYSGNGPKASGGDYCIQSNIVFSDEIPAS QDMPTNTTDITYVGDNATYSVPMVTSSEDANSPNVTVTAF WAWPNTTETDFKCKWTLTSGTPSGCENISGAFASNRFTD ITVSGLGTA PKTLIIITRTATNATTTTHKVFISKAPGSGL NDIFEAKQIEWEHHHHHHH
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151	gp350 B95-8 (2-425) WDN162-4AAA Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLESVDVYFQDVFGTMWCHHAEMQNPVYLIPETV PYIKAAA CNSTNITAVVRAQGLDVTLP LSLP TSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNN SILYVFYSGNGPKASGGDYCIQSNIVFSDEIPASQ DMPNTNTDITYVGD NATYSVPMVTS EDANSPNVTVTAFW AWPNN TETDFKCKWLTLSGT PSCENISGAFASNRTPDI TVSGLGTAPKTLII TRTATNATTTTHKVI FSKAPGSGLN DIFEAQKIEWHEHHHHHH
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153	gp350 B95-8 (2-425) D296R/I160R Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVITINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLSDVYFQDVFGTMWCHHAEMQNVPVYLIPETV PYRKWDNCNSTNITAVVRAQGLDVTLP LSLPTSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSRLRTPRPVSRF LGNNSILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTNTDITYVGDNATYSVPMVTSEDANSPNVTVTAFW AWPNNTEITDFKCKWTLTSGTPSGCENISGAFASNRTFDI TVSGLTAPKTLIIITRTATNATTTTHKVI FSKAPGSGLN DIFEAQKIEWHEHHHHHH
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155	gp350 B95-8 (2-425) D296R/P158R Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVITINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLSDVYFQDVFGTMWCHHAEMQNVPVYLIPETV

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157	gp350 B95-8 (2-425) D296R/P158N*/I160T Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLPHTKAVYQPR GAFGGS ENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSL ESVDVYFQDVFGTMWCHHAEMQNVPVLI PETV NYTKWDNCNSTNITAVVRAQGLDVTLP LSLPTS AQDSNF SVKTEMLGNEIDIECIMEDGEISQVLP GDNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNN SILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTTDDITYVGDNATYSVPMVTS EDANSPNVTVTAFW AWPNNTE TDFKCKWLTSGTPSGCENISGAFASNRTFDI TVSGLGTAPKTLII TRTATNATTTTHKVIFSKAPGSGLN DIFEAQKIEWHEHHHHHH
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SEQ ID NO Description		Sequence
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159	gp350 B95-8 (2-425) D296R/P158R/I160R Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEFFPYPTCN VCTADVNVTINFDVGGKKHQLDLDFGQLTPHTKAVYQPR GAFGGSENATNLFLELLGAGELALTMRSKKLPINVTG EEQQVSLSEVDVYFQDVFGTMWCHHAEMQNPVYLIPETV RYRKWDNCNSTNITAVVRAQGLDVTLP LSLPTSAQDSNF SVKTEMLGNEIDIECIMEDGEISQVLPDGNKFNITCSGY ESHVPSGGILTSTSPVATPIPGTGYAYSLRLTPRPVSRF LGNNSILYVFYSGNGPKASGGRYCIQSNIVFSDEIPASQ DMPNTNTDITYVDNATYSVPMVTSEDANSPNVTVAFW AWPNNTEITDFKCKWLTSTGTPSGCENISGAFASNRTFDI TVSGLGTAPKTLII TRTATNATTTTHKVI FSKAPGSLN DIFEAQKIEWEHHHHHHH
160	NAM encoding SEQ ID NO: 159	gaagctgccctgctcgtgtgccagtacaccatccagagc ctgatccacctgaccggcgaggacccccggcttcttcaac gtggaaatccccgagttccctcttaccctacctgcaac gtgtgcaccgcccagctgaacgtgacctcaacttcgac gtggggcggaagaagcaccagctggacctggatttcggc cagctgacctctcacaccaaggccgtgtatcagcccaga ggccctttggggcgagcgagaacgccaccaatctgttt ctgctggaactcctaggcgccggcgagctggccctgacc atgagaagcaagaactgcccatcaatgtgaccacaggc gaggaacagcaggtgtccctggaaagcgtggacgtgtac tttcaagacgtgttcggcaccatgtggtgccaccacgcc gagatgcagaaccccgtagctgatccccgagacagtgc cggtagccgaagtgggacaactgcaacagcaccaacatc accgcgctcgtgcccccgaggactggatgtgacactg cctctgagcctgcctaccagcggccaggaagcaacttc agcgtgaaaaaccgagatgctgggcaacgagatcgacatc gagtgcacatggaagatggcgagatcagccaggtgctg cccggcgacaacaagtccaacatcacatgcagcggctac gagagccacgtgccatctggcggcatcctgaccagcaca agcccagtgggcacacccatccctggcacaggctacgcc tacagcctgagactgacccccagaccgtgtccagattc ctgggcaacaacagcatcctgtacgtgttctacagcggc aacggccccaaggcctctggcggcggtagtgcacccag agcaacatcgtgttcagcgacgagatccccgccagccag gacatgccaccaataaccaccgacatcacgtacgtgggc gacaatgccacctacagcgtgccaatggtcacctccgag gacgccaacagccccaacgtgaccgtgacagccttttg gcctggcctaacaacaccgagacagacttcaagtgaag tggacctgacctccggcacccctagcggctgcgagaat atcagcggagccttcggcagcaacccggacctcgatc accgtgtctggcctgggcaccgcccccaagacctgatc atcaccagaaccgcccacaaatgccaccaccacaaccac aaagtgccttcagcaaggcccccggtctggcctgaac gacatttttgaggccagaagattgagtggcatgaacat caccaccaccacat
161	E7 clone Heavy Chain aa sequence	QVQLVQSGADVRKPGASVKVSKASTYIFTGYYIH WVRQAPGRGLEWLGWIHPNSGGTTYSQMFGGRVT MTRDRSITTSYMELSR LQSDDTAIYYCATLRFVEYSF DSWGQGLVTVSS
162	E7 clone Heavy Chain nucleotide sequence	CAGGTGCAGTGGTGCAGTCTGGCGCAGACGTGA GGAAGCCAGGAGCCTCCGTGAAGGTGTCTTGTA GGCCAGCACCTACATCTTACAGGCTACTATATCC ACTGGGTGAGGCAGGCACCAGGAAGGGCCTGGA

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SEQ ID NO	Description	Sequence
		GTGGCTGGGCTGGATTCAACCTAACTCTGGCGGCA CCACATACAGCCAGATGTTTCAGGGCAGAGTGAC CATGACACGGGACAGATCCATCACCACATCTTATA TGGAGCTGAGCCGGCTGCAGTCCGACGATACCGC CATCTACTATTGCGCCACACTGAGATTCGTGGAGT ATTCTTTGATAGCTGGGGCCAGGGCACCTGGTG ACAGTGAGCTCC
163	E7 clone Heavy Chain CDR1	TYIFTGY
164	E7 clone Heavy Chain CDR2	HPNSGG
165	E7 clone Heavy Chain CDR3	LRFVEYSFDS
166	E7 clone Kappa Chain aa sequence	EIVLTQSPGTLSSLSPGERATLSCRASQSISSSTYLAWYQ QIPGQAPRLLIYGASSRAAGIPDRFSGGGSGTDFTLTI SRLEPEDFAVYYCQQYGSSSPRSPGQGTKLEIK
167	E7 clone Kappa Chain nucleotide sequence	GAGATCGTGCTGACCCAGAGCCAGGCACACTGA GCCTGTCCCCAGGAGAGAGGGCCACCCTGTCTGT AGAGCCTCTCAGAGCATCAGCTCCACATACCTGGC CTGGTATCAGCAGATCCAGGACAGGCACCTAGG CTGCTGATCTACGGAGCCTCTAGCAGGGCAGCAG GCATCCCCGACCGCTTCTCCGGCGGAGGCTCTGGC ACCGACTTCACCTGACAACTCTCTCGGCTGGAGCC TGAGGACTTCGCCGTGTACTATTGCCAGCAGTATG GCTCCTCTCCAAGGTCCTTTGGCCAGGGCACAAG CTGGAGATCAAG
168	E7 clone Kappa Chain CDR1	RASQSISSSTYLA
169	E7 clone Kappa Chain CDR2	GASSRAA
170	E7 clone Kappa Chain CDR3	QQYGSSPRS
171	B8 clone Heavy Chain aa sequence	EVQLLESQGGALVQPGGSLRLSCAASGFTFKTYAMSW VRQVPKGLEWVSAISGSGTASYVADSVKGRFTLSR DNSKNTLYLQLSSLRDEDTGVYYCARRFLDWFGMD VWGLGTTVTVSS
172	B8 clone Heavy Chain nucleotide sequence	GAGGTGCAGCTGCTGGAGAGCGGCGGCCCTTGG TGCAGCCAGGAGGCAGCCTGCGGCTGTCTGTGCC GCCTCTGGCTTCACCTTTAAGACATACGCCATGTC CTGGGTGAGGCAGGTGCCTGGCAAGGGCCTGGAG TGGGTGTCTGCCATCTCCGGCTCTGGCACCGCCTC TTACTATGCCGACAGCGTGAAGGGCAGGTTACCC TGAGCCGCGATAACTCCAAGAATACACTGTATCTG CAGCTGAGCTCCCTGCGGGACGAGGATACCGGCG TGTAATATTGCGCCCGGAGATTCTGGACTGGTTT GGCATGGACGTGTGGGCTTGGGCACACAGTGA CAGTGCTCTAGC
173	B8 clone Heavy Chain CDR1	GFTFKTY
174	B8 clone Heavy Chain CDR2	SGSGTA
175	B8 clone Heavy Chain CDR3	RFLDWFGMDV
176	B8 clone Kappa Chain aa sequence	DIVMTQSPLSLPVTPGEPASISCLSSQSLQSNQNYV DWYLQKPGQSPQLLIYLGSNRASGVDPDRFSGSGSGT DFTLKISRVEAEDVGVYYCMVTLHPPTFGQAKVEI K
177	B8 clone Kappa Chain nucleotide sequence	GACATCGTGATGACCCAGTCCCCTCTGTCTTGCC AGTGACACCCGGCGAGCCTGCCTCTATCAGCTGTC TGAGCTCCAGAGCCTGCTGCAGTCCAACGGCTAC AATTATGTGGATTGGTACCTGCAGAAGCCAGGCC AGTCCCCCAGCTGCTGATCTATCTGGGCTCTAAC AGGGCCAGCGGCGTGCCCGACAGATTCTCCGGCT CTGGCAGCGGACCGACTTCACCTGAAGATCTCT CGGGTGAGGCAGAGGACGTGGGCGTGTAATATT GCATGGTGACCCCTGCACCCACCTACATTCTGGCCAG GGAGCCAAGGTGGAGATCAAG
178	B8 clone Kappa Chain CDR1	LSSQSLQSNQNYVD

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SEQ ID NO	Description	Sequence
179	B8 clone Kappa Chain CDR2	LGSNRAS
180	B8 clone Kappa Chain CDR3	MVTLHPPT
181	B3 clone Heavy Chain aa sequence	EGQLVQSGGGLVQPGGSLTLSCVSGFTFKNYEMN WVRQAPGKGLEWVSYSISGGIAIFHADSVKGRFTVS RDNAKNLLYLQMNSLRVEDTAVYYCARDENNVR PFDHWGQGTLTVTSS
182	B3 clone Heavy Chain nucleotide sequence	GAGGGACAGCTGGTGCAGTCCGGCGGAGGCCTGG TGCAGCCAGGAGGCTCCCTGACCTGTCTTGTGAG GTGAGCGGCTTACCTTCAAGAACTACGAGATGA ATTGGGTGCGGCAGGCACCTGGCAAGGCCTGGA GTGGGTGTCTTATATCAGCTCCGGCGGAATCGCAA TCTTCCACGCAGATTCCGTGAAGGGCAGGTTTACC GTGTCTCGCGACAACGCCAAGAATCTGCTGTACCT GCAGATGAACAGCCTGCGGGTGGAGGACACAGCC GTGTACTATTGCGCCAGGGATGAGAACACGTGC GGCGCCCTTCGACCACTGGGACAGGGCACCCCT GGTGACAGTGTCTAGC
183	B3 clone Heavy Chain CDR1	GFTFKNY
184	B3 clone Heavy Chain CDR2	SSGGIA
185	B3 clone Heavy Chain CDR3	DENNVRPFDH
186	B3 clone Lambda Chain aa sequence	QSVLTQPPSASGSPGQSVTISCTGSSSDVGAYDFVSW FQYYPGQAPKLIIEVNKRPSGVPARFSGSKSGNTAS LTVSGLQAEDEADYFCFSYGGTTNLRVFGGKLT
187	B3 clone Lambda Chain nucleotide sequence	CAGTCTGTGCTGACCCAGCCACCTAGCGCCTCCGG CTCTCCCGGCCAGAGCGTGACCATCTCCTGTACAG GCAGCTCCTCTGACGTGGCGCCTACGATTTCGTG AGCTGGTTTCAGCAGTATCCAGGCCAGGCCCCCAA GCTGATCATCTACGAGGTGAACAAGCGCCTTCCG GCGTGCCAGCCAGATTGAGCGGCTCCAAGTCTGGC AATACCGCCTCTCTGACAGTGAGCGGCTGCAGGC AGAGGACGAGGCAGATTACTTCTGCTTTCTTATG GCGGCACCACAAACCTGCGGGTGTTTGGCGGCGG CACCAGCTGACA
188	B3 clone Kappa Chain CDR1	TGSSSDVGAYDFVS
189	B3 clone Kappa Chain CDR2	EVNKRPS
190	B3 clone Kappa Chain CDR3	FSYGGTTNLRV
191	C7 clone Heavy Chain aa sequence	EVQLVESGGNLVQPGASLRSLCTASRFNFKYAMH WVRQTPGKGLEWVS AISWDSTYIDYNSVKGRFTIS RDNTRNSLYLQMNSLTAEATLYYCAKCEDYLRLC SAYDIWGHGTMVTSS
192	C7 clone Heavy Chain nucleotide sequence	GAGGTGACAGCTGGTGGAGAGCGGCGGAAACCTGG TGCAGCCAGGAGCCTCTCTGAGGCTGAGCTGTACC GCCTCCCGCTTCAACTTTAATAAGTACGCAATGCA CTGGGTGCGGCAGACCCCTGGCAAGGCCTGGAG TGGGTGTCTGCCATCAGCTGGGACTCCACATACAT CGATTATGGCAACTCCGTGAAGGGCAGGTTTACC ATCTCTCGGGACAACACAAGAAATAGCCTGTATCT GCAGATGAATTCCCTGACCGCCGAGGATACAGCC CTGTACTATTGCGCCAAGTGTGAGGACTACCTGCG GCTGTGCTCTGCCTATGATATCTGGGCCCACGGCA CCATGGTGACAGTGAGCTCC
193	C7 clone Heavy Chain CDR1	RFNFKNY
194	C7 clone Heavy Chain CDR2	SWDSTY
195	C7 clone Heavy Chain CDR3	CEDYLRLC SAYDI
196	C7 clone Kappa Chain aa sequence	DIVMTQSPLSLPVP TPGESASISCRSSQSLLSNGKNYL SWYLQKPGQSPQLLIDLSNRASGVSDRFGSGSGT DFTLKISRVEADVG VYYCMQAVQTPI TFGQGTRLA IK

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SEQ ID NO	Description	Sequence
197	C7 clone Kappa Chain nucleotide sequence	GACATCGTGATGACCCAGTCTCCTCTGAGCCTGCC CGTGACACCTGGCGAGTCTGCCAGCATCTCCTGTC GGAGCTCCCAGAGCCTGCTGCACTCCAACGGCAA GAATTACCTGTCTTGGTATCTGCAGAAGCCAGGCC AGAGCCCCAGCTGCTGATCGATCTGGGCTCCAAC AGGGCCTCCGGCGTGTCTGACAGATTCTCTGGCAG CGGCTCCGGCACCAGACTTACCCCTGAAGATCAGCA GGGTGGAGCCGACGATGTGGCGTGTAATTG CATGCAGGCCGTGCAGACCCCAATCACATTCCGGCC AGGGAACCCGCTGGCCATCAAG
198	C7 clone Kappa Chain CDR1	RSSQSLHNSNGKNYLS
199	C7 clone Kappa Chain CDR2	LGSNRAS
200	C7 clone Kappa Chain CDR3	MQAVQTPIT
201	A9 clone Heavy Chain aa sequence	QVQLVQSGAELKTPGASVKVSKASGYFTGYIYH WVRQAPGEGLEWTGWINPNSGATRYGQKFGSRVTL TSDTSSSTVYMEVSNLTSDDSAVYVCARELSYSIRGT GPLGYWGLGTLVTVSS
202	A9 clone Heavy Chain nucleotide sequence	CAGGTGCAGCTGGTGCAGTCCGGCGCAGAGCTGA AGACCCAGGAGGCCAGCGTGAAGGTGTCTCTGTAA GGCCTCTGGCTACACCTTACAGGCTACTATATCC ACTGGGTGCGGCAGGCACCAGGAGAGGGCTGGA GTGGACCGGCTGGATCAACCTAATAGCGCGCC ACAAGATACGGCCAGAAGTTTACGGGCCGCGTGA CCCTGACAAGCGACACCAGCTCCTCTACAGTGTAT ATGGAGGTGTCCAACCTGACCTCCGACGATTCTGC CGTGTACTATTGCGCCCGGAGCTGTCTTACAGCA TCAGAGGAACAGGACCCTGGGATATTGGGGCCT GGGCACCCTGGTGACAGTGAGCTCC
203	A9 clone Heavy Chain CDR1	GYTFTGY
204	A9 clone Heavy Chain CDR2	NPNSGA
205	A9 clone Heavy Chain CDR3	ELSYSIRGTGPLGY
206	A9 clone Kappa Chain aa sequence	EIVLTQSPGTLSSLSPGERATLSCRASQSVASKYLAWY QQKPGQAPRLLIYGASRATGIPDRFSGSGSGTDFTL TISRLEPEDFAVYYCQYYGSSPLTFGQGTKVEIK
207	A9 clone Kappa Chain nucleotide sequence	GAGATCGTGCTGACCCAGTCTCCAGGCACACTGTC CCTGTCTCCAGGAGAGAGGGCCACCCTGTCTTGTA GAGCCAGCCAGTCCGTGGCCAGCAAGTACCTGGC CTGGTATCAGCAGAAGCCAGGACAGGCACCTAGG CTGCTGATCTACGAGCCAGCTCCAGGGCAACCG GCATCCCCGACCGCTTCTCTGGCAGCGGCTCCGGC ACAGACTTCACCTGACAATCTCCAGGCTGGAGCC TGAGGACTTCGCCGTGTAATTGCCAGTACTATG GCTCTAGCCCACTGACCTTTGGCCAGGGCACAAG GTGGAGATCAAG
208	A9 clone Kappa Chain CDR1	RASQSVASKYLA
209	A9 clone Kappa Chain CDR2	GASSRAT
210	A9 clone Kappa Chain CDR3	QYYGSSPLT
211	A2 clone Heavy Chain aa sequence	EVQLAESGGGVVHPGSLRLSCTASGFTFSRHSMTW VRQAPGKLEWVAVISHDGSHKFIYDSVKGRFSISR DNAKNTLYLQMSLSGADTAVYYCVKDISSRSYGY LAGDSWGQGS LVTVSS
212	A2 clone Heavy Chain nucleotide sequence	GAGGTGCAGCTGGCCGAGTCTGGCGGAGGAGTGG TGCACCCAGGAGGCTCCCTGAGGCTGTCTTGTAAC GCCAGCGGCTTACATTTCTAGGCACAGCATGCA CTGGGTGCGCCAGGCACCTGGCAAGGGCCTGGAG TGGGTGGCCGTGATCTCCACGACGGCTCTCACAA GTTCTACGTGGATTCCGTGAAGGCGCGTTTAGCA TCTCCAGAGACAACGCCAAGAATACCTGTATCTG CAGATGAGCTCCCTGTCTGGCGCCGACACAGCCGT GTACTATTGCGTGAAGGATATCTCTAGCAGGAGCT

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SEQ ID NO	Description	Sequence
		ACGGCTATCTGGCAGGCGATAGCTGGGGACAGGG CTCCCTGGTGACCGTGTCCTCT
213	A2 clone Heavy Chain CDR1	GFTFSRH
214	A2 clone Heavy Chain CDR2	SHDGS
215	A2 clone Heavy Chain CDR3	DISSRSYGYLAGDS
216	A2 clone Kappa Chain aa sequence	DIQMTQSPSSLSASVGDIIITITCRASQSVVTVLNWYQ QKPGGAPRLLIYTTSKLQSGVPSRFRSGSGTFLTLTI NGLRPEDFATYYCQSYGTPPFTFGPGTRVEIN
217	A2 clone Kappa Chain nucleotide sequence	GATATTGAGATGACTCAGTCCCCAAGCAGCCTGAG CGCTCCGTGGGCGACATCATCACCATCACATGCA GGGCTCTCAGAGCGTGGTGACCTACCTGAAGTGG TATCAGCAGAAAGCCAGGAGGAGCACCTAGGCTGC TGATCTACACCACATCCAAGCTGCAGTCTGGCGTG CCATCCAGATTCTCCGGCTCTGGCAGCGCACCT GTTTACCCCTGACAATCAATGGCTGCGGCCCCGAGG ATTTGCGCACATACTATTGTGAGCAGCTATGGA ACCCCCCTTTACTTTTGACCAGGCACAGAGT GGAGATTAAC
218	A2 clone Kappa Chain CDR1	RASQSVVTVLN
219	A2 clone Kappa Chain CDR2	TTSKLQS
220	A2 clone Kappa Chain CDR3	QQSYGTPPFT
221	E1 clone Heavy Chain aa sequence	QVHLQQWAGLVKPSETLSLTCVQGGPFSGYWS WIRQPPGKLEWIGIEINHSNTHYNPSLKSRTISVD TSGNYFSLKLTSTVTAADAAYFCARGQQLLRNYYY YSGMDVWGQGTITVTVSS
222	E1 clone Heavy Chain nucleotide sequence	CAGGTGCACCTGCAGCAGTGGGGAGCAGGCCTGG TGAAGCCATCCGAGACACTGTCTCTGACATGTGCA GTGCAGGGAGGACCTTCTCTGGCTACTATTGGAG CTGATCAGGCAGCCACCTGGCAGGGCCTGGAG TGGATCGCGAGATCAACCACAGCGCAATACCC ACTACAACCCCTCTCTGAAGAGCCGGTGACCATC AGCGTGACACATCCGCAATTACTTCTCCCTGAA GCTGACCTCTGTGACAGCGCGGATGCCGCGTGT ATTTTTCGCGCCGGGGCCAGCAGCTGCTGAGAAAC TACTATTACTATTCCGCATGGACGTGTGGGGACA GGGAACACAGTGACAGTGAGCTCC
223	E1 clone Heavy Chain CDR1	GGPFSGY
224	E1 clone Heavy Chain CDR2	NHSGN
225	E1 clone Heavy Chain CDR3	GQQLLRNYYYYSGMDV
226	E1 clone Kappa Chain aa sequence	EIVLTQSPGTLSLSPGERATLSCRASQSVTSTYLA WYQKLGQPPRLLIIFGASNRATGIPDRFSGSGTDFTLT ITRLEPEDFAVYYCQRYGGSITFGQGRLEIK
227	E1 clone Kappa Chain nucleotide sequence	GAGATCGTGCTGACACAGTCCCCAGGCACCTGA GCCTGTCCCCAGGAGAGCGGGCCACTGTCTGT AGAGCCTCTCAGAGCGTGACCTCTACATACCTGGC CTGGTATCAGCAGAAGCTGGGCCAGCCCCCTAGG CTGCTGATCTTCGGCGCTCTAACAGGGCCACAGG CATCCCTGACCGCTTCTCCGGCTCTGGCAGCGGCA CCGACTTCACCCCTGACAATCACCAGACTGGAGCCC GAGGACTTCGCCGTGTACTATTGCCAGCGGTACGG CGGCAGCATCACATTTGGCCAGGGCACAGACTG GAGATCAAG
228	E1 clone Kappa Chain CDR1	RASQSVTSTYLA
229	E1 clone Kappa Chain CDR2	GASNRAT
230	E1 clone Kappa Chain CDR3	QRYGGSIT
231	C10 clone Heavy Chain aa sequence	EVQLVQSGGQVQPRRLRLSCAASGFTFSNYGMH WVRQVPGKGLQWVAIIWYDGSNKHAAASVQGRFRI

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SEQ ID NO	Description	Sequence
		SRDNSKNTVYLQMDGLRAEDTGMYYCVRDATTAT TEGTSQYYFDLWGQALVTVSS
232	C10 clone Heavy Chain nucleotide sequence	GAGGTGCAGCTGGTGCAGTCCGGCGGAGGAGTGG TGCAGCCACGGAGATCTCTGAGGCTGAGCTGTGCC GCCTCCGGCTTACCTTTTCTAACTACGGAATGCA CTGGGTGCGCCAGGTGCTTGGCAAGGCGCTGCAG TGGGTGGCCATCATCTGGTACGACGGCTCCAATAA GCACTATGCCGCTCTGTGCAGGGCAGGTTCCGCA TCTCTCGGATAACAGCAAGAATACCGTGTATCTG CAGATGGACGGCCTGCGGGCCGAGGATACAGGCA TGTAATTATGCGTGAGAGACGCCACCACAGCCACC ACAGAGGGCACCCAGCCAGTACTATTTTGATCTGTG GGGACAGGCGCCCTGGTGACAGTGAGCTCC
233	C10 clone Heavy Chain CDR1	GFTFSNY
234	C10 clone Heavy Chain CDR2	WYDGSN
235	C10 clone Heavy Chain CDR3	DATTATTEGTSQYYFDL
236	C10 clone Kappa Chain aa sequence	DIVMTQSPDSLAVSLGERATINCKSSQTLTYTNSKN YLAWYQQKVGQPPRLLIYWASTRESGVPDRFSGSGS GTDFTLTISLLAEDVAVYYCQQYYTTPFTFGGGTK VEVK
237	C10 clone Kappa Chain nucleotide sequence	GACATCGTGATGACCCAGAGCCCGATTCCCTGGC CGTGTCTCTGGGAGAGAGGGCAACAATCAACTGT AAGAGCTCCCAGACCCCTGCTGTACACATCCAACCTC TAAGAATTACCTGGCCTGGTATCAGCAGAAAGTG GGACAGCCACCTAGGCTGCTGATCTATTGGGCTC TACCAGGGAGAGCGGCTGCCAGACAGATTACAGC GGCTCCGGCTCTGGCAGAGACTTCAACCTGACAAAT CTCTAGCCTGCTGGCCGAGGACGTGGCCGTGTACT ATTGCCAGCAGTACTATACCACACCCCTGACCTTC GGCGGCGGCACAAAGGTGGAGGTGAAG
238	C10 clone Kappa Chain CDR1	KSSQTLTYTNSKNYLA
239	C10 clone Kappa Chain CDR2	WASTRES
240	C10 clone Kappa Chain CDR3	QYYTTPFT
241	B12 clone Heavy Chain aa sequence	EVQLVESGGGVVHPGKSLTISCEASGFTFNDHGIHW VRRAPGKLEWLALISKDGSKEYSTDVSKGRFTVSR DNSRNTVFLQMKSLTTEDTAIYYCAKMDMQCSSPSC STMDSYFAMDVWGQTTIVISS
242	B12 clone Heavy Chain nucleotide sequence	GAGGTGCAGCTGGTGGAGTCCGGCGGAGGAGTGG TGCACCCTGGCAAGTCTCTGACCCTGAGCTGTGAG GCCAGCGGCTTACCTTCAACGACCACGGCATCCA CTGGGTGCGGAGAGCACTTGGCAAGGCGCTGGAG TGGCTGGCCCTGATCTCTAAGGACGGCAGCAAGG AGTACAGCACCGATTCCGTGAAGGGCCGGTTTAC AGTGTCCAGGGATAACTCTCGCAATACCGTGTTC TGCAGATGAAGTCTCTGACCACAGAGACACAGC CATCTACTATTGCGCCAAGGATATGGGCCAGTGCA GCTCCCCCTCCTGTCTTACCATGGACAGCTATTTC GCAATGGACGTGTGGGACAGGGAACCACAGTGA TCGTGTCTAGC
243	B12 clone Heavy Chain CDR1	GFTFNDH
244	B12 clone Heavy Chain CDR2	SKDGSK
245	B12 clone Heavy Chain CDR3	DMGQCSSPSCSTMDSYFAMDV
246	B12 clone Kappa Chain aa sequence	DIVMTQSPSLPVPVTPGEPASISCRSSQNLRHNNGYNY LNWYLQKPGQSPQLLIYLGSIKASGVPDRFSGSGSGT DFTLKISRVEAEDVGVYYCMQALQTPPWTFGQGTK VDFK
247	B12 clone Kappa Chain nucleotide sequence	GACATCGTGATGACCCAGTCCCTCTGTCTCTGCC AGTGACACCCGGCGAGCCTGCCTCTATCAGCTGTC GGAGCTCCCAGAACCTGAGACACAACAATGGCTA CAACTATCTGAATTGGTACCTGCAGAAGCCAGGCC

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SEQ ID NO	Description	Sequence
		AGTCTCCCCAGCTGCTGATCTATCTGGGCAGCATC AGGGCCTCCGGCGTGCCCGACCGCTTCTCCGGCTC TGGCAGCGGCACCGACTTCACCCCTGAAGATCAGC CGGGTGAGGCAGAGGACGTGGGCGTGTACTATT GCATGCAGGCCCTGCAGACCCCCCTTGGACATTC GGCCAGGGCACCAGGTGGACTTCAAG
248	B12 clone Kappa Chain CDR1	RSSQNLRHNNGYNYLN
249	B12 clone Kappa Chain CDR2	LGSIRAS
250	B12 clone Kappa Chain CDR3	MQALQTPPWT
251	G7 clone Heavy Chain aa sequence	QVQLVQSGAEVKKPGASVKVSKASGYTFTSHYMH WVRQAPGQGLEWMGIIISPTGDFNTYAQKFGQGRVTL TRDTSSTDYMEVTSLSRSEDYAVYYCARDCSAWAP DYWGQGLTVTVSS
252	G7 clone Heavy Chain nucleotide sequence	CAGGTGCAGCTGGTGCAGTCCGGCGCAGAGGTGA AGAAGCCAGGAGCCAGCGTGAAGGTGTCTGTAA GGCCTCTGGCTACACCTTCACATCTCACTATATGC ACTGGGTGCGGCAGGCACAGGACAGGGCCTGGA GTGGATGGGCATCATCAGCCCTACAGGCGACTTCA CCAACTACGCCCAGAAGTTTCAGGGCCGGGTGAC CCTGACAAGAGACACCTCTACAAGCACCGATTAT ATGGAGGTGACATCCCTGAGGTCTGAGGATACCG CCGTGTACTATTGCGCAAGGGACTGTTCCGCCTGG GCCCCGATTACTGGGGACAGGGCACACTGGTGA CCGTGAGCTCC
253	G7 clone Heavy Chain CDR1	GYTFTSH
254	G7 clone Heavy Chain CDR2	SPTGDF
255	G7 clone Heavy Chain CDR3	DCSAWAPDY
256	G7 clone Kappa Chain aa sequence	QSALTRPPSVSRCPGQSITISCSGTSDDVGHNDNHVSW YQQHPGRAPKLMVYEVRNRPSTDRFSGSKSGNT ASLTISGLQAEDATYYCCSYTTTHRYIFGGGKLT
257	G7 clone Kappa Chain nucleotide sequence	CAGTCTGCCCTGACAAGGCCCCCTTCTGTGAGCCG CTGCCCTGGACAGAGCATCAATCTCCTGTTCTG GCACCACTCCGACGTGGGCCACGATAACCACGT GTCCTGGTACCAGCAGCACCAGGAAGGGCACCC AAGCTGATGGTGTATGAGGTGCGGAACAGACCAA GCGGCGTGTCCGACAGGTTACGCGCTCCAAGTCT GGCAATACAGCCTCTCTGACCATCAGCGGCCTGCA GGCAGAGGATGAGGCAACCTACTATTGCTGTTCTT ACACCACAACCCACCGGTATATCTTTGGCGGCGGC ACAAAGCTGACC
258	G7 clone Kappa Chain CDR1	SGTSSDVGHNDHVS
259	G7 clone Kappa Chain CDR2	EVRNRPS
260	G7 clone Kappa Chain CDR3	CSYTTTHRYI
261	E11 clone Heavy Chain aa sequence	QVQLLGSGPGLVKPSETLSLTCTVSGASISSPGYYWG FIRQSPGKLEWIGSMVSGGTTYNP SLKSRVTISMD MSNNQFSLRLNSVTAADTALYYCARGSRQLVRRATI DYWGQGALFTVSP
262	E11 clone Heavy Chain nucleotide sequence	CAGGTGCAGCTGTGGGCAGCGGCCAGGCCTGG TGAAGCCTTCTGAGACACTGAGCCTGACCTGTACA GTGTCTGGCGCCAGCATCAGCTCCCCAGGCTACTA TTGGGGCTTCATCAGGCAGAGCCCAGGCAAGGGC CTGGAGTGGATCGGCTCCATGGTGTCTGGCGGCAC CACATACTATAACCCTAGCCTGAAGTCCCGGGTGA CAATCTCCATGGACATGTCTAACAATCAGTTCAGC CTGAGGCTGAATTCCGTGACCGCCGCGATACAGC CCTGTACTATTGCGCAAGGGCTCCCGCCAGCTGG TGCGGAGAGCAACCATCGACTACTGGGGACAGGG CGCCTGTTTACAGTGTCTCCC
263	E11 clone Heavy Chain CDR1	GASISSPGY

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SEQ ID NO	Description	Sequence
264	E11 clone Heavy Chain CDR2	VSGGT
265	E11 clone Heavy Chain CDR3	GSRLVRRATIDY
266	E11 clone Lambda Chain aa sequence	QSVLTGPPSVSAGPGQQVFISCSGNSSNIGNNYVSWY QQLPGTAPKLLIYDSNKRPSGIPDRFSGSKSGTSATLG ITGLQTGDEADYYCGTWDSSLSAGVFGGGTKLT
267	E11 clone Lambda Chain nucleotide sequence	CAGTCTGTGCTGACCGGACCACCTCCGTGTCTGC CGGACCAGGACAGCAGGTGTTTCATCAGCTGTTCG GCAACAGCTCCAATATCGGCAACAATTACGTGTCT TGGTATCAGCAGCTGCCAGGCACAGCCCCCAAGC TGCTGATCTACGACTCTAACAAGCGGCCTAGCGGC ATCCCAGATAGATTCTCTGGCAGCAAGTCCGGCAC CAGCGCCACACTGGGCATCACCGCCTGCAGACA GGCGACGAGGCAGATTACTATTGCGGAACCTGGG ACTCTAGCCTGTCCGCGCGCTGTTTGGAGGAGGA ACCAAGCTGACA
268	E11 clone Lambda Chain CDR1	SGNSSNIGNNYVS
269	E11 clone Lambda Chain CDR2	DSNKRPS
270	E11 clone Lambda Chain CDR3	GTWDSSLSAGV
271	G5 clone Heavy Chain aa sequence	EVQLVESGGGLVKKPGESLRLSCAASGFTFSSYSMSW VRQAPGKGLEWVSCITSSGHTYYADSVKGRFAISRD NGKNSLYLQMNLRADTAVYFCAKELGAHSGLFY NGVPDYWGQGNPVTVSS
272	G5 clone Heavy Chain nucleotide sequence	GAGGTGCAGCTGGTGGAGTCCGGCGGAGGCCTGG TGAAGCCAGGCGAGTCTCTGAGGCTGAGCTGTGC CGCCTCCGGCTTCACTTTAGCTCCTACAGCATGT CCTGGGTGCGCCAGGCACCTGGCAAGGGCCTGGA GTGGGTGTCTGCATCACCTCTAGCGGCCACACAT ACTATGCCGACTCTGTGAAGGGCCGGTTCGCCATC AGCGGGATAACGGCAAGAATAGCCTGTACCTGC AGATGAACAATCTGCGGGCCGAGGACACCGCCGT GTATTTTGTGCAAAGGAGCTGGGAGCACACTCTG GCCTGTTTACAAACGGCGTGTGATTATTGGGGC CAGGGCAATCCCGTGACAGTGTCTCT
273	G5 clone Heavy Chain CDR1	GFTFSSY
274	G5 clone Heavy Chain CDR2	TSSGH
275	G5 clone Heavy Chain CDR3	ELGAHSGLFYNGVPDY
276	G5 clone Lambda Chain aa sequence	QSALTQPASVSGSPGQSITISCTGTSSDVGGYNYVSW YQQHPGKAPKLMIEVSNRPSGVSNRPSGSKGNTAS LTISGLRGEADYYCSSYTSSTLVVFGGGTKLT
277	G5 clone Lambda Chain nucleotide sequence	CAGTCCGCCCTGACCCAGCCAGCCTCCGTGTCTGG CAGCCCCGGCCAGTCTATCACAATCAGCTGTACCG GCACAAGCTCCGACGTGGCGGCTACAACCTACGT GAGCTGGTACCAGCAGCAGCCAGGCAAGGCACCT AAGCTGATGATCTATGAGGTGTCCAACAGGCCAA GCGGCGTGTCCAATAGATTCTCCGGCTCTAAGGGC AATACCGCCTCCCTGACAATCTCTGGCCTGAGGGG AGAGGACGAGGCAGATTACTATTGCTCTAGCTAC ACCTCCTCTAGCACACTGGTGGTGTGTCGCGCGG CACCAGCTGACA
278	G5 clone Lambda Chain CDR1	TGTSSDVGGYNYVS
279	G5 clone Lambda Chain CDR2	EVSNRPS
280	G5 clone Lambda Chain CDR3	SSYTSSSTLVV
281	H8 clone Heavy Chain aa	QVHLQQWAGLVKPSSETLSLTCVQGGPFSGYYWS

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SEQ ID NO	Description	Sequence
	sequence	WIRQPPGKGLEWIGEINHSNTHYNPSLKSRTISVD TSGNYFSLKLTSTVTAADAAYVFCARGQQLLRNYYY YSGMDVWGQGTTVTVSS
282	H8 clone Heavy Chain nucleotide sequence	CAGGTGCACCTGCAGCAGTGGGGAGCAGGCCTGG TGAAGCCATCCGAGACACTGTCTCTGACATGTGCA GTGCAGGGAGGACCCTTCTCTGGCTACTATTGGAG CTGGATCAGGCAGCCACCTGGCAAGGCCTGGAG TGGATCGGCAGATCAACCACAGCGCAATACCC ACTACAACCCCTCTCTGAAGAGCCGGTGACCATC AGCGTGGACACATCCGGCAATTACTTCTCCCTGAA GCTGACCTCTGTGACAGCGCCGATGCCGCCGTGT ATTTTTCGCCCGGGGCCAGCAGCTGCTGAGAAAC TACTATTACTATTCCGGCATGGACGTGTGGGACA GGGAACCACAGTGACAGTGAGCTCC
283	H8 clone Heavy Chain CDR1	GGPFSGY
284	H8 clone Heavy Chain CDR2	NHSGN
285	H8 clone Heavy Chain CDR3	GQQLLRNYYYYSGMDV
286	H8 clone Lambda Chain aa sequence	QSALTQPASVSGSPGQSITISCTETSRDVGDNVYVSW YQQHPGPAPKLIMYEVHKRPSGISNRFSGSKSGTTAS LTISGLQADDEGDYCYSSYTDKNITYVFGSGTQVT
287	H8 clone Lambda Chain nucleotide sequence	CAGTCTGCCCTGACCCAGCCAGCCTCTGTGAGCGG CTCCCCTGGCCAGTCCATCACAATCTCTTGTAACCG AGACATCTCGGGACGTGGGCGATTACAATATGT GAGCTGGTACCAGCAGCACCCAGGACCTGCACCA AAGCTGATCATGTATGAGGTGCACAAGCGGCCCT CTGGCATCAGCAATAGATTCTCTGGCAGCAAGTCC GGCACCACAGCCAGCCTGACCATCTCCGGCCTGCA GGCAGACGATGAGGGCGACTACTATTGCAGCTCC TACACCGATAAGAACACATACGTGTTCCGCAGCG GCACCCAGGTGACA
288	H8 clone Lambda Chain CDR1	TETSRDVGDNVYVS
289	H8 clone Lambda Chain CDR2	EVHKRPS
290	H8 clone Lambda Chain CDR3	SSYTDKNITYV
291	gp350 B95-8 (2-425) D296R/I160R Avi-His	EAALLVCQYTIQSLIHLTGEDPGFFNVEIPEPPFYPTC NVCTADVNTINFDVGGKKHQLDLDFGQLTPHTKA VYQPRGAFGSEENATNLFLELLGAGELALTMRSKK LPINVTTGEEQVSLESVDVYFQDVFGTMWCHHAE MQNPVYLIPETVPYIDWDCNSTNI TAVVRAQGLDV TLPLSLPTSQDSNFSVKTEMLGNEIDIECIMEDGEIS QVLPDGNKFNITCSGYESHVPSSGILTSTSPVATPIPG TGYAYSLRLTPRPVSRFLGNNSILYVFYSGNGPKASG GRYCIQSNIVFSDEIPASQDMPNTNTDITYVGDNATY SVPMTSEDANSPNVTVAFWAWPNNTETDFKCKW TLTSGTPSGCENISGAFASNRTFDITVSGLTAPKTLII TRTATNATTTTHKVIFSKAPGSGLNDIFEAQKIEWHE HHHHHH
292	NAM encoding SEQ ID NO: 281	GAAGCTGCCCTGCTCGTGTGCCAGTACACCATCCA GAGCCTGATCCACCTGACCGGCGAGGACCCCGGC TTCTTCAACGTGGAATCCCCGAGTTCCCTTCTA CCCTACCTGCAACGTGTGCACCGCCGACGTGAACG TGACCATCAACTTCGACGTGGGCGGCAAGAAGCA CCAGCTGGACCTGGATTTCGCCAGCTGACCCCTC ACACCAAGGCCGTGTATCAGCCAGAGGCGCCTTT GGCGGCAGCGAGAACGCCACCAATCTGTTTCTGCT GGAACCTCCTAGGCGCCGGCAGCTGGCCCTGACC ATGAGAAGCAAGAACTGCCCCATCAATGTGACCA CAGGCGAGGAACAGCAGGTGTCCCTGGAAAGCGT GGACGTGTACTTTCAAGACGTGTTTCGGCACCATGT GGTGCCACCAAGCGAGATGCGAACCCTGTGTA CCTGATCCCCGAGACAGTGCCCTACATCGATTGGG ACAACTGCAACAGCACCAACATCACGCCCTCGT CGGGGCCAGGGACTGGATGTGACACTGCCTCTG

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SEQ ID NO	Description	Sequence
		AGCCTGCCTACCAGCGCCAGGACAGCAACTTCA GCGTGAAAACCGAGATGCTGGGCAACGAGATCGA CATCGAGTGCATCATGGAAGATGGCGAGATCAGC CAGGTGCTGCCCGGCGACAACAAGTTCAACATCA CATGCAGCGGCTACGAGAGCCACGTGCCATCTGG CGGCATCCTGACCAGCACAGCCAGTGGCCACA CCCATCCCTGGCACAGGTACGCCACAGCCTGAG ACTGACCCCCAGACCCGTGTCCAGATTCTGGGCA ACAACAGCATCCTGTACGTGTTCTACAGCGGCAAC GGCCCCAAGGCTCTGGCGCGGTACTGTATCCA GAGCAACATCGTGTTCAGCGACGAGATCCCCGCC AGCCAGGACATGCCACCAATACCACCGACATCA CGTACGTGGCGACAATGCCACCTACAGCGTGCC AATGGTCACCTCCGAGGACGCCAACGCCCAAC GTGACCGTGACAGCCTTTTGGGCTGGCCTAACAA CACCAGACAGACTTCAAGTGCAAGTGGACCTG ACCTCCGGCACCCCTAGCGGCTGCGAGAATATCA GCGGAGCCTTCGCCAGCAACCGACCTTCGATATC ACCGTGTCTGGCCTGGGCACGCCCCCAAGACCT GATCATCACCAGAACCGCCACAAATGCCACCACC ACAACCCACAAAGTGATCTTCAGCAAGGCCCCCG GCTCTGGCCTGAACGACATTTTGAGGCCCAGAAG ATTGAGTGGCATGAACATCACCACCACCACCAT
293	Macaque Light chain backbone nucleotide sequence	tcgcgcgttt cggatgatgac ggtgaaaacc tctgacacat gcagctcccg gagacggtca cagcttgtct gtaagcggat gccgggagca gacaagcccg tcaggggcgc tcagcgggtg ttggcgggtg tcggggctgg cttactatg cgccatcaga gcagattgta ctgagagtgc accatatgag gtgtgaaata cgcacagat gcgtaaggag aaaataccgc atcagattgg tacattatg ctattggcca ttgcatacgt tgtatccata tcataatatg tacatttata ttggctcatg tccaacatta ccgcatgtt gacattgatt attgactagt tattaatagt aatcaattac ggggtcatta gttcatagcc catatatgga gttccgcgtt acataactta cggtaaatgg cccgcctggc tgaccgcccc acgacccccg cccattgacg tcaataatga cgtatgttcc catagtaacg ccaataggga ctttccattg acgtcaatgg gtggagtatt tacggttaac tgcccacttg gcagtagatc aagtgtatca tatgccaagt acgcccccta ttgacgtcaa tgacggtaaa tggcccgcct ggcattatgc ccagtagatg accttatggg actttcctac ttggcagtag atctacgtat tagtcatcgc tattaccatg gtgatgcggt tttggcagta catcaatggg cgtggatagc ggtttgactc acggggattt ccaagtctcc accccatgta cgtcaatggg agtttgtttt ggcacccaaa tcaacgggac tttccaaaat gtcgtaacaa ctcccgccca ttgacgcaaa tgggcggtag gcgtgtacgg tgggaggtct atataagcag agctcgttta gtgaaccgtc agatcgccctg gagacgccat ccacgctgtt ttgacctcca tctctccttc tagaagacac cgggacggat ccagcctcca tcggctcgca tctctccttc acgcgcgcgc cgcctacct gaggcgcgca tccacgcgag ttgagtcgag ttctgcgcgc tcccgcctgt ggtgcctcct gaactgcgtc cgcgctcag gtaagttaa agctcaggtc gagacggggt ctttgtccgg cgtcccttg gagcctacct agactcagcc ggctctccac gctttgcctg acctgcttg ctcaactcta gttaacggtg gagggcagtg tagtctgagc agtacacgtt gctgcgcgc gcgcaccag acataatagc tgacagacta acagactggt cttttccatg ggtcttttct gcagtcccg tcgtcgacac gtgtgatcag atatcgccgc cgtctagac caccatggga tggatcatga tcaccccttt tctagtagca actgcaaccg gtgtacattc ccagtctgcc ctgactcagc ctccctctgt gtctgggtct cctggacagt cggtcacct ctctgcact ggaaccagea gtgacgttga tggttataac tatgtctcct ggtaccaaca acatccaggc aaagccccc aactcatgat ttatggtgtc agcaatcggc cctcagggtt ctctgatcgc ttctctggct ccaagtctgg caacacggcc tccctgacca tctctgggct ccaggctgag gacgaggctg attattactg ttgttcatct acaaccagtt acacttacat cttcggaact gggaccaagg tcacagtact aggtcagccc aaggctgccc cctcggtcac tctcttcccg cctctctctg aggagcttca agccaacaag gccacactag tgtgtctgat cagtgaactc taccgggag ccgtggaagt ggcttggaa gcagatggca gcgctgtcaa cgcgggagtg gagaccacca aacctccaa acagagcaac aacaagtacg cggccagcag ctacctgagc ctgacgtcag accagtggaa gtcccacaag agctacagct gccaggtcac gcacgaaggg agcacgctgg agaagacagt ggcacctgca gaattgtcat agggatccag atctgctgtg cttctagtt gccagccatc tgttgttttg cctcccccg tgcttctctt gacctggaa ggtgccactc cactgtctct ttcctaataa aatgaggaaa ttgcatcgca ttgtctgagt aggtgtcatt ctattctggg ggttgggtg gggcaggaca gcaaggggga ggattgggaa gacaatagca ggcatgctgg ggatgcggtg ggtctatgg gtaccaggt gctgaagaat tgaccgggt cctcctgggc cagaaagaag caggcacatc ccttctctg tgacacaccc tgtccacgcc cctggttctt agttccagcc cactcatag gacactcata gctcaggag gctcgcctt caatcccacc cgttaaagta catggagcgg tctctccttc

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SEQ ID NO	Description	Sequence
		cctcaccagc ccaccaaac aaacctagcc tccaagagt ggaagaaatt aaagcaagat aggcatttaa gtgcagaggg agagaaaatg cctccaacat gtgaggaagt aatgagagaa atcatagaat ttaaggcca tgatttaagg ccatcatggc cttaatcttc cgcttctcgc ctcactgact cgctgcgctc ggcgcttcgg ctgcggcgag cggtatcagc tcactcaaaag gcggtaatatc ggttatccac agaatacagg gataacgcag gaaagaacat gtgagcaaaa ggccagcaaa agggcaggaa ccgtaaaaag gccgcgttgc tggcggtttt ccataggctc cgccccctg acgagcatca caaaaatcga cgctcaagtc agaggtggcg aaaccgcaca ggactataaa gataccaggc gtttccccc ggaagctccc tcgtgcgctc tcctgttccg acctgcgcgc ttaccggata cctgtccgcc tttctccctt cgggaagcgt ggcgctttct catagctcac gctgtaggta tctcagttcg gtgtaggctg ttcgctccaa gctgggctgt gtgcacgaac ccccggttca gccgcaccgc tgcgctttat ccggtaaacta tcgtcttgag tccaaccggg taagacacga cttatcgcca ctggcagcag ccaactggtaa caggattagc agagcgaggt atgtaggcgg tgctacagag ttctgaagt ggtggcctaa ctacggctac actagaagaa cagtatttgg tatctgcgct ctgctgaagc cagttacctt cggaaaaaga gttggtagct ctgatccgg caaacaaacc accgtggtg gcggtggttt tttgtttgc aagcagcaga ttacgcgcag aaaaaaagga tctcaagaag atcctttgat cttttctacg gggcttgacg ctcagtgga cgaaaactca cgtaaaggga ttttggtcat gagattatca aaaaggatct tcacctagat ccttttaaat taaaaatgaa gttttaatc aatctaaagt atatatgagt aaacttggtc tgacagttac caatgcttaa tcagttaggc acctatctca gcgatctgtc tatttcgttc atccatagtt gcctgactcg gggggggggg gcgctgaggt ctgctcgtg aagaagtggt tgctgactca taccaggcct gaatcgcccc atcatccagc cagaaagtga gggagccacg gttgatgaga gctttgtgt aggtggacca gttggtgatt ttgaactttt gctttgccac ggaacggtct gcgttgctcg gaagatcgct gatctgatcc ttcaactcag caaaagtctg atttattcaa caaagccgcc gtcccgtcaa gtcagcgtaa tgctctgcca gtgttacaac caattaaacca attctgatta gaaaaactca tcgagcatca aatgaaactg caatttattc atatcaggat tatcaatacc atatttttga aaaagccggt tctgtaatga aggagaaaac tcaccagggc agttccatag gatggcaaga tccgtgtatc ggctgctgat tccgactcgt ccaacatcaa tacaacctat taatttcccc tcgtcaaaaa taaggttatc aagtgagaaa tcaccatgag tgacgactga atccggtgag aatggcaaaa gcttatgcat ttctttccag acttggtcaa caggccagcc attacgctcg tcatcaaaat cactcgcatc aaccaaacgc ttattcattc gtgattgcgc ctgagcgaga cgaataacgc gatcgctggt aaaaggacaa ttacaacacg gaatcgaatg caaccggcgc aggaacactg ccagcgcatc aacaatattt tcacctgaat caggatattc ttctaatacc tggaatgctg ttttccccgg gatcgcaagt gtgagtaacc atgcatcatc aggagtacgg ataaaaatgt tgatggctcg aagaggcata aattccgtca gccagtttag tctgaccatc tcatctgtaa catcattggc aacgctacct ttgccatggt tcagaaacaa ctctggcgca tcgggcttcc catacaatcg atagattgtc gcacctgatt gcccgacatt atcgcgagcc catttatacc catataaatc agcatccatg ttggaattta atcgcgccct cgagcaagac gtttcccggt gaatatggtt cataaacacc cttgtattac tgtttatgta agcagacagt tttattgttc atgatgatatt atttttatct tgtgcaatgt aacatcagag attttgagac acaacgtggc tttccccccc ccccatat tgaagcattt atcagggtta ttgtctcatg agcgatatac tatttgaatg tatttagaaa aataaacaaa taggggttcc gcgcacattt cccgaaaaag tgccacctga cgtctaagaa accattatta tcatgacatt aacctataaa aataggcgta tcacgaggcc ctttcgtc

bolded/underlined letters indicate amino acids that differ from wild-type.

50

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1: Design of gp350 probes that knock out CR2-binding property. Selected residues substantially involved in CR2-binding, but less in antibody (H02) recognition, are defined by structures of gp350-CR2 and gp350-H02 Fab complexes. CR2-knock out (KO) mutations are designed by substitution of one or more residues among the selected residues. Experimentally tested mutations include Q122N (glycan addition at 122); D163N/164bS (glycan addition at 163 and one residue insertion between 164 and 165); D296R; W162A/D163A/N164A; D296R/I160R; D296R/P158R; D296R/P158N/I160T (glycan addition at 158); D296R/P158R/I160R.

FIG. 2: CR2- and antibody-binding profile of gp350 mutants. Binding of CR2-Ig (chimeric molecule of CR2 SCR1-2 domain and IgG Fc region) and a set of gp350-

specific monoclonal antibodies were measured by biolayer interferometry using Octet instrument. CR2-Ig or antibodies were immobilized on biosensors through protein A or anti-mouse Fc antibody (for 2L10) and assessed binding properties of various recombinant gp350 mutants. Graph shows percent binding of each mutant relative to its parental (wild-type) gp350. D296R mutation greatly (but not completely) eliminated CR2-binding while retaining binding of all the monoclonal antibodies. An additional mutation on D296R mutant completely abrogated the CR2-binding property of gp350. Among these variants, D296R/I160R retained binding of most monoclonal antibodies and was therefore chosen for further characterization. 2L10 is a murine non-CR2-binding site targeting antibody, used as a control.

FIGS. 3A-3F: Gating strategy of IgG+ B cell subset in human PBMCs. FIG. 3A: Lymphocytes were gated by SSC and FSC. FIG. 3: B cells were gated by CD19 and excluding

65

81

non-B cell surface markers. FIG. 3C: Surface IgG+ B cells were further gated by IgG and excluding IgM. SSC and FSC denote side- and forward-scatter, respectively. FIGS. 3D-3F: staining profile of IgG+ B cells with various EBV gp350 probes. There were three gp350 variants used to co-stain with anti-CD21 (complement receptor 2, CR2); WT, wild-type gp350 (FIG. 3D); D296R (FIG. 3E); and D296R/1160R (FIG. 3F). All the mutant probes were designed to reduce the CR2-binding property of gp350. WT and D296R gp350 probes bind to virtually all the IgG+ B cells while the D296R/1160R gp350 probe completely knocked down non-specific gp350 staining of B cells.

FIGS. 4A and 4B: Identification of EBV gp350-specific IgG+ B cells in individuals who have high serum neutralization titers. Cryopreserved PMBCs were stained with cell lineage differentiation markers as well as gp350 D296R/1160R probe. The gp350 probe were conjugated with two different fluorochromes (PE and BV785) to eliminate false positive events and increase confidence for specificity of stained B cells. Plots shown were IgG+ B cells gated as in previous figures. Rectangles indicate IgG+B cells stained with gp350 D296R/1160R in two different colors.

FIGS. 5A-5F: Isolation of gp350-specific B cells from Immunized Macaques. FIG. 5A shows rhesus macaque immunization schedule with EBV gp350 nanoparticles followed by B cell sorting. FIG. 5B shows animals Cy137 and Cy651 displayed the highest neutralization titers and were selected for B cell sorting. FIGS. 5C-5F show the results of B cell sorting strategy for non-human primate B cells.

FIGS. 6A and 6B: Identification and analysis of Vaccine-elicited anti-gp350 mAbs. FIG. 6A shows gp350-specific B cell sorting. FIG. 6B shows the neutralization analysis of select antibodies (72A1 antibody from ATCC is shown for reference).

FIG. 7: Clonotype analysis of immunoglobulin heavy chain sequences. Approximately 200 B cells were isolated from each of the two rhesus macaques analyzed. The B cell receptor sequences indicated low clonal expansion in the B cells isolated as judged by the sequence similarity of each B cell receptor.

FIGS. 8A and 8B: Neutralization potency of humanized mAb H02. FIG. 8A shows that humanized H02, as well as murinized H02, neutralize EBV equivalent to the parental macaque H02. FIG. 8B shows that the neutralizing potency of H02 (human, mouse and macaque) is about 50-fold higher than 72A1 antibody (shown for reference).

FIGS. 9A-9C: Targeting the receptor (CR2)-binding site of gp350: Flow cytometry based CR2 epitope mapping. FIG. 9A is a schematic of the competition-based FACS assay. FIG. 9B shows flow cytometry plots of gp350-binding to B cells in the presence and absence of control antibodies. FIG. 9C shows flow cytometry plots of rhesus macaque antibodies described in this disclosure (two non-neutralizing antibodies F11 and G12 are shown for reference).

FIGS. 10A-10C: Analysis of Vaccine-elicited Anti-gp350 mAbs: characterization of antibodies.

FIG. 10A shows the neutralization potency of select antibodies. FIG. 10B shows the affinity of antibodies to gp350. FIG. 10C shows the CDR H3 isoelectric point of select antibodies, indicating that gp350-specific neutralizing antibodies tend to have a higher isoelectric point than typical antibodies (Ebola GP-elected antibodies are shown for reference).

FIG. 11 shows the location of mutations entered into the heavy chain residues of claim A9; and

FIG. 12 shows the location of mutations entered into the light chain residues of claim A9.

82

FIG. 13 shows the design of the CR2-binding knockout gp350 probe, including specific residues that were substituted to knock down CR2-binding.

FIG. 14 shows the binding properties of the gp350 mutants. In this figure, # indicates no binding.

FIG. 15 shows the B cell staining profile of the new CR2-binding knock-out gp350 probe (D296R/K161D).

FIG. 16 shows the EBV neutralization potency of the two best human monoclonal antibodies (from clones A9 and E11). In these two clones, mutations were introduced in a structure-guided potency improving mutation analysis. For the mAb of clone A9, the introduced mutations were (referring to Kabat numbering) in the heavy chain were: Q1R, Y32R+Y98I, N53F, T70I, T70F, L96R, Y98R, Y98W, I100F, I100W, and Y102E; in the light chain were Y32E, G57D, T56Q, and T56E. For the mAb of clone E11, the introduced mutations were (referring to Kabat numbering) in the heavy chain were: V100F, V100I, V100W, V100R, V100Y, Q98R, Q98Y, and Y58R (residues 30-33 and 53-56 were also targeted for mutagenesis); in the light chain, residues 1, 27, 27a, 30, 32, 49, 50, 52, 53, 56, and 93-95 were targeted for mutagenesis.

DETAILED DESCRIPTION

I. Definitions

The terms “EBV gp350 protein” and “gp350” as used herein, refer to various Epstein Barr Virus polypeptides. The EBV gp350 proteins described herein may be isolated from a variety of sources, such as from human tissue types or from another source, or prepared by recombinant or synthetic methods. The term “EBV gp350 protein” refers to each individual gp350 polypeptide disclosed herein. All disclosures in this specification which refer to the “EBV gp350 protein” refer to each of the polypeptides individually as well as jointly. For example, descriptions of the preparation of, purification of, derivation of, formation of antibodies to or against, formation of gp350 binding oligopeptides to or against, administration of, compositions containing, treatment of a disease with, etc., pertain to each polypeptide of the disclosure individually. The term “EBV gp350 protein” also includes variants of the gp350 polypeptides disclosed herein.

A “native sequence EBV gp350 protein” comprises a polypeptide having the same amino acid sequence as the corresponding EBV gp350 protein derived from nature. Such native sequence EBV gp350 proteins can be isolated from nature or can be produced by recombinant or synthetic means. The term “native sequence EBV gp350 protein” specifically encompasses naturally-occurring truncated or secreted forms of the specific EBV gp350 protein (e.g., an extracellular domain sequence), naturally-occurring variant forms (e.g., alternatively spliced forms) and naturally-occurring allelic variants of the polypeptide. The native sequence EBV gp350 proteins disclosed herein may be mature or full-length native sequence polypeptides comprising the full-length amino acids sequences.

“EBV gp350 protein variant” means an EBV gp350 protein, preferably an active EBV gp350 protein, as defined herein having at least about 80% amino acid sequence identity with a full-length native sequence EBV gp350 protein sequence as disclosed herein, an extracellular domain of an EBV gp350 protein, as disclosed herein or any other fragment of a full-length EBV gp350 protein sequence as disclosed herein (such as those encoded by a nucleic acid that represents only a portion of the complete coding

sequence for a full-length EBV gp350 protein). Such EBV gp350 protein variants include, for instance, EBV gp350 proteins wherein one or more amino acid residues are added or deleted, at the N- or C-terminus of the full-length native amino acid sequence. Ordinarily, an EBV gp350 protein variant will have at least about 80% amino acid sequence identity, alternatively at least about 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% amino acid sequence identity, to a full-length native sequence EBV gp350 protein sequence as disclosed herein, or any other specifically defined fragment of a full-length EBV gp350 protein sequence as disclosed herein. Optionally, gp350 variant polypeptides will have no more than one conservative amino acid substitution as compared to the native EBV gp350 protein sequence, alternatively no more than 2, 3, 4, 5, 6, 7, 8, 9, or 10 conservative amino acid substitution as compared to the native EBV gp350 protein sequence.

“Percent (%) amino acid sequence identity” with respect to the EBV gp350 protein sequences identified herein is defined as the percentage of amino acid residues in a candidate sequence that are identical to the amino acid residues in the specific EBV gp350 protein sequence, after aligning the sequences and introducing gaps, if necessary, to achieve the maximum percent sequence identity, and not considering any conservative substitutions as part of the sequence identity. Alignment for purposes of determining percent amino acid sequence identity can be achieved in various ways that are within the skill in the art, for instance, using publicly available computer software such as BLAST, BLAST-2, ALIGN or Megalign (DNASTAR) software. Those skilled in the art can determine appropriate parameters for measuring alignment, including any algorithms needed to achieve maximal alignment over the full length of the sequences being compared. For purposes herein, % amino acid sequence identity values may be generated using the sequence comparison computer program ALIGN-2, wherein the complete source code for the ALIGN-2 program is provided in U.S. Pat. No. 7,160,985, which is incorporated herein by reference. The ALIGN-2 sequence comparison computer program was authored by Genentech, Inc. and the source code thereof has been filed with user documentation in the U.S. Copyright Office, Washington D.C., 20559, where it is registered under U.S. Copyright Registration No. TXU510087. The ALIGN-2 program is publicly available through Genentech, Inc., South San Francisco, California or may be compiled from the source code. All sequence comparison parameters are set by the ALIGN-2 program and do not vary.

In situations where ALIGN-2 is employed for amino acid sequence comparisons, the % amino acid sequence identity of a given amino acid sequence A to, with, or against a given amino acid sequence B (which can alternatively be phrased as a given amino acid sequence A that has or comprises a certain % amino acid sequence identity to, with, or against a given amino acid sequence B) is calculated as 100 times the fraction X/Y, where X is the number of amino acid residues scored as identical matches by the sequence alignment program ALIGN-2 in that program’s alignment of A and B, and where Y is the total number of amino acid residues in B. Where the length of amino acid sequence A is not equal to the length of amino acid sequence B, the % amino acid sequence identity of A to B will not equal the % amino acid sequence identity of B to A.

“gp350 variant polynucleotide” or “gp350 variant nucleic acid sequence” means a nucleic acid molecule which encodes an EBV gp350 protein, preferably an active EBV

gp350 protein, as defined herein and which has at least about 80% nucleic acid sequence identity with a nucleic acid sequence encoding a full-length native sequence EBV gp350 protein sequence as disclosed herein, an extracellular domain of an EBV gp350 protein, as disclosed herein or any other fragment of a full-length EBV gp350 protein sequence as disclosed herein (such as those encoded by a nucleic acid that represents only a portion of the complete coding sequence for a full-length EBV gp350 protein). Ordinarily, a gp350 variant polynucleotide will have at least about 80% nucleic acid sequence identity, alternatively at least about 81%, 82%, 83%, 84%, 85%, 86%, 87%, 88%, 89%, 90%, 91%, 92%, 93%, 94%, 95%, 96%, 97%, 98%, or 99% nucleic acid sequence identity with a nucleic acid sequence encoding a full-length native sequence EBV gp350 protein sequence, an extracellular domain of an EBV gp350 protein or any other fragment of a full-length EBV gp350 protein sequence. Variants do not encompass the native nucleotide sequence.

Ordinarily, gp350 variant polynucleotides are at least about 5 nucleotides in length, alternatively at least about 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80, 85, 90, 95, 100, 105, 110, 115, 120, 125, 130, 135, 140, 145, 150, 155, 160, 165, 170, 175, 180, 185, 190, 195, 200, 210, 220, 230, 240, 250, 260, 270, 280, 290, 300, 310, 320, 330, 340, 350, 360, 370, 380, 390, 400, 410, 420, 430, 440, 450, 460, 470, 480, 490, 500, 510, 520, 530, 540, 550, 560, 570, 580, 590, 600, 610, 620, 630, 640, 650, 660, 670, 680, 690, 700, 710, 720, 730, 740, 750, 760, 770, 780, 790, 800, 810, 820, 830, 840, 850, 860, 870, 880, 890, 900, 910, 920, 930, 940, 950, 960, 970, 980, 990, or 1000 nucleotides in length, wherein in this context the term “about” means the referenced nucleotide sequence length plus or minus 10% of that referenced length.

“Isolated,” when used to describe the various EBV gp350 proteins disclosed herein, means a polypeptide that has been identified and separated and/or recovered from a component of its natural environment. Contaminant components of its natural environment are materials that would typically interfere with diagnostic or therapeutic uses for the polypeptide, and may include enzymes, hormones, and other proteinaceous or non-proteinaceous solutes. The polypeptide may be purified (1) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator, or (2) to homogeneity by SDS-PAGE under non-reducing or reducing conditions using Coomassie blue or silver stain. Isolated polypeptide includes polypeptide in situ within recombinant cells, since at least one component of the EBV gp350 protein natural environment will not be present. Ordinarily, however, isolated polypeptide will be prepared by at least one purification step.

An “isolated” EBV gp350 protein-encoding nucleic acid or other polypeptide-encoding nucleic acid is a nucleic acid molecule that is identified and separated from at least one contaminant nucleic acid molecule with which it is ordinarily associated in the natural source of the polypeptide-encoding nucleic acid. An isolated polypeptide-encoding nucleic acid molecule is other than in the form or setting in which it is found in nature. Isolated polypeptide-encoding nucleic acid molecules therefore are distinguished from the specific polypeptide-encoding nucleic acid molecule as it exists in natural cells. However, an isolated polypeptide-encoding nucleic acid molecule includes polypeptide-encoding nucleic acid molecules contained in cells that ordinarily

express the polypeptide where, for example, the nucleic acid molecule is in a chromosomal location different from that of natural cells.

The term "control sequences" refers to DNA sequences necessary for the expression of an operably linked coding sequence in a host organism. The control sequences that are suitable for prokaryotes, for example, include a promoter, optionally an operator sequence, and a ribosome binding site. Eukaryotic cells are known to utilize promoters, polyadenylation signals, and enhancers.

Nucleic acid is "operably linked" when it is placed into a functional relationship with another nucleic acid sequence. For example, DNA for a presequence or secretory leader is operably linked to DNA for a polypeptide if it is expressed as a preprotein that participates in the secretion of the polypeptide; a promoter or enhancer is operably linked to a coding sequence if it affects the transcription of the sequence; or a ribosome binding site is operably linked to a coding sequence if it is positioned to facilitate translation. Generally, "operably linked" means that the DNA sequences being linked are contiguous, and, in the case of a secretory leader, contiguous and in reading phase. However, enhancers do not have to be contiguous. Linking is accomplished by ligation at convenient restriction sites. If such sites do not exist, the synthetic oligonucleotide adaptors or linkers are used in accordance with conventional practice.

"Stringency" of hybridization reactions is readily determinable by one of ordinary skill in the art, and generally is an empirical calculation dependent upon probe length, washing temperature, and salt concentration. In general, longer probes require higher temperatures for proper annealing, while shorter probes need lower temperatures. Hybridization generally depends on the ability of denatured DNA to reanneal when complementary strands are present in an environment below their melting temperature. The higher the degree of desired homology between the probe and hybridizable sequence, the higher the relative temperature which can be used. As a result, it follows that higher relative temperatures would tend to make the reaction conditions more stringent, while lower temperatures less so. For additional details and explanation of stringency of hybridization reactions, see Ausubel et al., *Current Protocols in Molecular Biology*, Wiley Interscience Publishers (1995).

"Stringent conditions" or "high stringency conditions" as defined herein, may be identified by those that: (1) employ low ionic strength and high temperature for washing, for example 0.015 M sodium chloride/0.0015 M sodium citrate/0.1% sodium dodecyl sulfate at 50° C.; (2) employ during hybridization a denaturing agent, such as formamide, for example, 50% (v/v) formamide with 0.1% bovine serum albumin/0.1% Ficoll/0.1% polyvinylpyrrolidone/50 mM sodium phosphate buffer at pH 6.5 with 750 mM sodium chloride, 75 mM sodium citrate at 42° C.; or (3) overnight hybridization in a solution that employs 50% formamide, 5×SSC (0.75 M NaCl, 0.075 M sodium citrate), 50 mM sodium phosphate (pH 6.8), 0.1% sodium pyrophosphate, 5×Denhardt's solution, sonicated salmon sperm DNA (50 µg/ml), 0.1% SDS, and 10% dextran sulfate at 42° C., with a 10 minute wash at 42° C. in 0.2×SSC (sodium chloride/sodium citrate) followed by a 10 minute high-stringency wash consisting of 0.1×SSC containing EDTA at 55° C.

"Moderately stringent conditions" may be identified as described by Sambrook et al., *Molecular Cloning: A Laboratory Manual*, New York: Cold Spring Harbor Press, 1989, and include the use of washing solution and hybridization conditions (e.g., temperature, ionic strength and % SDS) less stringent than those described above. An example of mod-

erately stringent conditions is overnight incubation at 37° C. in a solution comprising: 20% formamide, 5×SSC (150 mM NaCl, 15 mM trisodium citrate), 50 mM sodium phosphate (pH 7.6), 5×Denhardt's solution, 10% dextran sulfate, and 20 mg/ml denatured sheared salmon sperm DNA, followed by washing the filters in 1×SSC at about 37-50° C. The skilled artisan will recognize how to adjust the temperature, ionic strength, etc. as necessary to accommodate factors such as probe length and the like.

The term "epitope tagged" used herein refers to a chimeric polypeptide comprising an EBV gp350 protein or anti-gp350 antibody fused to a "tag polypeptide." The tag polypeptide has enough residues to provide an epitope against which an antibody can be made, yet is short enough such that it does not interfere with activity of the polypeptide to which it is fused. The tag polypeptide preferably also is fairly unique so that the antibody does not substantially cross-react with other epitopes. Suitable tag polypeptides generally have at least six amino acid residues and usually between about 8 and 50 amino acid residues (preferably, between about 10 and 20 amino acid residues).

"Active" or "activity" for the purposes herein refers to form(s) of an EBV gp350 protein which retain a biological and/or an immunological activity of native or naturally-occurring gp350, wherein "biological" activity refers to a biological function (either inhibitory or stimulatory) caused by a native or naturally-occurring gp350 other than the ability to induce the production of an antibody against an antigenic epitope possessed by a native or naturally-occurring gp350 and an "immunological" activity refers to the ability to induce the production of an antibody against an antigenic epitope possessed by a native or naturally-occurring gp350.

The term "antagonist" is used in the broadest sense, and includes any molecule that partially or fully blocks, inhibits, or neutralizes a biological activity of a native EBV gp350 protein disclosed herein. In a similar manner, the term "agonist" is used in the broadest sense and includes any molecule that mimics a biological activity of a native EBV gp350 protein disclosed herein. Suitable agonist or antagonist molecules specifically include agonist or antagonist antibodies or antibody fragments, fragments or amino acid sequence variants of native EBV gp350 proteins, peptides, antisense oligonucleotides, small organic molecules, etc. Methods for identifying agonists or antagonists of an EBV gp350 protein may comprise contacting an EBV gp350 protein with a candidate agonist or antagonist molecule and measuring a detectable change in one or more biological activities normally associated with the EBV gp350 protein.

"Treating" or "treatment" or "alleviation" refers to both therapeutic treatment and prophylactic or preventative measures, wherein the object is to prevent or slow down (lessen) the targeted pathologic condition or disorder. Those in need of treatment include those already with the disorder as well as those prone to have the disorder or those in whom the disorder is to be prevented. A subject or mammal is successfully "treated" for an EBV gp350 protein-expressing viral infection if, after receiving a therapeutic amount of an anti-gp350 antibody or gp350 binding oligopeptide according to the methods of this disclosure, the patient shows observable and/or measurable reduction in or absence of one or more of the following: reduction in the number of infected cells or absence of the infected cells; reduction in the number of infected cells; inhibition (i.e., slow to some extent and preferably stop) of EBV infection including the spread of infection into neurological tissues; inhibition (i.e., slow to some extent and preferably stop) of infection spread; inhi-

bition, to some extent, and/or relief to some extent, of one or more of the symptoms associated with the viral infection; reduced morbidity and mortality, and improvement in quality of life issues. To the extent the anti-gp350 antibody or gp350 binding oligopeptide may prevent growth or infection and/or kill existing infected cells, it may be cytostatic and/or cytotoxic. Reduction of these signs or symptoms may also be felt by the patient. The above parameters for assessing successful treatment and improvement in the EBV-associated diseases and disorders are readily measurable by routine procedures familiar to a medical provider.

"Chronic" administration refers to administration of the agent(s) in a continuous mode as opposed to an acute mode, to maintain the initial therapeutic effect (activity) for an extended period of time. "Intermittent" administration is treatment that is not consecutively done without interruption, but rather is cyclic in nature.

"Mammal" for purposes of the treatment of, alleviating the symptoms of or diagnosis of a viral infection refers to any animal classified as a mammal, including humans, domestic and farm animals, and zoo, sports, or pet animals, such as dogs, cats, cattle, horses, sheep, pigs, goats, rabbits, etc. Preferably, the mammal is human.

Administration "in combination with" one or more further therapeutic agents includes simultaneous (concurrent) and consecutive administration in any order.

"Carriers" as used herein include pharmaceutically acceptable carriers, excipients, or stabilizers which are non-toxic to the cell or mammal being exposed thereto at the dosages and concentrations employed. Often the physiologically acceptable carrier is an aqueous pH buffered solution. Examples of physiologically acceptable carriers include buffers such as phosphate, citrate, and other organic acids; antioxidants including ascorbic acid; low molecular weight (less than about 10 residues) polypeptide; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, arginine or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; sugar alcohols such as mannitol or sorbitol; salt-forming counterions such as sodium; and/or nonionic surfactants such as TWEEN®, polyethylene glycol (PEG), and PLURONICS®.

By "solid phase" or "solid support" is meant a non-aqueous matrix to which an antibody or gp350 binding oligopeptide of this disclosure can adhere or attach. Examples of solid phases encompassed herein include those formed partially or entirely of glass (e.g., controlled pore glass), polysaccharides (e.g., agarose), polyacrylamides, polystyrene, polyvinyl alcohol and silicones. Depending on the context, the solid phase can comprise the well of an assay plate or a lateral flow assay device; in others, it is a purification column (e.g., an affinity chromatography column). This term also includes a discontinuous solid phase of discrete particles, such as those described in U.S. Pat. No. 4,275,149.

A "liposome" is a small vesicle composed of various types of lipids, phospholipids and/or surfactant which is useful for delivery of a drug (such as an EBV gp350 protein, an antibody thereto or a gp350 binding oligopeptide) to a mammal. The components of the liposome are commonly arranged in a bilayer formation, similar to the lipid arrangement of biological membranes.

A "small" molecule or "small" organic molecule is defined herein to have a molecular weight below about 500 Daltons.

An "effective amount" of a polypeptide, antibody or gp350 binding oligopeptide, or an agonist or antagonist thereof as disclosed herein is an amount sufficient to carry out a specifically stated purpose. An "effective amount" may be determined empirically and in a routine manner, in relation to the stated purpose.

The term "therapeutically effective amount" refers to an amount of an antibody, polypeptide, or gp350 binding oligopeptide, or other drug effective to "treat" a disease or disorder in a subject or mammal. In the case of an EBV infection, the therapeutically effective amount of the drug may reduce the number of infected cells; inhibit (i.e., slow to some extent and preferably stop) spread of the infection into other cells, such as lymphatic or neurological cells organs; and/or relieve to some extent one or more of the symptoms associated with the infection. See the definition herein of "treating." To the extent the drug may prevent growth and/or kill existing infected cells, it may be cytostatic, cytotoxic, anti-inflammatory, immunomodulatory, and/or immunosuppressing.

A "growth inhibitory amount" of an anti-gp350 antibody or gp350 binding oligopeptide is an amount capable of inhibiting the growth of a cell, especially virus infected cell, either in vitro or in vivo. A "growth inhibitory amount" of an anti-gp350 antibody or gp350 binding oligopeptide for purposes of inhibiting infected cell growth may be determined empirically and in a routine manner.

A "cytotoxic amount" of an anti-gp350 antibody or gp350 binding is an amount capable of causing the destruction of a cell, especially virus infected cell, either in vitro or in vivo. A "cytotoxic amount" of an anti-gp350 antibody or gp350 binding oligopeptide for purposes of inhibiting cell growth may be determined empirically and in a routine manner.

The term "antibody" is used in the broadest sense and specifically covers, for example, single anti-gp350 monoclonal antibodies (including agonist, antagonist, and neutralizing antibodies), anti-gp350 antibody compositions with polypeptidic specificity, polyclonal antibodies, single chain anti-gp350 antibodies, and fragments of anti-gp350 antibodies (see below) as long as they exhibit the desired biological or immunological activity or specificity. The term "immunoglobulin" (Ig) is used interchangeable with antibody herein.

An "isolated antibody" is one which has been identified and separated and/or recovered from a component of its natural environment. Contaminant components of its natural environment are materials which would interfere with diagnostic or therapeutic uses for the antibody, and may include enzymes, hormones, and other proteinaceous or nonproteinaceous solutes. The antibody may be purified (1) to greater than 95% by weight of antibody as determined by the Lowry method, and most preferably more than 99% by weight, (2) to a degree sufficient to obtain at least 15 residues of N-terminal or internal amino acid sequence by use of a spinning cup sequenator, or (3) to homogeneity by SDS-PAGE under reducing or nonreducing conditions using Coomassie blue or silver stain. Isolated antibody includes the antibody in situ within recombinant cells because at least one component of the antibody's natural environment will not be present. Ordinarily, however, isolated antibody will be prepared by at least one purification step.

The basic 4-chain antibody unit is a heterotetrameric glycoprotein composed of two identical light (L) chains and two identical heavy (H) chains (an IgM antibody consists of 5 of the basic heterotetramer unit along with an additional polypeptide called J chain, and therefore contain 10 antigen binding sites, while secreted IgA antibodies can polymerize

to form polyvalent assemblages comprising 2-5 of the basic 4-chain units along with J chain). In the case of IgGs, the 4-chain unit is generally about 150,000 daltons. Each L chain is linked to a H chain by one covalent disulfide bond, while the two H chains are linked to each other by one or more disulfide bonds depending on the H chain isotype. Each H and L chain also has regularly spaced intrachain disulfide bridges. Each H chain has at the N-terminus, a variable domain (VH) followed by three constant domains (CH) for each of the α and γ chains and four CH domains for μ and ϵ isotypes. Each L chain has at the N-terminus, a variable domain (VL) followed by a constant domain (CL) at its other end. The VL is aligned with the VH and the CL is aligned with the first constant domain of the heavy chain (CH1). Particular amino acid residues are believed to form an interface between the light chain and heavy chain variable domains. The pairing of a VH and VL together forms a single antigen-binding site. For the structure and properties of the different classes of antibodies, see, e.g., Basic and Clinical Immunology, 8th edition, Daniel P. Stites, Abba I. Terr and Tristram G. Parslow (eds.), Appleton & Lange, Norwalk, C T, 1994, page 71 and Chapter 6.

The L chain from any vertebrate species can be assigned to one of two clearly distinct types, called kappa and lambda, based on the amino acid sequences of their constant domains. Depending on the amino acid sequence of the constant domain of their heavy chains (CH), immunoglobulins can be assigned to different classes or isotypes. There are five classes of immunoglobulins: IgA, IgD, IgE, IgG, and IgM, having heavy chains designated α , δ , ϵ , γ , and μ , respectively. The γ and α classes are further divided into subclasses based on relatively minor differences in CH sequence and function, e.g., humans express the following subclasses: IgG1, IgG2, IgG3, IgG4, IgA1, and IgA2.

The term "variable" refers to the fact that certain segments of the variable domains differ extensively in sequence among antibodies. The V domain mediates antigen binding and defines specificity of an antibody for its antigen. However, the variability is not evenly distributed across the 110-amino acid span of the variable domains. Instead, the V regions consist of relatively invariant stretches called framework regions (FRs) of 15-30 amino acids separated by shorter regions of extreme variability called "hypervariable regions" that are each 3-30, or more typically, 9-12 amino acids long. The variable domains of native heavy and light chains each comprise four FRs, largely adopting a β -sheet configuration, connected by three hypervariable regions, which form loops connecting, and in some cases forming part of, the β -sheet structure. The hypervariable regions in each chain are held together in close proximity by the FRs and, with the hypervariable regions from the other chain, contribute to the formation of the antigen-binding site of antibodies (see Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD. (1991)). The constant domains are not involved directly in binding an antibody to an antigen, but exhibit various effector functions, such as participation of the antibody in antibody dependent cellular cytotoxicity (ADCC).

The term "monoclonal antibody" as used herein refers to an antibody obtained from a population of substantially homogeneous antibodies, i.e., the individual antibodies comprising the population are identical except for possible naturally-occurring mutations that may be present in minor amounts. Monoclonal antibodies are highly specific, being directed against a single antigenic site. Furthermore, in contrast to polyclonal antibody preparations which include

different antibodies directed against different determinants (epitopes), each monoclonal antibody is directed against a single determinant on the antigen. In addition to their specificity, the monoclonal antibodies are advantageous in that they may be synthesized uncontaminated by other antibodies. The modifier "monoclonal" is not to be construed as requiring production of the antibody by any particular method. For example, useful monoclonal antibodies of this disclosure may be prepared by the hybridoma methodology first described by Kohler et al., Nature, 256: 495 (1975), or may be made using recombinant DNA methods in bacterial, eukaryotic animal or plant cells (see, e.g., U.S. Pat. No. 4,816,567). The "monoclonal antibodies" may also be isolated from phage antibody libraries using the techniques described in Clackson et al., Nature, 352:624-628 (1991) and Marks et al., J. Mol. Biol., 222:581-597 (1991), for example.

The monoclonal antibodies herein include "chimeric" antibodies in which a portion of the heavy and/or light chain is identical with, or homologous to, corresponding sequences in antibodies derived from a particular species or belonging to a particular antibody class or subclass, while the remainder of the chain(s) is identical with or homologous to corresponding sequences in antibodies derived from another species or belonging to another antibody class or subclass, as well as fragments of such antibodies, so long as they exhibit the desired biological activity (see U.S. Pat. No. 4,816,567; and Morrison et al., Proc. Natl. Acad. Sci. USA, 81:6851-6855 (1984)). Chimeric antibodies of interest herein include "primatized" antibodies comprising variable domain antigen-binding sequences derived from a non-human primate (e.g. Old World Monkey, Ape, etc.), and human constant region sequences.

An "intact" antibody is one which comprises an antigen-binding site as well as a CL and at least heavy chain constant domains, CH1, CH2 and CH3. The constant domains may be native sequence constant domains (e.g. human native sequence constant domains) or amino acid sequence variant thereof. Preferably, the intact antibody has one or more effector functions.

"Antibody fragments" comprise a portion of an intact antibody, preferably the antigen binding or variable region of the intact antibody. Examples of antibody fragments include Fab, Fab', F(ab')₂, and Fv fragments; diabodies; linear antibodies (see U.S. Pat. No. 5,641,870, Example 2; Zapata et al., Protein Eng. 8(10): 1057-1062 [1995]); single-chain antibody molecules; and multispecific antibodies formed from antibody fragments.

Papain digestion of antibodies produces two identical antigen-binding fragments, called "Fab" fragments, and a residual "Fc" fragment, a designation reflecting the ability to crystallize readily. The Fab fragment consists of an entire L chain along with the variable region domain of the H chain (VH), and the first constant domain of one heavy chain (CH1). Each Fab fragment is monovalent with respect to antigen binding, i.e., it has a single antigen-binding site. Pepsin treatment of an antibody yields a single large F(ab')₂ fragment which roughly corresponds to two disulfide-linked Fab fragments having divalent antigen-binding activity and is still capable of cross-linking antigen. Fab' fragments differ from Fab fragments by having additional few residues at the carboxy terminus of the CH1 domain including one or more cysteines from the antibody hinge region. Fab'-SH is the designation herein for Fab' in which the cysteine residue(s) of the constant domains bear a free thiol group. F(ab')₂ antibody fragments originally were produced as pairs of Fab'

fragments which have hinge cysteines between them. Other chemical couplings of antibody fragments are also known.

The Fc fragment comprises the carboxy-terminal portions of both H chains held together by disulfides. The effector functions of antibodies are determined by sequences in the Fc region, which region is also the part recognized by Fc receptors (FcR) found on certain types of cells.

"Fv" is the minimum antibody fragment which contains a complete antigen-recognition and -binding site. This fragment consists of a dimer of one heavy- and one light-chain variable region domain in tight, non-covalent association. From the folding of these two domains emanate six hypervariable loops (3 loops each from the H and L chain) that contribute the amino acid residues for antigen binding and confer antigen binding specificity to the antibody. However, even a single variable domain (or half of an Fv comprising only three CDRs specific for an antigen) can recognize and bind antigen, although at a lower affinity than the entire binding site.

"Single-chain Fv" (also abbreviated as "sFv" or "scFv") are antibody fragments that comprise the VH and VL antibody domains connected into a single polypeptide chain. Preferably, the sFv polypeptide further comprises a polypeptide linker between the VH and VL domains which enables the sFv to form the desired structure for antigen binding. For a review of sFv, see Pluckthun in *The Pharmacology of Monoclonal Antibodies*, vol. 113, Rosenburg and Moore eds., Springer-Verlag, New York, pp. 269-315 (1994); Borrebaeck 1995, *infra*.

The term "diabodies" refers to small antibody fragments prepared by constructing sFv fragments (see preceding paragraph) with short linkers (about 5-10 residues) between the VH and VL domains such that inter-chain but not intra-chain pairing of the V domains is achieved, resulting in a bivalent fragment, i.e., fragment having two antigen-binding sites. Bispecific diabodies are heterodimers of two "crossover" sFv fragments in which the VH and VL domains of the two antibodies are present on different polypeptide chains. Diabodies are described more fully in, for example, EP 404,097; WO 93/11161; and Hollinger et al., *Proc. Natl. Acad. Sci. USA*, 90:6444-6448 (1993).

"Humanized" forms of non-human (e.g., rodent) antibodies are chimeric antibodies that contain minimal sequence derived from the non-human antibody. For the most part, humanized antibodies are human immunoglobulins (recipient antibody) in which residues from a hypervariable region of the recipient are replaced by residues from a hypervariable region of a non-human species (donor antibody) such as mouse, rat, rabbit or non-human primate having the desired antibody specificity, affinity, and capability. In some instances, framework region (FR) residues of the human immunoglobulin are replaced by corresponding non-human residues. Furthermore, humanized antibodies may comprise residues that are not found in the recipient antibody or in the donor antibody. These modifications are made to further refine antibody performance. In general, the humanized antibody will comprise substantially all at least one, and typically two, variable domains, in which all or substantially all the hypervariable loops correspond to those of a non-human immunoglobulin and all or substantially all the FRs are those of a human immunoglobulin sequence. The humanized antibody optionally also will comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin. For further details, see Jones et al., *Nature* 321:522-25 (1986); Riechmann et al., *Nature* 332:323-329 (1988); and Presta, *Curr. Op. Struct. Biol.* 2:593-96 (1992).

A "species-dependent antibody," e.g., a mammalian anti-human IgE antibody, is an antibody which has a stronger binding affinity for an antigen from a first mammalian species than it has for a homologue of that antigen from a second mammalian species. Normally, the species-dependent antibody "binds specifically" to a human antigen (i.e., has a binding affinity (Kd) value of no more than about 1×10^{-7} M, preferably no more than about 1×10^{-8} and most preferably no more than about 1×10^{-9} M) but has a binding affinity for a homologue of the antigen from a second non-human mammalian species which is at least about 50-fold, or at least about 500-fold, or at least about 1000-fold, weaker than its binding affinity for the human antigen. The species-dependent antibody can be of any of the various types of antibodies as defined above, but preferably is a humanized or human antibody.

The term "variable domain residue numbering as in Kabat" or "amino acid position numbering as in Kabat," and variations thereof, refers to the numbering system used for heavy chain variable domains or light chain variable domains of the compilation of antibodies in Kabat et al., *Sequences of Proteins of Immunological Interest*, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD. (1991). Using this numbering system, the actual linear amino acid sequence may contain fewer or additional amino acids corresponding to a shortening of, or insertion into, a FR or CDR of the variable domain. For example, a heavy chain variable domain may include a single amino acid insert (residue 52a according to Kabat) after residue 52 of H2 and inserted residues (e.g. residues 82a, 82b, and 82c, etc. according to Kabat) after heavy chain FR residue 82. The Kabat numbering of residues may be determined for a given antibody by alignment at regions of homology of the sequence of the antibody with a "standard" Kabat numbered sequence.

The phrase "substantially similar," or "substantially the same," as used herein, denotes a sufficiently high degree of similarity between two numeric values (generally one associated with an antibody of the disclosure and the other associated with a reference/comparator antibody) such that one of skill in the art would consider the difference between the two values to be of little or no biological and/or statistical significance within the context of the biological characteristic measured by the values (e.g., Kd values). The difference between the two values is preferably less than about 50%, preferably less than about 40%, preferably less than about 30%, preferably less than about 20%, preferably less than about 10% as a function of the value for the reference/comparator antibody.

"Binding affinity" generally refers to the strength of the sum of noncovalent interactions between a single binding site of a molecule (e.g., an antibody) and its binding partner (e.g., an antigen). Unless indicated otherwise, as used herein, "binding affinity" refers to intrinsic binding affinity which reflects a 1:1 interaction between members of a binding pair (e.g., antibody and antigen). The affinity of a molecule X for its partner Y can generally be represented by the dissociation constant (Kd). Affinity can be measured by common methods known in the art, including those described herein. Low-affinity antibodies generally bind antigen slowly and tend to dissociate readily, whereas high-affinity antibodies generally bind antigen faster and tend to remain bound longer. A variety of methods of measuring binding affinity are known in the art, any of which can be used for purposes of this disclosure. Illustrative embodiments are described in the following.

The “Kd” or “Kd value” according to this disclosure is measured by a radiolabeled antigen binding assay (RIA) performed with the Fab version of an antibody of interest and its antigen as described by the following assay that measures solution binding affinity of Fabs for antigen by equilibrating Fab with a minimal concentration of (125I)-labeled antigen in the presence of a titration series of unlabeled antigen, then capturing bound antigen with an anti-Fab antibody-coated plate (Chen, et al., (1999) *J. Mol Biol* 293:865-881). To establish conditions for the assay, microtiter plates (Dyner) are coated overnight with 5 mcg/ml of a capturing anti-Fab antibody (Cappel Labs) in 50 mM sodium carbonate (pH 9.6), and subsequently blocked with 2% (w/v) bovine serum albumin in PBS for two to five hours at room temperature (approximately 23° C.). In a non-adsorbant plate, 100 pM or 26 pM [125I]-antigen are mixed with serial dilutions of a Fab of interest (e.g., consistent with assessment of an anti-VEGF antibody, Fab-12, in Presta et al., (1997) *Cancer Res.* 57:4593-4599). The Fab of interest is then incubated overnight; however, the incubation may continue for a longer period (e.g., 65 hours) to ensure that equilibrium is reached. Thereafter, the mixtures are transferred to the capture plate for incubation at room temperature (e.g., for one hour). The solution is then removed and the plate washed eight times with 0.1% Tween-20 in PBS. When the plates have dried, 150 microliter/well of scintillant (MicroScint-20; Packard) is added, and the plates are counted on a Topcount gamma counter (Packard) for ten minutes. Concentrations of each Fab that give less than or equal to 20% of maximal binding are chosen for use in competitive binding assays. The Kd or Kd value may also be measured by using surface plasmon resonance assays using a BIAcore™-2000 or a BIAcore™-3000 (BIAcore, Inc., Piscataway, NJ) at 25° C. with immobilized antigen CM5 chips at approx. 10 response units (RU). Briefly, carboxymethylated dextran biosensor chips (CM5, BIAcore Inc.) are activated with N-ethyl-N'-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC) and N-hydroxysuccinimide (NHS) according to the supplier's instructions. Antigen is diluted with 10 mM sodium acetate, pH 4.8, into 5 mcg/ml (approx. 0.2 uM) before injection at a flow rate of 5 microliter/minute to achieve approximately 10 response units (RU) of coupled protein. Following the injection of antigen, 1M ethanolamine is injected to block unreacted groups. For kinetics measurements, two-fold serial dilutions of Fab (0.78 nM to 500 nM) are injected in PBS with 0.05% Tween 20 (PBST) at 25° C. at a flow rate of approximately 25 microliter/min. Association rates (kon) and dissociation rates (koff) are calculated using a simple one-to-one Langmuir binding model (BIAcore Evaluation Software version 3.2) by simultaneous fitting the association and dissociation sensorgram. The equilibrium dissociation constant (Kd) is calculated as the ratio koff/kon. See, e.g., Chen, Y., et al., (1999) *J. Mol Biol* 293:865-881. If the on-rate exceeds 10⁶ M⁻¹ S⁻¹ by the surface plasmon resonance assay above, then the on-rate can be determined by using a fluorescent quenching technique that measures the increase or decrease in fluorescence emission intensity (excitation=295 nm; emission=340 nm, 16 nm band-pass) at 25° C. of a 20 nM anti-antigen antibody (Fab form) in PBS, pH 7.2, in the presence of increasing concentrations of antigen as measured in a spectrometer, such as a stop-flow equipped spectrometer (Aviv Instruments) or a 8000-series SLM-Aminco spectrophotometer (ThermoSpectronic) with a stirred cuvette.

An “on-rate” or “rate of association” or “association rate” or “kon” according to this disclosure can also be determined

with the same surface plasmon resonance technique described above using a BIAcore™-2000 or a BIAcore™-3000 (BIAcore, Inc., Piscataway, NJ) at 25° C. with immobilized antigen CM5 chips at approx. 10 response units (RU). Briefly, carboxymethylated dextran biosensor chips (CM5, BIAcore Inc.) are activated with N-ethyl-N'-(3-dimethylaminopropyl)-carbodiimide hydrochloride (EDC) and N-hydroxysuccinimide (NHS) according to the supplier's instructions. Antigen is diluted with 10 mM sodium acetate, pH 4.8, into 5 mcg/ml (approx. 0.2 uM) before injection at a flow rate of 5 microliter/minute to achieve approximately 10 response units (RU) of coupled protein. Following the injection of 1M ethanolamine to block unreacted groups. For kinetics measurements, two-fold serial dilutions of Fab (0.78 nM to 500 nM) are injected in PBS with 0.05% Tween 20 (PBST) at 25° C. at a flow rate of approximately 25 microliter/min. Association rates (kon) and dissociation rates (koff) are calculated using a simple one-to-one Langmuir binding model (BIAcore Evaluation Software version 3.2) by simultaneous fitting the association and dissociation sensorgram. The equilibrium dissociation constant (Kd) was calculated as the ratio koff/kon. See, e.g., Chen, Y., et al., (1999) *J. Mol Biol* 293:865-81. However, if the on-rate exceeds 10⁶ M⁻¹ S⁻¹ by the surface plasmon resonance assay above, then the on-rate is preferably determined by using a fluorescent quenching technique that measures the increase or decrease in fluorescence emission intensity (excitation=295 nm; emission=340 nm, 16 nm band-pass) at 25° C. of a 20 nM anti-antigen antibody (Fab form) in PBS, pH 7.2, in the presence of increasing concentrations of antigen as measured in a spectrometer, such as a stop-flow equipped spectrophotometer (Aviv Instruments) or a 8000-series SLM-Aminco spectrophotometer (ThermoSpectronic) with a stirred cuvette.

The phrase “substantially reduced,” or “substantially different”, as used herein, denotes a sufficiently high degree of difference between two numeric values (generally one associated with an antibody of the disclosure and the other associated with a reference/comparator antibody) such that one of skill in the art would consider the difference between the two values to be of statistical significance within the context of the biological characteristic measured by the values (e.g., Kd values, HAMA response). The difference between the two values is preferably greater than about 10%, preferably greater than about 20%, preferably greater than about 30%, preferably greater than about 40%, preferably greater than about 50% as a function of the value for the reference/comparator antibody.

An “antigen” is a predetermined antigen to which an antibody can selectively bind. The target antigen may be polypeptide, carbohydrate, nucleic acid, lipid, hapten or other naturally occurring or synthetic compound. Preferably, the target antigen is an EBV gp350 polypeptide. An “acceptor human framework” for the purposes herein is a framework comprising the amino acid sequence of a VL or VH framework derived from a human immunoglobulin framework, or from a human consensus framework. An acceptor human framework “derived from” a human immunoglobulin framework or human consensus framework may comprise the same amino acid sequence thereof, or may contain pre-existing amino acid sequence changes. Where pre-existing amino acid changes are present, preferably no more than 5 and preferably 4 or less, or 3 or less, pre-existing amino acid changes are present. Where pre-existing amino acid changes are present in a VH, preferably those changes are only at three, two, or one of positions 71H, 73H and 78H; for instance, the amino acid residues at those positions may

be 71A, 73T and/or 78A. The VL acceptor human framework may be identical in sequence to the VL human immunoglobulin framework sequence or human consensus framework sequence.

Antibodies of this disclosure may be able to compete for binding to the same epitope as is bound by a second antibody. Monoclonal antibodies are considered to share the "same epitope" if each blocks binding of the other by 40% or greater at the same antibody concentration in a standard in vitro antibody competition binding analysis.

A "human consensus framework" is a framework which represents the most commonly occurring amino acid residue in a selection of human immunoglobulin VL or VH framework sequences. Generally, the selection of human immunoglobulin VL or VH sequences is from a subgroup of variable domain sequences. Generally, the subgroup of sequences is a subgroup as in Kabat et al., supra. For the VL, the subgroup may be subgroup kappa I as in Kabat et al. For the VH, the subgroup is subgroup III as in Kabat et al.

A "VH subgroup III consensus framework" comprises the consensus sequence obtained from the amino acid sequences in variable heavy subgroup III of Kabat et al.

A "VL subgroup I consensus framework" comprises the consensus sequence obtained from the amino acid sequences in variable light kappa subgroup I of Kabat et al.

An "unmodified human framework" is a human framework which has the same amino acid sequence as the acceptor human framework, e.g. lacking human to non-human amino acid substitution(s) in the acceptor human framework.

An "altered hypervariable region" for the purposes herein is a hypervariable region comprising one or more (e.g. one to about 16) amino acid substitution(s) therein.

An "un-modified hypervariable region" for the purposes herein is a hypervariable region having the same amino acid sequence as a non-human antibody from which it was derived, i.e. one which lacks one or more amino acid substitutions therein.

The term "hypervariable region", "HVR", "HV" or "CDR", when used herein refers to the regions of an antibody variable domain which are hypervariable in sequence and/or form structurally defined loops. Generally, antibodies comprise six hypervariable regions; three in the VH (H1, H2, H3), and three in the VL (L1, L2, L3). Several hypervariable region delineations are in use and are encompassed herein. The Kabat Complementarity Determining Regions (CDRs) are based on sequence variability and are the most commonly used. The "contact" hypervariable regions are based on an analysis of the available complex crystal structures. The residues from each of these hypervariable regions are noted below. Unless otherwise denoted, Kabat numbering is employed. Hypervariable region locations are generally: amino acids 24-34 (HVR-L1), amino acids 49-56 (HVR-L2), amino acids 89-97 (HVR-L3), amino acids 26-35A (HVR-H1), amino acids 49-65 (HVR-H2), and amino acids 93-102 (HVR-H3). Hypervariable regions may also comprise "extended hypervariable regions" as follows: amino acids 24-36 (L1), and amino acids 46-56 (L2) in the VL, numbered according to Kabat et al., supra for each of these definitions.

A "human antibody" is one which possesses an amino acid sequence which corresponds to that of an antibody produced by a human and/or has been made using any of the techniques for making human antibodies as disclosed herein. This definition of a human antibody specifically excludes a humanized antibody comprising non-human antigen-binding residues.

An "affinity matured" antibody is one with one or more alterations in one or more CDRs thereof which result in an improvement in the affinity or binding specificity of the antibody for antigen, compared to a parent antibody which does not possess those alteration(s). Preferred affinity matured antibodies will have nanomolar or even picomolar affinities for the target antigen. Affinity matured antibodies are produced by procedures known in the art. Marks et al. *Bio/Technology* 10:779-783 (1992) describes affinity maturation by VH and VL domain shuffling. Random mutagenesis of CDR and/or framework residues is described by: Barbas et al. *Proc Nat. Acad. Sci. USA* 91:3809-3813 (1994); Schier et al. *Gene* 169:147-55 (1995); Yelton et al. *J. Immunol.* 155:1994-2004 (1995); Jackson et al., *J. Immunol.* 154(7):3310-19 (1995); and Hawkins et al, *J. Mol. Biol.* 226:889-96 (1992).

A "blocking" antibody or an "antagonist" antibody is one which inhibits or reduces biological activity of the antigen it binds. Preferred blocking antibodies or antagonist antibodies substantially or completely inhibit the biological activity of the antigen.

A "gp350 binding oligopeptide" is an oligopeptide that binds, preferably specifically, to an EBV gp350 protein. gp350 binding oligopeptides may be chemically synthesized using known oligopeptide synthesis methodology or may be prepared and purified using recombinant technology. gp350 binding oligopeptides are usually at least about 5 amino acids in length, alternatively at least about 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100 amino acids in length or more, wherein such oligopeptides are capable of binding, preferably specifically, to an EBV gp350 protein. gp350 binding oligopeptides of this disclosure preferably comprise or consist of at least one complementarity determining region (CDR) of the antibodies of this disclosure. gp350 binding oligopeptides may be identified without undue experimentation using well known techniques. In this regard, techniques for screening oligopeptide libraries for oligopeptides that are capable of specifically binding to a polypeptide target are known in the art (see, e.g., U.S. Pat. Nos. 5,556,762, 5,750,373, 4,708,871, 4,833,092, 5,223,409, 5,403,484, 5,571,689, 5,663,143; PCT Publication Nos. WO 84/03506 and WO84/03564; Geysen et al., *Proc. Natl. Acad. Sci. U.S.A.*, 81:3998-4002 (1984); Geysen et al., *Proc. Natl. Acad. Sci. U.S.A.*, 82:178-182 (1985); Geysen et al., in *Synthetic Peptides as Antigens*, 130-149 (1986); Geysen et al., *J. Immunol. Meth.*, 102:259-274 (1987); Schoofs et al., *J. Immunol.*, 140:611-616 (1988), Cwirla, S. E. et al. (1990) *Proc. Natl. Acad. Sci. USA*, 87:6378; Lowman, H. B. et al. (1991) *Biochemistry*, 30:10832; Clackson, T. et al. (1991) *Nature*, 352: 624; Marks, J. D. et al. (1991), *J. Mol. Biol.*, 222:581; Kang, A. S. et al. (1991) *Proc. Natl. Acad. Sci. USA*, 88:8363, and Smith, G. P. (1991) *Current Opin. Biotechnol.*, 2:668).

An antibody, oligopeptide or other organic molecule "which binds" an antigen of interest, e.g. an EBV polypeptide antigen target, is one that binds the antigen with sufficient affinity such that the antibody or oligopeptide is useful as a diagnostic and/or therapeutic agent in targeting a viral particle, or a cell or a tissue expressing the antigen, and does not significantly cross-react with other proteins, such as other herpes virus proteins. The extent of binding of the antibody or oligopeptide to a "non-target" protein will often

be less than about 10% of the binding of the antibody or oligopeptide to its target protein as determined by fluorescence activated cell sorting (FACS) analysis or radioimmuno-precipitation (RIA). Regarding the binding of an antibody or oligopeptide to a target molecule, the terms “specific binding” or “specifically binds to” or “is specific for” a particular polypeptide or an epitope on a particular polypeptide target means binding that is measurably different from a non-specific interaction. Specific binding can be measured, for example, by determining binding of a molecule compared to binding of a control molecule, which generally is a molecule of similar structure that does not have binding activity. For example, specific binding can be determined by competition with a control molecule that is similar to the target, for example, an excess of non-labeled target. In this case, specific binding is indicated if the binding of the labeled target to a probe is competitively inhibited by excess unlabeled target. The term “specific binding” or “specifically binds to” or “is specific for” a particular polypeptide or an epitope on a particular polypeptide target as used herein can be exhibited, for example, by a molecule having a K_d for the target of at least about 10^{-4} M, alternatively at least about 10^{-5} M, alternatively at least about 10^{-6} M, alternatively at least about 10^{-7} M, alternatively at least about 10^{-8} M, alternatively at least about 10^{-9} M, alternatively at least about 10^{-10} M, alternatively at least about 10^{-11} M, alternatively at least about 10^{-12} M, or greater. The term “specific binding” may refer to binding where a molecule binds to a particular polypeptide or epitope on a particular polypeptide without substantially binding to any other polypeptide or polypeptide epitope.

An antibody or oligopeptide that “inhibits the growth of infected cells expressing an EBV gp350 protein” or a “growth inhibitory” antibody or oligopeptide is one which results in measurable growth inhibition of infected cells expressing or overexpressing the appropriate EBV gp350 protein. The EBV gp350 protein may be a transmembrane polypeptide expressed on the surface of an infected cell or may be a polypeptide that is produced and secreted by an infected cell. Preferred growth inhibitory anti-gp350 antibodies or oligopeptides inhibit growth of gp350-expressing cells by greater than 20%, preferably from about 20% to about 50%, and even more preferably, by greater than 50% (e.g., from about 50% to about 100%) as compared to the appropriate control, the control typically being cells not treated with the antibody or oligopeptide being tested. Growth inhibition can be measured at an antibody concentration of about 0.1 to 30 mcg/ml or about 0.5 nM to 200 nM in cell culture, where the growth inhibition is determined 1-10 days after exposure of the cells to the antibody. Growth inhibition of cells in vivo can be determined in various ways such as is described in the Examples section below. The antibody is growth inhibitory in vivo if administration of the anti-gp350 antibody at about 1 µg/kg to about 100 mg/kg body weight results in reduction in infected cells or inhibited EBV proliferation within about 1 day to 3 months from the first administration of the antibody, preferably within about 1 to 5 days.

Antibody “effector functions” refer to those biological activities attributable to the Fc region (a native sequence Fc region or amino acid sequence variant Fc region) of an antibody, and vary with the antibody isotype. Examples of antibody effector functions include: C1q binding and complement dependent cytotoxicity; Fc receptor binding; antibody-dependent cell-mediated cytotoxicity (ADCC);

phagocytosis; down regulation of cell surface receptors (e.g., B cell receptor); and B cell activation.

“Antibody-dependent cell-mediated cytotoxicity” or “ADCC” refers to a form of cytotoxicity in which secreted Ig bound onto Fc receptors (FcRs) present on certain cytotoxic cells (e.g., Natural Killer (NK) cells, neutrophils, and macrophages) enable these cytotoxic effector cells to bind specifically to an antigen-bearing target cell and subsequently kill the target cell with cytotoxins. The antibodies “arm” the cytotoxic cells and are absolutely required for such killing. The primary cells for mediating ADCC, NK cells, express FcγRIII only, whereas monocytes express FcγRI, FcγRII and FcγRIII. FcR expression on hematopoietic cells is summarized in Table 3 on page 464 of Ravetch and Kinet, *Annu. Rev. Immunol.* 9:457-92 (1991). To assess ADCC activity of a molecule of interest, an in vitro ADCC assay, such as that described in U.S. Pat. No. 5,500,362 or 5,821,337 may be performed. Useful effector cells for such assays include peripheral blood mononuclear cells (PBMC) and Natural Killer (NK) cells. Alternatively, or additionally, ADCC activity of the molecule of interest may be assessed in vivo, e.g., in an animal model such as that disclosed in Clynes et al. (USA) 95:652-56 (1998).

“Fc receptor” or “FcR” describes a receptor that binds to the Fc region of an antibody. The preferred FcR is a native sequence human FcR. Moreover, a preferred FcR is one which binds an IgG antibody (a gamma receptor) and includes receptors of the FcγRI, FcγRII and FcγRIII subclasses, including allelic variants and alternatively spliced forms of these receptors. FcγRII receptors include FcγRIIA (an “activating receptor”) and FcγRIIB (an “inhibiting receptor”), which have similar amino acid sequences that differ primarily in the cytoplasmic domains thereof. Activating receptor FcγRIIA contains an immunoreceptor tyrosine-based activation motif (ITAM) in its cytoplasmic domain. Inhibiting receptor FcγRIIB contains an immunoreceptor tyrosine-based inhibition motif (ITIM) in its cytoplasmic domain. (see review M. in Daeron, *Annu. Rev. Immunol.* 15:203-234 (1997)). FcRs are reviewed in Ravetch and Kinet, *Annu. Rev. Immunol.* 9:457-492 (1991); Capel et al., *Immunomethods* 4:25-34 (1994); and de Haas et al., *J. Lab. Clin. Med.* 126:330-41 (1995). Other FcRs, including those to be identified in the future, are encompassed by the term “FcR” herein. The term also includes the neonatal receptor, FcRn, which is responsible for the transfer of maternal IgGs to the fetus (Guyer et al., *J. Immunol.* 117:587 (1976) and Kim et al., *J. Immunol.* 24:249 (1994)).

“Human effector cells” are leukocytes which express one or more FcRs and perform effector functions. Preferably, the cells express at least FcγRIII and perform ADCC effector function. Examples of human leukocytes which mediate ADCC include peripheral blood mononuclear cells (PBMC), natural killer (NK) cells, monocytes, cytotoxic T cells and neutrophils; with PBMCs and NK cells being preferred. The effector cells may be isolated from a native source, e.g., from blood.

“Complement dependent cytotoxicity” or “CDC” refers to the lysis of a target cell in the presence of complement. Activation of the classical complement pathway is initiated by the binding of the first component of the complement system (C1q) to antibodies (of the appropriate subclass) which are bound to their cognate antigen. To assess complement activation, a CDC assay, e.g., as described in Gazzano-Santoro et al., *J. Immunol. Methods* 202:163 (1996), may be performed.

An antibody or oligopeptide which “induces cell death” is one which causes a viable cell to become nonviable. The cell

is one which expresses an EBV gp350 protein or is infected with EBV. Cell death *in vitro* may be determined in the absence of complement and immune effector cells to distinguish cell death induced by antibody-dependent cell-mediated cytotoxicity (ADCC) or complement dependent cytotoxicity (CDC). Thus, the assay for cell death may be performed using heat inactivated serum (i.e., in the absence of complement) and in the absence of immune effector cells. To determine whether the antibody or oligopeptide can induce cell death, loss of membrane integrity as evaluated by uptake of propidium iodide (PI), trypan blue (see Moore et al. *Cytotechnology* 17:1-11 (1995)) or 7AAD can be assessed relative to untreated cells.

A "gp350-expressing cell" is a cell which expresses an endogenous or transfected EBV gp350 protein which may include expression either on the cell surface or in a secreted form.

As used herein, the term "immunoadhesin" designates antibody-like molecules which combine the binding specificity of a heterologous protein (an "adhesin") with the effector functions of immunoglobulin constant domains. Structurally, the immunoadhesins comprise a fusion of an amino acid sequence with the desired binding specificity which is other than the antigen recognition and binding site of an antibody (i.e., is "heterologous"), and an immunoglobulin constant domain sequence. The adhesin part of an immunoadhesin molecule typically is a contiguous amino acid sequence comprising at least the binding site of a receptor or a ligand. The immunoglobulin constant domain sequence in the immunoadhesin may be obtained from any immunoglobulin, such as IgG-1, IgG-2, IgG-3, or IgG-4 subtypes, IgA (including IgA-1 and IgA-2), IgE, IgD, or IgM.

The word "label" when used herein refers to a detectable compound or composition which is conjugated directly or indirectly to an antibody or oligopeptide to generate a "labeled" antibody or oligopeptide. The label may be detectable by itself (e.g. radioisotope labels or fluorescent labels) or, in the case of an enzymatic label, may catalyze chemical alteration of a substrate compound or composition which is detectable.

The term "cytotoxic agent" as used herein refers to a substance that inhibits or prevents the function of cells and/or causes destruction of cells. The term is intended to include radioactive isotopes (e.g., At211, I131, I125, Y90, Re186, Re188, Sm153, Bi212, P32 and radioactive isotopes of Lu), chemotherapeutic agents, enzymes and fragments thereof such as nucleolytic enzymes, antibiotics, immune suppressants, and toxins such as small molecule toxins or enzymatically active toxins of bacterial, fungal, plant or animal origin, including fragments and/or variants thereof. An antiviral agent causes destruction of virus-infected cells.

A "growth inhibitory agent" when used herein refers to a compound or composition which inhibits growth of a cell, especially an EBV-infected cell, either *in vitro* or *in vivo*. Thus, the growth inhibitory agent may be one which significantly reduces the percentage of EBV-infected cells in S phase. Examples of growth inhibitory agents include agents that block cell cycle progression (at a place other than S phase), such as agents that induce G1 arrest and M-phase arrest. Classical M-phase blockers include the vincas (vincristine and vinblastine), taxanes, and topoisomerase II inhibitors such as doxorubicin, epirubicin, daunorubicin, etoposide, and bleomycin. Those agents that arrest G1 also spill over into S-phase arrest, for example, DNA alkylating

agents such as tamoxifen, prednisone, dacarbazine, mechlorethamine, cisplatin, methotrexate, 5-fluorouracil, and ara-C.

The term "cytokine" is a generic term for proteins released by one cell population which act on another cell as intercellular mediators. Examples of such cytokines are lymphokines, monokines, and traditional polypeptide hormones. Included among the cytokines are growth hormone such as human growth hormone, N-methionyl human growth hormone, and bovine growth hormone; parathyroid hormone; thyroxine; insulin; proinsulin; relaxin; prolaxin; glycoprotein hormones such as follicle stimulating hormone (FSH), thyroid stimulating hormone (TSH), and luteinizing hormone (LH); hepatic growth factor; fibroblast growth factor; prolactin; placental lactogen; tumor necrosis factor- α and - β ; mullerian-inhibiting substance; mouse gonadotropin-associated peptide; inhibin; activin; vascular endothelial growth factor; integrin; thrombopoietin (TPO); nerve growth factors such as NGF- β ; platelet-growth factor; transforming growth factors (TGFs) such as TGF- α and TGF- β ; insulin-like growth factor-I and -II; erythropoietin (EPO); osteoinductive factors; interferons such as interferon- α , - β , and - γ ; colony stimulating factors (CSFs) such as macrophage-CSF (M-CSF); granulocyte-macrophage-CSF (GM-CSF); and granulocyte-CSF (G-CSF); interleukins (ILs) such as IL-1, IL-1a, IL-2, IL-3, IL-4, IL-5, IL-6, IL-7, IL-8, IL-9, IL-11, IL-12; a tumor necrosis factor such as TNF- α or TNF- β ; and other polypeptide factors including LIF and kit ligand (KL). As used herein, the term cytokine includes proteins from natural sources or from recombinant cell culture and biologically active equivalents of the native sequence cytokines.

The term "package insert" is used to refer to instructions customarily included in commercial packages of therapeutic products, that contain information about the indications, usage, dosage, administration, contraindications and/or warnings concerning the use of such therapeutic products.

II. Compositions and Methods

A. Anti-gp350 Antibodies

This disclosure provides anti-gp350 antibodies which may find use herein as therapeutic and/or diagnostic agents. Exemplary antibodies include polyclonal, monoclonal, humanized, bispecific, and heteroconjugate antibodies.

1. Polyclonal Antibodies

Polyclonal antibodies are preferably raised in animals by multiple subcutaneous or intraperitoneal injections of the relevant antigen and an adjuvant. It may be useful to conjugate the relevant antigen (especially when synthetic peptides are used) to a protein that is immunogenic in the species to be immunized. For example, the antigen can be conjugated to keyhole limpet hemocyanin (KLH), serum albumin, bovine thyroglobulin, or soybean trypsin inhibitor, using a bifunctional or derivatizing agent, e.g., maleimido-benzoyl sulfosuccinimide ester (conjugation through cysteine residues), N-hydroxysuccinimide (through lysine residues), glutaraldehyde, succinic anhydride, SOCl₂, or R1N=C=NR, where R and R1 are different alkyl groups.

Animals are immunized against the antigen, immunogenic conjugates, or derivatives by combining, e.g., 100 mcg or 5 mcg of the protein or conjugate (for rabbits or mice, respectively) with 3 volumes of Freund's complete adjuvant and injecting the solution intradermally at multiple sites. One month later, the animals are boosted with $\frac{1}{5}$ to $\frac{1}{10}$ the original amount of peptide or conjugate in Freund's complete adjuvant by subcutaneous injection at multiple sites.

Seven to fourteen days later, the animals are bled and the serum is assayed for antibody titer. Animals are boosted until the titer plateaus. Conjugates also can be made in recombinant cell culture as protein fusions. Also, aggregating agents such as alum are suitably used to enhance the immune response.

2. Monoclonal Antibodies

Monoclonal antibodies may be made using the hybridoma method first described by Kohler et al., *Nature*, 256:495 (1975), or may be made by recombinant DNA methods (U.S. Pat. No. 4,816,567).

In the hybridoma method, a mouse or other appropriate host animal, is immunized as described above to elicit lymphocytes that produce or are capable of producing antibodies that will specifically bind to the protein used for immunization. Alternatively, lymphocytes may be immunized in vitro. After immunization, lymphocytes are isolated and then fused with a myeloma cell line using a suitable fusing agent, such as polyethylene glycol, to form a hybridoma cell (Goding, *Monoclonal Antibodies: Principles and Practice*, pp. 59-103 (Academic Press, 1986)).

The hybridoma cells thus prepared are seeded and grown in a suitable culture medium which medium preferably contains one or more substances that inhibit the growth or survival of the unfused, parental myeloma cells (also referred to as fusion partner). For example, if the parental myeloma cells lack the enzyme hypoxanthine guanine phosphoribosyl transferase (HGPRT or HPRT), the selective culture medium for the hybridomas typically will include hypoxanthine, aminopterin, and thymidine (HAT medium), which substances prevent the growth of HGPRT-deficient cells.

Preferred fusion partner myeloma cells are those that fuse efficiently, support stable high-level production of antibody by the selected antibody-producing cells, and are sensitive to a selective medium that selects against the unfused parental cells. Preferred myeloma cell lines are murine myeloma lines, such as those derived from MOPC-21 and MPC-11 mouse tumors available from the Salk Institute Cell Distribution Center, San Diego, California USA, and SP-2 and derivatives e.g., X63-Ag8-653 cells available from the American Type Culture Collection, Manassas, Virginia, USA. Human myeloma and mouse-human heteromyeloma cell lines also have been described for producing human monoclonal antibodies (Kozbor, *J. Immunol.*, 133:3001 (1984); and Brodeur et al., *Monoclonal Antibody Production Techniques and Applications*, pp. 51-63 (Marcel Dekker, Inc., New York, 1987)).

Culture medium in which hybridoma cells are growing is assayed for production of monoclonal antibodies directed against the antigen. Preferably, the binding specificity of monoclonal antibodies produced by hybridoma cells is determined by immunoprecipitation or by an in vitro binding assay, such as radioimmunoassay (RIA) or enzyme-linked immunosorbent assay (ELISA). The binding affinity of the monoclonal antibody can, for example, be determined by the Scatchard analysis described in Munson et al., *Anal. Biochem.*, 107:220 (1980).

Once hybridoma cells that produce antibodies of the desired specificity, affinity, and/or activity are identified, the clones may be subcloned by limiting dilution procedures and grown by standard methods (Goding, *Monoclonal Antibodies: Principles and Practice*, pp. 59-103 (Academic Press, 1986)). Suitable culture media for this purpose include, for example, D-MEM or RPMI-1640 medium. In addition, the hybridoma cells may be grown in vivo as ascites tumors in an animal e.g., by i.p. injection of the cells into mice.

The monoclonal antibodies secreted by the subclones are suitably separated from the culture medium, ascites fluid, or serum by conventional antibody purification procedures such as, for example, affinity chromatography (e.g., using protein A or protein G-Sepharose) or ion-exchange chromatography, hydroxylapatite chromatography, gel electrophoresis, dialysis, etc.

DNA encoding the monoclonal antibodies is readily isolated and sequenced using conventional procedures (e.g., by using oligonucleotide probes that are capable of binding specifically to genes encoding the heavy and light chains of murine antibodies). The hybridoma cells serve as a preferred source of such DNA. Once isolated, the DNA may be placed into expression vectors, which are then transfected into host cells such as *E. coli* cells, simian COS cells, Chinese Hamster Ovary (CHO) cells, or myeloma cells that do not otherwise produce antibody protein, to obtain the synthesis of monoclonal antibodies in the recombinant host cells. Review articles on recombinant expression in bacteria of DNA encoding the antibody include Skerra et al., *Curr. Opinion in Immunol.*, 5:256-262 (1993) and Pluckthun, *Immunol. Revs.* 130:151-188 (1992).

Monoclonal antibodies or antibody fragments may be isolated from antibody phage libraries generated using the techniques described in McCafferty et al., *Nature*, 348:552-554 (1990). Clackson et al., *Nature*, 352:624-628 (1991) and Marks et al., *J. Mol. Biol.*, 222:581-597 (1991) describe the isolation of murine and human antibodies, respectively, using phage libraries. Subsequent publications describe the production of high affinity (nM range) human antibodies by chain shuffling (Marks et al., *Bio/Technology*, 10:779-783 (1992)), as well as combinatorial infection and in vivo recombination as a strategy for constructing very large phage libraries (Waterhouse et al., *Nuc. Acids. Res.* 21:2265-66 (1993)). Thus, these techniques are viable alternatives to traditional monoclonal antibody hybridoma techniques for isolation of monoclonal antibodies.

The DNA that encodes the antibody may be modified to produce chimeric or fusion antibody polypeptides, for example, by substituting human heavy chain and light chain constant domain (CH and CL) sequences for the homologous murine sequences (U.S. Pat. No. 4,816,567; and Morrison, et al., *Proc. Natl Acad. Sci. USA*, 81:6851 (1984)), or by fusing the immunoglobulin coding sequence with all or part of the coding sequence for a non-immunoglobulin polypeptide (heterologous polypeptide). The non-immunoglobulin polypeptide sequences can substitute for the constant domains of an antibody, or they are substituted for the variable domains of one antigen-combining site of an antibody to create a chimeric bivalent antibody comprising one antigen-combining site having specificity for an antigen and another antigen-combining site having specificity for a different antigen.

3. Human and Humanized Antibodies

The anti-gp350 antibodies of this disclosure may further comprise humanized antibodies or human antibodies. Humanized forms of non-human (e.g., murine) antibodies are chimeric immunoglobulins, immunoglobulin chains or fragments thereof (such as Fv, Fab, Fab', F(ab')₂ or other antigen-binding subsequences of antibodies) which contain minimal sequence derived from non-human immunoglobulin. Humanized antibodies include human immunoglobulins (recipient antibody) in which residues from a complementary determining region (CDR) of the recipient are replaced by residues from a CDR of a non-human species (donor antibody) such as mouse, rat or rabbit having the desired specificity, affinity and capacity. In some instances, Fv

framework residues of the human immunoglobulin are replaced by corresponding non-human residues. Humanized antibodies may also comprise residues which are found neither in the recipient antibody nor in the imported CDR or framework sequences. In general, the humanized antibody will comprise substantially all of at least one, and typically two, variable domains, in which all or substantially all of the CDR regions correspond to those of a non-human immunoglobulin and all or substantially all of the FR regions are those of a human immunoglobulin consensus sequence. The humanized antibody optimally also will comprise at least a portion of an immunoglobulin constant region (Fc), typically that of a human immunoglobulin (Jones et al., *Nature*, 321:522-25 (1986); Riechmann et al., *Nature*, 332:323-29 (1988); and Presta, *Curr. Op. Struct. Biol.*, 2:593-96 (1992)).

Methods for humanizing non-human antibodies are known in the art. Generally, a humanized antibody has one or more amino acid residues introduced into it from a source which is non-human. These non-human amino acid residues are often referred to as "import" residues, which are typically taken from an "import" variable domain. Humanization can be essentially performed by substituting rodent CDRs or CDR sequences for the corresponding sequences of a human antibody. Accordingly, such "humanized" antibodies are chimeric antibodies (U.S. Pat. No. 4,816,567), wherein substantially less than an intact human variable domain has been substituted by the corresponding sequence from a non-human species. In practice, humanized antibodies are typically human antibodies in which some CDR residues and possibly some FR residues are substituted by residues from analogous sites in rodent antibodies.

The choice of human variable domains, both light and heavy, to be used in making the humanized antibodies is important to reduce antigenicity and HAMA response (human anti-mouse antibody) when the antibody is intended for human therapeutic use. According to the "best-fit" method, the sequence of the variable domain of a rodent antibody is screened against the entire library of known human variable domain sequences. The human V domain sequence which is closest to that of the rodent is identified and the human framework region (FR) within it accepted for the humanized antibody (Sims et al., *J. Immunol.* 151:2296 (1993); Chothia et al., *J. Mol. Biol.*, 196:901 (1987)). Another method uses a framework region derived from the consensus sequence of all human antibodies of a subgroup of light or heavy chains. The same framework may be used for several different humanized antibodies (Carter et al., *Proc. Natl. Acad. Sci. USA*, 89:4285 (1992); Presta et al., *J. Immunol.* 151:2623 (1993)).

It is further important that antibodies be humanized with retention of high binding affinity for the antigen and other favorable biological properties. To achieve this goal, humanized antibodies may be prepared by a process of analysis of the parental sequences and various conceptual humanized products using three-dimensional models of the parental and humanized sequences. Three-dimensional immunoglobulin models are commonly available and are familiar to those skilled in the art. Computer programs are available which illustrate and display probable three-dimensional conformational structures of selected candidate immunoglobulin sequences. Inspection of these displays permits analysis of the likely role of the residues in the functioning of the candidate immunoglobulin sequence, i.e., the analysis of residues that influence the ability of the candidate immunoglobulin to bind its antigen. In this way, FR residues can be selected and combined from the recipient and import sequences so that the desired antibody characteristic, such as

increased affinity or specificity for the target antigen(s), is achieved. In general, the hypervariable region residues are directly and most substantially involved in influencing antigen binding.

Various forms of humanized anti-gp350 antibodies are contemplated. For example, the humanized antibody may be an antibody fragment, such as a Fab, which is optionally conjugated with one or more cytotoxic agent(s) to generate an immunoconjugate. Alternatively, the humanized antibody may be an intact antibody, such as an intact IgG1 antibody.

As an alternative to humanization, human antibodies can be generated. For example, it is now possible to produce transgenic animals (e.g., mice) that are capable, upon immunization, of producing a full repertoire of human antibodies in the absence of endogenous immunoglobulin production. For example, it has been described that the homozygous deletion of the antibody heavy-chain joining region (JH) gene in chimeric and germ-line mutant mice results in complete inhibition of endogenous antibody production. Transfer of the human germ-line immunoglobulin gene array into such germ-line mutant mice will result in the production of human antibodies upon antigen challenge. See, e.g., Jakobovits et al., *Proc. Natl. Acad. Sci. USA*, 90:2551 (1993); Jakobovits et al., *Nature*, 362:255-58 (1993); Bruggemann et al., *Year in Immuno.* 7:33 (1993); U.S. Pat. Nos. 5,545,806, 5,569,825, 5,591,669 (all of GenPharm); 5,545,807; and WO 97/17852.

Alternatively, phage display technology (McCafferty et al., *Nature* 348:552-53 (1990)) can be used to produce human antibodies and antibody fragments in vitro, from immunoglobulin variable (V) domain gene repertoires from unimmunized donors. Using this technique, antibody V domain genes are cloned in-frame into either a major or minor coat protein gene of a filamentous bacteriophage, such as M13 or fd, and displayed as functional antibody fragments on the surface of the phage particle. Because the filamentous particle contains a single-stranded DNA copy of the phage genome, selections based on the functional properties of the antibody also result in selection of the gene encoding the antibody exhibiting those properties. Thus, the phage mimics some of the properties of the B-cell. Phage display can be performed in a variety of formats, reviewed in, e.g., Johnson, Kevin S. and Chiswell, David J., *Current Opinion in Structural Biology* 3:564-71 (1993). Several sources of V-gene segments can be used for phage display. Clackson et al., *Nature*, 352:624-28 (1991) isolated a diverse array of anti-oxazolone antibodies from a small random combinatorial library of V genes derived from the spleens of immunized mice. A repertoire of V genes from unimmunized human donors can be constructed and antibodies to a diverse array of antigens (including self-antigens) can be isolated essentially following the techniques described by Marks et al., *J. Mol. Biol.* 222:581-97 (1991), or Griffith et al., *EMBO J.* 12:725-34 (1993). See, also, U.S. Pat. Nos. 5,565,332 and 5,573,905.

As discussed above, human antibodies may also be generated in vitro in activated B cells (see, for example, U.S. Pat. Nos. 5,567,610 and 5,229,275).

4. Antibody Fragments

In certain circumstances, there are advantages to using antibody fragments, rather than whole antibodies. The smaller size of the fragments allows for rapid clearance, and may lead to improved access to EBV-infected cells or organs in a mammal.

Various techniques have been developed to produce antibody fragments. Traditionally, these fragments were derived via proteolytic digestion of intact antibodies (see, e.g.,

Morimoto et al., Journal of Biochemical and Biophysical Methods 24:107-117 (1992); and Brennan et al., Science, 229:81 (1985)). However, these fragments can now be produced directly by recombinant host cells. Fab, Fv and ScFv antibody fragments can all be expressed in and secreted from *E. coli*, thus allowing the facile production of large amounts of these fragments. Antibody fragments can be isolated from the antibody phage libraries discussed above. Alternatively, Fab'-SH fragments can be directly recovered from *E. coli* and chemically coupled to form F(ab')₂ fragments (Carter et al., Bio/Technology 10:163-67 (1992)). Using another approach, F(ab')₂ fragments can be isolated directly from recombinant host cell culture. Fab and F(ab')₂ fragment with increased in vivo half-life comprising a salvage receptor binding epitope residues are described in U.S. Pat. No. 5,869,046. Other techniques to produce antibody fragments will be apparent to the skilled practitioner. The antibody of choice may also be a single chain Fv fragment (scFv). See WO 93/16185; U.S. Pat. Nos. 5,571,894; and 5,587,458. Fv and sFv are the only species with intact combining sites that are devoid of constant regions; thus, they are suitable for reduced nonspecific binding during in vivo use. sFv fusion proteins may be constructed to yield fusion of an effector protein at either the amino or the carboxy terminus of an sFv. See Antibody Engineering, ed. Borrebaeck, supra. The antibody fragment may also be a "linear antibody", e.g., as described in U.S. Pat. No. 5,641,870. Such linear antibody fragments may be monospecific or bispecific.

5. Bispecific Antibodies

Bispecific antibodies have binding specificities for at least two different epitopes. Exemplary bispecific antibodies may bind to two different epitopes of a gp350 protein. Other such antibodies may combine a gp350 binding site with a binding site for another protein. Alternatively, an anti-gp350 arm may be combined with an arm which binds to a triggering molecule on a leukocyte such as a T-cell receptor molecule (e.g. CD3) (see, e.g., Baeuerle, et al., Curr. Opin. Mol. Ther. 11(1):22-30 (2009)), or Fc receptors for IgG (FcγR), such as FcγRI (CD64), FcγRII (CD32) and FcγRIII (CD16), to focus and localize cellular defense mechanisms to the gp350-expressing cell. Bispecific antibodies may also be used to localize cytotoxic agents to EBV-infected cells which express gp350. These antibodies possess a gp350-binding arm and an arm which binds the cytotoxic agent (e.g., saporin, anti-interferon-α, *vinca* alkaloid, ricin A chain, methotrexate or radioactive isotope hapten). Bispecific antibodies can be prepared as full length antibodies or antibody fragments (e.g., F(ab')₂ bispecific antibodies).

WO 96/16673 describes a bispecific anti-ErbB2/anti-FcγRIII antibody and U.S. Pat. No. 5,837,234 discloses a bispecific anti-ErbB2/anti-FcγRI antibody. A bispecific anti-ErbB2/Fcα antibody is shown in WO98/02463. U.S. Pat. Nos. 5,821,337 and 6,407,213 teach bispecific anti-ErbB2/anti-CD3 antibodies. Additional bispecific antibodies that bind an epitope on the CD3 antigen and a second epitope have been described in U.S. Pat. No. 5,078,998 (anti-CD3/tumor cell antigen); U.S. Pat. No. 5,601,819 (anti-CD3/IL-2R; anti-CD3/CD28; anti-CD3/CD45); U.S. Pat. No. 6,129,914 (anti-CD3/malignant B cell antigen); 7,112,324 (anti-CD3/CD19); U.S. Pat. No. 6,723,538 (anti-CD3/CCR5); U.S. Pat. No. 7,235,641 (anti-CD3/EpCAM); U.S. Pat. No. 7,262,276 (anti-CD3/ovarian tumor antigen); and U.S. Pat. No. 5,731,168 (anti-CD3/CD4IgG).

Methods for making bispecific antibodies are known in the art. Traditional production of full length bispecific antibodies is based on the co-expression of two immunoglobulin

heavy chain-light chain pairs, where the two chains have different specificities (Millstein et al., Nature 305:537-39 (1983)). Because of the random assortment of immunoglobulin heavy and light chains, these hybridomas (quadromas) produce a potential mixture of 10 different antibody molecules, of which only one has the correct bispecific structure. Purification of the correct molecule, which is usually done by affinity chromatography steps, is rather cumbersome, and the product yields are low. Similar procedures are disclosed in WO 93/08829, and in Traunecker et al., EMBO J. 10:3655-59 (1991).

Using a different approach, antibody variable domains with the desired binding specificities (antibody-antigen combining sites) are fused to immunoglobulin constant domain sequences. Preferably, the fusion is with an Ig heavy chain constant domain, comprising at least part of the hinge, CH2, and CH3 regions. It is preferred to have the first heavy-chain constant region (CH1) containing the site necessary for light chain bonding, present in at least one of the fusions. DNAs encoding the immunoglobulin heavy chain fusions and, if desired, the immunoglobulin light chain, are inserted into separate expression vectors, and are co-transfected into a suitable host cell. This provides for greater flexibility in adjusting the mutual proportions of the three polypeptide fragment when unequal ratios of the three polypeptide chains used in the construction provide the optimum yield of the desired bispecific antibody. It is, however, possible to insert the coding sequences for two or all three polypeptide chains into a single expression vector when the expression of at least two polypeptide chains in equal ratios results in high yields or when the ratios have no significant effect on the yield of the desired chain combination.

Preferably, the bispecific antibodies are composed of a hybrid immunoglobulin heavy chain with a first binding specificity in one arm, and a hybrid immunoglobulin heavy chain-light chain pair (providing a second binding specificity) in the other arm. This asymmetric structure facilitates the separation of the desired bispecific compound from unwanted immunoglobulin chain combinations, as the presence of an immunoglobulin light chain in only one half of the bispecific molecule provides for a facile way of separation. This approach is disclosed in WO 94/04690. For further details of generating bispecific antibodies see, for example, Suresh et al., Methods in Enzymology 121:210 (1986).

Using another approach described in U.S. Pat. No. 5,731,168, the interface between a pair of antibody molecules can be engineered to maximize the percentage of heterodimers which are recovered from recombinant cell culture. The preferred interface comprises at least a part of the CH3 domain. In this method, one or more small amino acid side chains from the interface of the first antibody molecule are replaced with larger side chains (e.g., tyrosine or tryptophan). Compensatory "cavities" of identical or similar size to the large side chain(s) are created on the interface of the second antibody molecule by replacing large amino acid side chains with smaller ones (e.g., alanine or threonine). This provides a mechanism for increasing the yield of the heterodimer over other unwanted end-products such as homodimers.

Bispecific antibodies include cross-linked or "heteroconjugate" antibodies. For example, one of the antibodies in the heteroconjugate can be coupled to avidin, the other to biotin. Such antibodies have been proposed to target immune system cells to unwanted cells (U.S. Pat. No. 4,676,980), and for treatment of HIV infection (WO 91/00360, WO 92/200373, and EP 03089). Heteroconjugate antibodies may

be made using any convenient cross-linking methods. Suitable cross-linking agents are well known in the art, and are disclosed in U.S. Pat. No. 4,676,980, along with several cross-linking techniques.

Techniques for generating bispecific antibodies from antibody fragments have also been described in the literature. For example, bispecific antibodies can be prepared using chemical linkage. Brennan et al., *Science* 229:81 (1985) describe a procedure wherein intact antibodies are proteolytically cleaved to generate F(ab')₂ fragments. These fragments are reduced in the presence of the dithiol complexing agent, sodium arsenite, to stabilize vicinal dithiols and prevent intermolecular disulfide formation. The Fab' fragments generated are then converted to thionitrobenzoate (TNB) derivatives. One of the Fab'-TNB derivatives is then reconverted to the Fab'-thiol by reduction with mercaptoethylamine and is mixed with an equimolar amount of the other Fab'-TNB derivative to form the bispecific antibody. The bispecific antibodies produced can be used as agents for the selective immobilization of enzymes.

Recent progress has facilitated the direct recovery of Fab'-SH fragments from *E. coli*, which can be chemically coupled to form bispecific antibodies. Shalaby et al., *J. Exp. Med.* 175: 217-25 (1992) describe the production of a fully humanized bispecific antibody F(ab')₂ molecule. Each Fab' fragment was separately secreted from *E. coli* and subjected to directed chemical coupling in vitro to form the bispecific antibody. The bispecific antibody thus formed could bind to cells overexpressing the ErbB2 receptor and normal human T cells, as well as trigger the lytic activity of human cytotoxic lymphocytes against human breast tumor targets. Various techniques for making and isolating bispecific antibody fragments directly from recombinant cell culture have also been described. For example, bispecific antibodies have been produced using leucine zippers. Kostelny et al., *J. Immunol.* 148(5):1547-53 (1992). The leucine zipper peptides from the Fos and Jun proteins were linked to the Fab' portions of two different antibodies by gene fusion. The antibody homodimers were reduced at the hinge region to form monomers and then re-oxidized to form the antibody heterodimers. This method can also be utilized to produce antibody homodimers. The "diabody" technology described by Hollinger et al., *Proc. Natl. Acad. Sci. USA* 90:6444-48 (1993) has provided an alternative mechanism for making bispecific antibody fragments. The fragments comprise a VH connected to a VL by a linker which is too short to allow pairing between the two domains on the same chain. Accordingly, the VH and VL domains of one fragment are forced to pair with the complementary VL and VH domains of another fragment, thereby forming two antigen-binding sites. Another strategy for making bispecific antibody fragments using single-chain Fv (sFv) dimers has also been reported. See Gruber et al., *J. Immunol.*, 152:5368 (1994).

Antibodies with more than two valencies are contemplated. For example, trispecific antibodies can be prepared. Tutt et al., *J. Immunol.* 147:60 (1991).

6. Heteroconjugate Antibodies

Heteroconjugate antibodies are also within the scope of this disclosure. Heteroconjugate antibodies are composed of two covalently joined antibodies. Such antibodies have been proposed to target immune system cells to unwanted cells (see U.S. Pat. No. 4,676,980), and for treatment of HIV infection (see WO 91/00360; WO 92/200373; EP 03089). It is contemplated that the antibodies may be prepared in vitro using known methods in synthetic protein chemistry, including those involving crosslinking agents. For example, immunotoxins may be constructed using a disulfide

exchange reaction or by forming a thioether bond. Examples of suitable reagents for this purpose include iminothiolate and methyl-4-mercaptobutyrimidate and those disclosed in U.S. Pat. No. 4,676,980.

7. Multivalent Antibodies

A multivalent antibody may be internalized (and/or catabolized) faster than a bivalent antibody by a cell expressing an antigen to which the antibodies bind. The antibodies of this disclosure may be multivalent antibodies (which are other than of the IgM class) with three or more antigen binding sites (e.g. tetravalent antibodies), which can be readily produced by recombinant expression of nucleic acid encoding the polypeptide chains of the antibody. The multivalent antibody can comprise a dimerization domain and three or more antigen binding sites. The preferred dimerization domain comprises (or consists of) an Fc region or a hinge region. In this scenario, the antibody will comprise an Fc region and three or more antigen binding sites amino-terminal to the Fc region. The preferred multivalent antibody herein comprises (or consists of) three to about eight, but preferably four, antigen binding sites. The multivalent antibody comprises at least one polypeptide chain (and preferably two polypeptide chains), wherein the polypeptide chain(s) comprise two or more variable domains. For instance, the polypeptide chain(s) may comprise VD1-(X1)_n-VD2-(X2)_n-Fc, wherein VD1 is a first variable domain, VD2 is a second variable domain, Fc is one polypeptide chain of an Fc region, X1 and X2 represent an amino acid or polypeptide, and n is 0 or 1. For instance, the polypeptide chain(s) may comprise: VH-CH1-flexible linker-VH-CH1-Fc region chain; or VH-CH1-VH-CH1-Fc region chain. The multivalent antibody preferably further comprises at least two (and preferably four) light chain variable domain polypeptides. The multivalent antibody may comprise from about two to about eight light chain variable domain polypeptides. The light chain variable domain polypeptides contemplated here comprise a light chain variable domain and, optionally, further comprise a CL domain.

8. Effector Function Engineering

It may be desirable to modify the antibody of the disclosure with respect to effector function, e.g., to enhance antigen-dependent cell-mediated cytotoxicity (ADCC) and/or complement dependent cytotoxicity (CDC) of the antibody. This may be achieved by introducing one or more amino acid substitutions in an Fc region of the antibody. Alternatively or additionally, cysteine residue(s) may be introduced in the Fc region, thereby allowing interchain disulfide bond formation in this region. The homodimeric antibody thus generated may have improved internalization capability and/or increased complement-mediated cell killing and antibody-dependent cellular cytotoxicity (ADCC). See Caron et al., *J. Exp. Med.* 176:1191-95 (1992) and Shopes, B. *J. Immunol.* 148:2918-22 (1992). Homodimeric antibodies with enhanced anti-viral activity may also be prepared using heterobifunctional cross-linkers as described in Wolff et al., *Cancer Research* 53:2560-2565 (1993). Alternatively, an antibody can be engineered which has dual Fc regions and may thereby have enhanced complement lysis and ADCC capabilities. See Stevenson et al., *Anti-Cancer Drug Design* 3:219-30 (1989).

To increase the serum half-life of the antibody, a salvage receptor binding epitope may be incorporated into the antibody (especially an antibody fragment) as described in U.S. Pat. No. 5,739,277. The term "salvage receptor binding epitope" refers to an epitope of the Fc region of an IgG

molecule (e.g., IgG1, IgG2, IgG3, or IgG4) that is responsible for increasing the in vivo serum half-life of the IgG molecule.

9. Immunconjugates

The disclosure also pertains to immunoconjugates comprising an antibody conjugated to a cytotoxic agent such as a chemotherapeutic agent, a growth inhibitory agent, a toxin (e.g., an enzymatically active toxin of bacterial, fungal, plant, or animal origin, or fragments thereof), or a radioactive isotope (i.e., a radioconjugate). Chemotherapeutic agents useful in the generation of such immunoconjugates have been described above. Enzymatically active toxins and fragments thereof that can be used include diphtheria A chain, nonbinding active fragments of diphtheria toxin, exotoxin A chain (from *Pseudomonas aeruginosa*), ricin A chain, abrin A chain, modeccin A chain, alpha-sarcin, *Aleu-rites fordii* proteins, dianthin proteins, *Phytolaca americana* proteins (PAPI, PAPII, and PAP-S), *Momordica charantia* inhibitor, curcin, crotin, *Sapaonaria officinalis* inhibitor, gelonin, mitogellin, restrictocin, phenomycin, enomycin, and the tricothecenes.

This disclosure further contemplates an immunoconjugate formed between an antibody and a compound with nucleolytic activity (e.g., a ribonuclease or a DNA endonuclease such as a deoxyribonuclease; DNase).

For selective destruction of an EBV-infected cell, the antibody may comprise a radioactive atom. A variety of radioactive isotopes are available to produce radioconjugated anti-gp350 antibodies. Examples include At211, I131, I125, Y90, Re186, Re188, Sm153, Bi212, P32, Pb212 and radioactive isotopes of Lu. When the conjugate is used for diagnosis, it may comprise a radioactive atom for scintigraphic studies, for example tc99m or I123, or a spin label for nuclear magnetic resonance (NMR) imaging (also known as magnetic resonance imaging, MRI), such as iodine-123 again, iodine-131, indium-111, fluorine-19, carbon-13, nitrogen-15, oxygen-17, gadolinium, manganese or iron.

The radio- or other labels may be incorporated in the conjugate in known ways. For example, the peptide may be biosynthesized or may be synthesized by chemical amino acid synthesis using suitable amino acid precursors involving, for example, fluorine-19 in place of hydrogen. Labels such as tc99m or I123, Re186, Re188 and In111 can be attached via a cysteine residue in the peptide. Yttrium-90 can be attached via a lysine residue. The IODOGEN method (Fraker et al (1978) Biochem. Biophys. Res. Commun. 80: 49-57 can be used to incorporate iodine-123. "Monoclonal Antibodies in Immunosciintigraphy" (Chatal, CRC Press 1989) describes other methods in detail.

Conjugates of the antibody and cytotoxic agent may be made using a variety of bifunctional protein coupling agents such as N-succinimidyl-3-(2-pyridyldithio) propionate (SPDP), succinimidyl-4-(N-maleimidomethyl) cyclohexane-1-carboxylate, iminothiolane (IT), bifunctional derivatives of imidoesters (such as dimethyl adipimidate HCL), active esters (such as disuccinimidyl suberate), aldehydes (such as glutaraldehyde), bis-azido compounds (such as bis (p-azidobenzoyl) hexanediamine), bis-diazonium derivatives (such as bis-(p-diazoniumbenzoyl)-ethylenediamine), diisocyanates (such as tolyene 2,6-diisocyanate), and bis-active fluorine compounds (such as 1,5-difluoro-2,4-dinitrobenzene). For example, a ricin immunotoxin can be prepared as described in Vitetta et al., Science 238:1098 (1987). Carbon-14-labeled 1-isothiocyanatobenzyl-3-methyldiethylene triaminopentaacetic acid (MX-DTPA) is an exemplary chelating agent for conjugation of radionucle-

otide to the antibody. See WO94/11026. The linker may be a "cleavable linker" facilitating release of the cytotoxic drug in the cell. For example, an acid-labile linker, peptidase-sensitive linker, photolabile linker, dimethyl linker or disulfide-containing linker (Chari et al., Cancer Research 52:127-131 (1992); U.S. Pat. No. 5,208,020) may be used.

The compounds of the disclosure expressly contemplate, but are not limited to, an ADC prepared with cross-linker reagents: BMPS, EMCS, GMBS, HBVS, LC-SMCC, MBS, MPBH, SBAP, SIA, SIAB, SMCC, SMPB, SMPH, sulfo-EMCS, sulfo-GMBS, sulfo-KMUS, sulfo-MBS, sulfo-SIAB, sulfo-SMCC, and sulfo-SMPB, and SVSB (succinimidyl-(4-vinylsulfone) benzoate) which are commercially available from Pierce Biotechnology, Inc., Rockford, IL).

Alternatively, a fusion protein comprising the anti-EBV gp350 antibody and cytotoxic agent may be made, e.g., by recombinant techniques or peptide synthesis. The length of DNA may comprise respective regions encoding the two portions of the conjugate either adjacent one another or separated by a region encoding a linker peptide which does not destroy the desired properties of the conjugate.

The antibody may also be conjugated to a "receptor" (such streptavidin) for utilization in pre-targeting of viral infected cells, wherein the antibody-receptor conjugate is administered to the patient, followed by removal of unbound conjugate from the circulation using a clearing agent and then administration of a "ligand" (e.g., avidin) which is conjugated to a cytotoxic agent (e.g., a radionucleotide).

10. Immunoliposomes

The anti-gp350 antibodies disclosed herein may also be formulated as immunoliposomes. A "liposome" is a small vesicle composed of various types of lipids, phospholipids and/or surfactant which is useful for delivery of a drug to a mammal. The components of the liposome are commonly arranged in a bilayer formation, similar to the lipid arrangement of biological membranes. Liposomes containing the antibody are prepared by methods known in the art, such as described in Epstein et al., Proc. Natl. Acad. Sci. USA 82:3688 (1985); Hwang et al., Proc. Natl. Acad. Sci. USA 77:4030 (1980); U.S. Pat. Nos. 4,485,045 and 4,544,545; and WO97/38731 published Oct. 23, 1997. Liposomes with enhanced circulation time are disclosed in U.S. Pat. No. 5,013,556.

Particularly useful liposomes can be generated by the reverse phase evaporation method with a lipid composition comprising phosphatidylcholine, cholesterol and PEG-derivatized phosphatidylethanolamine (PEG-PE). Liposomes are extruded through filters of defined pore size to yield liposomes with the desired diameter. Fab' fragments of the antibody of this disclosure can be conjugated to the liposomes as described in Martin et al., J. Biol. Chem. 257:286-288 (1982) via a disulfide interchange reaction. A chemotherapeutic agent is optionally contained within the liposome. See Gabizon et al., J. National Cancer Inst. 81(19):1484 (1989).

B. gp350 Binding Oligopeptides

gp350 binding oligopeptides of this disclosure are oligopeptides that bind, preferably specifically, to an EBV gp350 protein. gp350 binding oligopeptides may be chemically synthesized using known oligopeptide synthesis methodology or may be prepared and purified using recombinant technology. gp350 binding oligopeptides are usually at least about 5 amino acids in length, alternatively at least about 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71,

72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, or 100 amino acids in length or more, wherein such oligopeptides that are capable of binding, preferably specifically, to an EBV gp350 protein. gp350 binding oligopeptides may be identified without undue experimentation using well known techniques. In this regard, it is noted that techniques for screening oligopeptide libraries for oligopeptides that are capable of specifically binding to a polypeptide target are well known in the art (see, e.g., U.S. Pat. Nos. 5,556,762, 5,750,373, 4,708,871, 4,833,092, 5,223,409, 5,403,484, 5,571,689, 5,663,143; PCT Publication Nos. WO 84/03506 and WO84/03564; Geysen et al., Proc. Natl. Acad. Sci. U.S.A., 81:3998-4002 (1984); Geysen et al., Proc. Natl. Acad. Sci. U.S.A., 82:178-182 (1985); Geysen et al., in Synthetic Peptides as Antigens, 130-149 (1986); Geysen et al., J. Immunol. Meth., 102:259-274 (1987); Schoofs et al., J. Immunol., 140:611-616 (1988), Cwirla, S. E. et al. (1990) Proc. Natl. Acad. Sci. USA, 87:6378; Lowman, H. B. et al. (1991) Biochemistry, 30:10832; Clackson, T. et al. (1991) Nature, 352: 624; Marks, J. D. et al. (1991), J. Mol. Biol., 222:581; Kang, A. S. et al. (1991) Proc. Natl. Acad. Sci. USA, 88:8363, and Smith, G. P. (1991) Current Opin. Biotechnol., 2:668).

Bacteriophage (phage) display is one known technique used to screen large oligopeptide libraries to identify member(s) of those libraries which are capable of specifically binding to a polypeptide target. Phage display is a technique by which variant polypeptides are displayed as fusion proteins to the coat protein on the surface of bacteriophage particles (Scott, J. K. and Smith, G. P. (1990) Science 249:386). The utility of phage display lies in the fact that large libraries of selectively randomized protein variants (or randomly cloned cDNAs) can be rapidly and efficiently sorted for those sequences that bind to a target molecule with high affinity. Display of peptide (Cwirla, S. E. et al. (1990) Proc. Natl. Acad. Sci. USA, 87:6378) or protein (Lowman, H. B. et al. (1991) Biochemistry, 30:10832; Clackson, T. et al. (1991) Nature, 352: 624; Marks, J. D. et al. (1991), J. Mol. Biol., 222:581; Kang, A. S. et al. (1991) Proc. Natl. Acad. Sci. USA, 88:8363) libraries on phage have been used for screening millions of polypeptides or oligopeptides for ones with specific binding properties (Smith, G. P. (1991) Current Opin. Biotechnol., 2:668). Sorting phage libraries of random mutants requires a strategy for constructing and propagating a large number of variants, a procedure for affinity purification using the target receptor, and a means of evaluating the results of binding enrichments. U.S. Pat. Nos. 5,223,409, 5,403,484, 5,571,689, and 5,663,143.

Although most phage display methods have used filamentous phage, lambdoid phage display systems (WO 95/34683; U.S. Pat. No. 5,627,024), T4 phage display systems (Ren et al., Gene, 215:439 (1998); Zhu et al., Cancer Research, 58(15): 3209-14 (1998); Jiang et al., Infection & Immunity, 65(11): 4770-77 (1997); Ren et al., Gene, 195(2):303-11 (1997); Ren, Protein Sci., 5: 1833 (1996); Efimov et al., Virus Genes, 10:173 (1995)) and T7 phage display systems (Smith and Scott, Methods in Enzymology, 217: 228-57 (1993); U.S. Pat. No. 5,766,905) are also known.

Many other improvements and variations of the basic phage display concept have now been developed. These improvements enhance the ability of display systems to screen peptide libraries for binding to selected target molecules and to display functional proteins with the potential of screening these proteins for desired properties. Combinatorial reaction devices for phage display reactions have been developed (WO 98/14277) and phage display libraries

have been used to analyze and control bimolecular interactions (WO 98/20169; WO 98/20159) and properties of constrained helical peptides (WO 98/20036). WO 97/35196 describes a method of isolating an affinity ligand in which a phage display library is contacted with one solution in which the ligand will bind to a target molecule and a second solution in which the affinity ligand will not bind to the target molecule, to selectively isolate binding ligands. WO 97/46251 describes a method of biopanning a random phage display library with an affinity purified antibody and then isolating binding phage, followed by a micropanning process using microplate wells to isolate high affinity binding phage. The use of *Staphylococcus aureus* protein A as an affinity tag has also been reported (Li et al. (1998) Mol Biotech., 9:187). WO 97/47314 describes the use of substrate subtraction libraries to distinguish enzyme specificities using a combinatorial library which may be a phage display library. A method for selecting enzymes suitable for use in detergents using phage display is described in WO 97/09446. Additional methods of selecting specific binding proteins are described in U.S. Pat. Nos. 5,498,538, 5,432, 018, and WO 98/15833.

Methods of generating peptide libraries and screening these libraries are also disclosed in U.S. Pat. Nos. 5,723,286, 5,432,018, 5,580,717, 5,427,908, 5,498,530, 5,770,434, 5,734,018, 5,698,426, 5,763,192, and 5,723,323.

C. Screening for Anti-Gp350 Antibodies and Gp350 Binding Oligopeptides with the Desired Properties

Techniques for generating antibodies or oligopeptides that bind to EBV gp350 proteins have been described above. One may further select antibodies or oligopeptides with certain biological characteristics, as desired.

The growth inhibitory effects of an anti-gp350 antibody of this disclosure may be assessed by methods known in the art, e.g., using cells which express an EBV gp350 protein either endogenously or following transfection with the gp350 gene. For example, appropriate EBV infected cells may be treated with an anti-gp350 monoclonal antibody or oligopeptide of this disclosure at various concentrations for a few days (e.g., 2-7) and stained with crystal violet or MTT or analyzed by some other colorimetric assay. Another method of measuring proliferation would be by comparing 3H-thymidine uptake by the cells treated in the presence or absence of an anti-gp350 antibody, or gp350 binding oligopeptide of the disclosure. After treatment, the cells are harvested and the amount of radioactivity incorporated into the DNA quantitated in a scintillation counter. Appropriate positive controls include treatment of a selected cell line with a growth inhibitory antibody known to inhibit growth of that cell line. Growth inhibition of infected cells in vivo can be determined in various ways known in the art. Preferably, the anti-gp350 antibody, or gp350 binding oligopeptide will inhibit cell proliferation of an EBV infected cell in vitro or in vivo by about 25-100% compared to the untreated infected cell, more preferably, by about 30-100%, and even more preferably by about 50-100% or 70-100%. Growth inhibition can be measured at an antibody concentration of about 0.5 to 30 µg/ml or about 0.5 nM to 200 nM in cell culture, where the growth inhibition is determined 1-10 days after exposure of the cells to the antibody. The antibody is growth inhibitory in vivo if administration of the anti-gp350 antibody at about 1 µg/kg to about 100 mg/kg body weight results in reduction in cell growth or proliferation within about 5 days to 3 months from the first administration of the antibody, preferably within about 5 to 30 days.

To select for an anti-gp350 antibody, gp350 binding oligopeptide which induces cell death, loss of membrane

integrity as indicated by, e.g., propidium iodide (PI), trypan blue or 7AAD uptake may be assessed relative to control. A PI uptake assay can be performed in the absence of complement and immune effector cells. EBV gp350 protein-expressing cells are incubated with medium alone or medium containing the appropriate anti-gp350 antibody (e.g., at about 10 g/ml), gp350 binding oligopeptide. The cells are incubated for a 3-day time period. Following each treatment, cells are washed and aliquoted into 35 mm strainer-capped 12x75 tubes (1 ml per tube, 3 tubes per treatment group) for removal of cell clumps. Tubes then receive PI (10 g/ml). Samples may be analyzed using a FACSCAN® flow cytometer and FACSCONVERT® CellQuest software (Becton Dickinson). Those anti-gp350 antibodies, or gp350 binding oligopeptides that induce statistically significant levels of cell death as determined by PI uptake may be selected as cell death-inducing anti-gp350 antibodies or gp350 binding oligopeptides.

To screen for antibodies or oligopeptides which bind to an epitope on an EBV gp350 protein bound by an antibody of interest, a routine cross-blocking assay such as that described in Antibodies, A Laboratory Manual, Cold Spring Harbor Laboratory, Ed Harlow and David Lane (1988), can be performed. This assay can be used to determine if a test antibody or oligopeptide binds the same site or epitope as a known anti-gp350 antibody. Alternatively or additionally, epitope mapping can be performed by methods known in the art. For example, the antibody sequence can be mutagenized such as by alanine scanning, to identify contact residues. The mutant antibody is initially tested for binding with polyclonal antibody to ensure proper folding. In a different method, peptides corresponding to different regions of an EBV gp350 protein can be used in competition assays with the test antibodies or with a test antibody and an antibody with a characterized or known epitope.

D. Anti-Gp350 Antibody and EBV Gp350 Binding Oligopeptide Variants

In addition to the anti-gp350 antibodies described herein, it is contemplated that anti-gp350 antibody variants can be prepared. Anti-gp350 antibody variants can be prepared by introducing appropriate nucleotide changes into the encoding DNA, and/or by synthesis of the desired antibody or polypeptide. Those skilled in the art will appreciate that amino acid changes may alter post-translational processes of the anti-gp350 antibody, such as changing the number or position of glycosylation sites or altering the membrane anchoring characteristics.

Variations in the anti-gp350 antibodies described herein can be made, for example, using any of the techniques and guidelines for conservative and non-conservative mutations set forth, for instance, in U.S. Pat. No. 5,364,934. Variations may be a substitution, deletion, or insertion of one or more codons encoding the antibody or polypeptide that results in a change in the amino acid sequence as compared with the native sequence antibody or polypeptide. Optionally, the variation is by substitution of at least one amino acid with any other amino acid in one or more of the domains of the anti-gp350 antibody. Guidance in determining which amino acid residue may be inserted, substituted or deleted without adversely affecting the desired activity may be found by comparing the sequence of the anti-gp350 antibody with that of homologous known protein molecules and minimizing the number of amino acid sequence changes made in regions of high homology. Amino acid substitutions can be the result of replacing one amino acid with another amino acid having similar structural and/or chemical properties, such as the replacement of a leucine with a serine, i.e., conservative

amino acid replacements. Insertions or deletions may optionally be in the range of about 1 to 5 amino acids. The variation allowed may be determined by systematically making insertions, deletions, or substitutions of amino acids in the sequence and testing the resulting variants for activity exhibited by the full-length or mature native sequence.

Anti-gp350 antibody fragments are provided herein. Such fragments may be truncated at the N-terminus or C-terminus, or may lack internal residues, for example, when compared with a full length native antibody. Certain fragments lack amino acid residues that are not essential for a desired biological activity of the anti-gp350 antibody.

Anti-gp350 antibody fragments may be prepared by any of a number of conventional techniques. Desired peptide fragments may be chemically synthesized. An alternative approach involves generating antibody or polypeptide fragments by enzymatic digestion, e.g., by treating the protein with an enzyme known to cleave proteins at sites defined by particular amino acid residues, or by digesting the DNA with suitable restriction enzymes and isolating the desired fragment. Yet another suitable technique involves isolating and amplifying a DNA fragment encoding a desired antibody or polypeptide fragment, by polymerase chain reaction (PCR). Oligonucleotides that define the desired termini of the DNA fragment are employed at the 5' and 3' primers in the PCR. Preferably, anti-gp350 antibody fragments share at least one biological and/or immunological activity with the native anti-gp350 antibodies disclosed herein.

Conservative substitutions of interest are shown in Table 1 under the heading of preferred substitutions. If such substitutions result in a change in biological activity, then more substantial changes, denominated exemplary substitutions in Table 1, or as further described below in reference to amino acid classes, are introduced and the products screened.

TABLE 1

Original Residue	Exemplary Substitutions	Preferred Substitutions
Ala (A)	Val; Leu; Ile	Val
Arg (R)	Lys; Gln; Asn	Lys
Asn (N)	Gln; His; Asp, Lys; Arg	Gln
Asp (D)	Glu; Asn	Glu
Cys (C)	Ser; Ala	Ser
Gln (Q)	Asn; Glu	Asn
Glu (E)	Asp; Gln	Asp
Gly (G)	Ala	Ala
His (H)	Asn; Gln; Lys; Arg	Arg
Ile (I)	Leu; Val; Met; Ala; Phe; Norleucine	Leu
Leu (L)	Norleucine; Ile; Val; Met; Ala; Phe	Ile
Lys (K)	Arg; Gln; Asn	Arg
Met (M)	Leu; Phe; Ile	Leu
Phe (F)	Trp; Leu; Val; Ile; Ala; Tyr	Tyr
Pro (P)	Ala	Ala
Ser (S)	Thr	Thr
Thr (T)	Val; Ser	Ser
Trp (W)	Tyr; Phe	Tyr
Tyr (Y)	Trp; Phe; Thr; Ser	Phe
Val (V)	Ile; Leu; Met; Phe; Ala; Norleucine	Leu

Substantial modifications in function or immunological identity of the anti-gp350 antibody are accomplished by selecting substitutions that differ significantly in their effect on maintaining (a) the structure of the polypeptide backbone in the area of the substitution, for example, as a sheet or helical conformation, (b) the charge or hydrophobicity of the molecule at the target site, or (c) the bulk of the side chain. Naturally occurring residues are divided into groups based on common side-chain properties:

- (1) hydrophobic: Norleucine, Met, Ala, Val, Leu, Ile;
- (2) neutral hydrophilic: Cys, Ser, Thr; Asn; Gln
- (3) acidic: Asp, Glu;
- (4) basic: His, Lys, Arg;
- (5) residues that influence chain orientation: Gly, Pro; and
- (6) aromatic: Trp, Tyr, Phe.

Non-conservative substitutions will entail exchanging a member of one of these classes for another class. Such substituted residues also may be introduced into the conservative substitution sites or, more preferably, into the remaining (non-conserved) sites.

The variations can be made using methods known in the art such as oligonucleotide-mediated (site-directed) mutagenesis, alanine scanning, and PCR mutagenesis. Site-directed mutagenesis (Carter et al., *Nucl. Acids Res.*, 13:4331 (1986); Zoller et al., *Nucl. Acids Res.*, 10:6487 (1987)), cassette mutagenesis (Wells et al., *Gene*, 34:315 (1985)), restriction selection mutagenesis (Wells et al., *Philos. Trans. R. Soc. London SerA*, 317:415 (1986)) or other known techniques can be performed on the cloned DNA to produce the anti-gp350 antibody variant DNA.

Scanning amino acid analysis can also be employed to identify one or more amino acids along a contiguous sequence. Among the preferred scanning amino acids are relatively small, neutral amino acids. Such amino acids include alanine, glycine, serine, and cysteine. Alanine is typically a preferred scanning amino acid among this group because it eliminates the side-chain beyond the beta-carbon and is less likely to alter the main-chain conformation of the variant (Cunningham and Wells, *Science*, 244:1081-85 (1989)). Alanine is also typically preferred because it is the most common amino acid. Further, it is frequently found in both buried and exposed positions (Creighton, *The Proteins*, (W.H. Freeman & Co., N.Y.); Choithia, *J. Mol. Biol.*, 150:1 (1976)). If alanine substitution does not yield adequate amounts of variant, an isoteric amino acid can be used.

Any cysteine residue not involved in maintaining the proper conformation of the anti-gp350 antibody also may be substituted, generally with serine, to improve the oxidative stability of the molecule and prevent aberrant crosslinking. Conversely, cysteine bond(s) may be added to the anti-gp350 antibody to improve its stability (particularly where the antibody is an antibody fragment such as an Fv fragment).

A particularly preferred type of substitutional variant involves substituting one or more hypervariable region residues of a parent antibody (e.g., a humanized or human antibody). Generally, the resulting variant(s) selected for further development will have improved biological properties relative to the parent antibody from which they are generated. A convenient way for generating such substitutional variants involves affinity maturation using phage display. Briefly, several hypervariable region sites (e.g., 6-7 sites) are mutated to generate all possible amino substitutions at each site. The antibody variants thus generated are displayed in a monovalent fashion from filamentous phage particles as fusions to the gene III product of M13 packaged within each particle. The phage-displayed variants are then screened for their biological activity (e.g., binding affinity) as herein disclosed. To identify candidate hypervariable region sites for modification, alanine scanning mutagenesis can be performed to identify hypervariable region residues contributing significantly to antigen binding. Alternatively, or additionally, it may be beneficial to analyze a crystal structure of the antigen-antibody complex to identify contact points between the antibody and EBV gp350 protein. Such contact residues and neighboring residues are candidates for

substitution according to the techniques elaborated herein. Once such variants are generated, the panel of variants is subjected to screening as described herein and antibodies with superior properties in one or more relevant assays may be selected for further development.

Nucleic acid molecules encoding amino acid sequence variants of the anti-gp350 antibody are prepared by a variety of methods known in the art. These methods include, but are not limited to, isolation from a natural source (in the case of naturally occurring amino acid sequence variants) or preparation by oligonucleotide-mediated (or site-directed) mutagenesis, PCR mutagenesis, and cassette mutagenesis of an earlier prepared variant or a non-variant version of the anti-gp350 antibody.

E. Modifications of Anti-gp350 Antibodies

Covalent modifications of anti-gp350 antibodies and EBV gp350 proteins are included within the scope of this disclosure. One type of covalent modification includes reacting targeted amino acid residues of an anti-gp350 antibody with an organic derivatizing agent that can react with selected side chains or the N- or C-terminal residues of the anti-gp350 antibody. Derivatization with bifunctional agents is useful, for instance, for crosslinking anti-gp350 antibody to a water-insoluble support matrix or surface for use in purifying anti-gp350 antibodies, or detection of gp350 protein in biological samples, or EBV diagnostic assays. Commonly used crosslinking agents include, e.g., 1,1-bis(diazotacetyl)-2-phenylethane, glutaraldehyde, N-hydroxysuccinimide esters, for example, esters with 4-azidosalicylic acid, homobifunctional imidoesters, including disuccinimidyl esters such as 3,3'-dithiobis (succinimidyl)propionate, bifunctional maleimides such as bis-N-maleimido-1,8-octane and agents such as methyl-3-[(p-azidophenyl)dithio]propionimide.

Other modifications include deamidation of glutaminyl and asparaginyl residues to the corresponding glutamyl and aspartyl residues, respectively, hydroxylation of proline and lysine, phosphorylation of hydroxyl groups of seryl or threonyl residues, methylation of the α -amino groups of lysine, arginine, and histidine side chains (T. E. Creighton, *Proteins: Structure and Molecular Properties*, W.H. Freeman & Co., San Francisco, pp. 79-86 (1983)), acetylation of the N-terminal amine, and amidation of any C-terminal carboxyl group.

Another type of covalent modification of the anti-gp350 antibody included within the scope of this disclosure comprises altering the native glycosylation pattern of the antibody or polypeptide. "Altering the native glycosylation pattern" is intended for purposes herein to mean deleting one or more carbohydrate moieties found in native sequence anti-gp350 antibody (either by removing the underlying glycosylation site or by deleting the glycosylation by chemical and/or enzymatic means), and/or adding one or more glycosylation sites that are not present in the native sequence anti-gp350 antibody. In addition, the phrase includes qualitative changes in the glycosylation of the native proteins, involving a change in the nature and proportions of the various carbohydrate moieties present.

Glycosylation of antibodies and other polypeptides is typically either N-linked or O-linked. N-linked refers to the attachment of the carbohydrate moiety to the side chain of an asparagine residue. The tripeptide sequences asparagine-X-serine and asparagine-X-threonine, where X is any amino acid except proline, are the recognition sequences for enzymatic attachment of the carbohydrate moiety to the asparagine side chain. Thus, the presence of either of these tripeptide sequences in a polypeptide creates a potential

glycosylation site. O-linked glycosylation refers to the attachment of one of the sugars N-acetyl galactosamine, galactose, or xylose to a hydroxyamino acid, most commonly serine or threonine, although 5-hydroxyproline or 5-hydroxylysine may also be used.

Addition of glycosylation sites to the anti-gp350 antibody is conveniently accomplished by altering the amino acid sequence such that it contains one or more of the above-described tripeptide sequences (for N-linked glycosylation sites). The alteration may also be made by the addition of, or substitution by, one or more serine or threonine residues to the sequence of the original anti-gp350 antibody (for O-linked glycosylation sites). The anti-gp350 antibody amino acid sequence may optionally be altered through changes at the DNA level, particularly by mutating the DNA encoding the anti-gp350 antibody at preselected bases such that codons are generated that will translate into the desired amino acids.

Another means of increasing the number of carbohydrate moieties on the anti-gp350 antibody is by chemical or enzymatic coupling of glycosides to the polypeptide. Such methods are described in the art, e.g., in WO 87/05330 published 11 Sep. 1987, and in Aplin and Wriston, *CRC Crit. Rev. Biochem.*, pp. 259-306 (1981).

Removal of carbohydrate moieties present on the anti-gp350 antibody may be accomplished chemically or enzymatically or by mutational substitution of codons encoding for amino acid residues that serve as targets for glycosylation. Chemical deglycosylation techniques are known in the art and described, for instance, by Hakimuddin, et al., *Arch. Biochem. Biophys.*, 259:52 (1987) and by Edge et al., *Anal. Biochem.*, 118:131 (1981). Enzymatic cleavage of carbohydrate moieties on polypeptides can be achieved using a variety of endo- and exo-glycosidases as described by Thotakura et al., *Meth. Enzymol.*, 138:350 (1987).

Another type of covalent modification of anti-gp350 antibody comprises linking the antibody or polypeptide to one of a variety of nonproteinaceous polymers, e.g., polyethylene glycol (PEG), polypropylene glycol, or polyoxyalkylenes, in the manner set forth in U.S. Pat. Nos. 4,640,835; 4,496,689; 4,301,144; 4,670,417; 4,791,192 or 4,179,337. The antibody or polypeptide also may be entrapped in microcapsules prepared, for example, by coacervation techniques or by interfacial polymerization (for example, hydroxymethylcellulose or gelatin-microcapsules and poly(methylmethacrylate) microcapsules, respectively), in colloidal drug delivery systems (for example, liposomes, albumin microspheres, microemulsions, nano-particles and nanocapsules), or in macroemulsions. Such techniques are disclosed in Remington's *Pharmaceutical Sciences*, 16th edition, Oslo, A., Ed., (1980).

The anti-gp350 antibody of this disclosure may also be modified in a way to form chimeric molecules comprising an anti-gp350 antibody fused to another, heterologous polypeptide or amino acid sequence.

Such a chimeric molecule may comprise a fusion of the anti-gp350 antibody with a tag polypeptide which provides an epitope to which an anti-tag antibody can selectively bind. The epitope tag is generally placed at the amino- or carboxyl-terminus of the anti-gp350 antibody. The presence of such epitope-tagged forms of the anti-gp350 antibody can be detected using an antibody against the tag polypeptide. Also, provision of the epitope tag enables the anti-gp350 antibody to be readily purified by affinity purification using an anti-tag antibody or another type of affinity matrix that binds to the epitope tag. Various tag polypeptides and their respective antibodies are well known in the art. Examples

include poly-histidine (poly-his) or poly-histidine-glycine (poly-his-gly) tags; the flu HA tag polypeptide and its antibody 12CA5 [Field et al., *Mol. Cell. Biol.*, 8:2159-2165 (1988)]; the c-myc tag and the 8F9, 3C7, 6E10, G4, B7 and 9E10 antibodies thereto (Evan et al., *Molecular and Cellular Biology*, 5:3610-16 (1985)); and the Herpes Simplex virus glycoprotein D (gD) tag and its antibody (Paborsky et al., *Protein Engineering*, 3(6):547-553 (1990)). Other tag polypeptides include the Flag-peptide (Hopp et al., *BioTechnology*, 6:1204-10 (1988)); the KT3 epitope peptide (Martin et al., *Science*, 255:192-194 (1992)); an α -tubulin epitope peptide (Skinner et al., *J. Biol. Chem.*, 266:15163-15166 (1991)); and the T7 gene 10 protein peptide tag (Lutz-Freyermuth et al., *Proc. Natl. Acad. Sci. USA*, 87:6393-6397 (1990)).

Alternatively, the chimeric molecule may comprise a fusion of the anti-gp350 antibody with an immunoglobulin or a region of an immunoglobulin. For a bivalent form of the chimeric molecule (also referred to as an "immunoadhesin"), such a fusion could be to the Fc region of an IgG molecule. The Ig fusions preferably include the substitution of a soluble (transmembrane domain deleted or inactivated) form of an anti-gp350 antibody in place of at least one variable region within an Ig molecule. Preferably, the immunoglobulin fusion includes the hinge, CH2 and CH3, or the hinge, CH1, CH2 and CH3 regions of an IgG1 molecule. For the production of immunoglobulin fusions, see also U.S. Pat. No. 5,428,130.

F. Preparation of Anti-Gp350 Antibodies and EBV Gp350 Binding Oligopeptides

The description below relates primarily to production of anti-gp350 antibodies and EBV gp350 binding oligopeptides by culturing cells transformed or transfected with a vector containing anti-gp350 antibody-, or EBV gp350 binding oligopeptide-encoding nucleic acid. It is, of course, contemplated that alternative methods, which are well known in the art, may be employed to prepare anti-gp350 antibodies and EBV gp350 binding oligopeptides. For instance, the appropriate amino acid sequence, or portions thereof, may be produced by direct peptide synthesis using solid-phase techniques (e.g., Stewart et al., *Solid-Phase Peptide Synthesis*, W.H. Freeman Co., San Francisco, C A (1969); Merrifield, *J. Am. Chem. Soc.*, 85:2149-54 (1963)). In vitro protein synthesis may be performed using manual techniques or by automation. Automated synthesis may be accomplished, for instance, using an Applied Biosystems Peptide Synthesizer (Foster City, CA) using manufacturer's instructions. Various portions of the anti-gp350 antibody may be chemically synthesized separately and combined using chemical or enzymatic methods to produce the desired anti-gp350 antibody.

1. Isolation of DNA Encoding Anti-Gp350 Antibody

DNA encoding anti-gp350 antibody may be obtained from a cDNA library prepared from tissue believed to possess the anti-gp350 antibody mRNA and to express it at a detectable level. Accordingly, human anti-gp350 antibody DNA can be conveniently obtained from a cDNA library prepared from human tissue. The anti-gp350 antibody-encoding gene may also be obtained from a genomic library or by known synthetic procedures (e.g., automated nucleic acid synthesis).

Libraries can be screened with probes (such as oligonucleotides of at least about 20-80 bases) designed to identify the gene of interest or the protein encoded by it. Screening the cDNA or genomic library with the selected probe may be conducted using standard procedures, such as described in Sambrook et al., *Molecular Cloning: A Labo-*

ratory Manual (New York: Cold Spring Harbor Laboratory Press, 1989). An alternative means to isolate the gene encoding anti-gp350 antibody is to use PCR methodology [Sambrook et al., supra; Dieffenbach et al., PCR Primer: A Laboratory Manual (Cold Spring Harbor Laboratory Press, 1995)].

Techniques for screening a cDNA library are well known in the art. The oligonucleotide sequences selected as probes should be of sufficient length and sufficiently unambiguous that false positives are minimized. The oligonucleotide is preferably labeled such that it can be detected upon hybridization to DNA in the library being screened. Methods of labeling are well known in the art, and include the use of radiolabels like ³²P-labeled ATP, biotinylation or enzyme labeling. Hybridization conditions, including moderate stringency and high stringency, are provided in Sambrook et al., supra.

Sequences identified in such library screening methods can be compared and aligned to other known sequences deposited and available in public databases such as GenBank or other private sequence databases. Sequence identity (at either the amino acid or nucleotide level) within defined regions of the molecule or across the full-length sequence can be determined using methods known in the art and as described herein.

Nucleic acids having protein coding sequences may be obtained by screening selected cDNA or genomic libraries using the deduced amino acid sequence disclosed herein for the first time, and, if necessary, using conventional primer extension procedures as described in Sambrook et al., supra, to detect precursors and processing intermediates of mRNA that may not have been reverse-transcribed into cDNA.

2. Selection and Transformation of Host Cells

Host cells are transfected or transformed with expression or cloning vectors described herein for anti-gp350 antibody production and cultured in conventional nutrient media modified as appropriate for inducing promoters, selecting transformants, or amplifying the genes encoding the desired sequences. The culture conditions, such as media, temperature, pH and the like, can be selected by the skilled artisan without undue experimentation. In general, principles, protocols, and practical techniques for maximizing the productivity of cell cultures can be found in Mammalian Cell Biotechnology: a Practical Approach, M. Butler, ed. (IRL Press, 1991) and Sambrook et al., supra.

Methods of eukaryotic cell transfection and prokaryotic cell transformation are known to the ordinarily skilled artisan, for example, CaCl₂, CaPO₄, liposome-mediated, and electroporation. Depending on the host cell used, transformation is performed using standard techniques appropriate to such cells. The calcium treatment employing calcium chloride, as described in Sambrook et al., supra, or electroporation is generally used for prokaryotes. Infection with *Agrobacterium tumefaciens* is used for transformation of certain plant cells, as described by Shaw et al., Gene, 23:315 (1983) and WO 89/05859 published 29 Jun. 1989. For mammalian cells without such cell walls, the calcium phosphate precipitation method of Graham and van der Eb, Virology, 52:456-457 (1978) can be employed. General aspects of mammalian cell host system transfections have been described in U.S. Pat. No. 4,399,216. Transformations into yeast are typically carried out according to the method of Van Solingen et al., J. Bact., 130:946 (1977) and Hsiao et al., Proc. Natl. Acad. Sci. (USA), 76:3829 (1979). However, other methods for introducing DNA into cells, such as by nuclear microinjection, electroporation, bacterial protoplast fusion with intact cells, or polycations, e.g., polybrene,

polyornithine, may also be used. For various techniques for transforming mammalian cells, see Keown et al., Methods in Enzymology, 185:527-537 (1990) and Mansour et al., Nature, 336:348-352 (1988).

Suitable host cells for cloning or expressing the DNA in the vectors herein include prokaryote, yeast, or higher eukaryote cells. Suitable prokaryotes include but are not limited to eubacteria, such as Gram-negative or Gram-positive organisms, for example, Enterobacteriaceae such as *E. coli*. Various *E. coli* strains are publicly available, such as *E. coli* K12 strain MM294 (ATCC 31,446); *E. coli* X1776 (ATCC 31,537); *E. coli* strain W3110 (ATCC 27,325) and K5 772 (ATCC 53,635). Other suitable prokaryotic host cells include Enterobacteriaceae such as *Escherichia*, e.g., *E. coli*, *Enterobacter*, *Erwinia*, *Klebsiella*, *Proteus*, *Salmonella*, e.g., *Salmonella typhimurium*, *Serratia*, e.g., *Serratia marcescans*, and *Shigella*, as well as Bacilli such as *B. subtilis* and *B. licheniformis* (e.g., *B. licheniformis* 41P disclosed in DD 266,710 published 12 Apr. 1989), *Pseudomonas* such as *P. aeruginosa*, and *Streptomyces*. These examples are illustrative rather than limiting. Strain W3110 is one particularly preferred host or parent host because it is a common host strain for recombinant DNA product fermentations. Preferably, the host cell secretes minimal amounts of proteolytic enzymes. For example, strain W3110 may be modified to effect a genetic mutation in the genes encoding proteins endogenous to the host, with examples of such hosts including *E. coli* W3110 strain 1A2, which has the complete genotype tonA; *E. coli* W3110 strain 9E4, which has the complete genotype tonA ptr3; *E. coli* W3110 strain 27C7 (ATCC 55,244), which has the complete genotype tonA ptr3 phoA E15 (argF-lac)169 degP ompT kanr; *E. coli* W3110 strain 37D6, which has the complete genotype tonA ptr3 phoA E15 (argF-lac)169 degP ompT rbs7 ilvG kanr; *E. coli* W3110 strain 40B4, which is strain 37D6 with a non-kanamycin resistant degP deletion mutation; and an *E. coli* strain having mutant periplasmic protease disclosed in U.S. Pat. No. 4,946,783. Alternatively, in vitro methods of cloning, e.g., PCR or other nucleic acid polymerase reactions, are suitable.

Full length antibody, antibody fragments, and antibody fusion proteins can be produced in bacteria, in particular when glycosylation and Fc effector function are not needed, such as when the therapeutic antibody is conjugated to a cytotoxic agent (e.g., a toxin) and the immunoconjugate by itself shows effectiveness in EBV or EBV-infected cell destruction. Full length antibodies have greater half-life in circulation. Production in *E. coli* is faster and more cost efficient. For expression of antibody fragments and polypeptides in bacteria, see, e.g., U.S. Pat. No. 5,648,237 (Carter et al.), U.S. Pat. No. 5,789,199 (Joly et al.), and U.S. Pat. No. 5,840,523 (Simmons et al.) which describes translation initiation region (TIR) and signal sequences for optimizing expression and secretion, these patents incorporated herein by reference. After expression, the antibody is isolated from the *E. coli* cell paste in a soluble fraction and can be purified through, e.g., a protein A or G column depending on the isotype. Final purification can be carried out similar to the process for purifying antibody expressed, e.g., in CHO cells.

In addition to prokaryotes, eukaryotic microbes such as filamentous fungi or yeast are suitable cloning or expression hosts for anti-gp350 antibody-encoding vectors. *Saccharomyces cerevisiae* is a commonly used lower eukaryotic host microorganism. Others include *Schizosaccharomyces pombe* (Beach and Nurse, Nature, 290: 140 [1981]; EP 139,383 published 2 May 1985); *Kluyveromyces* hosts (U.S.

Pat. No. 4,943,529; Fleer et al., Bio/Technology, 9:968-975 (1991)) such as, e.g., *K. lactis* (MW98-8C, CBS683, CBS4574; Louvencourt et al., J. Bacteriol., 154(2):737-742 [1983]), *K. fragilis* (ATCC 12,424), *K. bulgaricus* (ATCC 16,045), *K. wickerhamii* (ATCC 24,178), *K. waltii* (ATCC 56,500), *K. drosophilae* (ATCC 36,906; Van den Berg et al., Bio/Technology, 8:135 (1990)), *K. thermotolerans*, and *K. marxianus*; *yarrowia* (EP 402,226); *Pichia pastoris* (EP 183,070; Sreekrishna et al., J. Basic Microbiol., 28:265-278 [1988]); *Candida*; *Trichoderma reesia* (EP 244,234); *Neurospora crassa* (Case et al., Proc. Natl. Acad. Sci. USA, 76:5259-5263 [1979]); *Schwanniomyces* such as *Schwanniomyces occidentalis* (EP 394,538 published 31 Oct. 1990); and filamentous fungi such as, e.g., *Neurospora*, *Penicillium*, *Tolypocladium* (WO 91/00357 published 10 Jan. 1991), and *Aspergillus* hosts such as *A. nidulans* (Ballance et al., Biochem. Biophys. Res. Commun., 112:284-289 [1983]; Tilburn et al., Gene, 26:205-221 [1983]; Yelton et al., Proc. Natl. Acad. Sci. USA, 81: 1470-1474 [1984]) and *A. niger* (Kelly and Hynes, EMBO J., 4:475-479 [1985]). Methylophilic yeasts are suitable herein and include, but are not limited to, yeast capable of growth on methanol selected from the genera consisting of *Hansenula*, *Candida*, *Kloeckera*, *Pichia*, *Saccharomyces*, *Torulopsis*, and *Rhodotorula*. A list of specific species that are exemplary of this class of yeasts may be found in C. Anthony, The Biochemistry of Methylophilic Yeasts, 269 (1982).

Suitable host cells for the expression of glycosylated anti-gp350 antibody are derived from multicellular organisms. Examples of invertebrate cells include insect cells such as *Drosophila* S2 and *Spodoptera* Sf9, as well as plant cells, such as cell cultures of cotton, corn, potato, soybean, *petunia*, tomato, and tobacco. Numerous baculoviral strains and variants and corresponding permissive insect host cells from hosts such as *Spodoptera frugiperda* (caterpillar), *Aedes aegypti* (mosquito), *Aedes albopictus* (mosquito), *Drosophila melanogaster* (fruitfly), and *Bombyx mori* have been identified. A variety of viral strains for transfection are publicly available, e.g., the L-1 variant of *Autographa californica* NPV and the Bm-5 strain of *Bombyx mori* NPV, and such viruses may be used as the virus herein according to this disclosure, particularly for transfection of *Spodoptera frugiperda* cells.

However, interest has been greatest in vertebrate cells, and propagation of vertebrate cells in culture (tissue culture) has become a routine procedure. Examples of useful mammalian host cell lines are monkey kidney CV1 line transformed by SV40 (COS-7, ATCC CRL 1651); human embryonic kidney line (293 or 293 cells subcloned for growth in suspension culture, Graham et al., J. Gen. Virol. 36:59 (1977)); baby hamster kidney cells (BHK, ATCC CCL 10); Chinese hamster ovary cells/-DHFR (CHO, Urlaub et al., Proc. Natl. Acad. Sci. USA 77:4216 (1980)); mouse sertoli cells (TM4, Mather, Biol. Reprod. 23:243-251 (1980)); monkey kidney cells (CV1 ATCC CCL 70); African green monkey kidney cells (VERO-76, ATCC CRL-1587); human cervical carcinoma cells (HELA, ATCC CCL 2); canine kidney cells (MDCK, ATCC CCL 34); buffalo rat liver cells (BRL 3A, ATCC CRL 1442); human lung cells (W138, ATCC CCL 75); human liver cells (Hep G2, HB 8065); mouse mammary tumor (MMT 060562, ATCC CCL51); TRI cells (Mather et al., Annals N.Y. Acad. Sci. 383:44-68 (1982)); MRC 5 cells; FS4 cells; and a human hepatoma line (Hep G2).

Host cells are transformed with the above-described expression or cloning vectors for anti-gp350 antibody production and cultured in conventional nutrient media modi-

fied as appropriate for inducing promoters, selecting transformants, or amplifying the genes encoding the desired sequences.

3. Selection and Use of a Replicable Vector

The nucleic acid (e.g., cDNA or genomic DNA) encoding anti-gp350 antibody may be inserted into a replicable vector for cloning (amplification of the DNA) or for expression. Various vectors are publicly available. The vector may, for example, be in the form of a plasmid, cosmid, viral particle, or phage. The appropriate nucleic acid sequence may be inserted into the vector by a variety of procedures. In general, DNA is inserted into an appropriate restriction endonuclease site(s) using techniques known in the art. Vector components generally include, but are not limited to, one or more of a signal sequence, an origin of replication, one or more marker genes, an enhancer element, a promoter, and a transcription termination sequence. Construction of suitable vectors containing one or more of these components employs standard ligation techniques which are known to the skilled artisan.

The anti-gp350 monoclonal antibodies may be produced recombinantly not only directly, but also as a fusion polypeptide with a heterologous polypeptide, which may be a signal sequence or other polypeptide having a specific cleavage site at the N-terminus of the mature protein or polypeptide. In general, the signal sequence may be a component of the vector, or it may be a part of the anti-gp350 antibody-encoding DNA that is inserted into the vector. The signal sequence may be a prokaryotic signal sequence selected, for example, from the group of the alkaline phosphatase, penicillinase, lpp, or heat-stable enterotoxin II leaders. For yeast secretion, the signal sequence may be, e.g., the yeast invertase leader, alpha factor leader (including *Saccharomyces* and *Kluyveromyces* a-factor leaders, the latter described in U.S. Pat. No. 5,010,182), or acid phosphatase leader, the *C. albicans* glucoamylase leader (EP 362,179), or the signal described in WO 90/13646 published 15 Nov. 1990. In mammalian cell expression, mammalian signal sequences may be used to direct secretion of the protein, such as signal sequences from secreted polypeptides of the same or related species, as well as viral secretory leaders.

Both expression and cloning vectors contain a nucleic acid sequence that enables the vector to replicate in one or more selected host cells. Such sequences are well known for a variety of bacteria, yeast, and viruses. The origin of replication from the plasmid pBR322 is suitable for most Gram-negative bacteria, the 2 μ plasmid origin is suitable for yeast, and various viral origins (SV40, polyoma, adenovirus, VSV or BPV) are useful for cloning vectors in mammalian cells.

Expression and cloning vectors will typically contain a selection gene, also termed a selectable marker. Typical selection genes encode proteins that (a) confer resistance to antibiotics or other toxins, e.g., ampicillin, neomycin, methotrexate, or tetracycline, (b) complement auxotrophic deficiencies, or (c) supply critical nutrients not available from complex media, e.g., the gene encoding D-alanine racemase for *Bacilli*.

An example of suitable selectable markers for mammalian cells are those that enable the identification of cells competent to take up the anti-gp350 antibody-encoding nucleic acid, such as DHFR or thymidine kinase. An appropriate host cell when wild-type DHFR is employed is the CHO cell line deficient in DHFR activity, prepared and propagated as described by Urlaub et al., Proc. Natl. Acad. Sci. USA, 77:4216 (1980). A suitable selection gene for use in yeast is

the *trp1* gene present in the yeast plasmid YRp7 [Stinchcomb et al., *Nature*, 282:39 (1979); Kingsman et al., *Gene*, 7:141 (1979); Tschemper et al., *Gene*, 10:157 (1980)]. The *trp1* gene provides a selection marker for a mutant strain of yeast lacking the ability to grow in tryptophan, for example, ATCC No. 44076 or PEP4-1 (Jones, *Genetics*, 85:12 (1977)).

Expression and cloning vectors usually contain a promoter operably linked to the anti-gp350 antibody-encoding nucleic acid sequence to direct mRNA synthesis. Promoters recognized by a variety of potential host cells are well known. Promoters suitable for use with prokaryotic hosts include the β -lactamase and lactose promoter systems (Chang et al., *Nature*, 275:615 (1978); Goeddel et al., *Nature*, 281:544 (1979)), alkaline phosphatase, a tryptophan (*trp*) promoter system (Goeddel, *Nucleic Acids Res.*, 8:4057 (1980)), and hybrid promoters such as the *tac* promoter (deBoer et al., *Proc. Natl. Acad. Sci. USA*, 80:21-25 (1983)). Promoters for use in bacterial systems also will contain a Shine-Dalgarno sequence operably linked to the DNA encoding anti-gp350 antibody.

Examples of suitable promoting sequences for use with yeast hosts include the promoters for 3-phosphoglycerate kinase (Hitzeman et al., *J. Biol. Chem.*, 255:2073 (1980)) or other glycolytic enzymes (Hess et al., *J. Adv. Enzyme Reg.*, 7:149 (1968); Holland, *Biochemistry*, 17:4900 (1978)), such as enolase, glyceraldehyde-3-phosphate dehydrogenase, hexokinase, pyruvate decarboxylase, phosphofructokinase, glucose-6-phosphate isomerase, 3-phosphoglycerate mutase, pyruvate kinase, triosephosphate isomerase, phosphoglucose isomerase, and glucokinase.

Other yeast promoters, which are inducible promoters having the additional advantage of transcription controlled by growth conditions, are the promoter regions for alcohol dehydrogenase 2, isocytochrome C, acid phosphatase, degradative enzymes associated with nitrogen metabolism, metallothionein, glyceraldehyde-3-phosphate dehydrogenase, and enzymes responsible for maltose and galactose utilization. Suitable vectors and promoters for use in yeast expression are further described in EP 73,657.

Anti-gp350 antibody transcription from vectors in mammalian host cells is controlled, for example, by promoters obtained from the genomes of viruses such as polyoma virus, fowlpox virus (UK 2,211,504), adenovirus (such as Adenovirus 2), bovine papilloma virus, avian sarcoma virus, cytomegalovirus, a retrovirus, hepatitis-B virus and Simian Virus 40 (SV40), from heterologous mammalian promoters, e.g., the actin promoter or an immunoglobulin promoter, and from heat-shock promoters, provided such promoters are compatible with the host cell systems.

Transcription of a DNA encoding the anti-gp350 antibody by higher eukaryotes may be increased by inserting an enhancer sequence into the vector. Enhancers are cis-acting elements of DNA, usually about from 10 to 300 bp, that act on a promoter to increase its transcription. Many enhancer sequences are now known from mammalian genes (globin, elastase, albumin, α -fetoprotein, and insulin). Typically, however, one will use an enhancer from a eukaryotic cell virus. Examples include the SV40 enhancer on the late side of the replication origin (bp 100-270), the cytomegalovirus early promoter enhancer, the polyoma enhancer on the late side of the replication origin, and adenovirus enhancers. The enhancer may be spliced into the vector at a position 5' or 3' to the anti-gp350 antibody coding sequence, but is preferably located at a site 5' from the promoter.

Expression vectors used in eukaryotic host cells (yeast, fungi, insect, plant, animal, human, or nucleated cells from

other multicellular organisms) will also contain sequences necessary for the termination of transcription and for stabilizing the mRNA. Such sequences are commonly available from the 5' and, occasionally 3', untranslated regions of eukaryotic or viral DNAs or cDNAs. These regions contain nucleotide segments transcribed as polyadenylated fragments in the untranslated portion of the mRNA encoding anti-gp350 antibody.

Still other methods, vectors, and host cells suitable for adaptation to the synthesis of anti-gp350 antibody in recombinant vertebrate cell culture are described in Gething et al., *Nature*, 293:620-25 (1981); Mantei et al., *Nature*, 281:40-46 (1979); EP 117,060; and EP 117,058.

4. Culturing the Host Cells

The host cells used to produce the anti-gp350 antibody of this disclosure may be cultured in a variety of media. Commercially available media such as Ham's F10 (Sigma), Minimal Essential Medium ((MEM), (Sigma), RPMI-1640 (Sigma), and Dulbecco's Modified Eagle's Medium ((DMEM), Sigma) are suitable for culturing the host cells. In addition, any of the media described in Ham et al., *Meth. Enz.* 58:44 (1979), Barnes et al., *Anal. Biochem.* 102:255 (1980), U.S. Pat. Nos. 4,767,704; 4,657,866; 4,927,762; 4,560,655; or 5,122,469; WO 90/03430; WO 87/00195; or U.S. Pat. Re. 30,985 may be used as culture media for the host cells. Any of these media may be supplemented as necessary with hormones and/or other growth factors (such as insulin, transferrin, or epidermal growth factor), salts (such as sodium chloride, calcium, magnesium, and phosphate), buffers (such as HEPES), nucleotides (such as adenosine and thymidine), antibiotics (such as gentamycin), trace elements (defined as inorganic compounds usually present at final concentrations in the micromolar range), and glucose or an equivalent energy source. Any other necessary supplements may also be included at appropriate concentrations that would be known to those skilled in the art. The culture conditions, such as temperature, pH, and the like, are those previously used with the host cell selected for expression, and will be apparent to the ordinarily skilled artisan.

5. Detecting Gene Amplification/Expression

Gene amplification and/or expression may be measured in a sample directly, for example, by conventional Southern blotting, Northern blotting to quantitate the transcription of mRNA [Thomas, *Proc. Natl. Acad. Sci. USA*, 77:5201-5205 (1980)], dot blotting (DNA analysis), or in situ hybridization, using an appropriately labeled probe, based on the sequences provided herein. Alternatively, antibodies may be employed that can recognize specific duplexes, including DNA duplexes, RNA duplexes, and DNA-RNA hybrid duplexes or DNA-protein duplexes. The antibodies in turn may be labeled and the assay may be carried out where the duplex is bound to a surface, so that upon the formation of duplex on the surface, the presence of antibody bound to the duplex can be detected.

Gene expression, alternatively, may be measured by immunological methods, such as immunohistochemical staining of cells or tissue sections and assay of cell culture or body fluids, to quantitate directly the expression of gene product. Antibodies useful for immunohistochemical staining and/or assay of sample fluids may be either monoclonal or polyclonal, and may be prepared in any mammal. Conveniently, the antibodies may be prepared against a native sequence EBV gp350 protein or against a synthetic peptide based on the DNA sequences provided herein or against exogenous sequence fused to gp350 DNA and encoding a specific antibody epitope.

6. Purification of Anti-Gp350 Antibodies and Gp350 Binding Oligopeptides

Forms of anti-gp350 antibody may be recovered from culture medium or from host cell lysates. If membrane-bound, it can be released from the membrane using a suitable detergent solution (e.g. Triton-X 100) or by enzymatic cleavage. Cells employed in expression of anti-gp350 antibody can be disrupted by various physical or chemical means, such as freeze-thaw cycling, sonication, mechanical disruption, or cell lysing agents.

It may be desired to purify anti-gp350 antibody from recombinant cell proteins or polypeptides. The following procedures are exemplary of suitable purification procedures: by fractionation on an ion-exchange column; ethanol precipitation; reverse phase HPLC; chromatography on silica or on a cation-exchange resin such as DEAE; chromatofocusing; SDS-PAGE; ammonium sulfate precipitation; gel filtration using, for example, Sephadex G-75; protein A Sepharose columns to remove contaminants such as IgG; and metal chelating columns to bind epitope-tagged forms of the anti-gp350 antibody and EBV gp350 protein. Various methods of protein purification may be employed and such methods are known in the art and described for example in Deutscher, *Methods in Enzymology*, 182 (1990); Scopes, *Protein Purification: Principles and Practice*, Springer-Verlag, New York (1982). The purification step(s) selected will depend, for example, on the nature of the production process used and the anti-gp350 antibody produced.

When using recombinant techniques, the antibody can be produced intracellularly, in the periplasmic space, or directly secreted into the medium. If the antibody is produced intracellularly, as a first step, the particulate debris, either host cells or lysed fragments, are removed, for example, by centrifugation or ultrafiltration. Carter et al., *Bio/Technology* 10:163-167 (1992) describe a procedure for isolating antibodies which are secreted to the periplasmic space of *E. coli*. Briefly, cell paste is thawed in the presence of sodium acetate (pH 3.5), EDTA, and phenylmethylsulfonyl fluoride (PMSF) over about 30 min. Cell debris can be removed by centrifugation. Where the antibody is secreted into the medium, supernatants from such expression systems are generally first concentrated using a commercially available protein concentration filter, for example, an Amicon or Millipore Pellicon ultrafiltration unit. A protease inhibitor such as PMSF may be included in any of the foregoing steps to inhibit proteolysis and antibiotics may be included to prevent the growth of adventitious contaminants.

The antibody composition prepared from the cells can be purified using, for example, hydroxylapatite chromatography, gel electrophoresis, dialysis, and affinity chromatography, with affinity chromatography being the preferred purification technique. The suitability of protein A as an affinity ligand depends on the species and isotype of any immunoglobulin Fc domain that is present in the antibody. Protein A can be used to purify antibodies that are based on human $\gamma 1$, $\gamma 2$ or $\gamma 4$ heavy chains (Lindmark et al., *J. Immunol. Meth.* 62:1-13 (1983)). Protein G is recommended for all mouse isotypes and for human $\gamma 3$ (Guss et al., *EMBO J.* 5:1567-1575 (1986)). The matrix to which the affinity ligand is attached is most often agarose, but other matrices are available. Mechanically stable matrices such as controlled pore glass or poly(styrenedivinyl)benzene allow for faster flow rates and shorter processing times than can be achieved with agarose. Where the antibody comprises a CH3 domain, the Bakerbond ABXTM resin (J. T. Baker, Phillipsburg, NJ) is useful for purification. Other techniques for protein purifi-

cation such as fractionation on an ion-exchange column, ethanol precipitation, Reverse Phase HPLC, chromatography on silica, chromatography on heparin SEPHAROSETM chromatography on an anion or cation exchange resin (such as a polyaspartic acid column), chromatofocusing, SDS-PAGE, and ammonium sulfate precipitation are also available depending on the antibody to be recovered.

Following any preliminary purification step(s), the mixture comprising the antibody of interest and contaminants may be subjected to low pH hydrophobic interaction chromatography using an elution buffer at a pH between about 2.5-4.5, preferably performed at low salt concentrations (e.g., from about 0-0.25M salt).

G. Pharmaceutical Formulations

Therapeutic formulations of the anti-gp350 antibodies or gp350 binding oligopeptides of this disclosure are prepared for storage by mixing the antibody, polypeptide, or oligopeptide having the desired degree of purity with optional pharmaceutically acceptable carriers, excipients or stabilizers (Remington's *Pharmaceutical Sciences* 16th edition, Osol, A. Ed. (1980)), in the form of lyophilized formulations or aqueous solutions. Acceptable carriers, excipients, or stabilizers are nontoxic to recipients at the dosages and concentrations employed, and include buffers such as acetate, Tris, phosphate, citrate, and other organic acids; antioxidants including ascorbic acid and methionine; preservatives (such as octadecyldimethylbenzyl ammonium chloride; hexamethonium chloride; benzalkonium chloride, benzethonium chloride; phenol, butyl or benzyl alcohol; alkyl parabens such as methyl or propyl paraben; catechol; resorcinol; cyclohexanol; 3-pentanol; and m-cresol); low molecular weight (less than about 10 residues) polypeptides; proteins, such as serum albumin, gelatin, or immunoglobulins; hydrophilic polymers such as polyvinylpyrrolidone; amino acids such as glycine, glutamine, asparagine, histidine, arginine, or lysine; monosaccharides, disaccharides, and other carbohydrates including glucose, mannose, or dextrans; chelating agents such as EDTA; tonics such as trehalose and sodium chloride; sugars such as sucrose, mannitol, trehalose or sorbitol; surfactant such as polysorbate; salt-forming counter-ions such as sodium; metal complexes (e.g., Zn-protein complexes); and/or non-ionic surfactants such as TWEEN®, PLURONICS® or polyethylene glycol (PEG). The antibody preferably comprises the antibody at a concentration of between 5-200 mg/ml, preferably between 10-100 mg/ml.

The formulations herein may also contain more than one active compound as necessary for the indication being treated, preferably those with complementary activities that do not adversely affect each other. For example, in addition to an anti-gp350 antibody or gp350 binding oligopeptide, it may be desirable to include in the one formulation, an additional antibody, e.g., a second anti-gp350 antibody which binds a different epitope on the EBV gp350 protein. Alternatively, or additionally, the composition may further comprise a cytokine, an anti-inflammatory agent, or an interferon. Such molecules are suitably present in combination in amounts that are effective for the purpose intended.

The active ingredients may also be entrapped in microcapsules prepared, for example, by coacervation techniques or by interfacial polymerization, for example, hydroxymethylcellulose or gelatin-microcapsules and poly-(methylmethacrylate) microcapsules, respectively, in colloidal drug delivery systems (for example, liposomes, albumin microspheres, microemulsions, nano-particles and nanocapsules)

or in macroemulsions. Such techniques are disclosed in Remington's Pharmaceutical Sciences, 16th edition, Osol, A. Ed. (1980).

Sustained-release preparations may be prepared. Suitable examples of sustained-release preparations include semi-permeable matrices of solid hydrophobic polymers containing the antibody, which matrices are in the form of shaped articles, e.g., films, or microcapsules. Examples of sustained-release matrices include polyesters, hydrogels (for example, poly(2-hydroxyethyl-methacrylate), or poly(vinyl-alcohol)), polylactides (U.S. Pat. No. 3,773,919), copolymers of L-glutamic acid and γ ethyl-L-glutamate, non-degradable ethylene-vinyl acetate, degradable lactic acid-glycolic acid copolymers such as the LUPRON DEPOT® (injectable microspheres composed of lactic acid-glycolic acid copolymer and leuprolide acetate), and poly-D-(-)-3-hydroxybutyric acid.

The formulations to be used for in vivo administration must be sterile. This is readily accomplished by filtration through sterile filtration membranes.

H. Diagnosis and Treatment with Anti-Gp350 Antibodies or Gp350 Binding Oligopeptides

gp350 expression may be evaluated using an in vivo diagnostic assay, e.g., by administering a molecule (such as an anti-gp350 antibody or gp350 binding oligopeptide) which binds the molecule to be detected and is tagged with a detectable label (e.g., a radioactive isotope or a fluorescent label) and externally scanning the patient for localization of the label.

As described above, the anti-gp350 antibodies or oligopeptides of this disclosure have various non-therapeutic applications. The anti-gp350 antibodies or oligopeptides of this disclosure are useful for diagnosis and staging of EBV infections. The antibodies or oligopeptides are also useful for purification or immunoprecipitation of EBV gp350 protein from cells, for detection and quantitation of EBV gp350 protein in vitro, e.g., in an ELISA or a Western blot, to kill and eliminate gp350-expressing cells from a population of mixed cells as a step in the purification of other cells.

Currently, EBV infection prevention and treatment involves preventing transmission of the virus, vaccination, or administration of interferons. Anti-gp350 antibody or oligopeptide therapy (such as by passive immunotherapy) may be especially desirable in elderly patients or immunocompromised patients or pregnant patients who may not tolerate the side effects of vaccination or vaccine components or interferons, or who cannot mount an immunological response.

A conjugate comprising an anti-gp350 antibody or oligopeptide conjugated with a cytotoxic agent may be administered to the patient. Preferably, the immunoconjugate bound to the anti-gp350 antibody is internalized by the cell, resulting in increased therapeutic efficacy of the immunoconjugate in killing the infected cell to which it binds. Preferably, the cytotoxic agent targets or interferes with the nucleic acid in the infected cell. The anti-gp350 antibodies or oligopeptides or conjugates thereof are administered to a human patient, in accord with known methods, such as intravenous administration, e.g., as a bolus or by continuous infusion over a period of time, by intramuscular, intraperitoneal, intracerebrospinal, subcutaneous, intra-articular, intrasynovial, intrathecal, oral, topical, or inhalation routes. Intravenous or subcutaneous administration of the antibody or oligopeptide is preferred.

Other therapeutic regimens may be combined with the administration of the anti-gp350 antibody or oligopeptide. The combined administration includes co-administration,

using separate formulations or a single pharmaceutical formulation, and consecutive administration in either order, wherein preferably there is a time period while both (or all) active agents simultaneously exert their biological activities.

5 Preferably such combined therapy results in a synergistic therapeutic effect.

It may also be desirable to combine administration of the anti-gp350 antibody or antibodies or oligopeptides with administration of an antibody directed against another EBV antigen.

10 The therapeutic treatment methods of this disclosure may include the combined administration of an anti-gp350 antibody (or antibodies) or oligopeptides and an interferon.

For the prevention or treatment of EBV infection or EBV-associated disease, the dosage and mode of administration of these antibodies and therapeutic proteins will be chosen by the medical provider according to known criteria. The appropriate dosage of antibody or oligopeptide will depend on the type of disease to be treated, as defined above, the severity and course of the disease, whether the antibody or oligopeptide is administered for preventive or therapeutic purposes, previous therapy, the patient's clinical history and response to the antibody or oligopeptide and the discretion of the medical provider. The antibody or oligopeptide is suitably administered to the patient at one time or over a series of treatments. Preferably, the antibody or oligopeptide is administered by intravenous infusion or by subcutaneous injections. Depending on the type and severity of the disease, about 1 mcg/kg to about 50 mg/kg body weight (e.g., about 0.1-15 mg/kg/dose) of antibody can be an initial candidate dosage for administration to the patient, whether, for example, by one or more separate administrations, or by continuous infusion. A dosing regimen can comprise administering an initial loading dose of about 4 mg/kg, followed by a weekly maintenance dose of about 2 mg/kg of the anti-gp350 antibody. However, other dosage regimens may be useful. A typical daily dosage might range from about 1 mcg/kg to 100 mg/kg or more, depending on the factors mentioned above. For repeated administrations over several days or longer, depending on the condition, the treatment is sustained until a desired suppression of disease symptoms occurs. The progress of this therapy can be readily monitored by conventional methods and assays and based on criteria known to medical providers of skill in the art.

45 Aside from administration of the anti-gp350 antibody to a patient, this disclosure contemplates administration of the antibody by gene therapy. Such administration of nucleic acid encoding the antibody is encompassed by the expression "administering a therapeutically effective amount of an antibody." See, for example, WO96/07321 concerning the use of gene therapy to generate intracellular antibodies.

There are two major approaches to getting the nucleic acid (optionally contained in a vector) into the patient's cells; in vivo and ex vivo. For in vivo delivery, the nucleic acid is injected directly into the patient, usually at the site where the antibody is required. For ex vivo treatment, the patient's cells are removed, the nucleic acid is introduced into these isolated cells and the modified cells are administered to the patient either directly or, for example, encapsulated within porous membranes which are implanted into the patient (see, e.g., U.S. Pat. Nos. 4,892,538 and 5,283,187). There are a variety of techniques available for introducing nucleic acids into viable cells. The techniques vary depending upon whether the nucleic acid is transferred into cultured cells in vitro, or in vivo in the cells of the intended host. Techniques suitable for the transfer of nucleic acid into mammalian cells in vitro include the use of liposomes,

electroporation, microinjection, cell fusion, DEAE-dextran, the calcium phosphate precipitation method, etc. A commonly used vector for ex vivo delivery of the gene is a retroviral vector.

The currently preferred in vivo nucleic acid transfer techniques include transfection with viral vectors (such as adenovirus, Herpes simplex I virus, or adeno-associated virus) and lipid-based systems (useful lipids for lipid-mediated transfer of the gene are DOTMA, DOPE and DC-Chol, for example). For review of the currently known gene marking and gene therapy protocols see Anderson et al., *Science* 256:808-813 (1992). See also WO 93/25673 and the references cited therein.

The anti-gp350 antibodies of the disclosure can be in the different forms encompassed by the definition of "antibody" herein. Thus, the antibodies include full length or intact antibody, antibody fragments, native sequence antibody or amino acid variants, humanized, chimeric or fusion antibodies, immunoconjugates, and functional fragments thereof. In fusion antibodies, an antibody sequence is fused to a heterologous polypeptide sequence. The antibodies can be modified in the Fc region to provide desired effector functions. As discussed in more detail above, with the appropriate Fc regions, the naked antibody bound on the cell surface can induce cytotoxicity, e.g., via antibody-dependent cellular cytotoxicity (ADCC) or by recruiting complement in complement dependent cytotoxicity, or some other mechanism. Alternatively, where it is desirable to eliminate or reduce effector function, to minimize side effects or therapeutic complications, certain other Fc regions may be used.

These antibodies may include an antibody that competes for binding or binds substantially to, the same epitope as the antibodies of the disclosure. Antibodies having the biological characteristics of the present anti-gp350 antibodies of this disclosure are also contemplated, specifically including the in vivo targeting, and infection inhibiting or preventing, or cytotoxic characteristics.

The present anti-gp350 antibodies or oligopeptides are useful for treating an EBV infection or alleviating one or more symptoms of the infection in a mammal. The antibody or oligopeptide can bind to at least a portion of an infected cell that express EBV gp350 protein in the mammal. Preferably, the antibody or oligopeptide is effective to destroy or kill gp350-expressing cells or inhibit the growth of such cells, in vitro or in vivo, upon binding to EBV gp350 protein on the cell. Such an antibody includes a naked anti-gp350 antibody (not conjugated to any agent). Naked antibodies that have cytotoxic or cell growth inhibition properties can be further harnessed with a cytotoxic agent to render them even more potent in EBV or EBV-infected cell destruction. Cytotoxic properties can be conferred to an anti-gp350 antibody by, e.g., conjugating the antibody with a cytotoxic agent, to form an immunoconjugate as described herein. The cytotoxic agent or a growth inhibitory agent is preferably a small molecule.

This disclosure also provides a composition comprising an anti-gp350 antibody or oligopeptide of the disclosure, and a carrier. For the purposes of treating EBV infection, compositions can be administered to the patient in need of such treatment, wherein the composition can comprise one or more anti-gp350 antibodies present as an immunoconjugate or as the naked antibody. The compositions may comprise these antibodies or oligopeptides in combination with other therapeutic agents. The formulation may be a therapeutic formulation comprising a pharmaceutically acceptable carrier.

This disclosure also provides isolated nucleic acids encoding the anti-gp350 antibodies. Nucleic acids encoding both the H and L chains and especially the hypervariable region residues, chains which encode the native sequence antibody as well as variants, modifications and humanized versions of the antibody, are encompassed.

The disclosure also provides methods useful for treating an EBV infection or alleviating one or more symptoms of the infection in a mammal, comprising administering a therapeutically effective amount of an anti-gp350 antibody or oligopeptide of this disclosure to the mammal. The antibody or oligopeptide therapeutic compositions can be administered short term (acutely) or chronically, or intermittently as directed by a medical professional. Also provided are methods of inhibiting the growth of, and killing an EBV gp350 protein-expressing cell.

This disclosure also provides methods useful for treating or preventing post-transplant lymphoproliferative disorder (PTLD) in a mammal. In these methods, the antibodies or fragments thereof, are particularly effective for PTLD in EBV-seronegative persons (i.e., not previously infected with EBV) and given within a few days of solid organ or bone marrow transplant. In these methods, multiple doses of the antibodies, or functional fragments thereof, may be given over time (for example, in persons undergoing solid organ transplants, these proteins may be given within 72 hours of transplant and additional doses may be given at 1, 4, 6, 8, 12, and 16 weeks after transplant). Virtually all transplant recipients are screened for EBV and CMV serology prior to transplant. The antibodies, or functional fragments thereof, may also be administered to EBV-seropositive persons, but the rate of PTLD is lower in EBV seropositive persons than seronegative persons. In these methods, the antibodies, or functional fragments thereof, and an EBV vaccine may be co-administered. In these co-administration methods, the antibodies, or functional fragments thereof, and the vaccine (s) may be administered on the same day at different sites, as is done for combined treatment with vaccine and immunoglobulin for rabies, hepatitis B, or tetanus exposure. In these methods, a single dose of the antibodies, or functional fragments thereof, would likely be given because the recipient would be expected to produce antibody in response to the vaccine.

J. Articles of Manufacture and Kits

This disclosure also provides assay devices, kits, and articles of manufacture comprising at least one anti-gp350 antibody or oligopeptide of this disclosure, optionally linked to a label, such as a fluorescent or radiolabel. The articles of manufacture may contain materials useful for the detection, diagnosis, or treatment of EBV infection. A preferred device is a lateral flow assay device which provides for point-of-care detection and/or diagnosis of an EBV infection. The article of manufacture comprises a container and a label or package insert on or associated with the container. Suitable containers include, for example, bottles, vials, syringes, etc. The containers may be formed from a variety of materials such as glass or plastic. The container holds a composition which is effective for detecting or treating the EBV infection and may have a sterile access port (for example the container may be an intravenous solution bag or a vial having a stopper pierceable by a hypodermic injection needle). At least one active agent in the composition is an anti-gp350 antibody or oligopeptide of this disclosure. The label or package insert indicates that the composition is used for detecting or treating EBV infection. The label or package insert may further comprise instructions for using the antibody or oligopeptide composition, e.g., in the testing or

treating of the infected patient. Additionally, the article of manufacture may further comprise a second container comprising a pharmaceutically-acceptable buffer, such as bacteriostatic water for injection (BWI), phosphate-buffered saline, Ringer's solution and dextrose solution. It may further include other materials desirable from a commercial and user standpoint, including other buffers, diluents, filters, needles, and syringes.

Kits are also provided that are useful for various purposes, e.g., for EBV-infected cell killing assays, for purification, or immunoprecipitation of EBV gp350 protein from cells. For isolation and purification of EBV gp350 protein, the kit can contain an anti-gp350 antibodies or oligopeptides coupled to beads (e.g., sepharose beads). Kits can be provided which contain the antibodies or oligopeptides for detection and quantitation of EBV gp350 protein in vitro, e.g., in an ELISA or a Western blot. As with the article of manufacture, the kit comprises a container and a label or package insert on or associated with the container. The container holds a composition comprising at least one anti-gp350 antibody or oligopeptide of the disclosure. Additional containers may be included that contain, e.g., diluents and buffers, control antibodies. The label or package insert may provide a description of the composition as well as instructions for the intended in vitro or diagnostic use.

The anti-gp350 antibody or oligopeptide of this disclosure may also be provided as part of an assay device. Such assay devices include lateral flow assay devices. A common type of disposable lateral flow assay device includes a zone or area for receiving the liquid sample, a conjugate zone, and a reaction zone. These assay devices are commonly known as lateral flow test strips. They employ a porous material, e.g., nitrocellulose, defining a path for fluid flow capable of supporting capillary flow. Examples include those described in U.S. Pat. Nos. 5,559,041, 5,714,389, 5,120,643, and 6,228,660 all of which are incorporated herein by reference in their entireties. The anti-gp350 antibody or oligopeptide of this disclosure may also be used in a lateral flow assay device in conjunction with other antibodies to detect multiple EBV proteins or other herpesvirus proteins using a single biological sample from a subject or patient being tested on one portable, point-of-care device.

Another type of assay device is a non-porous assay device having projections to induce capillary flow. Examples of such assay devices include the open lateral flow device as disclosed in PCT International Publication Nos. WO 2003/103835, WO 2005/089082, WO 2005/118139, and WO 2006/137785, all of which are incorporated herein by reference in their entireties.

K. Anti-Gp350 Antibody Probes and B-Cell Probes

Anti-gp350 antibody and anti-gp350 B-cell probes of this disclosure are polypeptides that bind, preferably specifically, to an anti-gp350 antibody, or a B-cell expressing an anti-gp350 B cell receptor (BCR) (anti-gp350 B cell). Accordingly, the three-dimensional structure of such a probe resembles the three-dimensional structure of at least a portion of the EBV gp350 protein. The three-dimensional structure of an anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may resemble the three-dimensional structure of the CR2-binding region of an EBV gp350 protein. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence from the CR2-binding region of an EBV gp350 protein. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise at least 10, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, or at

least 350 contiguous amino acids from the amino acid sequence forming the CR2-binding region of an EBV gp350 protein, wherein the probe has the ability to bind an anti-gp350 antibody, or a B-cell expressing an anti-gp350 BCR, and wherein the binding of the B-cell is through the BCR. An anti-gp350 antibody probe, or anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 90%, at least 95%, at least 97% or at least 99% identical to the amino acid sequence of the CR2-binding region of EBV gp350, wherein the probe can bind an anti-gp350 antibody, or a B-cell expressing an anti-gp350 BCR, and wherein the binding of the B-cell is through the BCR.

Anti-gp350 antibody probes, and anti-gp350 B-cell probes, of this disclosure can be any size sufficient to bind to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise at least 10, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, or at least 450 amino acids in length. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise at least 10, at least 20, at least 30, at least 40, at least 50, at least 75, at least 100, at least 150, at least 200, at least 250, at least 300, at least 350, at least 400, or at least 450 contiguous amino acids from the amino acid sequence forming the CR2-binding region of and EBV gp350 protein, wherein the probe has the ability to bind an anti-gp350 antibody, or a B-cell expressing an anti-gp350 BCR, and wherein the binding of the B-cell is through the BCR.

Using an anti-gp350 antibody probe, or anti-gp350 B-cell probe, of this disclosure having a sequence identical to that of EBV gp35, to isolate gp350-specific B-cells might be problematic, because most B-cells, including memory B-cells, constitutively express CR2 on their surface. Consequently, probes having a three-dimensional structure, and/or a sequence, identical to the three-dimensional structure, and/or sequence, of the CR2-binding region of EBV gp350 will bind to almost all B-cells. Thus, anti-gp350 antibody probes, and anti-gp350 B-cell probes, of this disclosure may lack the ability to bind CR2. Such anti-gp350 antibody probes, and anti-gp350 B-cell probes, can be constructed, for example, by using the sequence of the CR2-binding region of an EBV gp350 protein to construct the probe, and then altering the sequence so that it no longer binds CR2. Thus, anti-gp350 antibody probes, or an anti-gp350 B-cell probes, of this disclosure may comprise a variant sequence of an EBV gp350 CR2 binding region. An anti-gp350 antibody probe, or anti-gp350 B-cell probe, of this disclosure comprises the sequence of an EBV gp350 CR2 binding region, wherein the sequence has been altered to reduce, or eliminate, binding of the probe to CR2. Such alterations include, but are not limited to, substitutions, insertions, and deletions of amino acid residues. Such probes preferentially bind to an anti-gp350 antibody, or an anti-gp350 BCR, but do not bind CR2. An anti-gp350 antibody probe, or anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to an EBV gp350 CR2 binding region, wherein the probe binds to an anti-gp350 antibody, or an anti-gp350 BCR, and wherein the probe does not bind CR2. An anti-gp350 antibody probe, or anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID

residue have binding properties (e.g., ionic charge, hydrophobicity, etc.) different from the amino acid residue being replaced.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 Cr2-binding region, wherein the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 has been substituted with an amino acid residue selected from the group consisting of methionine, leucine, isoleucine, glycine, alanine, valine, phenylalanine, tyrosine, tryptophan, lysine, arginine, histidine, glutamate, and aspartate, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 has been substituted with asparagine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid P158 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid P158 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99%

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein one or more amino acids has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein one or more amino acids has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein one or more amino acids selected from the group consisting of serine, threonine, cysteine, proline, asparagine, and glutamine, has been inserted between the

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein one or more amino acids has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein one or more amino acids has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 has been substituted with an amino acid residue selected from the group consisting of tyrosine, phenylalanine, tryptophan, serine, threonine, cysteine, proline, asparagine, glutamine, lysine, arginine, histidine, glycine, alanine, valine, leucine, methionine, and isoleucine, wherein one or more amino acids selected from the group consisting of serine, threonine, cysteine, proline, asparagine, and glutamine, has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid W162 of SEQ ID NO:124 has been substituted

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to the amino acid sequence of an EBV gp350 CR2-binding region, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 has substituted with one or more amino acid residues, wherein the amino acid corresponding to amino acid 1160 of SEQ ID NO:124 has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90%

least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid K161 of SEQ ID NO:124 has been deleted, or has been substituted with one or more amino acid residues, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:125, wherein the amino acid corresponding to amino acid K161 of SEQ ID NO:124 has been substituted with an amino acid residue selected from the group consisting of methionine, leucine, isoleucine, glycine, alanine, valine, phenylalanine, tyrosine, tryptophan, lysine, arginine, histidine, glutamate, serine, and aspartate, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:127 or SEQ ID NO:145, wherein the amino acid corresponding to amino acid Q122 of SEQ ID NO:124 is asparagine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:127 or SEQ ID NO:145.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:129 or SEQ ID NO:147, wherein the amino acid corresponding to amino acid D163 of SEQ ID NO:124 is asparagine, and wherein a serine residue has been inserted between the amino acid corresponding to amino acid N164 of SEQ ID NO:124 and the amino acid corresponding to C165 of SEQ ID NO:124, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:129 or SEQ ID NO:147.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:131 or SEQ ID NO:149, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 is arginine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:131 or SEQ ID NO:149.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:133 or SEQ ID NO:151, wherein the amino acids corresponding to amino acids W162-N164 of SEQ ID NO:124 have each been substituted with an alanine residue, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically

binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:133 or SEQ ID NO:151.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:135 or SEQ ID NO:153, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 is arginine, wherein the amino acid corresponding to amino acid I160 of SEQ ID NO:124 is arginine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:135 or SEQ ID NO:153.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:137 or SEQ ID NO:155, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 is arginine, wherein the amino acid corresponding to amino acid P158 of SEQ ID NO:124 is arginine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:137 or SEQ ID NO:155.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:139 or SEQ ID NO:157, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 is arginine, wherein the amino acid corresponding to amino acid P158 of SEQ ID NO:124 is asparagine, wherein the amino acid corresponding to amino acid I160 of SEQ ID NO:124 is threonine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:139 or SEQ ID NO:157.

An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise an amino acid sequence at least 80% identical, at least 85% identical, at least 90% identical, at least 95% identical, at least 97% identical, or at least 99% identical to SEQ ID NO:141 or SEQ ID NO:159, wherein the amino acid corresponding to amino acid D296 of SEQ ID NO:124 is arginine, wherein the amino acid corresponding to amino acid P158 of SEQ ID NO:124 is arginine, wherein the amino acid corresponding to amino acid I160 of SEQ ID NO:124 is arginine, wherein the probe lacks the ability to bind CR2, and wherein the probe specifically binds to an anti-gp350 antibody or an anti-gp350 B-cell. An anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of this disclosure may comprise SEQ ID NO:141 or SEQ ID NO:159.

Thus, this disclosure includes a nucleic acid molecule encoding an anti-gp350 antibody probe, or an anti-gp350 B-cell probe, disclosed herein. Such nucleic acid molecule includes DNA, RNA, mixtures thereof, and modified forms thereof. This disclosure also includes nucleic acid vectors,

145

such as, for example, plasmids, expression vectors, viral vectors, etc., as well as cells comprising such nucleic acid vectors. Methods of producing nucleic acid molecules and nucleic acid vectors of the invention have been described herein.

Anti-gp350 antibody and anti-gp350 B-cell probes of this disclosure can be produced chemically (e.g., oligopeptide synthesis methodology), or they can be produced using recombinant technology. For example, an anti-gp350 antibody probe, or anti-gp350 B-cell probe can be designed, by producing a nucleic acid molecule encoding the probe, and the nucleic acid molecule introduced into a cell where it can be expressed.

This disclosure includes methods of identifying anti-gp350 antibodies, comprising contacting a test solution containing antibodies with an anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of the disclosure, and isolating an antibody that specifically binds to the anti-gp350 antibody probe, or the anti-gp350 B-cell probe, thereby identifying an anti-gp350 antibody. The anti-gp350 antibody probe may be joined to an affinity tag such as, for example, a His tag, an epitope tag or a labeled tag (e.g., fluorescent label). The anti-gp350 antibody probe, or the anti-gp350 B-cell, may be immobilized on a surface.

This disclosure also includes methods of identifying an anti-gp350 B-cell (i.e., a B-cell expressing an anti-gp350 B-cell receptor), comprising contacting a test solution containing B-cells with an anti-gp350 antibody probe, or an anti-gp350 B-cell probe, of the disclosure, and isolating a B-cell that specifically binds to the anti-gp350 antibody probe, or the anti-gp350 B-cell probe, thereby identifying a B-cell expressing an anti-gp350 B-cell receptor. The anti-gp350 antibody probe, or the anti-gp350 B-cell probe, may be joined to an affinity tag such as, for example, a His tag, an epitope tag or a labeled tag (e.g., fluorescent label). The anti-gp350 antibody probe, or the anti-gp350 B-cell probe, may be immobilized on a surface.

All patent and literature references cited in the present specification are hereby incorporated by reference in their entirety.

146

The foregoing disclosure is sufficient to enable one skilled in the art to practice the invention. The present invention is not to be limited in scope by the constructs described, because the described embodiments are intended as illustrations of certain aspects of the invention and any constructs that are functionally equivalent are within the scope of this invention.

Example

The inventors isolated and characterized a set of human monoclonal antibodies from two blood bank donors who have extraordinarily high serum neutralizing antibody titers to EBV. Using the receptor-binding null gp350 probes described in this invention they successfully identified gp350-specific B cells in peripheral blood mononuclear cells without having issues with non-targeting memory B cells that ubiquitously express CR2 (i.e. gp350 receptor) on their surface. Of those monoclonal antibodies, antibodies named A9 and E11 derived from 2 independent donors possessed the highest neutralization activity against EBV, and hence, they were structurally characterized. Structures of antigen-binding fragment (Fab) of A9 and E11 were solved to resolution of 2.4 and 1.8 Å, respectively. Additionally, the structure of Fab A9 in complex with gp350 receptor-binding domain was solved to 3.5 Å. Based on the structural information they designed a set of mutations on both antibodies A9 and E11 that potentially improve affinity and therefore, enhance neutralization potency.

These highly potent neutralizing monoclonal antibodies are useful in prophylaxis to prevent EBV infection in EBV-naïve transplantation recipients who revive organs/bone marrow from EBV-positive donors. These EBV-naïve transplantation recipients are at high risk for developing lymphoproliferative disease caused by EBV. Also, patients with certain genetic disorders (e.g. XLP1) are at high risk of developing fatal mononucleosis after primary infection with the virus. Currently, there are no such countermeasures that specifically target EBV.

These antibodies are also useful as therapeutics to treat infectious mononucleosis or to prevent reactivation of virus in persistently EBV infected individuals.

SEQUENCE LISTING

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Sequence total quantity: 293
SEQ ID NO: 1          moltype = AA  length = 121
FEATURE               Location/Qualifiers
source                1..121
                     mol_type = protein
                     organism = Macaca mulatta

SEQUENCE: 1
QVQLQESGPG LVKPAETLSL TCTVSGGSFS SYWWGWIRQS PGKGLEWIGH ISSGGNNYLN 60
PSLKSRLVTL SLDTSKNQFSL KLSVTAADS AVYYCARAPR IIVRGYFDQ WQGVLTVS 120
S 121

SEQ ID NO: 2          moltype = DNA  length = 363
FEATURE               Location/Qualifiers
source                1..363
                     mol_type = genomic DNA
                     organism = Macaca mulatta

SEQUENCE: 2
caggtgcagc tgcaggagtc gggcccagga ctggtgaagc ctgcggagac cctgtccctc 60
acctgcactg tctctggtgg ctctttcagc agttactggt ggggctggat ccgtcagtc 120
ccagggaagg gactggagtg gattgggcat atcagtagtg gtggaacaa ctacctaat 180
ccgtccctca agagtcgagt caccctgtca ctagacacgt ccaagaacca gttctccctg 240
aagctgaact ctgtgaccgc cgcggaactcg gccgtgtatt actgtgccag agccccccgt 300
attgttgta gaggccgata ctttgaccaa tggggccagg gagtccctgt caccgtctcc 360
tca 363

SEQ ID NO: 3          moltype = AA  length = 8

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-continued

FEATURE	Location/Qualifiers	
source	1..8	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 3		
GGSFSSYW		8
SEQ ID NO: 4	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 4		
ISSGGNN		7
SEQ ID NO: 5	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 5		
APAPRIVVRG RYFDQW		16
SEQ ID NO: 6	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
source	1..107	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 6		
DIQMTQSPSS LSASVGDVTI	ITCRASQGIN IYLNWFQQR	60
GFSGSGSGTD FTLTISSLQP	EDPATYYCLQ CESYPLTFGG	107
	GTKVEIK	
SEQ ID NO: 7	moltype = DNA length = 321	
FEATURE	Location/Qualifiers	
source	1..321	
	mol_type = genomic DNA	
	organism = Macaca mulatta	
SEQUENCE: 7		
gacatccaga tgacccagtc	tccatcctcc ctgtctgcat	60
atcacttgcc gggcaagtca	gggtatcaat atctacctga	120
gggaaagccc ctaaaactcct	gatctatgct gcgaccactt	180
agattcagcg gcagtggatc	tgggacagat ttcactctca	240
gaagatttcg caacttatta	ctgtctacag tgtgaaagtt	300
gggaccaagg tggagatcaa	a	321
SEQ ID NO: 8	moltype = AA length = 6	
FEATURE	Location/Qualifiers	
source	1..6	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 8		
QGIIY		6
SEQ ID NO: 9	moltype = length =	
SEQUENCE: 9		
000		
SEQ ID NO: 10	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 10		
LQCESYPLT		9
SEQ ID NO: 11	moltype = AA length = 123	
FEATURE	Location/Qualifiers	
source	1..123	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 11		
QVQLQESGPG LVKPSSETLSL	TCTVSGGSIS GAYYYWSWIR	60
YINPSLKSRA TISKDTPKNQ	FSLKLSSTVA ADTAVYYCAR	120
VSS	GRRLGYSNWF DVWGPGVLVT	123
SEQ ID NO: 12	moltype = DNA length = 369	
FEATURE	Location/Qualifiers	
source	1..369	
	mol_type = genomic DNA	

-continued

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                                organism = Macaca mulatta
SEQUENCE: 12
caggtgcaac tgcaggagtc gggcccagga ctggtgaagc cttcggagac cctgtccctc 60
acctgcactg tctctggtgg ctccatcagc ggtgcttact actactggag ctggattcga 120
cagcccccgg ggaagggact ggactggatt ggatatatct atggaagttt tgggagtgcc 180
tactacaacc cctccctcaa gagtcgagcc accatttcaa aagacacgcc caagaaccag 240
ttctccctga aactgagctc tgtgaccgcc gcggacacgg ccgtgtatta ctgtgcgaga 300
ggaaggcgac taggtattc gaactgggtc gatgtctggg gcccgaggag cctggtcacc 360
gtctcctca                                     369

SEQ ID NO: 13      moltype = AA  length = 10
FEATURE           Location/Qualifiers
source           1..10
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 13
GGSIGGAYYY                                     10

SEQ ID NO: 14      moltype = AA  length = 8
FEATURE           Location/Qualifiers
source           1..8
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 14
IYGSFGSA                                           8

SEQ ID NO: 15      moltype = AA  length = 15
FEATURE           Location/Qualifiers
source           1..15
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 15
ARGRRLGYSN WFDVW                                   15

SEQ ID NO: 16      moltype = AA  length = 107
FEATURE           Location/Qualifiers
source           1..107
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 16
DIQMTQSPSS LSASVGDKVT ITCRTSQDVS SYLAWYQQKP GKAPQLLIYA ASSLQSGVPS 60
RFTGSGSGAE FTLTISSLQP EDFASYICQQ YKNLPLTFGG GTKVEIK             107

SEQ ID NO: 17      moltype = DNA  length = 321
FEATURE           Location/Qualifiers
source           1..321
                 mol_type = genomic DNA
                 organism = Macaca mulatta

SEQUENCE: 17
gacatccaga tgaccagtc tccatcttcc ctgtctgcat ctgtaggaga caaagtcacc 60
atcacttgtc ggacaagtca ggacgttagc agttatttag cctggatca gcagaaacca 120
gggaaagccc ctgagctctc gatctatgct gcatccagtt tgcaaaagtgg ggtcccatca 180
aggttcaccg gcagtggatc tggggcagaa ttcactctca ccatcagcag ccttcagcct 240
gaagattttg catcatatta ctgtcaacag tataaaaatc tcccgctcac ttgcgcgga 300
gggaccaaag tggagatcaa a                                     321

SEQ ID NO: 18      moltype = AA  length = 6
FEATURE           Location/Qualifiers
source           1..6
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 18
QDVSSY                                           6

SEQ ID NO: 19      moltype =      length =
SEQUENCE: 19
000

SEQ ID NO: 20      moltype = AA  length = 9
FEATURE           Location/Qualifiers
source           1..9
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 20
QQYKNLPLT                                           9

SEQ ID NO: 21      moltype = AA  length = 129
FEATURE           Location/Qualifiers
source           1..129

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mol_type = protein
organism = Macaca mulatta

SEQUENCE: 21
QVQLQESGPG LVKPSETLSL TCDVSGGSFS GDFYWSWIRQ PPGKGLDWIG NIHSSAGTK 60
YKPSLKSRVT ISKDTSKNQF SLKLSSVTAA DTAVYYCTRG PLSRIVAGFG RGINWFDVWG 120
PGVLVTVSS 129

SEQ ID NO: 22      moltype = DNA length = 387
FEATURE           Location/Qualifiers
source            1..387
                  mol_type = genomic DNA
                  organism = Macaca mulatta

SEQUENCE: 22
cagggtgcagc tgcaggagtc gggcccagga ctggtgaagc cttcgagac cctgtccctc 60
acctgcgatg tctctggtgg ctctctcagc ggtgatttct actggagctg gatccgccag 120
ccccaggga agggactgga ctggattggg aatatccatg gcagcagtc gggcaccaaa 180
tacaagccct cctcaagag tcgagtcacc atttcaaaag acacgtccaa gaaccagttc 240
tccttgaaac tgagctctgt gaccgcccgc gacacggccg tctattactg tacgagaggc 300
ccccttagta ggatagtagc tggttttggg aggggggatta actggttcga tgtctggggc 360
cgggagtcct tggtcaccgt ctctca 387

SEQ ID NO: 23      moltype = AA length = 9
FEATURE           Location/Qualifiers
source            1..9
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 23
GGSFSGDFY 9

SEQ ID NO: 24      moltype = AA length = 8
FEATURE           Location/Qualifiers
source            1..8
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 24
IHSSAGT 8

SEQ ID NO: 25      moltype = AA length = 22
FEATURE           Location/Qualifiers
source            1..22
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 25
TRGPLSRIVA GFGRGINWFD VW 22

SEQ ID NO: 26      moltype = AA length = 116
FEATURE           Location/Qualifiers
source            1..116
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 26
QPVLTPPTSL SASPGASVRL SCTLSSGINV GSYSIFWYQQ KPGSPPRYLL FYFSDSSKHQ 60
GSGVPSRFSK SKDTSANAGL LLISGLQSED EADYYCAIWH SSASVLFGGG TRLTVL 116

SEQ ID NO: 27      moltype = DNA length = 348
FEATURE           Location/Qualifiers
source            1..348
                  mol_type = genomic DNA
                  organism = Macaca mulatta

SEQUENCE: 27
cagcctgtgc tgaccagacc aacctccctc tcagcatctc cgggagcatc agtcagactc 60
agctgcacct tgagcagtggt catcaatggt ggtagttaca gcatattctg gtaccagcag 120
aagccaggga gtctccccgc gtaccttctg ttctatttct cagactcaag taagcaccag 180
ggctctggag tccccagccg tttctctgga tccaaggata cttcagccaa tgcagggctt 240
ttactgatct ctgggctcca gtctgaagat gaggctgact attactgtgc catatggcac 300
agcagcgctt ctgtgttatt cggaggaggg acccggtcta cagtacta 348

SEQ ID NO: 28      moltype = AA length = 14
FEATURE           Location/Qualifiers
source            1..14
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 28
TLSSGINVGS YSIF 14

SEQ ID NO: 29      moltype = AA length = 7
FEATURE           Location/Qualifiers
source            1..7
                  mol_type = protein

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organism = Macaca mulatta

SEQUENCE: 29
YFSDSSK 7

SEQ ID NO: 30      moltype = AA  length = 10
FEATURE           Location/Qualifiers
source            1..10
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 30
AIWHSSASVL 10

SEQ ID NO: 31      moltype = AA  length = 120
FEATURE           Location/Qualifiers
source            1..120
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 31
QLQLQESGPG LVKPSETLSL TCAVSGGSIS GYYWSWIRQP PGKGPEWIGF IDGNTVGTNY 60
NPSLKSRLVTL SKDTSKNQFS LKVSSVTAAD TAVYYCARKP LRRYFWFDVW GPGVLTVSS 120

SEQ ID NO: 32      moltype = DNA  length = 360
FEATURE           Location/Qualifiers
source            1..360
                  mol_type = genomic DNA
                  organism = Macaca mulatta

SEQUENCE: 32
cagctgcagc tgcaggagtc gggcccagga ctggtgaagc cttcggagac cctgtccctc 60
acctgcgctg tctctgggtg ctccatcagc ggttactact ggagttggat tcgccagccc 120
ccagggaagg gaccggagtg gattgggttt attgatggtg atactgtggg caccactac 180
aaccctctcc tcaagagtcg agtcaccctt tcaaaagaca cgtccaagaa tcagttctcc 240
ctgaagtgta gttctgtgac cgccgcggac acggccgtgt attactgtgc gaggaagccg 300
ctacgccgtt attctgtggt cgatgtctgg ggcccgggag tcctgggtcac cgtctctca 360

SEQ ID NO: 33      moltype = AA  length = 8
FEATURE           Location/Qualifiers
source            1..8
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 33
GGSIGGY 8

SEQ ID NO: 34      moltype = AA  length = 8
FEATURE           Location/Qualifiers
source            1..8
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 34
IDGNTVGT 8

SEQ ID NO: 35      moltype = AA  length = 14
FEATURE           Location/Qualifiers
source            1..14
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 35
ARKPLRRYFW FDVW 14

SEQ ID NO: 36      moltype = AA  length = 111
FEATURE           Location/Qualifiers
source            1..111
                  mol_type = protein
                  organism = Macaca mulatta

SEQUENCE: 36
QSVLTQPPSV SGDPGQRTVI SCTGSSSNIG AGYYVYWYQQ FPGTAPKLLI YQDNKRPSGV 60
SDRFGSGKSG TSASLTITGL QPGDEADYYC SAWDSSLSAV MFGRGTRLTV L 111

SEQ ID NO: 37      moltype = DNA  length = 333
FEATURE           Location/Qualifiers
source            1..333
                  mol_type = genomic DNA
                  organism = Macaca mulatta

SEQUENCE: 37
cagctctgtg tgacgcagcc gccctcagtg tctggggacc cggggcagag ggtcaccatc 60
tcgtgcactg ggagcagctc caacatcggg gcgggttatt atgtatactg gtaccagcag 120
ttcccaggaa cgcccccaa actcctcatc tatcaagata ataagcgacc ctcagggtt 180
tctgaccgat tctctggctc caagtctggt acctcagcct ccctgaccat cactgggctc 240
cagcctgggg atgaggctga ttattactgc tcagcatggg atagcagcct gagtgtctgt 300
atgttcggaa gaggcaccag gctgacagta cta 333

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mol_type = genomic DNA
organism = Macaca mulatta

SEQUENCE: 47
caggtgcccc tgactcagcc tccctctgtg tctgggtctc ctggacagtc ggteaccatc 60
tctgcactg gaaccagcag tgacatcggg gggtataact atgtctcctg gtaccaacaa 120
caccagcga aagcccccaa agtcacgatt tatgaggta gtaagcgcc ctcaggggtc 180
tctgatcgt tctctggttc caaatctggc aacatagcct ccctgaccat ctcggggtc 240
caggctgagg acgaggctga ttattactgc agctcatatg caggcagcaa cactttctta 300
tctggaggag ggacccggct gacagtacta 330

SEQ ID NO: 48      moltype = AA  length = 9
FEATURE           Location/Qualifiers
source           1..9
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 48
SSDIGGYNY 9

SEQ ID NO: 49      moltype =  length =
SEQUENCE: 49
000

SEQ ID NO: 50      moltype = AA  length = 10
FEATURE           Location/Qualifiers
source           1..10
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 50
SSYAGSNTFL 10

SEQ ID NO: 51      moltype = AA  length = 123
FEATURE           Location/Qualifiers
source           1..123
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 51
QVQLQESGPG LVKPSETLSL TCAVSGGSIS SNYWSWIRQP PGKGLEWIGR IYGSGGSTDY 60
NPSLKSRTVI STDTSKNQFS LKVSSVTAAD TAVYYCARVR IQWVQLRGWF DVWGPGLVLT 120
VSS 123

SEQ ID NO: 52      moltype = DNA  length = 369
FEATURE           Location/Qualifiers
source           1..369
                 mol_type = genomic DNA
                 organism = Macaca mulatta

SEQUENCE: 52
caggtgcagc tgcaggagtc gggcccagga ctggtgaagc cttcgagac cctgtccctc 60
acctgcgctg tctctggtgg ctccatcagc agtaactact ggagctggat ccgccagccc 120
ccagggaagg gactggagtg gattggacgt atctatggta gtggtgggag caccgactac 180
aaccctccc tcaagagtcg agtcaccatt tcaacagaca cgtccaagaa ccagttctcc 240
ctgaaggtga gctctgtgac cgcgcgggac acggccgtgt attactgtgc gagagtgcgg 300
atacagtggg tacagtgtcg aggtctggtc gatgtctggg gcccgggagt cctggtcacc 360
gtctctca 369

SEQ ID NO: 53      moltype = AA  length = 10
FEATURE           Location/Qualifiers
source           1..10
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 53
GGSISSNYWS 10

SEQ ID NO: 54      moltype = AA  length = 8
FEATURE           Location/Qualifiers
source           1..8
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 54
IYGSGGST 8

SEQ ID NO: 55      moltype = AA  length = 17
FEATURE           Location/Qualifiers
source           1..17
                 mol_type = protein
                 organism = Macaca mulatta

SEQUENCE: 55
ARVRIQWVQL RGWFDVW 17

SEQ ID NO: 56      moltype = AA  length = 107

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FEATURE Location/Qualifiers
 source 1..107
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 56
 YIQMTQSPSS LSASVGDVT FTGPASQSFS SSLAWYQQKP GKAPNLLIYS ASSLQCGVRS 60
 RFGSKSGTD FTLTISSLQP EDIASYYCQQ YYSYPFTFGP GTKLDIK 107

SEQ ID NO: 57 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 source 1..321
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 57
 tacatacaga tgacgcagtc tccatcctcc ctgtctgcat ctgtaggaga cacagtcacc 60
 ttcactggcc ccgcaagtca gagcttagt agtagtttag cctggatatc gcagaaacca 120
 gggaaagccc ctaacctcct gatctatagt gcatccagtt tgcaatgtgg ggttcgttcg 180
 aggttcagtg gcagtaagtc tgggacagat ttcactctca ccatcagcag cctgcagcct 240
 gaagatattg ctagtattata ctgtcaacag tattacagtt atccattcac ttctggcccc 300
 gggaccaaac tggatatcaa a 321

SEQ ID NO: 58 moltype = AA length = 6
 FEATURE Location/Qualifiers
 source 1..6
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 58
 QSFSSS 6

SEQ ID NO: 59 moltype = length =
 SEQUENCE: 59
 000

SEQ ID NO: 60 moltype = AA length = 9
 FEATURE Location/Qualifiers
 source 1..9
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 60
 QQYYSYPFT 9

SEQ ID NO: 61 moltype = AA length = 119
 FEATURE Location/Qualifiers
 source 1..119
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 61
 QVQLQESGPG LVRPSETLSL TCAVSGGSIS SNYWSWIRQP PGKGLEWIGY ISGSTGSTYQ 60
 NPSLKSRVTV SKDTSKNQFS LKLNSTVTAAD TAVYYCARSG RRGSSLDLWG RGLVTVSS 119

SEQ ID NO: 62 moltype = DNA length = 357
 FEATURE Location/Qualifiers
 source 1..357
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 62
 caggtgcagc tgcaggagtc gggcccagga ctggtgaggg cttcggagac cctatccctc 60
 acctgcgctg tctctggtgg ctccatcagc agtaactact ggagctggat tcgccagccc 120
 ccagggaagg ggctggagtg gattgggtat atctctggta gtactgggag cacctaccag 180
 aaccctccc tcaagagtcg agtcaccgtt tcaaaagaca cgtctaagaa ccagttctcc 240
 ctgaagctga attctgtgac cgccgaggac acggccgtgt attactgtgc gagaagtggg 300
 agaagaggca gctcattgga tttgtggggc cggggagttc tggtcaccgt ctectca 357

SEQ ID NO: 63 moltype = AA length = 8
 FEATURE Location/Qualifiers
 source 1..8
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 63
 GGSISSNY 8

SEQ ID NO: 64 moltype = AA length = 8
 FEATURE Location/Qualifiers
 source 1..8
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 64
 ISGSTGST 8

SEQ ID NO: 65	moltype = AA length = 13		
FEATURE	Location/Qualifiers		
source	1..13		
	mol_type = protein		
	organism = Macaca mulatta		
SEQUENCE: 65			
ARSGRRGSSL DLW			13
SEQ ID NO: 66	moltype = AA length = 111		
FEATURE	Location/Qualifiers		
source	1..111		
	mol_type = protein		
	organism = Macaca mulatta		
SEQUENCE: 66			
QSVLTQPPSV SGPDPGRVTI	SCTGSSSNIG	AGYYVWYQQ FPGTAPKLLI	YQDNKRPSGV 60
SDRFSGSKSG TSASLTITGL	QPGDEADYYC	SAWDSSLSAV FFGGGTRLTV	L 111
SEQ ID NO: 67	moltype = DNA length = 333		
FEATURE	Location/Qualifiers		
source	1..333		
	mol_type = genomic DNA		
	organism = Macaca mulatta		
SEQUENCE: 67			
cagctctgtgc tgacgcagcg	gccctcagtg	tctgggggacc	cggggcgagag ggtcaccatc 60
tcgtgcactg ggagcagctc	caacatcggg	gcgggttatt	atgtatactg gtaccagcag 120
ttcccaggaa cggcccccaa	actcctcatc	tatcaagata	ataagcgacc ctcagggggt 180
tctgaccgat tctctggctc	caagctctgt	acctcagcct	ccctgaccat cactggggctc 240
cagcctgggg atgaggctga	ttattactgc	tcagcatggg	atagcagcct gagtgcctgtg 300
ttcttcggag gagggaccgc	gctgacagta	cta	333
SEQ ID NO: 68	moltype = AA length = 9		
FEATURE	Location/Qualifiers		
source	1..9		
	mol_type = protein		
	organism = Macaca mulatta		
SEQUENCE: 68			
SSNIGAGYY			9
SEQ ID NO: 69	moltype = length =		
SEQUENCE: 69			
000			
SEQ ID NO: 70	moltype = AA length = 11		
FEATURE	Location/Qualifiers		
source	1..11		
	mol_type = protein		
	organism = Macaca mulatta		
SEQUENCE: 70			
SAWDSSLSAV F			11
SEQ ID NO: 71	moltype = AA length = 121		
FEATURE	Location/Qualifiers		
source	1..121		
	mol_type = protein		
	organism = Macaca mulatta		
SEQUENCE: 71			
QLQESGGLV KPSETLSLTC	TVSGGSISDT	YRWSWIRQSP	GKGLEWIAIY YGTTTSTNYN 60
PSLKSRLTIS KDTSKNQFSL	NLRSVTAADT	AVYYCARGDS	GGRSAHVFHF WQQLRLTVS 120
S			121
SEQ ID NO: 72	moltype = DNA length = 363		
FEATURE	Location/Qualifiers		
source	1..363		
	mol_type = genomic DNA		
	organism = Macaca mulatta		
SEQUENCE: 72			
cagctgcagg agtcggggccc	aggactgggt	aagccttcgg	agaccctgtc cctcacctgc 60
actgtctctg gtggctccat	cagcgatact	taccgggtgga	gctggattcg ccagtcceca 120
gggaagggac tggagtgga	tgccctacatc	tatggtacta	ctacgagcac caactacaac 180
ccctccctca agagtcgact	caccatttca	aaagacacgt	ccaagaacca gttctccttg 240
gaactgaggt ctgtgaccgc	cgcggacacg	gccgtgtatt	actgtgcgag aggggatagc 300
ggtggccgggt cagcgcgatg	ttttcatttc	tggggccaag	ggctcagggt caccgtctct 360
tca			363
SEQ ID NO: 73	moltype = AA length = 9		
FEATURE	Location/Qualifiers		
source	1..9		
	mol_type = protein		
	organism = Macaca mulatta		

-continued

SEQUENCE: 73
GGISIDTYR 9

SEQ ID NO: 74 moltype = AA length = 8
FEATURE Location/Qualifiers
source 1..8
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 74
IYGTITST 8

SEQ ID NO: 75 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 75
ARGDSGGRSA HVPFHW 16

SEQ ID NO: 76 moltype = AA length = 111
FEATURE Location/Qualifiers
source 1..111
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 76
QSVLTQPPSV FGDPGQRITI SCTGSSSNIG AGYVYVYQQ PPGTAPKLLI YQDNKRPSGV 60
SDRFGSGSKG SSASLTITGL QPGDEADYYC SAWDSSLVR VFGGGTRLTV L 111

SEQ ID NO: 77 moltype = DNA length = 333
FEATURE Location/Qualifiers
source 1..333
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 77
cagagtgttc tgacgcagcc gccctcagtg ttggtggacc cggggcagag gatcaccatc 60
tcgtgcactg ggagcagctc caacatcggg gcgggttatt atgtatactg gtaccagcag 120
ttcccaggaa cggcccccaa actcctcatc tatcaagata ataagcgacc ctcaggggtt 180
tctgaccgat tttctggctc caagtctggt tcctcagcct ccctgaccat cactgggctc 240
cagcctgggg atgaggctga ttattactgc tcagcatggg atagcagcct gagtgtacgg 300
gttttcggag gagggaccgc gctgacagta cta 333

SEQ ID NO: 78 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 78
SSNIGAGYY 9

SEQ ID NO: 79 moltype = length =
SEQUENCE: 79
000

SEQ ID NO: 80 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 80
SAWDSSLVR V 11

SEQ ID NO: 81 moltype = AA length = 122
FEATURE Location/Qualifiers
source 1..122
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 81
EVQLVESGGG LVQPGGSLRL SCAASGFTFS SYGMNWRQA PGKGLEWVSF ITNTGKTTY 60
ADSVRGRFTI SRDNAKSVS LQMSLRAED TAVYYCTRGR GRHWSSGVF DFWGQGLRVT 120
VS 122

SEQ ID NO: 82 moltype = DNA length = 368
FEATURE Location/Qualifiers
source 1..368
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 82
gaggtgcaac tgggtggagtc tggaggaggc ttggtccagc ctggagggtc cctgagactc 60
tcctgtgcag cctctggatt caccttcagt agttacggca tgaactgggt ccgccaggct 120

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ccgggaaagg ggctggagtg ggtctcattc attactaaca ctggtaaaac cacatactac 180
gctgactctg tgagggggccg attcaccatc tccagagaca acgccaagaa gtcgggtgtct 240
ctacaaatga gtagcctgag agccgaggac acggccgtct attactgtac taggggaaga 300
ggtagacacg gctgggtccag tgggtgtttt gatttctggg gccaaaggtct cagggtcacc 360
gtctcttc 368

```

```

SEQ ID NO: 83      moltype = AA length = 8
FEATURE           Location/Qualifiers
source            1..8
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 83
GFTFSSYG 8

```

```

SEQ ID NO: 84      moltype = AA length = 8
FEATURE           Location/Qualifiers
source            1..8
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 84
ITNTGKTT 8

```

```

SEQ ID NO: 85      moltype = AA length = 16
FEATURE           Location/Qualifiers
source            1..16
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 85
TGRGRHWSS GVPDFW 16

```

```

SEQ ID NO: 86      moltype = AA length = 111
FEATURE           Location/Qualifiers
source            1..111
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 86
QSVLTQPPSV FGDPGQRITI SCTGSSSNIG AGYYVYWYQQ FPGTAPKLLI YQDNKRPSGV 60
SDRFSGSKFG SSASLTITGV QRGDEGDYYC SAWDSSLVSR VLGGGTRLTV L 111

```

```

SEQ ID NO: 87      moltype = DNA length = 333
FEATURE           Location/Qualifiers
source            1..333
                  mol_type = genomic DNA
                  organism = Macaca mulatta

```

```

SEQUENCE: 87
cagctctgttc tgacgcagcc gccctcagtg ttcggggacc cggggcagag gatcaccatt 60
tcgtgcactg ggagcagctc caacatcggg gcgggttatt atgtatactg gtaccagcag 120
ttcccaggaa cggcccccaa actcctcatc tatcaagata ataagcgacc ctcaggggtt 180
tctgaccgat tctctggctc caagtgttgg tcctcagcct ccctgaccat cactggggtc 240
cagcgtgggg atgaggtgga ttattactgc tcagcatggg atagcagcct gagtgtacgg 300
gttttgggag gaggggaccg gctgacagta cta 333

```

```

SEQ ID NO: 88      moltype = AA length = 9
FEATURE           Location/Qualifiers
source            1..9
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 88
SSNIGAGYY 9

```

```

SEQ ID NO: 89      moltype = length =
SEQUENCE: 89
000

```

```

SEQ ID NO: 90      moltype = AA length = 11
FEATURE           Location/Qualifiers
source            1..11
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 90
SAWDSSLVSR V 11

```

```

SEQ ID NO: 91      moltype = AA length = 121
FEATURE           Location/Qualifiers
source            1..121
                  mol_type = protein
                  organism = Macaca mulatta

```

```

SEQUENCE: 91
EVQLVESGGG LVQPGGSLRL SCVASGFTFS DRYIDWVRQA PGKGLEWVST ISTGSGDTAL 60

```


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YDSVVKGRFT ISRDNAKNTL YLQMNSLRAE DTAVYYCARH SGTFYTHFDY WGQGVLTVS 120
S 121

SEQ ID NO: 92 moltype = DNA length = 363
FEATURE Location/Qualifiers
source 1..363
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 92
gaagtgcagt tgggtggagtc tgggggaggc ttggtacagc cggggggggtc cctgagactc 60
tcctgtgtag cctctggatt caccttcagt gaccgctaca tagactgggt ccgccaggct 120
ccagggaagg gcctgggagtg ggtctcaact attagcactg gtagtggtga taccgcattg 180
tactcagact ctgtcaaggg ccgattcacc atctccagag acaacgcca gaacacactg 240
tatctgcaaa tgaacagcct gagagccgaa gacacggctg tctattactg tgcgagacat 300
agtgggtactt tttaacacca ctttgactac tggggccagg gagtcctggt caccgtctcc 360
tca 363

SEQ ID NO: 93 moltype = AA length = 8
FEATURE Location/Qualifiers
source 1..8
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 93
GFTFSDRY 8

SEQ ID NO: 94 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 94
ISTGSGDTA 9

SEQ ID NO: 95 moltype = AA length = 14
FEATURE Location/Qualifiers
source 1..14
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 95
ARHSGTFYTH FDYW 14

SEQ ID NO: 96 moltype = AA length = 107
FEATURE Location/Qualifiers
source 1..107
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 96
DIQMTQSPSS LSASVGDVTV FTCRASRSIS SWLAWYQKP GRAPKVLIIK ASSLQSGVPS 60
RFGSGSGGTD FTLTISSLQS EDFATYYCQQ YSSRPPTFGQ GTKVEIR 107

SEQ ID NO: 97 moltype = DNA length = 321
FEATURE Location/Qualifiers
source 1..321
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 97
gacatccaga tgacccagtc tccatcctcc ctgtctgcat ctgtaggaga cacagtcacc 60
ttcacctgcc gggcgagtcg gactattagc agctgggttag cctgggtatca gcagaaacca 120
gggagagccc ctaaagtccct gatctataag gcgtccagtt tgcaaaagtg gggtccttca 180
agggttcagcg gcagtggtatc tgggacagac ttcactctca ccatcagcag cctacagtct 240
gaagattttg caacatatta ttgtcaacag tatagtagtc gccctccgac gttcggccaa 300
gggaccaagg tggaaatcag a 321

SEQ ID NO: 98 moltype = AA length = 6
FEATURE Location/Qualifiers
source 1..6
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 98
RSISSW 6

SEQ ID NO: 99 moltype = length =
SEQUENCE: 99
000

SEQ ID NO: 100 moltype = AA length = 9
FEATURE Location/Qualifiers
source 1..9
 mol_type = protein

-continued

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organism = Macaca mulatta

SEQUENCE: 100
QQYSSRPPT 9

SEQ ID NO: 101      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 101
QQVLQESGPG LLKPSETLSL TCAVSGGSFS SFWWSWLRQP PEKGLEWIGE INGDSGSTNY 60
NPSLKSRVTI SKDASKNQFS LKLTSTVTAD TAVFYCARVR RILRSLDVWG RGLVTVSS 119

SEQ ID NO: 102      moltype = DNA  length = 357
FEATURE            Location/Qualifiers
source             1..357
                   mol_type = genomic DNA
                   organism = Macaca mulatta

SEQUENCE: 102
cagggtgcagc tgcaggagtc gggcccagga ctgctgaagc cttcggagac cctgtccctc 60
acctgcgctg tctctggttg ctccctcagt agtttctggt ggagctggct ccgccagccc 120
ccagaaaagg gactggagtg gattggggag atcaatggtg atagtgggag caccaactac 180
aaccctctcc tcaagagtcg agtcaccatt tcaaaagacg cgtccaagaa ccagttctcc 240
ctgaaactga cctctgtgac cgccgcggac acggccggtt ttactgtgc gagagttcgg 300
cgaattctga ggtcattgga tgtctggggc cggggagttc tggtcaccgt ctctca 357

SEQ ID NO: 103      moltype = AA  length = 8
FEATURE            Location/Qualifiers
source             1..8
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 103
GGSFSSFW 8

SEQ ID NO: 104      moltype = AA  length = 8
FEATURE            Location/Qualifiers
source             1..8
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 104
INGDSGST 8

SEQ ID NO: 105      moltype = AA  length = 13
FEATURE            Location/Qualifiers
source             1..13
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 105
ARVRRILRSL DVW 13

SEQ ID NO: 106      moltype = AA  length = 107
FEATURE            Location/Qualifiers
source             1..107
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 106
DIQMTQSPFS LFAFVGDRVT ITCQASQGIS HLLAWYQKP GKAPKLLIYS ASTLQSGVPS 60
RFGSGGFGTE FTLTISSLQP EDFATYYCQQ HNSYPRTFGQ GTKVEIK 107

SEQ ID NO: 107      moltype = DNA  length = 321
FEATURE            Location/Qualifiers
source             1..321
                   mol_type = genomic DNA
                   organism = Macaca mulatta

SEQUENCE: 107
gacatccaga tgacccagtc tcctttttcc ttgtttgcac ttgtaggaga cagagtcacc 60
atcacttgcc aagccagtcg ggggtattagc cacttgtagc cttggtagca gcagaaacca 120
gggaaagccc ctaagctcct tatattattc gcatccactt tgcaaagtgg ggtcccatca 180
agggtcagcg gcagtggatt tgggacggaa ttcactctca ccatcagcag cctgcagcct 240
gaagattttg caacttatta ctgtcaacag cataatagtt accctcggac gttcggccaa 300
gggaccaagg tggaaatcaa a 321

SEQ ID NO: 108      moltype = AA  length = 6
FEATURE            Location/Qualifiers
source             1..6
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 108
QGISHL 6

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SEQ ID NO: 109      moltype =   length =
SEQUENCE: 109
000

SEQ ID NO: 110      moltype = AA length = 9
FEATURE             Location/Qualifiers
source              1..9
                   mol_type = protein
                   organism = Macaca mulatta

SEQUENCE: 110
QQHNSYPRT

SEQ ID NO: 111      moltype = DNA length = 5845
FEATURE             Location/Qualifiers
source              1..5845
                   mol_type = genomic DNA
                   organism = Macaca mulatta

SEQUENCE: 111
tcgcgcgcttt cggtgatgac ggtgaaaacc tctgacacat gcagctcccc gagacgggtca 60
cagcttgtct gtaagcggat gccggggagca gacaagcccc tcagggcgcg tcagcgggtg 120
ttggcgggtg tcggggctgg cttaaactatg cggcatcaga gcagattgta ctgagagtgc 180
accatattgc gtgtgaaata ccgcacagat gcgtaaggag aaaataccgc atcagattgg 240
ctattggcca ttgcatacgt tgtatccata tcataatatg tacatttata ttggctcatg 300
tccaacatta ccgcoatggt gacattgatt attgactagt tattaatagt aatcaattac 360
ggggtcatta gttcatagcc catatatgga gttccgcgtt acataactta cggtaaatgg 420
ccgcctctgg tgaccgcccc acgacccccg cccattgacg tcaataatga cgtatgttcc 480
catagtaacg ccaataggga ctttccattg acgtcaatgg gtggagtatt tacggtaaac 540
tgcccacttg gcagttacatc aagtgtatca tatgccaagt acgcccccta ttgacgtcaa 600
tgacggtaaa tggcccgctt ggcattatgc ccagttacat accttatggg actttcctac 660
ttggcagtac atctacgtat tagtcatcgc tattaccatg gtgatgcggt tttggcagta 720
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cgtcaatggg agtttgtttt ggccacaaaa tcaacggggac ttccaaaaat gtcgtaacaa 840
ctccgcccca ttgacgcaaa tgggcggtag gcgtgtacgg tgggaggtct atatagcag 900
agctcgttta gtgaaccgtc agatcgctg gagacgcat ccacgtggtt ttgacctcca 960
tagaagacac cgggaccgat ccagcctcca tcggctcgca tctctccttc acgccccgcg 1020
cgccctacct gaggccgcca tccacgcccg ttgagtcgcg tctgcccgc tcccgctctg 1080
ggtgcctcct gaactgcgtc cgcgctctag gtaagttaa agctcaggtc gagacggggc 1140
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accctgcttg ctcaactcta gttaccggtg gagggcagtg tagtctgagc agtactcgtt 1260
gctgcgcgcg gcgccaccag acataatagc tgacagacta acagactggt cctttccatg 1320
ggtcttttct gcagtcaccc tctgcgacac gtgtgatcag atatcgcggc cgctctagac 1380
caccatggga ttgctatgta tcatcctttt tctagtagca actgcaaccg gtgtacattc 1440
ccaggtgcag tgggtgcagc tgcacgagtc gggcccagga ctggtgaagc cttcggagac 1500
cctgtccctc acctgcgctg tctctggttg ctctatcagc agtagctact ggagctggat 1560
ccgccaggcc ccagggaagg gatgggagt gattgggtat gtctatggta gtggtcgtga 1620
caccacgcag aacccctccc tcaagagtcg agtcacccgt tcagtagaca cgtccaagaa 1680
ccagctctcc ctgaagctga gatctgtgac cgcgcgggac acggccgctgt attactgtgc 1740
gagcagcggc tggcctcctg ggttggaacta ctggggccag gtagtcacgg tcaccgtctc 1800
ctcagctagc ccaaggggcc ctatgtgttt tctctgtggc cctagcagca gaagcacatc 1860
tgaatctaca gccgcctctg gctgcctggt gaaagattac tccccgagc cgttgaccgt 1920
gtcttggaaat agcggctctc tgaccagcgg cgtgcacaca ttccagctg tgcgtcagag 1980
cagcggcctg tatctctgta gcagcgtggt gacagtgcca agcagctctc tgggcaccca 2040
gacctacgtg tgcaacgtga accacaagcc cagcaacacc aaggtggaca agcgggtgga 2100
aatcaagacc ttggggcgcg gtaagcaagcc tctacctgt cctccttcta ccagccctga 2160
actgtggggc ggacctagcg gtgtctctgt ccccccaag cccaaggaca cctgatgat 2220
cagcagaacc ccggaagtga cctgcgtggt ggtggatgtg tcccaggaag atcccgacgt 2280
gaagttcaat tggtagctga acggcgccga ggtgcacat gccagacaa agcccagaga 2340
gacacagtac aacagcacct accgggtggt gtctgtgctg accgtgacac accaggactg 2400
gctgaacggc aaagagtaca catgcaaggt gtccaacaag gccctgcctg ccccatccca 2460
gaaaaccatc agcaaggaca agggccagcc cagagaacct caggtgtaca cactgcccc 2520
cagcagagag gaactgacca agaactcaggt gtccctgacc tgtctggtga aaggcttcta 2580
ccccagcgac atcgtgtgtg aatgggagtc tagcggacag cccgagaaca cctacaagac 2640
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caaccactac acccagaagt cctctagcct gagccccgga aagtgtgatg gaacacgtgg 2820
gatccagatc tgcgtgtgct tctagtgtcc agccatctgt tgtttgcccc tcccccgctg 2880
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catcgcatgt tctgagttag gtcatctcta ttctgggggg tgggggtgggg caggacagca 3000
agggggagga ttgggaagac aatagcaggc atgctgggga tgcgggtggg tctatgggta 3060
cccagggtct gaagaattga cccggttctt cctggggcag aaagaagcag gcacatcccc 3120
ttctctgtga cacacctgt ccacgcccct ggttcttagt tccagccccca ctcataggac 3180
actcatagct caggaggggt ccgccttcaa tccccccgcg taaagtactt ggagcgggtc 3240
ctccctccct catcagccca ccaaaccaaa cctagcctcc aagagtggga agaaattaaa 3300
gcaagatagg ctattaagtg cagaggggaga gaaaatgcct ccaacatgtg aggaagtaat 3360
gagagaaatc atagaatttt aaggccatga tttaaggcca tcatggcctt aatcttccgc 3420
ttcctcgctc actgactcgc tgcgtcgggt cgttcggctg cggcgagcgg tatcagctca 3480
tccaaggcgc gtaatacggg tatccacaga atcaggggat aacgcaggaa agaacatgtg 3540
agcaaaaggc cagcaaaagg ccaggaacccg taaaaggccc gcgtgtgctg cgtttttcca 3600

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taggtccgcg cccctcgacg agcatcacaa aaatcgacgc tcaagtcaga ggtggcgaaa 3660
cccgacagga ctataaagat accaggcggt tccccctgga agctccctcg tgcgtctccc 3720
tgttccgacc ctgcccgtta ccggatacct gtcccgccttt ctcccctcgg gaagcgtggc 3780
gctttctcat agctcacgct gtaggtatct cagttcgggtg taggtcggtc gctccaagct 3840
gggctgtgtg caccgaacccc ccgttcagcc cgaccgctgc gccttatccg gtaactatcg 3900
tcttgagtcg aaccocggtaa gacacgactt atcgccactg gcagcagcca ctggtaacag 3960
gattagcaga gcgaggtatg tagggcggtgc tacagagttc ttgaagtggt ggcctaacta 4020
cggctacact agaagaacag tatttgggat ctgcgctctg ctgaagccag ttaccttcgg 4080
aaaaagagtt ggtagctctt gatccggcaa acaaacccac gctggtagcg gtgggttttt 4140
tgtttgcaag cagcagatta cgcgcagaaa aaaaggatct caagaagatc cttgatctt 4200
ttctacgggg tctgacgctc agtggaaacga aaactcacgt taagggattt tggcatgag 4260
attatcaaaa aggatcttca cctagatcct ttaaatataa aaatgaagtt ttaatacaat 4320
ctaaagtata tatgagtaaa cttggtctga cagttaccaa tgcttaatca gtgaggcacc 4380
tatctcagcg catcgctctat ttctgtcatc catagttgcc tgactcgggg gggggggggc 4440
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gggtattttg aacttttggc ttgccacgga acggtctgcg ttgtcgggaa gatgcgtgat 4620
ctgatccttc aactcagcaa aagttcgatt tattcaacaa agccgcgcgc ccgtcaagtc 4680
agcgtaatgc tttgccagtg ttacaaccaa ttaaccaatt ctgattagaa aaactcatcg 4740
agcatcaaat gaaactgcaa tttattcata tcaggattat caataccata tttttgaaaa 4800
agccgtttct gtaatgaagg agaaaaactca ccgaggcagt tccataggat ggcaagatcc 4860
tggtatcggt tcgcgattcc gactcgtcca acatcaatac aacctattaa ttccctcgc 4920
tcaaaaataa ggttatcaag tgagaaatca ccatgagtga cgactgaatc cggtgagaat 4980
ggcaaaagct tatgcatctt ttccagact gtgtcaacag gccagccatt acgctcgtca 5040
tcaaaatcac tcgcatcaac caaacggtta ttcattcgtg attgcgcctg agcgaagaca 5100
aatacgcgat cgtgttataa aggacaatta caaacaggaa tcgaatgcaa ccggcgccag 5160
aacactgccg gcgcatcaac aatattttca cctgaatcag gatattcttc taatacctgg 5220
aatgctgttt tcccggggat cgcagtggtg agtaaccatg catcatcagg agtacggata 5280
aaatgcttga tggctcggaag aggcataaat tccgacagcc agtttagtct gacctatcca 5340
tctgtaacat cattggcaac gctacctttg ccatgtttca gaaacaactc tggcgcatcg 5400
ggcttcccat acaatcgata gattgtcgca cctgattgcc cgacattatc gcgagcccat 5460
ttatacccat ataaatcagc atccatggtg gaatttaatc gcggcctcga gcaagacggt 5520
tcccgttgaa tatggctcat aacacccctt gtattactgt ttatgtaagc agacagtttt 5580
attgttcatt atgatattat tttatcttgt gcaatgtaac atcagagatt ttgagacaca 5640
acgtggcttt cccccccccc ccattattga agcattttatc aggggttattg tctcatgagc 5700
ggatacatat ttgaatgtat ttgaaaaaat aaacaaatag gggttccgcg cacatttccc 5760
cgaaaagtgc cactgacgtc ctaagaaacc attattatca tgacattaac ctataaaaat 5820
agcggtatca cgaggccctt tcgtc 5845

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SEQ ID NO: 112      moltype = AA length = 473
FEATURE
source             Location/Qualifiers
                    1..473
                    mol_type = protein
                    organism = Macaca mulatta

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SEQUENCE: 112
MGWSCILFLF VATAATGVHSQ VQLVQLHESG PGLVKPSETL SLTCAVSGGS ISSSYWSWIR 60
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SGWPPGLDYL QGGVTVTVSS ASTKGPSVFP LAPSSRSTSE STAALGCLVK DYFPEPVTVS 180
WNSGSLTSGV HTFFPAVLQSS GLYSLSSVVT VPSSSLGTQT YVCNVNHKPS NTKVKDKRVEI 240
KTCGGGSKPP TCPPCTSPEL LGGPSVFLFP PKPKDTLMIS RTPEVTCVVV DVSQEDPDVK 300
FNWYVNGAEV HHAQTKPRET QYNSTYRVVS VLTVTQDWL NGKEYTCKVS NKALPAPIQK 360
TTSKDKGQPR EPQVYTLPPS REELTKNQVS LTCLVKGFYP SDIVVEWESS GQPENYTKTT 420
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SEQ ID NO: 113      moltype = DNA length = 5100
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source             Location/Qualifiers
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                    mol_type = genomic DNA
                    organism = Macaca mulatta

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SEQUENCE: 113
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SEQ ID NO: 114      moltype = AA length = 229
FEATURE            Location/Qualifiers
source             1..229
                   mol_type = protein
                   organism = Macaca mulatta

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PRLVIYSGST RAAGIPDRFS GSRWGPDYNL TISNLESGDF GVIYCCQYEF FGQGTKVQVD 120
IKRTVAAPSV FIFPPSEQV KSGTVSVVCL LNNFYPREAS VKWKVDGALK TGNSQESVTE 180
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SEQ ID NO: 115      moltype = AA length = 235

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FEATURE Location/Qualifiers
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 organism = Macaca mulatta

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 GTGKTVTVLG QPKAAPSVTL FPPSSEELQA NKATLVCLIS DFYPGAVEVA WKADGSAVNA 180
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SEQ ID NO: 116 moltype = DNA length = 5842
 FEATURE Location/Qualifiers
 source 1..5842
 mol_type = genomic DNA
 organism = Macaca mulatta

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SEQ ID NO: 117      moltype = AA length = 472
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                   mol_type = protein
                   organism = Macaca mulatta

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NSGSLTSGVH TFPVAVLQSSG LYSLSVVTV PSSSLGTQTY VCNVNHKPSN TKVDRKVEIK 240
TCGGGSKPPT CPPCTSPELL GGPVSFLFPP KPKDTLMISR TPEVTCVVVD VSQEDPDVKF 300
NWYVNGAEVH HAQTKPRETQ YNSTYRVVSV LTVTHQDWLN GKEYTCKVSN KALPAPIQKT 360
ISKDKGQPRE PQVYTLPPSR EELTKNQVSL TCLVKGFYPS DIVVEWESSG QPENTYKTTT 420
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SEQ ID NO: 118      moltype = DNA length = 5118
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source             Location/Qualifiers
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                   organism = Macaca mulatta

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mol_type = other DNA
organism = Synthetic construct

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ccaggctgcc ctgactcagc ctccctctgt gtctgggtct cctggacagt cggtaaccat 1500
ctcctgcact ggaaccagca gtgacatcgg tgggtataac tatgtctcct ggtaccaaca 1560
acaccaggcg aaagccccc aagtcatgat ttatgaggtc agtaagcgcg cctcaggggt 1620
ctctgatcgc ttctctggtt ccaaatcttg caacatagcc tccctgacca tctctgggct 1680
ccaggctgag gacgaggtg attattactg cagctcatat gcaggcagca acactttctt 1740
attcggaggga gggaccggcg tgacagtcct aggtcagccc aaggtgcgcc cctcggctac 1800
tctgttcccg cctcctctct agggagttca agccaacaag gccacactgt tgtgtctcat 1860
aagtgacttc taccggggag ccgtgacagt ggccctggaag gcagatagca gcccgcgtca 1920
ggcgggagtg gagaccacca caccctccaa acaaagcaac acaagtagc cggccagcag 1980
ctatctgagc ctgacgcctg agcagtgga gttccacaga agctacagct gccaggtcac 2040
gcatgaaggg agcacctgg agaagacagt ggcccttaca gaatgttcat gaggatccag 2100

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atctgctgtg ccttctagtt gccagccatc tgttgtttgc cctccccccg tgccttccct 2160
gaccttgaa ggtgocactc ccactgtcct ttcctaataa aatgaggaaa ttgcctcgca 2220
ttgtctgagt aggtgtcatt ctattctggg ggttggggtg gggcaggaca gcaaggggga 2280
ggattgggaa gacaatagca ggcattgctg ggatgcggtg ggctctatgg gtaccaggt 2340
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gctcaggagg gctccgcctt caatcccacc cgctaaagta cttggagcgg tctctccctc 2520
cctcatcagc ccaccaaacc aaacctagcc tccaagagtg ggaagaaatt aaagcaagat 2580
aggctattaa gtgcagaggg agagaaaatg cctccaacat gtgaggaaat aatgagagaa 2640
atcatagaat tttaaggcca tgatttaagg ccactcatggc cttaatcttc cgcttccctg 2700
ctcactgact cgctgctctc ggtcgttcgg ctgcggcgag cggtatcagc tcaactcaag 2760
gcggttaaac ggttatccac agaatacagg gataacgcag gaaagaaacat gtgagcaaaa 2820
ggccagcaaa aggccaggaa ccgtaaaaag gccgcgttgc tggcggtttt ccataggctc 2880
cgccccctg acgagcatca caaaaatcga cgctcaagtc agagggtggc aaaccgcaca 2940
ggactataaa gataccaggc gtttccccct ggaagctccc tcgtgcgctc tctgttccg 3000
accctgcccg ttaccggata cctgtccgcc tttctccctt cgggaagcgt ggcgctttct 3060
catagctcac gctgtaggta tctcagttcg gtgtaggctg ttcgctccaa gctgggctgt 3120
gtgcacgaac ccccgttcca gcccgaccgc tgcgccttat ccggttaacta tcgtcttgag 3180
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gttggtagct tttgatccgg caaacaacc accgctggta gcggtgggtt tttgtttgc 3420
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gggtctgacg ctcagtgga cgaaaactca cgtaaaggga ttttggtcat gagattatca 3540
aaaaggatct tcacctagat ccttttaaat taaaatgaa gttttaaatc aatctaaagt 3600
atatatgagt aaactttggtc tgacagttac caatgcttaa tcagtgggc accatctca 3660
gcgactctgc tattctgttc atccatagtt gcctgactcg gggggggggg gcgctgaggt 3720
ctgcctcgtg aagaaggtgt tgctgactca taccaggcct gaatgcgcc atcatccagc 3780
cagaaagtga gggagccacg gttgatgaga gctttgtgt aggtggacca gttggtgatt 3840
tgaaactttt gctttgccac ggaacggctc gcgtgtcgg gaagatgcgt gatctgatcc 3900
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tgctctgcca gtgttacaac caattaacca attctgatta gaaaaactca tcgagcatca 4020
aatgaaactg caattttatt atatcaggat tatcaatacc atatttttga aaaagccgtt 4080
tctgtaatga aggagaaaac tcaccgaggc agttccatag gatggcaaga tccgtgtatc 4140
ggtctgcgat tccgactcgt ccaacatcaa tacaacctat taatttcccc tcgtcaaaaa 4200
taaggttatc aagtggagaa tcaccatgag tgacagctga atccggtgag aatggcaaaa 4260
gcttatgcat ttctttccag actgtttcaa caggccagcc attacgctcg tcatcaaat 4320
cactcgcac aaccaaaccg ttattcattc gtgattgcgc ctgagcgaga cgaataacgc 4380
gatcgctggt aaaaggacaa ttacaacag gaatcgaatg caaccggcgc aggaacactg 4440
ccagcgcac aacaatattt tcacctgaat caggatattc ttctaatacc tggaatgctg 4500
ttttccggg gatcgagtg gtgagtaacc atgcatcac aggagtacgg ataaaatgct 4560
tgatggctcg aagaggcata aatccgtca gccagtttag tctgaccatc tcatctgtaa 4620
catcattggc aacgatattt ttgccatgtt tcagaaacaa ctctggcgca tcgggcttcc 4680
catacaatgc atagattgtc gcaacctgatt gcccgacatt atcgcgagcc catttatacc 4740
catataaatc agcatccatg ttggaattta atcgcggcct cgagcaagac gtttcccggt 4800
gaatatggct cataaacacc cttgattacc tgtttatgta agcagacagt tttattgttc 4860
atgatgatat atttttatct tgtgcaatgt aacatcagag attttgagac acaacgtggc 4920
tttccccccc ccccatattt tgaagcattt atcagggtta ttgtotcatg agcgataca 4980
tatttgaatg tatttagaaa ataaacaaa taggggttcc gcgcacattt ccccgaaaag 5040
tgccacctga cgtctaagaa accattatta tcatgacatt aacctataaa aataggcgta 5100
tcacgaggcc ctttcgtc 5118

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SEQ ID NO: 123      moltype = AA length = 235
FEATURE
source             Location/Qualifiers
                   1..235
                   mol_type = protein
                   organism = Synthetic construct

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SEQUENCE: 123
MGWSCIIILFL VATATGVHSQ AALTQPPSVS GSPGQSVTIS CTGTSSDIGG YNVVSWYQQH 60
PGKAPKVMIIY EVSKRPSGVS DRFSGSKSGN IASLTISGLQ AEDEADYYCS SYAGSNTFLF 120
GGGTRLTLVLG QPKAAPSVTL FPPSSEELQA NKATLVCLIS DFYPGAVTVA WKADSSPVKA 180
GVETTPPSKQ SNNKYAASSY LSLTPEQWKS HRSYSCQVTH EGSTVEKTVA PTECS 235

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SEQ ID NO: 124      moltype = AA length = 907
FEATURE
source             Location/Qualifiers
                   1..907
                   mol_type = protein
                   organism = Human gammaherpesvirus 4

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SEQUENCE: 124
MEAALLVCQY TIQSLIHLTG EDPGFFNVEI PEPFFPYPTCN VCTADVNVTI NFDVGGKKHQ 60
LDLDFGQLTP HTKAVYQPRG AFGGSEENATN LFLLELLGAG ELALTMRSKK LPINVTGEE 120
QQVLSLESVDV YFQDVFGTMW CHHAEMQNPV YLIPETVPYI KWDNCNSTNI TAVVRAQGLD 180
VTLPLSLPTS AQDSNFSVKT EMLGNEIDIE CIMEDGEISQ VLPDGNKFNI TCSGYESHVP 240
SGGILTSTSP VATPIPGTGY AYSLRLTPRP VSRFLGNNSI LYVPYSGNGP KASGGDYCIQ 300
SNIVFSDEIP ASQDMPTNTT DITYVGDNAT YSVPMTSED ANSPNVTVTA FWAHPNNTET 360
DFKCKWTLTS GTPSGCENIS GAFASNRTFD ITVSGLGTA KTLII TRTAT NATTTTHKVI 420
FSKAPESTTT SPTSNTTGFA APNTTTLGLPS STHVPTNLTA PASTGPTVST ADVTSPTPAG 480
TTSASPVPPT SPSPRDNGTE SKAPDMTSPT SAVTTPTPNA TSPTS AVTTP TPNATSPTPA 540
VTTPTPNATS PTLGKTSPTS AVTTPTPNAT SPTLGKTSPT SAVTTPTPNA TSPTLGKTSPT 600

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TSAVTTPTPN	ATSPTVGETS	PQANTTNHTL	GGTSSTPVVT	SPPKNATSAV	TTGQHNTSS	660
STSSMSLRPS	SISETLSPST	SDNSTSHMPL	L TSAHPTGGE	NITQVTPAST	STHHVSTSSP	720
APRPGTTSQA	SGPGNSSTST	KPGEVNVTKG	TPPKNATSPQ	APSGQKTAVP	TVTSTGGKAN	780
STTGKHTTG	HGARTSTEPT	TDYGGDSTTP	RTRYNATTYL	PPSTSSELRP	RWTFTSPPV	840
TAQATVPVPP	TSQPRFSNLS	MLVLQWASLA	VLTL LLLLV	ADCAFRNLS	TSHTYTPPY	900
DDAETVY						907

SEQ ID NO: 125 moltype = AA length = 426
 FEATURE Location/Qualifiers
 source 1..426
 mol_type = protein
 organism = Synthetic construct

SEQUENCE: 125

EAALLVCQYT	IQSLIHLTGE	DPGFFNVEIP	EFPFYPTCNV	CTADVNVITN	FDVGGKKHQL	60
DLDGGLTPH	TKAVYQPRGA	FGGSENATNL	FLLELLGAGE	LALTMRSKKL	PINVTTGEEQ	120
QVSLESVDVY	PQDVFGTMWC	HHAEMQNPVY	LIPETVPYIK	WDNCNSTNIT	AVVRAQGLDV	180
TLPLSLPTSA	QDSNFSVKTE	MLGNEIDIEC	IMEDGEISQV	LPGDNKFNIT	CSGYESHVPS	240
GGILTSTSPV	ATPIPGTGYA	YSLRLTPRPV	SRFLGNNSIL	YVFYSGNGPK	ASGGDYCIQS	300
NIVFSDEIPA	SQDMPTNTTD	ITYVGDNATY	SVPMVTSEDA	NSPNVTVTAF	WAWPNNTETD	360
FKCKWLTLSG	TPSGCENISG	AFASNRTFDI	TVSGLGTAPK	TLIITRTATN	ATTTTHKVIF	420
SKAPGS						426

SEQ ID NO: 126 moltype = DNA length = 1281
 FEATURE Location/Qualifiers
 source 1..1281
 mol_type = other DNA
 organism = Synthetic construct

SEQUENCE: 126

gaagctgccc	tgctcgtgtg	ccagtagacc	atccagagcc	tgatccacct	gaccggcgag	60
gaccccggtc	tcttcaacgt	ggaaatcccc	gagttccctc	tctaccctac	ctgcaacgtg	120
tgacccgccc	acgtgaacgt	gacccatcaac	tgcagcgtgg	gcggcaagaa	gcaccagctg	180
gacctggatt	tcggccagct	gacccctcac	accaaaggccg	tgatcacgcc	cagaggcgcc	240
tttgccggca	gcgagaacgc	caccaatctg	tttctgctgg	aactcctagg	cgccggcgag	300
ctggccctga	ccatgagaag	caagaaactg	cccatcaatg	tgaccacagg	cgaggaaacg	360
caggtgtccc	tggaaagcgt	ggacgtgtac	tttcaagacg	tgctcgccac	catgtggtgc	420
caccacgccc	agatgcagaa	ccccgtgtac	ctgatccccg	agacagtgcc	ctacatcaag	480
tgggacaact	gcaacagcac	caacatcacc	gccgtcgtgc	gggcccaggg	actggatgtg	540
acactgcctc	tgagcctgcc	taccagcgcc	caggacagca	acttcagcgt	gaaaaccgag	600
atgctgggca	acgagatcga	catcgagtgc	atcatggaga	atggcgagat	cagccagggtg	660
ctgcccggcg	acaacaagtt	caacatcaca	tgacgcggct	acgagagcca	cgtgccatct	720
ggcgccatcc	tgaccagcac	aagcccagtg	gccacaccca	tccctggcac	aggctacgcc	780
tacagcctga	gactgacccc	cagaccctgt	tccagattcc	tgggcaacaa	cagcatcctg	840
tacgtgttct	acagcggcaa	cgcccccagg	gcctctggcg	gcgattactg	tatccagagc	900
aacatcgtgt	tcagcgacga	gatccccgcc	agccaggaca	tgcccaccaa	taccaccgac	960
atcacgtacg	tgggcgacaa	tgccacctac	agcgtgccaa	tggtcacctc	cgaggacgcc	1020
aaacgcccc	acgtgaccgt	gacagccttt	tgggcctggc	ctaacaacac	cgagacagac	1080
ttcaagtgc	agtggaacct	gacctccggc	acccctagcg	gctgcgagaa	tatcagcgga	1140
gccttcgcca	gcaaccggac	cttcgatatc	accgtgtctg	gcctgggcac	cgcccccaag	1200
accctgatca	tcaccagAAC	gccacaaat	gccaccacca	caaccacaa	agtgatcttc	1260
agcaaggccc	ccggtctctg	c				1281

SEQ ID NO: 127 moltype = AA length = 426
 FEATURE Location/Qualifiers
 source 1..426
 mol_type = protein
 organism = Synthetic construct

SEQUENCE: 127

EAALLVCQYT	IQSLIHLTGE	DPGFFNVEIP	EFPFYPTCNV	CTADVNVITN	FDVGGKKHQL	60
DLDGGLTPH	TKAVYQPRGA	FGGSENATNL	FLLELLGAGE	LALTMRSKKL	PINVTTGEEQ	120
NVSLESVDVY	PQDVFGTMWC	HHAEMQNPVY	LIPETVPYIK	WDNCNSTNIT	AVVRAQGLDV	180
TLPLSLPTSA	QDSNFSVKTE	MLGNEIDIEC	IMEDGEISQV	LPGDNKFNIT	CSGYESHVPS	240
GGILTSTSPV	ATPIPGTGYA	YSLRLTPRPV	SRFLGNNSIL	YVFYSGNGPK	ASGGDYCIQS	300
NIVFSDEIPA	SQDMPTNTTD	ITYVGDNATY	SVPMVTSEDA	NSPNVTVTAF	WAWPNNTETD	360
FKCKWLTLSG	TPSGCENISG	AFASNRTFDI	TVSGLGTAPK	TLIITRTATN	ATTTTHKVIF	420
SKAPGS						426

SEQ ID NO: 128 moltype = DNA length = 1278
 FEATURE Location/Qualifiers
 source 1..1278
 mol_type = other DNA
 organism = Synthetic construct

SEQUENCE: 128

gaagctgccc	tgctcgtgtg	ccagtagacc	atccagagcc	tgatccacct	gaccggcgag	60
gaccccggtc	tcttcaacgt	ggaaatcccc	gagttccctc	tctaccctac	ctgcaacgtg	120
tgacccgccc	acgtgaacgt	gacccatcaac	tgcagcgtgg	gcggcaagaa	gcaccagctg	180
gacctggatt	tcggccagct	gacccctcac	accaaaggccg	tgatcacgcc	cagaggcgcc	240
tttgccggca	gcgagaacgc	caccaatctg	tttctgctgg	aactcctagg	cgccggcgag	300
ctggccctga	ccatgagaag	caagaaactg	cccatcaatg	tgaccacagg	cgaggaaacg	360
aacgtgtccc	tggaaagcgt	ggacgtgtac	tttcaagacg	tgctcgccac	catgtggtgc	420

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caccacgccc agatgcagaa ccccggtgtac ctgatccccg agacagtgcc ctacatcaag 480
tgggacaaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcacgtg gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccaggtg 660
ctgcccgggc acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcgccatcc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtacgcc 780
tacagcctga gactgacccc cagaccctg tccagattcc tgggcaacaa cagcctctg 840
tacgtgttct acagcgccaa cggccccaag gcctctggcg gcgattactg tatccagagc 900
aacatcggtg tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tggggcgacaa tggcacctac agcgtgccaa tggtcacctc cgaggacgcc 1020
aacagcccca acgtgacgtg gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtgcg agtgagacct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac ctctgatata accgtgtctg gcctgggcac cgcccccaag 1200
accctgatca tcaccagaac cgccacaat gccaccacca caaccacaaa agtgatcttc 1260
agcaaggccc ccggtctc 1278

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SEQ ID NO: 129      moltype = AA  length = 427
FEATURE            Location/Qualifiers
source              1..427
                    mol_type = protein
                    organism = Synthetic construct

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SEQUENCE: 129
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY PQDVFGTMWC HHAEMQNPVY LIPETVPYIK WNNCSNSTNI TAVVRAQGLD 180
VTLPLSLPTS AQDSNFSVKTE EMLGNEIDIE CIMEDGEISQ VLPDGNKFNI TCSGYESHVP 240
SGGILTSTSP VATPIPGTGY AYSRLRLTPRP VSRFLGNNSI LYVFSYSGNGP KASGGDYCIQ 300
SNIVFSDEIP ASQDMPNTNT DITYVGDNAT YSVPMTSED ANSPNVTVTA FWAWPNNTET 360
DPKCKWTLTS GTPSGCENIS GAFASNRTFD ITVSGLGATP KTLIITRTAT NATTTTHKVI 420
FSKAPGS 427

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SEQ ID NO: 130      moltype = DNA  length = 1281
FEATURE            Location/Qualifiers
source              1..1281
                    mol_type = other DNA
                    organism = Synthetic construct

```

```

SEQUENCE: 130
gaagctgccc tgctcgtgtg ccagtacacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtc tcttcaacgt ggaatcccc gagttcccc tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtatcagcc cagaggcgcc 240
tttgccggca ccgagaacgc caccaatctg tttctgctgg aactcctagg cgcggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacg 360
caggtgtccc tggaaagcgt ggaagtgatc tttcaagacg tgttcggcac catgtgggtg 420
caccacgccc agatgcagaa ccccggtgtac ctgatccccg agacagtgcc ctacatcaag 480
tggaacaaca cctgcaacag caccacatc accgcccgtc tgccggccca gggactggat 540
gtgacactgc ccttgagcct gcctaccagc gccaggaca gcaacttcag cgtgaaaacc 600
gagatgctgg gcaacgagat cgacatcgag tgcacatgag aagatggcga gatcagccag 660
gtgctgcccc gcgacaacaa gtccaacatc acatgcagcg gctacgagag ccacgtgcc 720
tctggcgcca tcttgaccag cacaagccca gtggccacac ccattccctg caccagctac 780
gcctacagcc tgagactgac ccccgagccc gtgtccagat tccctgggcaa caacagcatc 840
ctgtacgtgt tctacagcgg caacggcccc aaggcctctg gcggcgatta ctgtatccag 900
agcaacatcg tgttcagcga cgagatcccc gccagccagg acatgcccac caataccacc 960
gacatcagct acgtggcgga caatgccacc tacagcgtgc caatggtcac ctccaggagc 1020
gccaacagcc ccaacgtgac cgtgacagcc ttttgggcct ggcctaacaa caccgagaca 1080
gacttcaagt gcaagtggac cctgacctcc ggcacccccta gcggctgcga gaatacagc 1140
ggagccttcg ccagcaaccg gaccttcgat atcaccgtgt ctggcctggg caccgcccc 1200
aagaccctga tcatcaccag aaccgccaca aatgccacca ccacaaccca caaagtgtatc 1260
ttcagcaagg ccccggtctc 1281

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SEQ ID NO: 131      moltype = AA  length = 426
FEATURE            Location/Qualifiers
source              1..426
                    mol_type = protein
                    organism = Synthetic construct

```

```

SEQUENCE: 131
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY PQDVFGTMWC HHAEMQNPVY LIPETVPYIK WNNCSNSTNI TAVVRAQGLD 180
VTLPLSLPTS AQDSNFSVKTE EMLGNEIDIE CIMEDGEISQ VLPDGNKFNI TCSGYESHVP 240
SGGILTSTSP VATPIPGTGY AYSRLRLTPRP VSRFLGNNSI LYVFSYSGNGP KASGGDYCIQ 300
SNIVFSDEIP ASQDMPNTNT DITYVGDNAT YSVPMTSED ANSPNVTVTA FWAWPNNTET 360
DPKCKWTLTS GTPSGCENIS GAFASNRTFD ITVSGLGATP KTLIITRTAT NATTTTHKVI 420
FSKAPGS 426

```

```

SEQ ID NO: 132      moltype = DNA  length = 1278
FEATURE            Location/Qualifiers
source              1..1278
                    mol_type = other DNA

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organism = Synthetic construct
SEQUENCE: 132
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtagcagcc cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtgcc ctacatcaag 480
tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccaggtg 660
ctgcccgggc acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcggcatcc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtctacgc 780
tacagcctga gactgacccc cagaccctgt tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cgcccccaag gcctctggcg gccggtactg tatccagagc 900
aacatcgtgt tcagcgcagc gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tgggcgacaa tgcccacctac agcgtgccaa tggtcacctc cgaggacgcc 1020
aacagcccca acgtgacgct gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtgca agtggaccct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac cttcgatatc accgtgtctg gcctgggcac cgcccccaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaaa agtgatcttc 1260
agcaaggccc ccggtctt 1278

```

```

SEQ ID NO: 133      moltype = AA  length = 426
FEATURE             Location/Qualifiers
source              1..426
                    mol_type = protein
                    organism = Synthetic construct

```

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SEQUENCE: 133
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITIN FVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY PQDVFGMTWC HHAEMQNPVY LIPETVPYIK AAACNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSTSPV ATP1PGTGYA YSLRLTPRPV SRLFNNNSIL YVFYSGNGPK ASGGDYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITVVDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRTFDI TVSGLGTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGS 426

```

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SEQ ID NO: 134      moltype = DNA  length = 1278
FEATURE             Location/Qualifiers
source              1..1278
                    mol_type = other DNA
                    organism = Synthetic construct

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SEQUENCE: 134
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtagcagcc cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtgcc ctacatcaag 480
gccgccgctc gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccaggtg 660
ctgcccgggc acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcggcatcc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtctacgc 780
tacagcctga gactgacccc cagaccctgt tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cgcccccaag gcctctggcg gcgattactg tatccagagc 900
aacatcgtgt tcagcgcagc gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tgggcgacaa tgcccacctac agcgtgccaa tggtcacctc cgaggacgcc 1020
aacagcccca acgtgacgct gacgaccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtgca agtggaccct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac cttcgatatc accgtgtctg gcctgggcac cgcccccaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaaa agtgatcttc 1260
agcaaggccc ccggtctt 1278

```

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SEQ ID NO: 135      moltype = AA  length = 426
FEATURE             Location/Qualifiers
source              1..426
                    mol_type = protein
                    organism = Synthetic construct

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```

SEQUENCE: 135
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITIN FVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY PQDVFGMTWC HHAEMQNPVY LIPETVPYRK WDNCSNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240

```

-continued

GGILTSSTSPV ATPIPGTGYA YSLRLTPRPV SRFGLNNSIL YVFYSGNGPK ASGGRYCIQS 300
 NIVFSDEIPA SQDMPTNTTD ITVVDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
 FKCKWLTLSG TPGSCENISG AFASNRFTDI TVSGLGTAPK TLIITRTATN ATTTTHKVIF 420
 SKAPGS 426

SEQ ID NO: 136 moltype = DNA length = 1278
 FEATURE Location/Qualifiers
 source 1..1278
 mol_type = other DNA
 organism = Synthetic construct

SEQUENCE: 136

gaagctgccc	tgctcgtgtg	ccagtaacacc	atccagagcc	tgatccacct	gaccggcgag	60
gaccccggtc	tcttcaacgt	ggaaatcccc	gagttccctt	tctaccctac	ctgcaacgtg	120
tgacccggcg	acgtgaacgt	gaccatcaac	ttcgacgtgg	gcggcaagaa	gcaccagctg	180
gacctggatt	tcggccagct	gacccctcac	accaaggccg	tgatcacgac	cagaggcgcc	240
tttggcgcca	gcgagaacgc	caccaatctg	tttctgctgg	aactcctagg	cgccggcgag	300
ctggccctga	ccatgagaag	caagaaactg	cccatcaatg	tgaccacagg	cgaggaaacag	360
caggtgtccc	tggaaagcgt	ggacgtgtac	tttcaagacg	tgttcggcac	catgtggtgc	420
caccacggcg	agatgcagaa	ccccgtgtac	ctgatccccg	agacagtggc	ctaccggaag	480
tgggacaaact	gcaacagcac	caacatcacc	gccgtcgtgc	gggcccagg	actggatgtg	540
acactgcctc	tgagcctgcc	taccagcgcc	caggacagca	acttcagcgt	gaaaaccgag	600
atgctgggca	acgagatcga	catcgagtgc	atcatggaag	atggcgagat	cagccagggtg	660
ctgcccggcg	acaacaagtt	caacatcaca	tgacgaggct	acgagagcca	cgtgccatct	720
ggcggcatcc	tgaccagcac	aagcccagtg	gccacaccca	tccctggcac	aggctacgcc	780
tacagcctga	gactgacccc	cagaccctgt	tccagattcc	tgggcaacaa	cagcatcctg	840
tacgtgttct	acagcgccaa	cggccccaag	gcctctggcg	gccggtagct	tatccagagc	900
aacatcgtgt	tcagcgacga	gatccccgcc	agccaggaca	tgcccaccaa	taccaccgac	960
atcacgtacg	tgggcgacaa	tgccacctac	agcgtgccaa	tggtcacctc	cgaggacgcc	1020
aacagcccca	acgtgaccgt	gacagccttt	tgggcctggc	ctaacaacac	cgagacagac	1080
ttcaagtgca	agtggaccct	gacctccggc	accctagcgc	gctgcgagaa	tatcagcgga	1140
gccttcgcca	gcaacccggc	cttcgatatc	accgtgtctg	gcctgggcac	cgcccccaag	1200
accctgatca	tcaccagaac	cgccacaaat	gccaccacca	caaccacaaa	agtgatcttc	1260
agcaaggccc	ccggtctct					1278

SEQ ID NO: 137 moltype = AA length = 426
 FEATURE Location/Qualifiers
 source 1..426
 mol_type = protein
 organism = Synthetic construct

SEQUENCE: 137

EAALLVCQYT	IQSLIHLTGE	DPGFFNVEIP	EPFFPYPTCNV	CTADVNVITN	FVGGKKHQL	60
DLDFGQLTPH	TKAVYQPRGA	FGGSENATNL	FLLELLGAGE	LALTMRSKKL	PINVTTGEEQ	120
QVSLESVDVY	FQDVFGTMWC	HHAEQMNPVY	LIPETVRYIK	WDNCNSTNIT	AVVRAQGLDV	180
TLPLSLPTSA	QDSNFSVKTE	MLGNEIDIEC	IMEDGEISQV	LPGDNKFNIT	CSGYESHVPS	240
GGILTSSTSPV	ATPIPGTGYA	YSLRLTPRPV	SRFLGNNSIL	YVFYSGNGPK	ASGGRYCIQS	300
NIVFSDEIPA	SQDMPTNTTD	ITVVDNATY	SVPMVTSEDA	NSPNVTVTAF	WAWPNNTETD	360
FKCKWLTLSG	TPGSCENISG	AFASNRFTDI	TVSGLGTAPK	TLIITRTATN	ATTTTHKVIF	420
SKAPGS						426

SEQ ID NO: 138 moltype = DNA length = 1278
 FEATURE Location/Qualifiers
 source 1..1278
 mol_type = other DNA
 organism = Synthetic construct

SEQUENCE: 138

gaagctgccc	tgctcgtgtg	ccagtaacacc	atccagagcc	tgatccacct	gaccggcgag	60
gaccccggtc	tcttcaacgt	ggaaatcccc	gagttccctt	tctaccctac	ctgcaacgtg	120
tgacccggcg	acgtgaacgt	gaccatcaac	ttcgacgtgg	gcggcaagaa	gcaccagctg	180
gacctggatt	tcggccagct	gacccctcac	accaaggccg	tgatcacgac	cagaggcgcc	240
tttggcgcca	gcgagaacgc	caccaatctg	tttctgctgg	aactcctagg	cgccggcgag	300
ctggccctga	ccatgagaag	caagaaactg	cccatcaatg	tgaccacagg	cgaggaaacag	360
caggtgtccc	tggaaagcgt	ggacgtgtac	tttcaagacg	tgttcggcac	catgtggtgc	420
caccacggcg	agatgcagaa	ccccgtgtac	ctgatccccg	agacagtggc	gtacatcaag	480
tgggacaaact	gcaacagcac	caacatcacc	gccgtcgtgc	gggcccagg	actggatgtg	540
acactgcctc	tgagcctgcc	taccagcgcc	caggacagca	acttcagcgt	gaaaaccgag	600
atgctgggca	acgagatcga	catcgagtgc	atcatggaag	atggcgagat	cagccagggtg	660
ctgcccggcg	tcaacaagtt	caacatcaca	tgacgaggct	acgagagcca	cgtgccatct	720
ggcggcatcc	tgaccagcac	aagcccagtg	gccacaccca	tccctggcac	aggctacgcc	780
tacagcctga	gactgacccc	cagaccctgt	tccagattcc	tgggcaacaa	cagcatcctg	840
tacgtgttct	acagcgccaa	cggccccaag	gcctctggcg	gccggtagct	tatccagagc	900
aacatcgtgt	tcagcgacga	gatccccgcc	agccaggaca	tgcccaccaa	taccaccgac	960
atcacgtacg	tgggcgacaa	tgccacctac	agcgtgccaa	tggtcacctc	cgaggacgcc	1020
aacagcccca	acgtgaccgt	gacagccttt	tgggcctggc	ctaacaacac	cgagacagac	1080
ttcaagtgca	agtggaccct	gacctccggc	accctagcgc	gctgcgagaa	tatcagcgga	1140
gccttcgcca	gcaacccggc	cttcgatatc	accgtgtctg	gcctgggcac	cgcccccaag	1200
accctgatca	tcaccagaac	cgccacaaat	gccaccacca	caaccacaaa	agtgatcttc	1260
agcaaggccc	ccggtctct					1278

SEQ ID NO: 139 moltype = AA length = 426

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FEATURE
source

Location/Qualifiers
1..426
mol_type = protein
organism = Synthetic construct

SEQUENCE: 139

EAALLVCQYT	IQSLIHLTGE	DPGFFNVEIP	EPFPYPTCNV	CTADVNVNTIN	FDVGGKKHQL	60
DLDFGQLTPH	TKAVYQPRGA	FGGSENATNL	FLLELLGAGE	LALTMRSKKL	PINVTTGEEQ	120
QVSLESVDVY	QDVFVGTMCW	HHAEMQNPVY	LIPETVNYTK	WDCNCSNTNIT	AVVRAQGLDV	180
TLPLSLPTSA	QDSNFSVKTE	MLGNEIDIEC	IMEDGEISQV	LPGDNKFNIT	CSGYESHVPS	240
GGILTSTSPV	ATPIPGTGYA	YSLRLTPRPV	SRFLGNNSIL	YVFYSGNGPK	ASGGRYCIQS	300
NIVFSDEIPA	SQDMPTNTTD	ITYVGDNATY	SVPMVTSEDA	NSPNVTVTAF	WAWPNNTETD	360
FKCKWLTLSG	TPSGCENISG	AFASNRTFDI	TVSGLGTAPK	TLIITRTATN	ATTTTHKVIF	420
SKAPGS						426

SEQ ID NO: 140

FEATURE
source

moltype = DNA length = 1278
Location/Qualifiers
1..1278
mol_type = other DNA
organism = Synthetic construct

SEQUENCE: 140

gaagctgccc	tgctcgtgtg	ccagtagacc	atccagagcc	tgatccacct	gaccggcgag	60
gaccccggt	tcttcaacgt	ggaatcccc	gagttcccc	tctaccctac	ctgcaacgtg	120
tgacccggcg	acgtgaacgt	gaccatcaac	ttcgacgtgg	gcggcaagaa	gcaccagctg	180
gacctggatt	tcggccagct	gacccctcac	accaaggccg	tgatccagcc	cagaggcgcc	240
tttggcgcca	gcgagaacgc	caccaatctg	tttctgctgg	aactcctagg	cgccggcgag	300
ctggccctga	ccatgagaag	caagaaactg	cccatcaatg	tgaccacagg	cgaggaaacag	360
caggtgtccc	tggaaagcgt	ggacgtgtac	tttcaagacg	tgctcggcac	catgtggtgc	420
caccacggcg	agatgcagaa	ccccgtgtac	ctgatccccg	agacagtga	ctacaccaag	480
tgggacaaact	gcaacagcac	caacatcacc	gccgtcgtgc	gggcccaggg	actggatgtg	540
acactgcctc	tgagcctgcc	taccagcgcc	caggacagca	acttcagcgt	gaaaaccgag	600
atgctgggca	acgagatcga	catcgagtgc	atcatggaag	atggcgagat	cagccagggtg	660
ctgcccggcg	acaacaagtt	caacatcaca	tgacgcggct	acgagagcca	cgtgccatct	720
ggcgcatcc	tgaccagcac	aagcccagtg	gccacaccca	tccttgccac	aggctacgcc	780
tacagcctga	gactgacccc	cagaccctg	tccagattcc	tgggcaacaa	cagcatcctg	840
tacgtgttct	acagcgccaa	cggccccaag	gcctctggcg	gccggtactg	tatccagagc	900
aacatcgtgt	tcagcgacga	gatccccgcc	agccaggaca	tgcccaccaa	taccaccgac	960
atcacgtacg	tgggcgacaa	tgccaccctac	agcgtgccaa	tggtcacctc	cgaggacgcc	1020
aacagcccca	acgtgaccgt	gacagccttt	tgggcctggc	ctaacaacac	cgagacagac	1080
ttcaagtcca	agtggaccct	gacctccggc	acccttagcg	gctgcgagaa	tatcagcgga	1140
gccttcgcga	gcaaccggac	cttcgatata	accgtgtctg	gcctggggcac	cgccccaacg	1200
accctgatca	taccacgaac	cgccacaaat	gccaccacca	caaccacaaa	agtgatcttc	1260
agcaaggccc	ccggtctct					1278

SEQ ID NO: 141

FEATURE
source

moltype = AA length = 426
Location/Qualifiers
1..426
mol_type = protein
organism = Synthetic construct

SEQUENCE: 141

EAALLVCQYT	IQSLIHLTGE	DPGFFNVEIP	EPFPYPTCNV	CTADVNVNTIN	FDVGGKKHQL	60
DLDFGQLTPH	TKAVYQPRGA	FGGSENATNL	FLLELLGAGE	LALTMRSKKL	PINVTTGEEQ	120
QVSLESVDVY	QDVFVGTMCW	HHAEMQNPVY	LIPETVRYRK	WDCNCSNTNIT	AVVRAQGLDV	180
TLPLSLPTSA	QDSNFSVKTE	MLGNEIDIEC	IMEDGEISQV	LPGDNKFNIT	CSGYESHVPS	240
GGILTSTSPV	ATPIPGTGYA	YSLRLTPRPV	SRFLGNNSIL	YVFYSGNGPK	ASGGRYCIQS	300
NIVFSDEIPA	SQDMPTNTTD	ITYVGDNATY	SVPMVTSEDA	NSPNVTVTAF	WAWPNNTETD	360
FKCKWLTLSG	TPSGCENISG	AFASNRTFDI	TVSGLGTAPK	TLIITRTATN	ATTTTHKVIF	420
SKAPGS						426

SEQ ID NO: 142

FEATURE
source

moltype = DNA length = 1278
Location/Qualifiers
1..1278
mol_type = other DNA
organism = Synthetic construct

SEQUENCE: 142

gaagctgccc	tgctcgtgtg	ccagtagacc	atccagagcc	tgatccacct	gaccggcgag	60
gaccccggt	tcttcaacgt	ggaatcccc	gagttcccc	tctaccctac	ctgcaacgtg	120
tgacccggcg	acgtgaacgt	gaccatcaac	ttcgacgtgg	gcggcaagaa	gcaccagctg	180
gacctggatt	tcggccagct	gacccctcac	accaaggccg	tgatccagcc	cagaggcgcc	240
tttggcgcca	gcgagaacgc	caccaatctg	tttctgctgg	aactcctagg	cgccggcgag	300
ctggccctga	ccatgagaag	caagaaactg	cccatcaatg	tgaccacagg	cgaggaaacag	360
caggtgtccc	tggaaagcgt	ggacgtgtac	tttcaagacg	tgctcggcac	catgtggtgc	420
caccacggcg	agatgcagaa	ccccgtgtac	ctgatccccg	agacagtgcg	gtaccggaag	480
tgggacaaact	gcaacagcac	caacatcacc	gccgtcgtgc	gggcccaggg	actggatgtg	540
acactgcctc	tgagcctgcc	taccagcgcc	caggacagca	acttcagcgt	gaaaaccgag	600
atgctgggca	acgagatcga	catcgagtgc	atcatggaag	atggcgagat	cagccagggtg	660
ctgcccggcg	acaacaagtt	caacatcaca	tgacgcggct	acgagagcca	cgtgccatct	720
ggcgcatcc	tgaccagcac	aagcccagtg	gccacaccca	tccttgccac	aggctacgcc	780
tacagcctga	gactgacccc	cagaccctg	tccagattcc	tgggcaacaa	cagcatcctg	840
tacgtgttct	acagcgccaa	cggccccaag	gcctctggcg	gccggtactg	tatccagagc	900

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aacatcgtgt tcagcgacga gateccccgc agccaggaca tgccccacca taccaccgac 960
atcacgtacg tgggogacaa tgccacctac agcgtgccaa tggtaacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtga agtggacct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac ctctgatatc accgtgtctg gcctgggcac cgccccaaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaa agtgatcttc 1260
agcaaggccc ccggtctc

```

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SEQ ID NO: 143      moltype = AA length = 447
FEATURE
source              Location/Qualifiers
                    1..447
                    mol_type = protein
                    organism = Synthetic construct

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SEQUENCE: 143
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EFPFYPTCNV CTADVNVNTIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY QDVFQGTMCW HHAEMQNPVY LIPETVPYIK WDNCNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSSTPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGDYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITYVGDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRTFDI TVSGLTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGSGLND IFEAQKIEWH EHHHHHHH

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SEQ ID NO: 144      moltype = DNA length = 1341
FEATURE
source              Location/Qualifiers
                    1..1341
                    mol_type = other DNA
                    organism = Synthetic construct

```

```

SEQUENCE: 144
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtc tcttcaacgt ggaatcccc gagttccct tctaccctac ctgcaacgtg 120
tgaccgcggc acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtatcagcc cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacg 360
caggtgtccc tggaaagcgt ggacgtgtac ttccaagacg tgttcggcac catgtgggtg 420
caccacggcg agatgcagaa ccccggtgac ctgatcccc agacagtggc ctacatcaag 480
tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccgagg actggatgtg 540
acaactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccagggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcgccatcc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtcacgcc 780
tacagcctga gactgacccc cagaccggtg tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cggccccaag gcctctggcg gcgattactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccccaa taccaccgac 960
atcacgtacg tgggogacaa tgccacctac agcgtgccaa tggtaacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtga agtggacct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac ctctgatatc accgtgtctg gcctgggcac cgccccaaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaa agtgatcttc 1260
agcaaggccc ccggtctctg cctgaacgac atttttgagg ccagagaagat tgagtggcat 1320
gaacatcacc accaccacca t

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SEQ ID NO: 145      moltype = AA length = 447
FEATURE
source              Location/Qualifiers
                    1..447
                    mol_type = protein
                    organism = Synthetic construct

```

```

SEQUENCE: 145
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EFPFYPTCNV CTADVNVNTIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
NVSLESVDVY QDVFQGTMCW HHAEMQNPVY LIPETVPYIK WDNCNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSSTPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGDYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITYVGDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRTFDI TVSGLTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGSGLND IFEAQKIEWH EHHHHHHH

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SEQ ID NO: 146      moltype = DNA length = 1341
FEATURE
source              Location/Qualifiers
                    1..1341
                    mol_type = other DNA
                    organism = Synthetic construct

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SEQUENCE: 146
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtc tcttcaacgt ggaatcccc gagttccct tctaccctac ctgcaacgtg 120
tgaccgcggc acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtatcagcc cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300

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```

ctggccctga ccatgagaag caaagaactg cccatcaatg tgaccacagg cgaggaacag 360
aacgtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacgccg agatgcagaa ccccggtgtac ctgatccccg agacagtgcc ctacatcaag 480
tggaacaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccagggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcggcatcc tgaccagcac aagcccagtg gccacaccca tccctggcac aggctacgcc 780
tacagcctga gactgacccc cagaccctgt tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cggcccagag gcctctggcg gcgattactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccacgcac 960
atcacgtacg tgggagacaa tgcacacctac agcgtgccaa tggtcacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtcca agtggaccct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac ctcgatatac accgtgtctg gcctggggcac cgcgcccaag 1200
accctgatca tcaccagaa cgcacaaaat gccaccacca caaccacaa agtgatcttc 1260
agcaaggccc ccggtcttgg ctggaacgac atttttgagg ccagaagat tgagtggcat 1320
gaacatcacc accaccacca t 1341

```

```

SEQ ID NO: 147      moltype = AA length = 448
FEATURE            Location/Qualifiers
source              1..448
                    mol_type = protein
                    organism = Synthetic construct

```

```

SEQUENCE: 147
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY QDVFGTMWC HHAEMQNPVY LIPETVPYIK WNNSCNSTNI TAVVRAQGLD 180
VTLPLSLPTS AQDSNFSVKT EMLGNEIDIE CIMEDGEISQ VLPGDNKFNI TCSGYESHVP 240
SGGILTSTSP VATPIPGTGY AYSRLRLTPRP VSRFLGNNSI LYVFSYNGNP KASGGDYCIQ 300
SNIVFSDEIP ASQDMPNTTT DITYVGDNAT YSVPMTSED ANSPNVTVTA FWAWPNNTTET 360
DFKCKWTLTS GTPSGCENIS GAFASNRTFD ITVSLGTAP KTLIITRTAT NATTTTHKVI 420
FSKAPGSLN DIFEAQKIEW HEHHHHHH 448

```

```

SEQ ID NO: 148      moltype = DNA length = 1344
FEATURE            Location/Qualifiers
source              1..1344
                    mol_type = other DNA
                    organism = Synthetic construct

```

```

SEQUENCE: 148
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtc tcttcaacgt ggaaatcccc gagttccccct tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gacctcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtatcagcc cagaggcgcc 240
tttgccggca gcgagaacgc caccaatctg tttctcgtgg aactcctagg cgcggcgag 300
ctggccctga ccatgagaag caagaaaactg cccatcaatg tgaccacagg cgaggaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacgccg agatgcagaa ccccggtgtac ctgatccccg agacagtgcc ctacatcaag 480
tggaacaaca gctgcaacag caccacatc accgcccgtc tgccggccca gggactggat 540
gtgacactgc cctcagacct gcctaccagc gcccaggaca gcaacttcag cgtgaaaacc 600
gagatgcttg gcaacgagat cgacatcgag tgcacatgag aagatggcga gatcagccag 660
gtgctgcccc gcgacaacaa gttcaacatc acatgcagcg gctacgagag ccacgtgcc 720
tctggcgcca tctcagacct cacaagccca gtggccacac ccatccctgg cacaggctac 780
gcctacagcc tgagactgac ccccgagccc gtgtccagat tccctgggcaa caacagcatc 840
ctgtacgtgt tctacagcgg caacggcccc aaggcctctg gcggcgatta ctgtatccag 900
agcaacatct tgttcagcga cgagatcccc gccagccagg acatgcccac caataaccac 960
gacatcacgt acgtggcgga caatgccacc tacagcgtgc caatggtcac ctccaggag 1020
gcccaacagc ccaacgtgac cgtgacagcc ttttgggcct ggcctaacaa caccgagaca 1080
gacttcaagt gcaagtggac cctgacctcc ggcaacccta gcggctgcga gaatacagc 1140
ggagccttcg ccagcaaccg gaccttcgat atcaccgtgt ctggcctggg caccgcccc 1200
aagaccctga tcatcaccag aaccgcccaca aatgccacca ccacaaccca caaagtgatc 1260
ttcagcaagg ccccggtctc tggcctgaac gacatttttg agggccagaa gattgagtgg 1320
catgaacatc accaccacca ccat 1344

```

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SEQ ID NO: 149      moltype = AA length = 447
FEATURE            Location/Qualifiers
source              1..447
                    mol_type = protein
                    organism = Synthetic construct

```

```

SEQUENCE: 149
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY QDVFGTMWC HHAEMQNPVY LIPETVPYIK WNNSCNSTNI TAVVRAQGLD 180
VTLPLSLPTS AQDSNFSVKT EMLGNEIDIE CIMEDGEISQ VLPGDNKFNI TCSGYESHVP 240
GGILTSTSPV ATPIPGTGYA YSRLRLTPRPV SRFNGNSIL YVFSYNGNP ASGGRYCIQS 300
NIVFSDEIPA SQDMPNTTTD ITYVGDNAT YSVPMTSED ANSPNVTVTA FWAWPNNTTET 360
FKCKWTLTSG TPGSGCENIS GAFASNRTFD ITVSLGTAP KTLIITRTAT NATTTTHKVI 420
SKAPGSLND IFEAQKIEWH EHHHHHHH 447

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SEQ ID NO: 150      moltype = DNA length = 1341
FEATURE            Location/Qualifiers
source              1..1341
                   mol_type = other DNA
                   organism = Synthetic construct

SEQUENCE: 150
gaagctgccc tgcctgctgtg ccagtaacacc atccagagcc tgateccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtatcagcc cagaggcgcc 240
tttgccggca gcgagaacgc caccatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccattgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtgcc ctacatcaag 480
tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccgagg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccaggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggtc acgagagcca cgtgccatct 720
ggcgccatcc tgaccagcac aagcccgagt gccacaccca tccctggcac aggtacgccc 780
tacagcctga gactgacccc cagaccggtg tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cggccccaag gcctctggcg gccggtactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tggcgacaaa tgccacctac agcgtgccaa tggtaacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtga agtgagacct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac ctctgatata accgtgtctg gcctggggcac cgcccccaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaaa agtgatcttc 1260
agcaaggccc ccggtctctg cctgaacgac atttttgagg ccagaagat tgagtggcat 1320
gaacatcacc accaccacca t                                     1341

SEQ ID NO: 151      moltype = AA length = 447
FEATURE            Location/Qualifiers
source              1..447
                   mol_type = protein
                   organism = Synthetic construct

SEQUENCE: 151
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EFPFYPTCNV CTADVNVNTIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY QDVFVGTMCW HHAEMQNPVY LIPETVPYIK AAACNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSTSPV ATPIPGTGYA YSLRLTPRPV SRFLGNSIL YVFYSGNGPK ASGGDYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITYVGDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTTETD 360
FKCKWLTLSG TPGSCENISG AFASNRTFDI TVSGLTAPK TLII TRTATN ATTTTHKVFIF 420
SKAPGSGGLND IFEAQKIEWH EHHHHHHH                                     447

SEQ ID NO: 152      moltype = DNA length = 1341
FEATURE            Location/Qualifiers
source              1..1341
                   mol_type = other DNA
                   organism = Synthetic construct

SEQUENCE: 152
gaagctgccc tgcctgctgtg ccagtaacacc atccagagcc tgateccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgtatcagcc cagaggcgcc 240
tttgccggca gcgagaacgc caccatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccattgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtgcc ctacatcaag 480
gcccgccgct gcaacagcac caacatcacc gccgtcgtgc gggcccgagg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccaggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggtc acgagagcca cgtgccatct 720
ggcgccatcc tgaccagcac aagcccgagt gccacaccca tccctggcac aggtacgccc 780
tacagcctga gactgacccc cagaccggtg tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cggccccaag gcctctggcg gccgattactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tggcgacaaa tgccacctac agcgtgccaa tggtaacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtga agtgagacct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac ctctgatata accgtgtctg gcctggggcac cgcccccaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaaa agtgatcttc 1260
agcaaggccc ccggtctctg cctgaacgac atttttgagg ccagaagat tgagtggcat 1320
gaacatcacc accaccacca t                                     1341

SEQ ID NO: 153      moltype = AA length = 447
FEATURE            Location/Qualifiers
source              1..447
                   mol_type = protein

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organism = Synthetic construct

SEQUENCE: 153
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EFPFYPTCNV CTADVNVNTIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY QDVFGTMWC HHAEMQNPVY LIPETVPYRK WDNCSNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSTSPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGRYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITYVGDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRFTDI TVSGLTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGSGLND IFEAQKIEWH EHHHHHHH 447

SEQ ID NO: 154      moltype = DNA length = 1341
FEATURE
source             Location/Qualifiers
                   1..1341
                   mol_type = other DNA
                   organism = Synthetic construct

SEQUENCE: 154
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgatcacgcc cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgtctg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtggc ctaccggaag 480
tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccagggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcgcatccc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtacgccc 780
tacagcctga gactgacccc cagaccggtg tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cggccccaag gcctctggcg gccggtactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tggggcgacaa tgcccacctac agcgtgccaa tggtcacctc cgaggacgcc 1020
aacagcccca acgtgacgct gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtcca agtggaccct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac cctcgatctc accgtgtctg gcctgggcac cgccccaaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaaa agtgatcttc 1260
agcaaggccc ccggtctctg cctgaacgac atttttgagg ccagagaagt tgagtggcat 1320
gaacatcacc accaccacca t 1341

SEQ ID NO: 155      moltype = AA length = 447
FEATURE
source             Location/Qualifiers
                   1..447
                   mol_type = protein
                   organism = Synthetic construct

SEQUENCE: 155
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EFPFYPTCNV CTADVNVNTIN FDVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY QDVFGTMWC HHAEMQNPVY LIPETVRYIK WDNCSNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSTSPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGRYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITYVGDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRFTDI TVSGLTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGSGLND IFEAQKIEWH EHHHHHHH 447

SEQ ID NO: 156      moltype = DNA length = 1341
FEATURE
source             Location/Qualifiers
                   1..1341
                   mol_type = other DNA
                   organism = Synthetic construct

SEQUENCE: 156
gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgacccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccg tgatcacgcc cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgtctg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaacag 360
caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tgttcggcac catgtggtgc 420
caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtggc gtacatcaag 480
tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccagggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggct acgagagcca cgtgccatct 720
ggcgcatccc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtacgccc 780
tacagcctga gactgacccc cagaccggtg tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggcaa cggccccaag gcctctggcg gccggtactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tggggcgacaa tgcccacctac agcgtgccaa tggtcacctc cgaggacgcc 1020

```

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aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtgca agtggaccct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac cttcgatata accgtgtctg gcctggggcac cgcccccagg 1200
accctgatca tcaccagagc cgccacaaat gccaccacca caaccacaa agtgatcttc 1260
agcaaggccc ccggtctctg cctgaacgac atttttgagg ccagagaagat tgagtggcat 1320
gaacatcacc accaccacca t 1341

```

```

SEQ ID NO: 157      moltype = AA length = 447
FEATURE            Location/Qualifiers
source             1..447
                  mol_type = protein
                  organism = Synthetic construct

```

```

SEQUENCE: 157
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITN FVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY PQDVFGTMWC HHAEMQNPVY LIPETVNYTK WDNCSNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSTSPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGRYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITVVDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRTFDI TVSGLGTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGSGLND IFEAQKIEWH EHHHHHHH 447

```

```

SEQ ID NO: 158      moltype = DNA length = 1341
FEATURE            Location/Qualifiers
source             1..1341
                  mol_type = other DNA
                  organism = Synthetic construct

```

```

SEQUENCE: 158
gaagctgccc tgctcgtgtg ccagtacacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgcaccgccc acgtgaaagt gaccatcaac ttgcagctgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccc tgatcacgac cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacag 360
caggtgtccc tggaaagcgt ggaagtgtac tttcaagacg tgctcgccac catgtgggtg 420
caccacgccc agatgcagaa ccccggtgac ctgatccccc agacagtga ctacaccaag 480
tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccagggtg 660
ctgcccggcg acaacaagtt caacatcaca tgcagcggtc acgagagcca cgtgccatct 720
ggcgccatcc tgaccagcac aagcccagtg gccacaccca tccctggcac aggtctacgc 780
tacagcctga gactgacccc cagaccctgt tccagattcc tgggcaacaa cagcatcctg 840
tacgtgttct acagcggaac cgcccccagg gcctctggcg gccggtactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcactgacg tggggcgaaa tgcacacctac agcgtgccaa tggtaacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtgca agtggaccct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcgcca gcaaccggac cttcgatata accgtgtctg gcctggggcac cgcccccagg 1200
accctgatca tcaccagagc cgccacaaat gccaccacca caaccacaa agtgatcttc 1260
agcaaggccc ccggtctctg cctgaacgac atttttgagg ccagagaagat tgagtggcat 1320
gaacatcacc accaccacca t 1341

```

```

SEQ ID NO: 159      moltype = AA length = 447
FEATURE            Location/Qualifiers
source             1..447
                  mol_type = protein
                  organism = Synthetic construct

```

```

SEQUENCE: 159
EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EPPFYPTCNV CTADVNVITN FVGGKKHQL 60
DLDFGQLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
QVSLESVDVY PQDVFGTMWC HHAEMQNPVY LIPETVRYRK WDNCSNSTNIT AVVRAQGLDV 180
TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
GGILTSTSPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGRYCIQS 300
NIVFSDEIPA SQDMPTNTTD ITVVDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNNTETD 360
FKCKWTLTSG TPGSCENISG AFASNRTFDI TVSGLGTAPK TLIITRTATN ATTTTHKVIF 420
SKAPGSGLND IFEAQKIEWH EHHHHHHH 447

```

```

SEQ ID NO: 160      moltype = DNA length = 1341
FEATURE            Location/Qualifiers
source             1..1341
                  mol_type = other DNA
                  organism = Synthetic construct

```

```

SEQUENCE: 160
gaagctgccc tgctcgtgtg ccagtacacc atccagagcc tgatccacct gaccggcgag 60
gaccccggtt tcttcaacgt ggaaatcccc gagttccctt tctaccctac ctgcaacgtg 120
tgcaccgccc acgtgaaagt gaccatcaac ttgcagctgg gcggcaagaa gcaccagctg 180
gacctggatt tcggccagct gacccctcac accaaggccc tgatcacgac cagaggcgcc 240
tttggcgcca gcgagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300
ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacag 360

```

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cagggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tggtcggcac catgtggtgc 420
caccacgccc agatgcagaa ccccggtgtac ctgatccccg agacagtgcg gtaccggaag 480
tgggacaact gcaacacgac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccagggtg 660
ctgcccgcg acaacaagtt caacatcaca tgcagcggtc acgagagcca cgtgccatct 720
ggcggcaccc tgaccagcac aagcccagtg gccacacca tccctggcac aggctacgcc 780
tacagcctga gactgacccc cagaccggtg tccagattcc tgggcaacaa cagcatcctg 840
tacgtgtttc acagcggcga cggccccaag gccctctggcg gccggtactg tatccagagc 900
aacatcgtgt tcagcgacga gatccccgcc agccaggaca tgcccaccaa taccaccgac 960
atcacgtacg tgggcgacaa tgccacctac agcgtgccaa tggtcacctc cgaggacgcc 1020
aacagcccca acgtgaccgt gacagccttt tgggcctggc ctaacaacac cgagacagac 1080
ttcaagtga agtgagacct gacctccggc acccctagcg gctgcgagaa tatcagcgga 1140
gccttcggca gcaaccggac actcgatcgc gcctggggcac cgccccaag 1200
accctgatca tcaccagaac cgccacaaat gccaccacca caaccacaa agtgatcttc 1260
agcaaggccc ccggtcttgg cctgaacgac atttttgagg cccagaagat tgagtggcat 1320
gaacatcacc accaccacca t 1341

```

```

SEQ ID NO: 161      moltype = AA  length = 119
FEATURE
source              Location/Qualifiers
                    1..119
                    mol_type = protein
                    organism = Macaca mulatta

```

```

SEQUENCE: 161
QVQLVQSGAD VRKPGASVKV SCKASTYIFT GYIHVVRQA PGRGLEWLGW IHPNSGGTTY 60
SQMFQGRVTM TRDRSITTSY MELSLRQSD TAIYYCATLR FVEYSFDSWG QGTLVTVSS 119

```

```

SEQ ID NO: 162      moltype = DNA  length = 357
FEATURE
source              Location/Qualifiers
                    1..357
                    mol_type = genomic DNA
                    organism = Macaca mulatta

```

```

SEQUENCE: 162
caggtgcagc tgggtgcagtc tggcgacagc gtgaggaagc caggagcctc cgtgaagggtg 60
tcttctaagg ccagcaccta catcttcaca ggctactata tccactgggtg gaggcaggca 120
ccaggaaggg gcctggagtg gctgggcttg attcacccta actctggcgg caccacatac 180
agccagatgt ttcagggcag agtgaccatg acacgggaca gatccatcac cacatcttat 240
atggagctga gccgggctga gtcgacgat accgccatct actattgcgc cacactgaga 300
tcgtggagtg attcttttga tagctggggc cagggcaccg tggtgacagt gagctcc 357

```

```

SEQ ID NO: 163      moltype = AA  length = 7
FEATURE
source              Location/Qualifiers
                    1..7
                    mol_type = protein
                    organism = Macaca mulatta

```

```

SEQUENCE: 163
TYIFTGY 7

```

```

SEQ ID NO: 164      moltype = AA  length = 6
FEATURE
source              Location/Qualifiers
                    1..6
                    mol_type = protein
                    organism = Macaca mulatta

```

```

SEQUENCE: 164
HPNSGG 6

```

```

SEQ ID NO: 165      moltype = AA  length = 10
FEATURE
source              Location/Qualifiers
                    1..10
                    mol_type = protein
                    organism = Macaca mulatta

```

```

SEQUENCE: 165
LRFVEYSFDS 10

```

```

SEQ ID NO: 166      moltype = AA  length = 108
FEATURE
source              Location/Qualifiers
                    1..108
                    mol_type = protein
                    organism = Macaca mulatta

```

```

SEQUENCE: 166
EIVLTQSPGT LSLSPGERAT LSCRASQSIG STYLAWYQQI PGQAPRLLIY GASSRAAGIP 60
DRFSGGSGT DFTLTISRLE PEDFAVYQC QYGSSPRSPG QGKLEIK 108

```

```

SEQ ID NO: 167      moltype = DNA  length = 324
FEATURE
source              Location/Qualifiers
                    1..324
                    mol_type = genomic DNA
                    organism = Macaca mulatta

```

```

SEQUENCE: 167

```

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gagatcgtgc tgacccagag cccaggcaca ctgagcctgt ccccaggaga gagggccacc 60
ctgtcctgta gagcctctca gagcatcagc tccacataacc tggcctggta tcagcagatc 120
ccaggacagg cacctaggct gctgatctac ggagcctcta gcagggcagc aggcaccccc 180
gaccgcttct ccggcggagg ctctggcacc gacttcaccc tgacaatctc tcggctggag 240
cctgaggact tcgcogtgta ctattgccag cagtatggct cctctccaag gtcctttggc 300
cagggcaciaa agctggagat caag 324

```

```

SEQ ID NO: 168      moltype = AA  length = 12
FEATURE            Location/Qualifiers
source             1..12
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 168
RASQSSISSTY LA 12

```

```

SEQ ID NO: 169      moltype = AA  length = 7
FEATURE            Location/Qualifiers
source             1..7
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 169
GASSRAA 7

```

```

SEQ ID NO: 170      moltype = AA  length = 9
FEATURE            Location/Qualifiers
source             1..9
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 170
QQYGSPPRS 9

```

```

SEQ ID NO: 171      moltype = AA  length = 119
FEATURE            Location/Qualifiers
source             1..119
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 171
EVQLLESFGA LVQPGGSLRL SCAASGFTFK TYAMSWVRQV PGKGLEWVSA ISGSGTASY 60
ADSVKGRFTL SRDNSKNTLY LQLSSLRDED TGVYYCARRF LDWFGMDVWG LGTTVTVSS 119

```

```

SEQ ID NO: 172      moltype = DNA  length = 357
FEATURE            Location/Qualifiers
source             1..357
                   mol_type = genomic DNA
                   organism = Macaca mulatta

```

```

SEQUENCE: 172
gaggtgcagc tgctggagag cggcggcgcc ctggtgcagc caggaggcag cctgcggctg 60
tcctgtgccg cctctggctt cacctttaag acatacgcca tgctcctggg gaggcagggtg 120
cctggcaagg gcctggagtg ggtgtctgcc atctccggct ctggcaccgc ctcttactat 180
gccgacagcg tgaagggcag gttcacctcg agccgcgata actccaagaa tacactgtat 240
ctgcagctga gctccctgcg ggacgaggat accggcgtgt actattgcgc ccggagattc 300
ctggactggt ttggcatgga cgtgtggggc ctgggcacca cagtgcagct gtctagc 357

```

```

SEQ ID NO: 173      moltype = AA  length = 7
FEATURE            Location/Qualifiers
source             1..7
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 173
GFTFKTY 7

```

```

SEQ ID NO: 174      moltype = AA  length = 6
FEATURE            Location/Qualifiers
source             1..6
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 174
SGSGTA 6

```

```

SEQ ID NO: 175      moltype = AA  length = 10
FEATURE            Location/Qualifiers
source             1..10
                   mol_type = protein
                   organism = Macaca mulatta

```

```

SEQUENCE: 175
RFLDWFGMDV 10

```

```

SEQ ID NO: 176      moltype = AA  length = 111
FEATURE            Location/Qualifiers

```

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```

source                1..111
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 176
DIVMTQSPLS LPVTPGEPAS ISCLSSQSLL QSNQYNYVDW YLQKPGQSPQ LLIYLGSNRA 60
SEVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMVTLHPP TFGQGAKVEI K 111

SEQ ID NO: 177        moltype = DNA length = 333
FEATURE              Location/Qualifiers
source                1..333
                      mol_type = genomic DNA
                      organism = Macaca mulatta

SEQUENCE: 177
gacatcgtga tgacccagtc cctctgtct ctgccagtga caccggcgga gcctgcctct 60
atcagctgtc tgagctccca gagcctgctg cagtccaacg gctacaatta tgtggattgg 120
tacctgcaga agccaggcca gtcccccag ctgctgatct atctgggctc taacaggggcc 180
agcggcgctg ccgacagatt ctccgctct gccagcggca ccgacttcac cctgaagatc 240
tctcgggtgg aggcagagga cgtgggcgtg tactattgca tggtgaccct gcacccacct 300
acattcggcc agggagccaa ggtggagatc aag 333

SEQ ID NO: 178        moltype = AA length = 16
FEATURE              Location/Qualifiers
source                1..16
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 178
LSSQSLLQSN GYNYVD 16

SEQ ID NO: 179        moltype = AA length = 7
FEATURE              Location/Qualifiers
source                1..7
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 179
LGSNRAS 7

SEQ ID NO: 180        moltype = AA length = 8
FEATURE              Location/Qualifiers
source                1..8
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 180
MVTLHPPT 8

SEQ ID NO: 181        moltype = AA length = 120
FEATURE              Location/Qualifiers
source                1..120
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 181
EGQLVQSGGG LVQPGGSLTL SCEVSGFTFK NYEMNWVRQA PGKGLEWVSY ISSGGIAIFH 60
ADSVKGRFTV SRDNAKNLLY LQMNLSLRVED TAVYYCARDE NNVRPFDPHW GQGTLVTVSS 120

SEQ ID NO: 182        moltype = DNA length = 360
FEATURE              Location/Qualifiers
source                1..360
                      mol_type = genomic DNA
                      organism = Macaca mulatta

SEQUENCE: 182
gagggacagc tgggtgcagtc cgggcgaggc ctgggtgcagc caggaggctc cctgaccctg 60
tcttgtgagg tgagcggcctt caccttcaag aactacgaga tgaattgggt gcggcaggca 120
cctggcaagg gcctggagtg ggtgtcttat atcagctccg gcggaatcgc aatcttcac 180
gcagattccg tgaagggcag gtttaccgtg tctcgcgaca acgccaagaa tctgctgtac 240
ctgcagatga acagcctgcg ggtggaggac acagccgtgt actattgcgc cagggatgag 300
aacaacgtgc ggcggccctt cgaccactgg ggacagggca ccctggtgac agtgtctagc 360

SEQ ID NO: 183        moltype = AA length = 7
FEATURE              Location/Qualifiers
source                1..7
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 183
GFTFKNY 7

SEQ ID NO: 184        moltype = AA length = 6
FEATURE              Location/Qualifiers
source                1..6
                      mol_type = protein
                      organism = Macaca mulatta

```


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SEQUENCE: 184
SSGGIA 6

SEQ ID NO: 185 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
mol_type = protein
organism = Macaca mulatta

SEQUENCE: 185
DENNVRRPFD H 11

SEQ ID NO: 186 moltype = AA length = 109
FEATURE Location/Qualifiers
source 1..109
mol_type = protein
organism = Macaca mulatta

SEQUENCE: 186
QSVLTQPPSA SGSPGQSVTI SCTGSSSDVG AYDFVSWFQQ YPGQAPKLII YEVNKRPSGV 60
PARFSGSKSG NTASLTVSGL QAEDEADYFC FSYGGTTNLR VFGGGTKLT 109

SEQ ID NO: 187 moltype = DNA length = 327
FEATURE Location/Qualifiers
source 1..327
mol_type = genomic DNA
organism = Macaca mulatta

SEQUENCE: 187
cagtctgtgc tgacccagcc acctagcgcc tccggctctc ccggccagag cgtgaccatc 60
tcctgtacag gcagctcctc tgacgtgggc gcctacgatt tcgtgagctg gtttcagcag 120
tatccaggcc aggcceccaa gctgacatc tacgaggtga acaagcggcc ttccggcgtg 180
ccagccagat tcagcggctc caagtctggc aataccgcct ctctgacagt gagcggcctg 240
caggcagagg acgaggcaga ttactctgc ttttcttatg gcggcaccac aaacctgcgg 300
gtgtttggcg gcggcaccaa gctgaca 327

SEQ ID NO: 188 moltype = AA length = 14
FEATURE Location/Qualifiers
source 1..14
mol_type = protein
organism = Macaca mulatta

SEQUENCE: 188
TGSSSDVGAY DFVS 14

SEQ ID NO: 189 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
mol_type = protein
organism = Macaca mulatta

SEQUENCE: 189
EVNKRPS 7

SEQ ID NO: 190 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
mol_type = protein
organism = Macaca mulatta

SEQUENCE: 190
FSYGGTTNLR V 11

SEQ ID NO: 191 moltype = AA length = 122
FEATURE Location/Qualifiers
source 1..122
mol_type = protein
organism = Macaca mulatta

SEQUENCE: 191
EVQLVESGGN LVQPGASLRL SCTASRFNFN KYAMHWVRQT PGKGLEWVSA ISWDSTYIDY 60
GNSVKGRFTI SRDNRNSLY LQMSLTAED TALYYCAKCE DYLR LCSAYD IWGHGTMVTV 120
SS 122

SEQ ID NO: 192 moltype = DNA length = 366
FEATURE Location/Qualifiers
source 1..366
mol_type = genomic DNA
organism = Macaca mulatta

SEQUENCE: 192
gagggtgcagc tgggtggagag cggcggaac ctggtgcagc caggagcctc tctgaggctg 60
agctgtaccg cctccgctt caactttaat aagtacgcaa tgcactgggt gcggcagacc 120
cctggcaagg gcctggagtg ggtgtctgcc atcagctggg actccacata catcgattat 180
ggcaactccg tgaagggcag gttcaccatc tctcgggaca acacaagaaa tagcctgtat 240
ctgcagatga attccctgac cgcgcaggat acagccctgt actattgcgc caagtgtgag 300
gactacctgc ggctgtgctc tgcctatgat atctggggcc acggcaccat ggtgacagtg 360

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agctcc 366

SEQ ID NO: 193      moltype = AA  length = 7
FEATURE            Location/Qualifiers
source              1..7
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 193
RFNFNKY 7

SEQ ID NO: 194      moltype = AA  length = 6
FEATURE            Location/Qualifiers
source              1..6
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 194
SWDSTY 6

SEQ ID NO: 195      moltype = AA  length = 13
FEATURE            Location/Qualifiers
source              1..13
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 195
CEDYLR LCSA YDI 13

SEQ ID NO: 196      moltype = AA  length = 112
FEATURE            Location/Qualifiers
source              1..112
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 196
DIVMTQSPLS LPVTPGESAS ISCRSSQSLL HSNKKNYLSW YLQKPGQSPQ LLIDLGSNRA 60
SGVSDRPSGS GSGTDFTLKI SRVEADDDGV YYCMQAVQTP ITFGQGTRLA IK 112

SEQ ID NO: 197      moltype = DNA  length = 336
FEATURE            Location/Qualifiers
source              1..336
                    mol_type = genomic DNA
                    organism = Macaca mulatta

SEQUENCE: 197
gacatcgtga tgacccagtc tctcttgagc ctgcccgtag cactcggcga gtctgccagc 60
atctcctgtc ggagctccca gagcctgctg cactccaacg gcaagaatta cctgtcttgg 120
tatctgcaga agccaggcca gagccccacg ctgctgatcg atctggggctc caacaggggcc 180
tcggcgctgt ctgacagatt ctctggcagc ggctccggca ccgacttcac cctgaagatc 240
agcagggtgg aggcgcagca tgtggggcgtg tactattgca tgcaggccgtg gcagacccca 300
atcacattcg gccagggaac ccgcctggcc atcaag 336

SEQ ID NO: 198      moltype = AA  length = 16
FEATURE            Location/Qualifiers
source              1..16
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 198
RSSQSLLHSN GKNYLS 16

SEQ ID NO: 199      moltype = AA  length = 7
FEATURE            Location/Qualifiers
source              1..7
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 199
LGSNRAS 7

SEQ ID NO: 200      moltype = AA  length = 9
FEATURE            Location/Qualifiers
source              1..9
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 200
MQAVQTPIT 9

SEQ ID NO: 201      moltype = AA  length = 123
FEATURE            Location/Qualifiers
source              1..123
                    mol_type = protein
                    organism = Macaca mulatta

SEQUENCE: 201
QVQLVQSGAE LKTPGASVKV SCKASGYTFT GYIHVVRQA PEGLEWTGW INPNSGATRY 60

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GQKFQGRVTL TSDTSSSTVY MEVSNLTSDD SAVYYCAREL SYSIRGTGPL GYWGLGTLVT 120
VSS 123

SEQ ID NO: 202 moltype = DNA length = 369
FEATURE Location/Qualifiers
source 1..369
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 202
caggtgcagc tgggtgcagtc cggtgcagag ctgaagaccc caggagccag cgtgaagggtg 60
tcctgtaagg cctctgggcta caccctcaca ggctactata tccactgggt gcggcaggca 120
ccaggagagg gcctggagtg gaccggctgg atcaacccta atagcggcgc cacaagatac 180
ggccagaagt ttcaggggccg cgtgaccctg acaagcgaca ccagctcctc tacagtgtat 240
atggagggtg ccaacctgac ctccgacgat tctgccgtgt actattgcgc ccgggagctg 300
tcttacagca tcagaggaac aggaccactg ggatattggg gcctgggcac cctggtgaca 360
gtgagctcc 369

SEQ ID NO: 203 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 203
GYTFTGY 7

SEQ ID NO: 204 moltype = AA length = 6
FEATURE Location/Qualifiers
source 1..6
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 204
NPNSGA 6

SEQ ID NO: 205 moltype = AA length = 14
FEATURE Location/Qualifiers
source 1..14
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 205
ELSYSIRGTG PLGY 14

SEQ ID NO: 206 moltype = AA length = 108
FEATURE Location/Qualifiers
source 1..108
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 206
EIVLTQSPGT LSLSPGERAT LSCRASQSV SKYLAWYQQK PGQAPRLLIY GASSRATGIP 60
DRFSGSGSGT DFTLTISRLE PEDFAVYQC YYGSSPLTFG QGTKVEIK 108

SEQ ID NO: 207 moltype = DNA length = 324
FEATURE Location/Qualifiers
source 1..324
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 207
gagatcgtgc tgacccagtc tccaggcaca ctgtccctgt ctccaggaga gagggccacc 60
ctgtcttgta gagccagcca gtccgtggcc agcaagtacc tggcctggta tcagcagaag 120
ccaggacagg cacctaggct gctgatctac ggagccagct ccaggggcaac cggcatcccc 180
gaccgcttct ctggcagcgg ctccggcaca gacttcaccc tgacaatctc caggctggag 240
cctgaggact tcgccgtgta ctattgccag tactatggct ctagcccact gacctttggc 300
cagggcacaa aggtggagat caag 324

SEQ ID NO: 208 moltype = AA length = 12
FEATURE Location/Qualifiers
source 1..12
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 208
RASQSVASKY LA 12

SEQ ID NO: 209 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 209
GASSRAT 7

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FEATURE	Location/Qualifiers	
source	1..11	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 218		
RASQSVVTYL N		11
SEQ ID NO: 219	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 219		
TTSKLQS		7
SEQ ID NO: 220	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
source	1..10	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 220		
QQSYGTPPFT		10
SEQ ID NO: 221	moltype = AA length = 124	
FEATURE	Location/Qualifiers	
source	1..124	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 221		
QVHLQQWAG LVKPSSETLSL TCAVQGGPFS GYWSWIRQP PGKGLEWIGE INHSGNTHYN		60
PSLKSRVTIS VDTSGNYFSL KLTSVTAADA AVYFCARGQQ LLRNYYYYSG MDVWQGQTTV		120
TVSS		124
SEQ ID NO: 222	moltype = DNA length = 372	
FEATURE	Location/Qualifiers	
source	1..372	
	mol_type = genomic DNA	
	organism = Macaca mulatta	
SEQUENCE: 222		
caggtgcacc tgcagcagtg gggagcaggc ctggtgaagc catccgagac actgtctctg		60
acatgtgcag tgcagggagg acccttctct ggctactatt ggagctggat caggcagcca		120
cctggcaagg gcctggagtg gatcggcgag atcaaccaca gcggcaatac ccactacaac		180
ccctctctga agagccgggt gaccatcagc gtggacacat ccggcaatta cttctcctg		240
aagctgacct ctgtgacagc cgccgatgcc gccgtgtatt tttgcgccc gggccagcag		300
ctgctgagaa actactatta ctattccggc atggacgtgt ggggacaggg aaccacagtg		360
acagtgcagct cc		372
SEQ ID NO: 223	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 223		
GGPFSGY		7
SEQ ID NO: 224	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
source	1..5	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 224		
NHSGN		5
SEQ ID NO: 225	moltype = AA length = 16	
FEATURE	Location/Qualifiers	
source	1..16	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 225		
GQQLLRNYYY YSGMDV		16
SEQ ID NO: 226	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
source	1..107	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 226		
EIVLTQSPGT LSLSPGERAT LSCRASQSVT STYLAWYQQK LGQPPRLIF GASNRATGIP		60
DRFSGSGSGT DFTLTITRLE PEDFAVYQC RYGGSTITGQ GTRLEIK		107

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SEQ ID NO: 227 moltype = DNA length = 321
 FEATURE Location/Qualifiers
 source 1..321
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 227
 gagatcgtgc tgacacagtc cccaggcacc ctgagcctgt ccccaggaga gcgggccaca 60
 ctgtcctgta gagcctctca gagcgtgacc tctacataacc tggcctggta tcagcagaag 120
 ctgggccagc cccctaggtt gctgatcttc ggcgccctcta acagggccac aggcatacct 180
 gaccgcttct ccggtctctg cagcggcacc gacttcaccc tgacaatcac cagactggag 240
 cccgaggact tcgccgtgta ctattgccag cggtacggcg gcagcatcac atttgccag 300
 ggcaccagac tggagatcaa g 321

SEQ ID NO: 228 moltype = AA length = 12
 FEATURE Location/Qualifiers
 source 1..12
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 228
 RASQSVTSTY LA 12

SEQ ID NO: 229 moltype = AA length = 7
 FEATURE Location/Qualifiers
 source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 229
 GASNRAT 7

SEQ ID NO: 230 moltype = AA length = 8
 FEATURE Location/Qualifiers
 source 1..8
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 230
 QRYGGSIT 8

SEQ ID NO: 231 moltype = AA length = 126
 FEATURE Location/Qualifiers
 source 1..126
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 231
 EVQLVQSGGG VVQPRRLRL SCAASGFTFS NYGMHWVRQV PGKGLQWVAI IWYDGSNKHY 60
 AASVQGRFRI SRDNSKNTVY LQMDGLRAED TGMYYCVRDA TTATTEGTSQ YYFDLWGQGA 120
 LVTVSS 126

SEQ ID NO: 232 moltype = DNA length = 378
 FEATURE Location/Qualifiers
 source 1..378
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 232
 gaggtgcagc tgggtgcagtc cggcggagga gtggtgcagc cacggagatc tctgaggctg 60
 agctgtgccg cctccggtt cacttttct aactacggaa tgcaactgggt gcgccagggt 120
 cctggcaagg gctgcagtg ggtggccatc atctggtagc acggctccaa taagcactat 180
 gccgcctctg tgcagggcag gttccgcac tctcgggata acagcaagaa taccgtgtat 240
 ctgcagatgg acggcctcgg ggcggaggat acaggcatgt actattgcgt gagagacgcc 300
 accacagcca ccacagaggg caccagccag tactattttg atctgtgggg acagggcgcc 360
 ctggtgacag tgagctcc 378

SEQ ID NO: 233 moltype = AA length = 7
 FEATURE Location/Qualifiers
 source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 233
 GFTFSNY 7

SEQ ID NO: 234 moltype = AA length = 6
 FEATURE Location/Qualifiers
 source 1..6
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 234
 WYDGSN 6

SEQ ID NO: 235 moltype = AA length = 17

-continued

FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 235		
DATTATTEGT SQYYFDL		17
SEQ ID NO: 236	moltype = AA length = 113	
FEATURE	Location/Qualifiers	
source	1..113	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 236		
DIVMTQSPDS LAVSLGERAT INCKSSQTLT YTSNSKNYLA WYQQKVGQPP RLLIWASTR		60
ESGVDPDRFSG SGSGTDFTLT ISSLLAEDVA VYQCQQYYTT PLTFGGGTKV EVK		113
SEQ ID NO: 237	moltype = DNA length = 339	
FEATURE	Location/Qualifiers	
source	1..339	
	mol_type = genomic DNA	
	organism = Macaca mulatta	
SEQUENCE: 237		
gacatcgtga tgaccacagag ccccgattcc ctggccgtgt ctctgggaga gagggcaaca		60
atcaactgta agagctccca gacctgctgt tacacatcca actctaagaa ttacctggcc		120
tggtatcagc agaaagtggg acagccacct aggtctgtga tctattgggc ctctaccagg		180
gagagcggcg tgccagacag attcagcggc tccggctctg gcacagactt caccctgaca		240
atctctagcc tgetggccga ggacgtggcc gtgtactatt gccagcagta ctataccaca		300
cccctgacct tcggcgccgg cacaaaggtg gaggtgaag		339
SEQ ID NO: 238	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
source	1..17	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 238		
KSSQTLTYTS NSKNYLA		17
SEQ ID NO: 239	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
source	1..7	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 239		
WASTRES		7
SEQ ID NO: 240	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
source	1..9	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 240		
QQYYTTPLT		9
SEQ ID NO: 241	moltype = AA length = 130	
FEATURE	Location/Qualifiers	
source	1..130	
	mol_type = protein	
	organism = Macaca mulatta	
SEQUENCE: 241		
EVQLVESGGG VVHPGKSLTL SCEASGFTFN DHGIHWVRRR PGKGLEWLAL ISKDGSKKEYS		60
TDSVKGRFTV SRDNSRNTVF LQMKSLTTED TAIYYCAKDM GQCSPSCST MDSYFAMDVW		120
GQGTTVIVSS		130
SEQ ID NO: 242	moltype = DNA length = 390	
FEATURE	Location/Qualifiers	
source	1..390	
	mol_type = genomic DNA	
	organism = Macaca mulatta	
SEQUENCE: 242		
gaggtgcagc tgggtgagtc cggcggagga gtggtgcacc ctggcaagtc tctgacctg		60
agctgtgagg ccagcggtt caccctcaac gaccacggca tccactgggt gcggagagca		120
cctggcaagg gcttgagtg gctggccctg atctctaagg acggcagcaa ggagtacagc		180
accgattccg tgaagggccg gttcacagtg tccagggata actctcgcaa taccgtgtt		240
ctgcagatga agtctctgac cacagaggac acagccatct actattgcgc caaggatatg		300
ggccagtgca gctccccctc ctgttctacc atggacagct atttcgcaat ggacgtgtg		360
ggacagggaa ccacagtgat cgtgtctagc		390
SEQ ID NO: 243	moltype = AA length = 7	
FEATURE	Location/Qualifiers	

-continued

source	1..7 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 243 GFTFNDH		7
SEQ ID NO: 244 FEATURE source	moltype = AA length = 6 Location/Qualifiers 1..6 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 244 SKDGSK		6
SEQ ID NO: 245 FEATURE source	moltype = AA length = 21 Location/Qualifiers 1..21 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 245 DMGQCSSPSC STMDSYFAMD V		21
SEQ ID NO: 246 FEATURE source	moltype = AA length = 113 Location/Qualifiers 1..113 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 246 DIVMTQSPLS LPVTPGEPAS ISCRSSQNLR HNNGYNYLW YLQKPGQSPQ LLIYLGSI SGVPDRFSGS GSGTDFTLKI SRVEAEDVGV YYCMQALQTP PWTFQGQTKV DFK		60 113
SEQ ID NO: 247 FEATURE source	moltype = DNA length = 339 Location/Qualifiers 1..339 mol_type = genomic DNA organism = Macaca mulatta	
SEQUENCE: 247 gacatcgtga tgaccocagtc cctctgtgtct ctgccagtga caccocggcga gcctgcctct atcagctgtc ggagctccca gaacctgaga cacaacaatg gctacaacta tctgaattgg tacctgcaga agccaggcca gtctcccccag ctgctgatct atctgggcag catcagggcc tccggcgtgc ccgaccgctt cctccgctct ggcagcggca ccgacttcac cctgaagatc agccgggtgg aggcagagga cgtgggcgtg tactattgca tgcaggccct gcagaccccc ccttggacat tcggccaggg caccaagggtg gacttcaag		60 120 180 240 300 339
SEQ ID NO: 248 FEATURE source	moltype = AA length = 16 Location/Qualifiers 1..16 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 248 RSSQNLRHNN GYNYLN		16
SEQ ID NO: 249 FEATURE source	moltype = AA length = 7 Location/Qualifiers 1..7 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 249 LGSIRAS		7
SEQ ID NO: 250 FEATURE source	moltype = AA length = 10 Location/Qualifiers 1..10 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 250 MQALQTPPWT		10
SEQ ID NO: 251 FEATURE source	moltype = AA length = 118 Location/Qualifiers 1..118 mol_type = protein organism = Macaca mulatta	
SEQUENCE: 251 QVQLVQSGAE VKKPGASVKV SCKASGYTFT SHYMHWVRQA PGQGLEWMGI ISPTGDFTNY AQKFGQGRVTL TRDTSTSTDY MEVTSLSRSED TAVYYCARD C SAWAPDYWGQ GTLVTVSS		60 118
SEQ ID NO: 252 FEATURE	moltype = DNA length = 354 Location/Qualifiers	

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source                1..354
                      mol_type = genomic DNA
                      organism = Macaca mulatta

SEQUENCE: 252
caggtgcagc tgggtgcagtc cggcgccagag gtgaagaagc caggagccag cgtgaagggtg 60
tcctgtaagg cctctggcta caccttcaca tctcactata tgcactgggt gcggcaggca 120
ccaggacagg gctctggagt gatgggcatc atcagcccta caggcgactt caccaactac 180
gcccagaagt ttcagggccg ggtgaccctg acaagagaca cctctacaag caccgattat 240
atggaggtga catcctgag gtctgaggat accgccgtgt actattgcgc aaggggactgt 300
tccgcctggg cccccgatta ctggggacag ggcacactgg tgaccgtgag ctcc 354

SEQ ID NO: 253        moltype = AA length = 7
FEATURE              Location/Qualifiers
source                1..7
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 253
GYTFTSH                                                       7

SEQ ID NO: 254        moltype = AA length = 6
FEATURE              Location/Qualifiers
source                1..6
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 254
SPTGDF                                                         6

SEQ ID NO: 255        moltype = AA length = 9
FEATURE              Location/Qualifiers
source                1..9
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 255
DCSAWAPDY                                                     9

SEQ ID NO: 256        moltype = AA length = 108
FEATURE              Location/Qualifiers
source                1..108
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 256
QSALTRPPSV SRCPGQSITI SCSGTSSDVG HDNHVSWYQQ HPGRAPKLMV YEVRNRPSGV 60
SDRFSGSKSG NTASLTISGL QAEDEATYYC CSYTTTHRYI FGGGTKLT 108

SEQ ID NO: 257        moltype = DNA length = 324
FEATURE              Location/Qualifiers
source                1..324
                      mol_type = genomic DNA
                      organism = Macaca mulatta

SEQUENCE: 257
cagtctgccc tgacaaggcc cccttctgtg agccgctgcc ctggacagag catcacaaac 60
tcctgttctg gcaccagctc cgacgtgggc cagcataacc acgtgtcctg gtaccagcag 120
caccacaggaa gggcacccaa gctgatgggt tatgagggtg ggaacagacc aagcggcgtg 180
tccgacaggt tcagcggctc caagtctggc aatacagcct ctctgacct cagcggcctg 240
caggcagagg atgaggaac ctactattgc tgttcttaca ccacaacca ccggtatatt 300
tttgccggcg gcacaaagct gacc 324

SEQ ID NO: 258        moltype = AA length = 14
FEATURE              Location/Qualifiers
source                1..14
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 258
SGTSSDVGHD NHVS                                             14

SEQ ID NO: 259        moltype = AA length = 7
FEATURE              Location/Qualifiers
source                1..7
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 259
EVRNRPS                                                       7

SEQ ID NO: 260        moltype = AA length = 10
FEATURE              Location/Qualifiers
source                1..10
                      mol_type = protein
                      organism = Macaca mulatta

SEQUENCE: 260

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-continued

CSYTTTHRYI 10

SEQ ID NO: 261 moltype = AA length = 123
 FEATURE Location/Qualifiers
 source 1..123
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 261
 QVQLLGSGPG LVKPSETLSL TCTVSGASIS SPGYWGFIR QSPGKGLEWI GSMVSGGTTY 60
 YNPSLKSRVT ISMDMSNNQF SLRLNSVTAA DTALYYCARG SRQLVRRATI DYWGQGALFT 120
 VSP 123

SEQ ID NO: 262 moltype = DNA length = 369
 FEATURE Location/Qualifiers
 source 1..369
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 262
 cagggtgcagc tgctgggagc cggcccaggc ctggtgaagc cttctgagac actgagcctg 60
 acctgtacag tgtctggcgc cagcatcagc tcccagcgt actattgggg cttcatcagg 120
 cagagcccag gcaagggcct ggagtgatc ggctccatgg tgtctggcgg caccacatac 180
 tataacccta gcctgaagtc ccgggtgaca atctccatgg acatgtctaa caatcagttc 240
 agcctgaggc tgaattccgt gaccgcccgc gatacagccc tgtactattg cgcaaggggc 300
 tcccgccagc tgggtgcggag agcaaccatc gactactggg gacagggcgc cctgtttaca 360
 gtgtctccc 369

SEQ ID NO: 263 moltype = AA length = 9
 FEATURE Location/Qualifiers
 source 1..9
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 263
 GASISSPGY 9

SEQ ID NO: 264 moltype = AA length = 5
 FEATURE Location/Qualifiers
 source 1..5
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 264
 VSGGT 5

SEQ ID NO: 265 moltype = AA length = 13
 FEATURE Location/Qualifiers
 source 1..13
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 265
 GSRQLVRRAT IDY 13

SEQ ID NO: 266 moltype = AA length = 108
 FEATURE Location/Qualifiers
 source 1..108
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 266
 QSVLTGPPSV SAGPGQQVFI SCSGNSSNIG NNYVSWYQQL PGTA PKLLIY DSNKRPSGIP 60
 DRFSGSKSGT SATLGITGLQ TGDEADYYCG TWDSSL SAGV FGGTKLT 108

SEQ ID NO: 267 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 267
 cagtctgtgc tgaccggacc accttccgtg tctgccggac caggacagca ggtgttcac 60
 agctgttccg gcaacagctc caatatcggc aacaattacg tgtcttggtg tcagcagctg 120
 ccaggcacag cccccaagct gctgatctac gactctaaca agcggcctag cggcatacca 180
 gatagattct ctggcagcaa gtccggcacc agcgccacac tgggcatcac cgccctgcag 240
 acaggcgacg aggcagatta ctattgcgga acctgggact ctagcctgtc cgccggcgctg 300
 tttggaggag gaaccaagct gaca 324

SEQ ID NO: 268 moltype = AA length = 13
 FEATURE Location/Qualifiers
 source 1..13
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 268
 SGNSSNIGNN YVS 13

-continued

SEQUENCE: 277
cagtcgcccc tgaccagccc agcctccgtg tctggcagcc cgggccagtc tatcacaatc 60
agctgtaccg gcacaagctc cgacgtgggc ggctacaact acgtgagctg gtaccagcag 120
caccagggca aggcacctaa gctgatgac tatgaggtgt ccaacaggcc aagcggcgtg 180
tccaatagat tctcggctc taagggaat accgcctccc tgacaatctc tggcctgagg 240
ggagaggacg aggcagatta ctattgctct agctacacct cctctagcac actggtggtg 300
tttgccggcg gcaccaagct gaca 324

SEQ ID NO: 278 moltype = AA length = 14
FEATURE Location/Qualifiers
source 1..14
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 278
TGTSSDVGGY NYVS 14

SEQ ID NO: 279 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 279
EVSNRPS 7

SEQ ID NO: 280 moltype = AA length = 11
FEATURE Location/Qualifiers
source 1..11
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 280
SSYTSSSTLV V 11

SEQ ID NO: 281 moltype = AA length = 124
FEATURE Location/Qualifiers
source 1..124
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 281
QVHLQQWAG LVKPSSETLSL TCAVQGGPFS GYYWSWIRQP PGKGLEWIGE INHSGNTHYN 60
PSLKSRVTIS VDTSGNYFSL KLTSVTAADA AVYFCARGQQ LLRNYYYYSG MDVWQGTTV 120
TVSS 124

SEQ ID NO: 282 moltype = DNA length = 372
FEATURE Location/Qualifiers
source 1..372
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 282
cagggtgcacc tgcagcagtg gggagcaggc ctggtgaagc catccgagac actgtctctg 60
acatgtgcag tgcagggagg acccttctct ggctactatt ggagctggat caggcagcca 120
cctggcaagg gcctggagtg gatcggcgag atcaaccaca gcggcaatac ccactacaac 180
ccctctctga agagccgggt gaccatcagc gtggacacat ccggcaatta cttctccctg 240
aagctgacct ctgtgacagc cgccgatgcc gccgtgtatt tttgcgcccg gggccagcag 300
ctgctgagaa actactatta ctattccggc atggacgtgt ggggacaggg aaccacagtg 360
acagtgagct cc 372

SEQ ID NO: 283 moltype = AA length = 7
FEATURE Location/Qualifiers
source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 283
GGPFSGY 7

SEQ ID NO: 284 moltype = AA length = 5
FEATURE Location/Qualifiers
source 1..5
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 284
NHSGN 5

SEQ ID NO: 285 moltype = AA length = 16
FEATURE Location/Qualifiers
source 1..16
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 285
GQQLLRNYYY YSGMDV 16

-continued

SEQ ID NO: 286 moltype = AA length = 108
 FEATURE Location/Qualifiers
 source 1..108
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 286
 QSALTQPASV SGSPGQSITI SCTETSRDVG DYNVSWYQQ HPGPAPKLIM YEVHKRPSGI 60
 SNRFGSGSKG TTASLTISGL QADDEGDYYC SSYTDKNTYV FGSGTQVT 108

SEQ ID NO: 287 moltype = DNA length = 324
 FEATURE Location/Qualifiers
 source 1..324
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 287
 cagtctgccc tgaccagacc agcctctgtg agcggctccc ctggccagtc catcacaaac 60
 tcttgtagcc agacatctcg ggacgtgggc gattacaact atgtgagctg gtaccagcag 120
 caccagagac ctgcacacaa gctgatcatg tatgaggtgc acaagcggcc ctctggcctc 180
 agcaatagat tctctggcag caagtccggc accacagcca gcctgacct ctcgggacct 240
 caggcagacg atgagggcga ctactattgc agctcctaca ccgataagaa cacatacgtg 300
 ttcggcagcg gcaccaggt gaca 324

SEQ ID NO: 288 moltype = AA length = 14
 FEATURE Location/Qualifiers
 source 1..14
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 288
 TETSRDVG DY NYVS 14

SEQ ID NO: 289 moltype = AA length = 7
 FEATURE Location/Qualifiers
 source 1..7
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 289
 EVHKRPS 7

SEQ ID NO: 290 moltype = AA length = 10
 FEATURE Location/Qualifiers
 source 1..10
 mol_type = protein
 organism = Macaca mulatta

SEQUENCE: 290
 SSYTDKNTYV 10

SEQ ID NO: 291 moltype = AA length = 447
 FEATURE Location/Qualifiers
 source 1..447
 mol_type = protein
 organism = Synthetic construct

SEQUENCE: 291
 EAALLVCQYT IQSLIHLTGE DPGFFNVEIP EFPFYPTCNV CTADVNVNIN FVVGKKHQL 60
 DLDGGLTPH TKAVYQPRGA FGGSENATNL FLLELLGAGE LALTMRSKKL PINVTTGEEQ 120
 QVSLESVDVY QDVFVTMWC HHAEMQNPVY LIPETVPYID WDNCSNNTIT AVVRAQGLDV 180
 TLPLSLPTSA QDSNFSVKTE MLGNEIDIEC IMEDGEISQV LPGDNKFNIT CSGYESHVPS 240
 GGLTSTSTSPV ATPIPGTGYA YSLRLTPRPV SRFLGNNSIL YVFYSGNGPK ASGGRYCIQS 300
 NIVFSDEIPA SQDMPTNTTD ITYVGDNATY SVPMVTSEDA NSPNVTVTAF WAWPNNTETD 360
 FKCKWTLTSG TPGSCENISG AFASNRTFDI TVSGLTAPK TLIITRTATN ATTTTHKVIF 420
 SKAPGSGLND IFEAQKIEWH EHHHHHH 447

SEQ ID NO: 292 moltype = DNA length = 1341
 FEATURE Location/Qualifiers
 source 1..1341
 mol_type = other DNA
 organism = Synthetic construct

SEQUENCE: 292
 gaagctgccc tgctcgtgtg ccagtagacc atccagagcc tgatccacct gaccggcgag 60
 gaccggcgct tcttcaacgt ggaatcccc gagttccctt tctaccctac ctgcaacgtg 120
 tgcaccggcg acgtgaacgt gaccatcaac ttcgacgtgg gcggcaagaa gcaccagctg 180
 gacctggatt tcggccagct gaccctcac accaaggccg tgtatcagcc cagaggcgcc 240
 tttggcgcca gccagaacgc caccaatctg tttctgctgg aactcctagg cgccggcgag 300
 ctggccctga ccatgagaag caagaaactg cccatcaatg tgaccacagg cgaggaaacag 360
 caggtgtccc tggaaagcgt ggacgtgtac tttcaagacg tggtcggcac catgtgggtg 420
 caccacggcg agatgcagaa ccccggtgac ctgatccccg agacagtgcc ctacatcgat 480
 tgggacaact gcaacagcac caacatcacc gccgtcgtgc gggcccaggg actggatgtg 540
 acactgcctc tgagcctgcc taccagcgcc caggacagca acttcagcgt gaaaaccgag 600
 atgctgggca acgagatcga catcgagtgc atcatggaag atggcgagat cagccaggtg 660

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ctgcccggcg	acaacaagtt	caacatcaca	tgcagcggct	acgagagcca	cgtgccatct	720
ggcggcatcc	tgaccagcac	aagcccagtg	gccacaccca	tccttgccac	aggctacggc	780
tacagcctga	gactgacccc	cagaccctgt	tccagattcc	tgggcaacaa	cagcatcctg	840
tacgtgttct	acagcggcaa	cggcccacag	gcctctggcg	gccggctactg	tatccagagc	900
aacatcgtgt	tcagcgacga	gatcccggcc	agccaggaca	tgcccaccaa	taccacggac	960
atcacgtacg	tggggacaaa	tgccacctac	agcgtgccaa	tggtcacctc	caggagacgc	1020
aacagcccca	acgtgaccgt	gacagccttt	tggggcctggc	ctaacaacac	cgagacagac	1080
ttcaagtgcg	agtggaacct	gacctccggc	acccctagcg	gctgcgagaa	tatcagcggg	1140
gccttcggca	gcaaccggac	cttcgatata	accgtgtctg	gcctggggcac	cgcccccaag	1200
acccgatgca	tcaccagaac	cgcacaaaat	gccaccacca	caaccacaaa	agtgatcttc	1260
agcaaggccc	ccggtctctg	ctggaacgac	atcttttgagg	cccagaagat	tgagtggcat	1320
gaacatcacc	accaccacca	t				1341

SEQ ID NO: 293 moltype = DNA length = 5118
 FEATURE Location/Qualifiers
 source 1..5118
 mol_type = genomic DNA
 organism = Macaca mulatta

SEQUENCE: 293

tgcgcgcttt	cggtgatgac	ggtgaaaacc	tctgacacat	gcagctcccc	gagacgggtca	60
cagcttgtct	gtaagcggat	gccggggagca	gacaagcccc	tcagggcgcg	tcagcgggtg	120
ttggcgggtg	tcggggctgg	cttaactatg	cggcatcaga	gcagattgta	ctgagagtgc	180
accatattcg	gtgtgaaata	ccgcacagat	gcgtaaggag	aaaataccgc	atcagattgg	240
ctattggcca	ttgcatacgt	tgtatccata	tcataatatg	tacatttata	ttggctcatg	300
ttcaacatta	ccgcactgtt	gacattgatt	attgactagt	tattaatagt	aatcaattac	360
ggggtcatta	gtttatagcc	catatatgga	gttccgcgtt	acataaactta	cggtaaatgg	420
ccgcctctgg	tgaccgcccc	acgacccccg	cccatcgacg	tcaataatga	cgtatgttcc	480
catagtaacg	ccaataggga	ctttccattg	acgtcaatgg	gtggagtatt	tacggtaaac	540
tgccactctg	cgagtacatc	aagtgtatca	tatgccaaat	acgcccccta	ttgacgtcaa	600
tgacggtaaa	tggcccgccct	ggcattatgc	ccagtcacatg	accttatggg	actttccctac	660
ttggcagtac	atctacgtat	tagtcatcgc	tattaccatg	gtgatgcggt	tttggcagta	720
catcaatggg	cgtggatagc	ggttttgactc	acgggggattt	ccaagtctcc	acccatttga	780
cgtcaatggg	agtttggttt	ggcaccaaaa	tcaacggggac	tttccaaaaat	gtcgtaaaca	840
ctccgcccga	ttgacgcaaa	tgggcggtag	gcgtgtacgg	tgggaggtct	atataagcag	900
agctcggtta	gtgaaccgtc	agatcgccctg	gagacgccc	ccacgctgtt	ttgacctcca	960
tagaagacac	cgggaccgat	ccagccctcca	tcggctcgca	tctctccttc	acgcgcccgc	1020
cgcctacact	gaggccgcca	tcacgcgcgg	ttgagtcgcg	ttctgcgcgc	tcccgcctgt	1080
ggtgcctcct	gaactgcgtc	cgcgcctctag	gtaagtttaa	agctcaggtc	gagacggggc	1140
cttttgcggg	cgctcccttg	gagccctacct	agactcagcc	ggctctccac	gctttgcctg	1200
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25

The invention claimed is:

1. An isolated antibody that binds Epstein Barr Virus (EBV) gp350 protein, comprising a variable heavy (VH) domain and a variable light (VL) domain, wherein:

- (a) the VH domain comprises the complementarity determining region 1 (CDR1), CDR2 and CDR3 sequences of SEQ ID NO: 1, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 6;
- (b) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 11, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 3;
- (c) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 21, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 26;
- (d) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 31, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 36;
- (e) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 41, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 46;
- (f) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 161, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 166, wherein the antibody is an engineered antibody;
- (g) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 171, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 176, wherein the antibody is an engineered antibody;
- (h) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 181, the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 186, wherein the antibody is an engineered antibody;
- (i) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 191, and the VL

domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 196, wherein the antibody is an engineered antibody;

- (j) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 201, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 206, wherein the antibody is an engineered antibody;
- (k) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 211, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 216, wherein the antibody is an engineered antibody;
- (l) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 221, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 226, wherein the antibody is an engineered antibody;
- (m) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 231, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 236, wherein the antibody is an engineered antibody;
- (n) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 241, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 246, wherein the antibody is an engineered antibody;
- (o) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 251, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 256, wherein the antibody is an engineered antibody;
- (p) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 261, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 266, wherein the antibody is an engineered antibody;
- (q) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 271, and the VL

245

domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 276, wherein the antibody is an engineered antibody; or

- (r) the VH domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 281, and the VL domain comprises the CDR1, CDR2 and CDR3 sequences of SEQ ID NO: 286, wherein the antibody is an engineered antibody.

2. The antibody of claim 1, which is conjugated to a solid support selected from a support formed partially or entirely of glass, a polysaccharide, a polyacrylamide, a polystyrene, a polyvinyl alcohol, a silicone, an assay plate, and a purification column.

3. A composition comprising the antibody of claim 1 and a carrier.

4. The composition of claim 3, wherein the carrier comprises a pharmaceutically acceptable carrier.

5. An article of manufacture comprising a container and the composition of claim 4 contained within the container.

6. The article of manufacture of claim 5, further comprising a label affixed to the container, or a package insert included with the container, referring to the use of the composition of matter for the therapeutic treatment of or the diagnostic detection of an EBV infection.

7. An expression vector comprising a nucleic acid molecule encoding the antibody of claim 1, operably linked to a promoter.

8. An isolated host cell comprising the expression vector of claim 7.

9. The isolated host cell of claim 8, which is a mammalian cell, a bacterial cell, or a yeast cell.

10. A process for producing an antibody, comprising culturing the host cell of claim 8 under conditions suitable for expression of the antibody and recovering the antibody from the cell culture.

246

11. A method of determining the presence of an EBV gp350 protein in a sample suspected of containing the protein, comprising exposing the sample to the antibody of claim 1 and detecting binding of the antibody to a protein in the sample, wherein binding of the antibody to a protein is indicative of the presence of EBV gp350 protein in the sample.

12. The method of claim 11, wherein the antibody is detectably labeled.

13. The method of claim 12, wherein the label is selected from a radioisotope and a fluorescent label.

14. The method of claim 11, wherein the antibody is conjugated to a solid support selected from a support formed partially or entirely of glass, a polysaccharide, a polyacrylamide, a polystyrene, a polyvinyl alcohol, a silicone, an assay plate, and a purification column.

15. The method of claim 11, wherein the sample comprises a cell suspected of containing an EBV gp350 protein, and the cell is an EBV-infected cell, an epithelial cell, a B lymphocyte, an oropharyngeal cell, a nasopharyngeal cell, or a cancer cell.

16. A method of diagnosing and treating the presence of an EBV infection in a mammal, comprising determining the level of expression of a gene encoding an EBV gp350 protein in a test sample of tissue cells obtained from the mammal and in a control sample of known normal cells of the same tissue origin, wherein a higher level of expression of the EBV gp350 protein in the test sample, as compared to the control sample, is indicative of the presence of an EBV infection in the mammal from which the test sample was obtained, and wherein the mammal determined to have the presence of the EBV infection is administered a therapeutically effective amount of the antibody of claim 1.

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